Differential Equations

Question1

Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x \left((f(\mathrm{t}))^2 + (f'(\mathrm{t}))^2\right)\mathrm{dt}$. Then the mean of $f(\log_{\mathrm{e}}(1)), f(\log_{\mathrm{e}}(2)), \ldots\ldots, f(\log_{\mathrm{e}}(625))$ is equal to _____ .

JEE Main 2026 (Online) 23rd January Morning Shift

Answer: 1565

Solution:

Let $y = f(x)$

$y^2 = 25 + \int_0^x \left(f(t)^2 + f'(t)^2\right)dt$

Differentiating w.r.t. x

$$2y\frac{dy}{dx} = y^2 + \left(\frac{dy}{dx}\right)^2$$
$$\Rightarrow \left(\frac{dy}{dx} - y\right)^2 = 0 \Rightarrow \frac{dy}{y} = dx$$
$$\Rightarrow \ln|y| = x + c$$

Such that

$$f(0)^2 = 25 + \int_0^0 \left(f^2(t) + f'(t)^2\right)dt$$
$$\Rightarrow f(0) = 5$$
$$\Rightarrow \ln 5 = c$$

$$\Rightarrow \ln|y| = x + \ln 5$$
$$\Rightarrow \quad y = 5e^x$$
$$f(\ln k) = 5e^{\ln k} = 5k$$
Mean of $\{5, 10, \ldots 625 \times 5\}$
$$= \frac{(5 + 10 + \ldots 5 \times 625)}{625} = \frac{5(625)(626)}{625 \times 2}$$
$$= 5 \times 313$$
$$= 1565$$

Question2

If the solution curve $y = f(x)$ of the differential equation

$$\left(x^2-4\right)y' - 2xy + 2x\left(4-x^2\right)^2 = 0, x > 2,$$

passes through the point $(3, 15)$, then the local maximum value of f is _____

JEE Main 2026 (Online) 23rd January Evening Shift

Answer: 16

Solution:

$\left(x^2-4\right)y' - 2xy + 2x\left(4-x^2\right)^2 = 0, x > 2$

rearranging terms.

$$\left(x^2-4\right)\frac{dy}{dx} - 2xy = -2x\left(x^2-4\right)^2$$
$$\frac{dy}{dx} - \frac{2x}{x^2-4}y = -2x\left(x^2-4\right)$$

this is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$

$P = -\frac{2x}{x^2-4}, Q = -2x\left(x^2-4\right)$

finding integrating factor (I.F.).

$$\text{I.F.} = e^{\int P dx} = e^{\int -\frac{2x}{x^2-4}dx}$$
$$\text{I.F.} = e^{-\ln\left(x^2-4\right)} = \frac{1}{x^2-4}$$

the solution is $y \cdot (I.F) = . \int Q \cdot (I.F)dx +.$

$$y \cdot \frac{1}{x^2-4} = \int -2x\left(x^2-4\right) \cdot \frac{1}{x^2-4}dx + C$$
$$\frac{y}{x^2-4} = \int -2x dx + C$$
$$\frac{y}{x^2-4} = -x^2 + C \Rightarrow y = \left(x^2-4\right)\left(C-x^2\right)$$

substitute point $(3, 15)$

$15 = \left(3^2-4\right)\left(C-3^2\right)$

$15 = 5(C-9) \Rightarrow 3 = C-9 \Rightarrow C = 12$
$f(x) = \left(x^2-4\right)\left(12-x^2\right)$
$f(x) = -x^4 + 16x^2 - 48$

finding critical points for local maximum.

$f'(x) = -4x^3 + 32x$
$f'(x) = 0 \Rightarrow -4x\left(x^2 - 8\right) = 0$
since $x > 2, x^2 = 8 \Rightarrow x = \sqrt{8} = 2\sqrt{2}$

$f'\left((2\sqrt{2})^-\right)$ =positive, $f'\left((2\sqrt{2})^+\right)$ =negative

so sign of $f'(x)$ changing positive to negative about $x = 2\sqrt{2}$ so $x = 2\sqrt{2}$ is point of local maxima

calculating local maximum value.

$f(2\sqrt{2}) = \left((2\sqrt{2})^2 - 4\right)\left(12 - (2\sqrt{2})^2\right)$
$= (8-4)(12-8)$
$= 4 \times 4 = 16$

the local maximum value is 16 .

Question3

Let f be a twice differentiable function such that

$f(x) = \int_0^x \tan(t-x)dt - \int_0^x f(t)\tan t dt, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Then $f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$ is equal to _______.

JEE Main 2026 (Online) 4th April Evening Shift

Answer: 5

Solution:

Let

$f(x) = \int_0^x \tan(t-x)\,dt - \int_0^x f(t)\tan t\,dt, \qquad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$

We have to find

$f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right).$

Step 1: Simplify the first integral

Consider

$\int_0^x \tan(t-x)\,dt.$

Put

$u = t - x \implies du = dt.$

When $t = 0$, $u = -x$, and when $t = x$, $u = 0$.

So,

$\int_0^x \tan(t - x)\, dt = \int_{-x}^0 \tan u\, du.$

Now,

$\int \tan u\, du = -\ln(\cos u).$

Hence,

$\int_{-x}^0 \tan u\, du = [-\ln(\cos u)]_{-x}^0 = -\ln(\cos 0) + \ln(\cos(-x)).$

Since $\cos 0 = 1$ and $\cos(-x) = \cos x$,

$\int_0^x \tan(t - x)\, dt = \ln(\cos x).$

Therefore the given equation becomes

$f(x) = \ln(\cos x) - \int_0^x f(t) \tan t\, dt.$

So,

$f(x) + \int_0^x f(t) \tan t\, dt = \ln(\cos x).$

Step 2: Differentiate the equation

Differentiate both sides with respect to x.

Using the Fundamental Theorem of Calculus,

$\frac{d}{dx}\left(\int_0^x f(t) \tan t\, dt\right) = f(x) \tan x.$

Also,

$\frac{d}{dx}\left(\ln(\cos x)\right) = -\tan x.$

Thus,

$f'(x) + f(x) \tan x = -\tan x.$

So,

$f'(x) = -(f(x) + 1) \tan x.$

Step 3: Find $f(x)$ explicitly

Now observe that $f(x) = -1$ satisfies the differential equation:

$f'(x) = 0, \qquad -(f(x) + 1) \tan x = 0.$

Also from the original equation, put $x = 0$:

$f(0) = \int_0^0 \tan(t - 0)\, dt - \int_0^0 f(t) \tan t\, dt = 0.$

So,

$f(0) = 0.$

Let us solve the differential equation properly.

From

$f'(x) + (f(x) + 1)\tan x = 0,$

put

$y(x) = f(x) + 1.$

Then

$y'(x) + y(x)\tan x = 0.$

This gives

$\frac{y'}{y} = -\tan x.$

Integrating,

$\ln|y| = \int -\tan x\, dx = \ln(\cos x) + C.$

Hence,

$y = C\cos x.$

Therefore,

$f(x) + 1 = C\cos x.$

Using $f(0) = 0$,

$0 + 1 = C\cos 0 = C.$

So $C = 1$, and hence

$f(x) + 1 = \cos x$

which gives

$f(x) = \cos x - 1.$

Step 4: Find derivatives

Since

$f(x) = \cos x - 1,$

we get

$f'(x) = -\sin x,$

and

$f''(x) = -\cos x.$

Now compute required values:

1. $f''\left(\frac{\pi}{6}\right)$

$f''\left(\frac{\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}.$

2. $f'\left(-\frac{\pi}{6}\right)$

$f'\left(-\frac{\pi}{6}\right) = -\sin\left(-\frac{\pi}{6}\right) = \frac{1}{2}.$

So,

$12f'\left(-\frac{\pi}{6}\right) = 12 \cdot \frac{1}{2} = 6.$

3. $f\left(\frac{\pi}{6}\right)$

$f\left(\frac{\pi}{6}\right) = \cos\left(\frac{\pi}{6}\right) - 1 = \frac{\sqrt{3}}{2} - 1.$

Step 5: Add them

$f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$

$= -\frac{\sqrt{3}}{2} + 6 + \left(\frac{\sqrt{3}}{2} - 1\right)$

$= 6 - 1 = 5.$

Hence, the required value is

$\boxed{5}$.

Question4

Let $y = y(x)$ be the solution of the differential equation $x\sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx, y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y(e^{12})}{e^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \le 6$, is _____ .

JEE Main 2026 (Online) 5th April Morning Shift

Answer: 11

Solution:

We are given the differential equation

$x\sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx$

with initial condition $y(1) = \frac{\pi}{2}$.

Step 1: Express the differential equation in standard form

$x\sin\left(\frac{y}{x}\right)\frac{dy}{dx} = y\sin\left(\frac{y}{x}\right) - x$

$\Rightarrow \frac{dy}{dx} = \frac{y\sin\left(\frac{y}{x}\right) - x}{x\sin\left(\frac{y}{x}\right)}$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{\sin\left(\frac{y}{x}\right)}$

Step 2: Substitution $y = vx$

Let $y = vx$, where v is a function of x.

Then

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

Substitute in the differential equation:

$v + x\frac{dv}{dx} = v - \frac{1}{\sin v}$

Simplify:

$x\frac{dv}{dx} = -\frac{1}{\sin v}$

$\Rightarrow \sin v\, dv = -\frac{dx}{x}$

Step 3: Integrate both sides

$\int \sin v\, dv = -\int \frac{dx}{x}$

$-\cos v = -\ln|x| + C$

$\Rightarrow \cos v = \ln|x| + C$

Now recall $v = \frac{y}{x}$, so:

$$\boxed{\cos\left(\frac{y}{x}\right) = \ln|x| + C}$$

Step 4: Apply the initial condition $y(1) = \frac{\pi}{2}$

$x = 1,\ y = \frac{\pi}{2} \Rightarrow \frac{y}{x} = \frac{\pi}{2}$

$\cos\left(\frac{\pi}{2}\right) = 0 = \ln 1 + C \Rightarrow C = 0$

So the particular solution is:

$$\boxed{\cos\left(\frac{y}{x}\right) = \ln x}$$

Step 5: Find α

We are told that

$\alpha = \cos\left(\frac{y(e^{12})}{e^{12}}\right)$

From the equation above:

$\cos\left(\frac{y}{x}\right) = \ln x$

Thus at $x = e^{12}$:

$\alpha = \ln(e^{12}) = 12$

$\boxed{\alpha = 12}$

Step 6: Equation of circle

Given:

$x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$

Substitute $\alpha = 12$:

$x^2 + y^2 - 2px + 2py + 14 = 0$

Step 7: Write in standard form of circle

We know:

$x^2 + y^2 - 2gx - 2fy + c = 0$

represents a circle with center (g, f) and radius $r = \sqrt{g^2 + f^2 - c}$.

Compare with the given equation:

$x^2 + y^2 - 2px + 2py + 14 = 0$

We can write $+2py = -2fy \Rightarrow f = -p$.

Hence:

$g = p, \quad f = -p, \quad c = 14$

Radius:

$r = \sqrt{g^2 + f^2 - c} = \sqrt{p^2 + (-p)^2 - 14} = \sqrt{2p^2 - 14}$

Step 8: Condition for $r \leq 6$

$\sqrt{2p^2 - 14} \leq 6$

$\Rightarrow 2p^2 - 14 \leq 36$

$\Rightarrow 2p^2 \leq 50$

$\Rightarrow p^2 \leq 25$

$\Rightarrow -5 \leq p \leq 5$

Step 9: Count the integral values

Integral values of p are:

$-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$

Hence, total number of integral values $= 11$.

$\boxed{11}$

Question5

Let $y = y(x)$ be the solution of the differential equation $(\tan x)^{1/2}\,\mathrm{d}y = \left(\sec^3 x - (\tan x)^{3/2} y\right)\mathrm{d}x, 0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If

$y\left(\frac{\pi}{3}\right)=\frac{4}{5}\alpha$, then α^4 equals

_____ .

JEE Main 2026 (Online) 5th April Evening Shift

Answer: 48

Solution:

$\frac{dy}{dx}+y\tan x=\frac{\sec^3 x}{\sqrt{\tan x}}$

$\mathrm{IF}=e^{\int \tan x\,dx}=\sec x$

$y\sec x=\int \frac{\sec^4 x}{\sqrt{\tan x}}\,dx$

$y\sec x=\int \frac{(1+\tan^2 x)}{\sqrt{\tan x}}\sec^2 x\,dx$

$\tan x=t$

$y\sec x=2\sqrt{\tan x}+\frac{2}{5}(\tan x)^{5/2}+c$

$y\left(\frac{\pi}{4}\right)=\frac{6\sqrt{2}}{5}$

$\frac{6\sqrt{2}}{5}\times\sqrt{2}=2+\frac{2}{5}+c\rightarrow\frac{12}{5}=\frac{12}{5}+c$

$\rightarrow c=0$

$y\left(\frac{\pi}{3}\right)=\frac{8}{5}\cdot 3^{1/4}\rightarrow\alpha=2\times 3^{1/4}$

$\therefore \alpha^4=48$

Question6

Let $y=y(x)$ be the solution of the differential equation $\left(x^2-x\sqrt{x^2-1}\right)dy+\left(y\left(x-\sqrt{x^2-1}\right)-x\right)dx=0, x\geq 1$. If $y(1)=1$, then the greatest integer less than $y(\sqrt{5})$ is _____ .

JEE Main 2026 (Online) 6th April Morning Shift

Answer: 3

Solution:

First write the given differential equation in the standard linear form.

$\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x - \sqrt{x^2 - 1}}$

Now simplify the right side by rationalising:

$\frac{1}{x - \sqrt{x^2 - 1}} = \frac{x + \sqrt{x^2 - 1}}{(x - \sqrt{x^2 - 1})(x + \sqrt{x^2 - 1})} = \frac{x + \sqrt{x^2 - 1}}{x^2 - (x^2 - 1)} = x + \sqrt{x^2 - 1}$

So the equation becomes:

$\frac{dy}{dx} + \frac{y}{x} = x + \sqrt{x^2 - 1}$

This is a linear differential equation with integrating factor (I.F.):

I.f $= e^{\ln x} = x$

Multiply the whole equation by x:

$y.x = \int \left(x + \sqrt{x^2 - 1}\right) x \, dx$

Now integrate term by term:

$= \frac{x^2}{2} + \frac{(x^2 - 1)^{3/2}}{3} + C$

Use the initial condition $y(1) = 1$ to find C.

Given $y(1) = 1$

Substitute $x = 1$ in the expression for yx:

$1 = \frac{1}{2} + C \rightarrow C = \frac{1}{2}$

So,

$yx = \frac{x^2}{2} + \frac{(x^2 - 1)^{3/2}}{3} + \frac{1}{2}$

Divide by x:

$y = \frac{x}{2} + \frac{(x^2 - 1)^{3/2}}{3x} + \frac{1}{2x}$

Now put $x = \sqrt{5}$:

$\left[y\left(\sqrt{5}\right)\right] = \left[\frac{\sqrt{5}}{2} + \frac{8}{3} + \frac{1}{2\sqrt{5}}\right] = 3$

Question7

Let $y = y(x)$ be the solution curve of the differential equation $\left(1 + x^2\right)\mathrm{d}y + \left(y - \tan^{-1} x\right)dx = 0, y(0) = 1$. Then the value of $y(1)$ is :

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.

$\frac{4}{\mathrm{e}^{\pi/4}} - \frac{\pi}{2} - 1$

B.

$\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$

C.

$\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$

D.

$\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$

Answer: B

Solution:

Given

$(1+x^2)\,dy + (y - \tan^{-1}x)\,dx = 0, \qquad y(0) = 1,$

divide by dx:

$(1+x^2)\frac{dy}{dx} + y - \tan^{-1}x = 0$

$\frac{dy}{dx} + \frac{1}{1+x^2}y = \frac{\tan^{-1}x}{1+x^2}.$

This is linear. Integrating factor:

$\mu(x) = \exp\left(\int \frac{1}{1+x^2}\,dx\right) = e^{\tan^{-1}x}.$

So

$\frac{d}{dx}\left(ye^{\tan^{-1}x}\right) = e^{\tan^{-1}x} \cdot \frac{\tan^{-1}x}{1+x^2}.$

Let $t = \tan^{-1}x$, then $dt = \frac{dx}{1+x^2}$. Hence

$ye^t = \int te^t\,dt + C = e^t(t-1) + C.$

Therefore

$y = t - 1 + Ce^{-t} = \tan^{-1}x - 1 + Ce^{-\tan^{-1}x}.$

Use $y(0) = 1$ and $\tan^{-1}0 = 0$:

$1 = 0 - 1 + C \implies C = 2.$

So

$y(1) = \frac{\pi}{4} - 1 + 2e^{-\pi/4} = \frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1.$

Correct option: B.

Question8

Let $y = y(x)$ be the solution of the differential equation $\sec x \frac{dy}{dx} - 2y = 2 + 3\sin x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

$-\frac{5}{2}$

B.

$-3\sqrt{2} - 7$

C.

$-\frac{5}{4}$

D.

$-3\sqrt{3} - 7$

Answer: A

Solution:

$\frac{dy}{dx} - 2\cos x \cdot y = \cos x(2 + 3\sin x)$
$I \cdot F = e^{-2\sin x}$

$G \cdot Sy \cdot e^{-2\sin x} = \int e^{-2\sin x} \cos x(2 + 3\sin x)dx$

Put $\begin{matrix} \sin x = t \\ \cos x \cdot dx = dt \end{matrix}$

$y=e^{-2\sin x}\left(\frac{2+3\sin x}{-2}-\frac{3}{4}\right)+c$

When $x=0, c=0$

When $x=\frac{\pi}{6}, y=-\frac{5}{2}$

Question9

Let the solution curve of the differential equation $xdy-ydx=\sqrt{x^2+y^2}dx, x>0, y(1)=0$, be $y=y(x)$. Then $y(3)$ is equal to

JEE Main 2026 (Online) 22nd January Morning Shift

Options:

A.

6

B.

4

C.

1

D.

2

Answer: B

Solution:

$$xdy-ydx=\sqrt{x^2+y^2}dx$$
$$\frac{xdy-ydx}{x^2}=\frac{\sqrt{x^2+y^2}}{x^2}dx$$

$$\int\frac{d(y/x)}{\sqrt{1+(y/x)^2}}=\int\frac{1}{x}dx$$

$$\log\left(\frac{y}{x}+\sqrt{1+\left(\frac{y}{x}\right)^2}\right)=\log x+\log C$$
$$y+\sqrt{x^2+y^2}=x^2$$
$$y-\sqrt{x^2+y^2}=-1$$
$$2y=x^2-1$$
$$y(3)=4$$

Question10

If $y=y(x)$ satisfies the differential equation $16(\sqrt{x+9\sqrt{x}})(4+\sqrt{9+\sqrt{x}})\cos y\,\mathrm{d}y=(1+2\sin y)\mathrm{d}x, x>0$ and $y(256)=\frac{\pi}{2}, y(49)=\alpha$, then $2\sin\alpha$ is equal to :

JEE Main 2026 (Online) 22nd January Evening Shift

Options:

A.

$2\sqrt{2}-1$

B.

$\sqrt{2}-1$

C.

$2(\sqrt{2}-1)$

D.

$3(\sqrt{2}-1)$

Answer: A

Solution:

$$\int\frac{\cos y}{1+2\sin y}dy=\int\frac{1}{16(\sqrt{x+9\sqrt{x}})(4+\sqrt{9+\sqrt{x}})}dx$$
$$\Rightarrow\frac{1}{2}\int\frac{2\cos y}{1+2\sin y}dy=\int\frac{1}{4t}dt$$

Put $4+\sqrt{9+\sqrt{x}}=t$

$\Rightarrow \dfrac{1}{2\sqrt{9+\sqrt{x}}}\cdot\dfrac{1}{2\sqrt{x}}dx=dt \Rightarrow \dfrac{1}{4\sqrt{x+9\sqrt{x}}}dx=dt$

$\Rightarrow \dfrac{1}{2}\log|1+2\sin y|=\dfrac{1}{4}\log|t|+\log c \Rightarrow \sqrt{1+2\sin y}=ct^{1/4} \Rightarrow \dfrac{1}{4}\log|t|+\log c$

$\Rightarrow \sqrt{1+2\sin y}=ct^{1/4} \Rightarrow \sqrt{1+2\sin y}=c(4+\sqrt{9+\sqrt{x}})^{1/4}$

$\left(256,\dfrac{\pi}{2}\right) \Rightarrow \sqrt{3}=c(\sqrt{3}) \Rightarrow c=1$

$(49,\alpha) \Rightarrow \sqrt{1+2\sin\alpha}=(1)8^{1/4}$

$\Rightarrow 1+2\sin\alpha=2\sqrt{2} \Rightarrow 2\sin\alpha=2\sqrt{2}-1$

Question11

Let $y=y(x)$ be the solution of the differential equation $x^4\,\mathrm{d}y+\left(4x^3y+2\sin x\right)\mathrm{d}x=0, x>0, y\left(\frac{\pi}{2}\right)=0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

92

B.

72

C.

64

D.

81

Answer: D

Solution:

$x^4 dy + \left(4x^3 y + 2\sin x\right)dx = 0$

$\dfrac{dy}{dx} + \dfrac{4x^3}{x^4}y = \dfrac{-2\sin x}{x^4}$

$\dfrac{dy}{dx} + \dfrac{4}{x}y = \dfrac{-2\sin x}{x^4}$

$\text{IF } = e^{\int \frac{4}{x}dx = e^{\ln x^4} = x^4}$

$x^4 \dfrac{dy}{dx} + 4x^3 y = -2\sin x$

$\dfrac{d}{dx}\left(x^4 y\right) = -2\sin x$

$\Rightarrow x^4 y = 2\cos x + c$

$y\left(\dfrac{\pi}{2}\right) = 0$

$\Rightarrow c = 0$

$\therefore x^4 y = \cos x$

$\Rightarrow y = \dfrac{2\cos x}{x^4}$

$y\left(\dfrac{\pi}{3}\right) = \dfrac{2\cos\left(\frac{\pi}{3}\right)}{\left(\frac{\pi}{3}\right)^4} = \dfrac{1}{\left(\frac{\pi}{3}\right)^4} = \dfrac{81}{\pi^4}$

$\pi^4 y\left(\dfrac{\pi}{3}\right) = 81$

Question12

Let $y = y(x)$ be the solution of the differential equation

$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, x \neq 0.$

If $y(2) = 0$, then $\tan(y(1))$ is equal to

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

$-\frac{7}{4}$

B.

$-\frac{3}{4}$

C.

$\frac{3}{4}$

D.

$\frac{7}{4}$

Answer: D

Solution:

Given equation is :

$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, \quad x \neq 0$

Divide by $x\cos^2 y$

$\sec^2 y\frac{dy}{dx} - \frac{\sin 2y}{x\cos^2 y} = x^2\left(2 - x^3\right)$

$\sec^2 y\frac{dy}{dx} - \frac{2\tan y}{x} = 2x^2 - x^5$

Substitute $\tan y = t$ and differentiate w.r.t x

$\sec^2 y\frac{dy}{dx} = \frac{dt}{dx}$

Equation (1) becomes linear differential equation.

$\frac{dt}{dx} - \frac{2}{x}t = 2x^2 - x^5$

Integrating factor $IF = e^{\int\left(-\frac{2}{x}\right)dx} = e^{-2\ln x} = e^{\ln\left(\frac{1}{x^2}\right)} = \frac{1}{x^2}$

Solution of linear differential equation :

$t \cdot IF = \int(IF)\left(2x^2 - x^5\right)dx$

$\Rightarrow t \cdot \frac{1}{x^2} = \int\frac{1}{x^2}\left(2x^2 - x^5\right)dx$

$\Rightarrow \frac{t}{x^2} = \int\left(2 - x^3\right)dx = 2x - \frac{x^4}{4} + C$

put value of $t = \tan y$

$\frac{\tan y}{x^2} = 2x - \frac{x^4}{4} + C$

using condition $y(2) = 0$ to find C.

$\frac{\tan(0)}{2^2} = 2 \times 2 - \frac{2^4}{4} + C \Longrightarrow C = 0$

So, solution of differential equation is

$\tan y = 2x^3 - \frac{x^6}{4}$

to calculate $\tan(y(1))$, put $x = 1$ in the solution.

$\tan(y(1)) = 2(1)^3 - \frac{(1)^6}{4}$

$= 2 - \frac{1}{4} = \frac{7}{4}$

Question13

Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cot x$, $x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then

$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

-3π

B.

3π

C.

$-\pi$

D.

π

Answer: C

Solution:

The given equation is:

$x\frac{dy}{dx} - y = x^2 \cot x$

To get it into the standard form $\frac{dy}{dx} + P(x)y = Q(x)$, we divide the entire equation by x :

$\frac{dy}{dx} - \frac{1}{x}y = x \cot x$

Here, $P(x) = -\frac{1}{x}$ and $Q(x) = x \cot x$.

Finding the Integrating Factor (I.F.)

The integrating factor is calculated as:

$\text{ I.F. } = e^{\int P(x)dx} = e^{\int -\frac{1}{x}dx} = e^{-\ln x} = e^{\ln\left(x^{-1}\right)} = \frac{1}{x}$

The solution to the differential equation is:

$$y \cdot (I.F.) = \int Q(x) \cdot (I.F.) dx$$
$$\frac{y}{x} = \int (x \cot x) \cdot \frac{1}{x} dx$$
$$\frac{y}{x} = \int \cot x dx$$
$$\frac{y}{x} = \ln(\sin x) + C$$
$$y = x \ln(\sin x) + Cx$$

Applying the Initial Condition

We are given $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$. Plugging these values in:

$\frac{\pi}{2} = \frac{\pi}{2} \ln\left(\sin\frac{\pi}{2}\right) + C\left(\frac{\pi}{2}\right)$

So, the particular solution is $y = x\ln(\sin x) + x$.

We need to find the value of $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$.

- Calculate $y\left(\frac{\pi}{6}\right)$:

$y\left(\frac{\pi}{6}\right) = \frac{\pi}{6}\ln\left(\sin\frac{\pi}{6}\right) + \frac{\pi}{6} = \frac{\pi}{6}\ln\left(\frac{1}{2}\right) + \frac{\pi}{6} = \frac{\pi}{6}(1 - \ln 2)$

- Calculate $y\left(\frac{\pi}{4}\right)$:

$y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}\ln\left(\sin\frac{\pi}{4}\right) + \frac{\pi}{4} = \frac{\pi}{4}\ln\left(\frac{1}{\sqrt{2}}\right) + \frac{\pi}{4} = \frac{\pi}{4}\ln\left(2^{-1/2}\right) + \frac{\pi}{4} = \frac{\pi}{4}\left(1 - \frac{1}{2}\ln 2\right)$

- Calculate $y\left(\frac{\pi}{4}\right)$:

$y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}\ln\left(\sin\frac{\pi}{4}\right) + \frac{\pi}{4} = \frac{\pi}{4}\ln\left(\frac{1}{\sqrt{2}}\right) + \frac{\pi}{4} = \frac{\pi}{4}\ln\left(2^{-1/2}\right) + \frac{\pi}{4} = \frac{\pi}{4}\left(1 - \frac{1}{2}\ln 2\right)$

Now, substitute these into the expression:

$$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right) = 6\left[\frac{\pi}{6}(1 - \ln 2)\right] - 8\left[\frac{\pi}{4}\left(1 - \frac{1}{2}\ln 2\right)\right] = \pi(1 - \ln 2) - 2\pi\left(1 - \frac{1}{2}\ln 2\right)$$
$$= \pi - \pi\ln 2 - 2\pi + \pi\ln 2 = -\pi$$

Question14

Let $y = y(x)$ be the solution curve of the differential equation

$(1 + \sin x)\dfrac{dy}{dx} + (y + 1)\cos x = 0,\ y(0) = 0$. If the curve $y = y(x)$ passes through the point $\left(\alpha, \dfrac{-1}{2}\right)$,

then a value of α is:

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

$\frac{\pi}{6}$

B.

$\frac{\pi}{4}$

C.

$\frac{\pi}{3}$

D.

$\frac{\pi}{2}$

Answer: D

Solution:

$\int \frac{dy}{y+1} = -\int \frac{\cos x}{1+\sin x}\, dx$

On the left side, $\int \frac{dy}{y+1} = \ln(y+1)$.

On the right side, take $1+\sin x$ as substitution: let $u = 1+\sin x$, then $du = \cos x\, dx$. So, $-\int \frac{\cos x}{1+\sin x}\, dx = -\int \frac{du}{u} = -\ln(1+\sin x)$.

So we get $\Rightarrow \ln(y+1) = -\ln(1+\sin x) + C$

Now use the initial condition $y(0) = 0$. Put $x = 0$ and $y = 0$ in the equation:

$\therefore y(0) = 0 \rightarrow \ln(0+1) = -\ln(1+\sin 0) + C$

That is $0 = -\ln(1) + C \Rightarrow C = 0$.

So $\Rightarrow \ln(y+1) = -\ln(1+\sin x)$

Taking antilog on both sides, $\Rightarrow y+1 = \frac{1}{1+\sin x}$

Given that the curve passes through $\left(\alpha, -\frac{1}{2}\right)$, substitute $y = -\frac{1}{2}$ and $x = \alpha$:

$-\frac{1}{2} + 1 = \frac{1}{1+\sin\alpha}$

So $\frac{1}{2} = \frac{1}{1+\sin\alpha} \Rightarrow 1+\sin\alpha = 2 \Rightarrow \sin\alpha = 1$

Now $\sin\alpha = 1$ gives $\Rightarrow \alpha = \frac{\pi}{2}$ (as per options)

Question15

If the curve $y = f(x)$ passes through the point $(1, e)$ and satisfies the differential equation $dy = y(2 + log_e x)dx$, $x > 0$, then $f(e)$ is equal to :

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

e^e

B.

e^{e^2}

C.

e^{2e}

D.

e^{2e}

Answer: C

Solution:

$\frac{dy}{dx} = y(2 + \ln x)$

To solve this, first separate the variables by dividing both sides by y:

$\int \frac{dy}{y} = \int (2 + \ln x)\, dx$

Now integrate both sides. The left side gives $\ln y$. For the right side, integrate term by term:

$\ln y = 2x + x \ln x - x + C$

Combine like terms $2x - x = x$:

$\ln y = x + x \ln x + C$

Since it passes through $(1, e)$, put $x = 1$ and $y = e$ in the equation. Then $\ln y = \ln e = 1$ and $\ln 1 = 0$:

$1 = 1 + 0 + C \to C = 0$

So the required relation becomes:

$\ln y = x + x \ln x$

Taking antilog (exponential) on both sides:

$f(x) = y = e^{x + x \ln x}$

Now substitute $x = e$. Also, $\ln e = 1$:

$\Rightarrow f(e) = e^{e+e} = e^{2e}$

Question16

Let $x = x(y)$ be the solution of the differential equation $2y^2 \frac{dx}{dy} - 2xy + x^2 = 0, y > 1, x(e) = e$.

Then $x(e^2)$ is equal to:

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

$\frac{3}{2}e^2$

B.

$\frac{2}{3}e^2$

C.

e^2

D.

$2e^2$

Answer: B

Solution:

$2y(ydx - xdy) + x^2dy = 0$

$\Rightarrow -2yx^2\, d\left(\frac{y}{x}\right) + x^2dy = 0$

$\Rightarrow -2y\, d\left(\frac{y}{x}\right) + dy = 0$

$\Rightarrow -2d\left(\frac{y}{x}\right) + \frac{1}{y}dy = 0$

$\Rightarrow -\frac{2y}{x} + \log_e y = C$

Given $x(e) = e$

$\Rightarrow C = -1$

$\therefore -\frac{2y}{x} + \log_e y = -1$

$\Rightarrow \frac{2y}{x} - \log_e y = 1$

Put $y = e^2$, we get

$\Rightarrow \frac{2e^2}{x} - 2 = 1 \to \frac{2e^2}{x} = 3 \to x = \frac{2e^2}{3}$

$\Rightarrow x(e^2) = \frac{2e^2}{3}$

Question17

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \left(1 + x + x^2\right)\left(1 - y + y^2\right), y(0) = \frac{1}{2}$. Then $(2y(1) - 1)$ is equal to

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

$\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$

B.

$\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

C.

$\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$

D.

$\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

Answer: C

Solution:

Given

$\frac{dy}{dx} = (1 + x + x^2)(1 - y + y^2), \qquad y(0) = \frac{1}{2}$

We will solve it by the method of separation of variables.

Since

$\frac{dy}{1-y+y^2} = (1 + x + x^2)\,dx$

Now simplify the denominator on the left:

$$1-y+y^2=y^2-y+1=\left(y-\tfrac{1}{2}\right)^2+\tfrac{3}{4}$$

So,

$$\int \frac{dy}{\left(y-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}=\int(1+x+x^2)\,dx$$

Using the standard result

$$\int \frac{du}{u^2+a^2}=\frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right)$$

we get

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=x+\frac{x^2}{2}+\frac{x^3}{3}+C$$

Now use the initial condition $y(0)=\frac{1}{2}$.

Then

$$2y-1=0$$

so

$$\frac{2}{\sqrt{3}}\tan^{-1}(0)=C \Rightarrow C=0$$

Hence,

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=x+\frac{x^2}{2}+\frac{x^3}{3}$$

At $x=1$,

$$x+\frac{x^2}{2}+\frac{x^3}{3}=1+\frac{1}{2}+\frac{1}{3}=\frac{11}{6}$$

Therefore,

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y(1)-1}{\sqrt{3}}\right)=\frac{11}{6}$$

So

$$\tan^{-1}\left(\frac{2y(1)-1}{\sqrt{3}}\right)=\frac{11\sqrt{3}}{12}$$

Taking tangent on both sides,

$$\frac{2y(1)-1}{\sqrt{3}}=\tan\left(\frac{11\sqrt{3}}{12}\right)$$

Hence,

$$2y(1)-1=\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$$

So the correct answer is

$$\boxed{\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)}$$

Therefore, the correct option is $\boxed{\text{C}}$.

Question18

Let $y = y(x)$ be the solution of the differential equation :

$$\frac{dy}{dx} + \left(\frac{6x^2+(3x^2+2x^3+4)e^{-2x}}{(x^3+2)(2+e^{-2x})}\right)y = 2 + e^{-2x}$$

$x \in (-1, 2)$, satisfying $y(0) = \frac{3}{2}$. If $y(1) = \alpha\left(2 + e^{-2}\right)$, then α is equal to :

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$\frac{13}{8}$

B.

$\frac{6}{13}$

C.

$\frac{12}{13}$

D.

$\frac{13}{12}$

Answer: D

Solution:

Given differential equation is

$$\frac{dy}{dx} + \left(\frac{6x^2+(3x^2+2x^3+4)e^{-2x}}{(x^3+2)(2+e^{-2x})}\right)y = 2 + e^{-2x}$$

with initial condition

$y(0) = \frac{3}{2}$.

We have to find $y(1) = \alpha(2 + e^{-2})$.

Step 1: Compare with linear differential equation

This is of the form

$\frac{dy}{dx} + P(x)y = Q(x)$,

where

$P(x) = \frac{6x^2+(3x^2+2x^3+4)e^{-2x}}{(x^3+2)(2+e^{-2x})}, \qquad Q(x) = 2 + e^{-2x}.$

To solve, we find the integrating factor:

I.F. $= e^{\int P(x)\,dx}$.

Step 2: Simplify $P(x)$

Notice that

$\frac{d}{dx}(x^3+2) = 3x^2,$

and

$\frac{d}{dx}(2+e^{-2x}) = -2e^{-2x}.$

Now consider

$\frac{d}{dx}\left[(x^3+2)(2+e^{-2x})\right].$

Using product rule,

$= (3x^2)(2+e^{-2x}) + (x^3+2)(-2e^{-2x})$

$= 6x^2 + 3x^2e^{-2x} - 2x^3e^{-2x} - 4e^{-2x}.$

This does not match directly. So let us try another observation.

Write numerator as

$6x^2 + \left(3x^2+2x^3+4\right)e^{-2x}.$

Now check

$\frac{d}{dx}\left[(x^3+2)(2+e^{-2x})\right] = 6x^2 + \left(3x^2-2x^3-4\right)e^{-2x}.$

Hence if we take

$\frac{d}{dx}\left[(x^3+2)(2+e^{-2x})^{-1}\right]$

it becomes complicated. So instead, let us look for a useful factorization.

Observe that

$6x^2 + \left(3x^2+2x^3+4\right)e^{-2x} = 3x^2(2+e^{-2x}) + (x^3+2)\cdot 2e^{-2x}.$

Therefore

$P(x) = \frac{3x^2(2+e^{-2x})+(x^3+2)\cdot 2e^{-2x}}{(x^3+2)(2+e^{-2x})}$

$= \frac{3x^2}{x^3+2} + \frac{2e^{-2x}}{2+e^{-2x}}.$

So,

$P(x) = \frac{3x^2}{x^3+2} + \frac{2e^{-2x}}{2+e^{-2x}}.$

Step 3: Find integrating factor

Now

$\int P(x)\,dx = \int \frac{3x^2}{x^3+2}\,dx + \int \frac{2e^{-2x}}{2+e^{-2x}}\,dx.$

For the first integral:

Let $u = x^3 + 2$, then $du = 3x^2 dx$.

So,

$\int \frac{3x^2}{x^3+2}\, dx = \ln(x^3 + 2).$

For the second integral, note that

$\frac{d}{dx}(2 + e^{-2x}) = -2e^{-2x}.$

Hence

$\int \frac{2e^{-2x}}{2+e^{-2x}}\, dx = -\ln(2 + e^{-2x}).$

Therefore,

$\int P(x)\, dx = \ln(x^3 + 2) - \ln(2 + e^{-2x}) = \ln\left(\frac{x^3+2}{2+e^{-2x}}\right).$

So the integrating factor is

$\text{I.F.} = e^{\int P(x)\, dx} = \frac{x^3+2}{2+e^{-2x}}.$

Step 4: Multiply the equation by the integrating factor

Multiplying throughout by

$\frac{x^3+2}{2+e^{-2x}},$

we get

$\frac{x^3+2}{2+e^{-2x}} \frac{dy}{dx} + \frac{x^3+2}{2+e^{-2x}} P(x)y = (x^3 + 2).$

So the left side becomes

$\frac{d}{dx}\left[y \cdot \frac{x^3+2}{2+e^{-2x}}\right].$

Hence

$\frac{d}{dx}\left[y \cdot \frac{x^3+2}{2+e^{-2x}}\right] = x^3 + 2.$

Integrating both sides,

$y \cdot \frac{x^3+2}{2+e^{-2x}} = \int (x^3 + 2)\, dx = \frac{x^4}{4} + 2x + C.$

Thus,

$y = \frac{2+e^{-2x}}{x^3+2}\left(\frac{x^4}{4} + 2x + C\right).$

Step 5: Use initial condition

Given

$y(0) = \frac{3}{2}.$

Substitute $x = 0$:

$\frac{3}{2} = \frac{2+e^0}{0^3+2}(0 + 0 + C) = \frac{3}{2}C.$

So,

$C = 1.$

Hence solution is

$y = \frac{2+e^{-2x}}{x^3+2}\left(\frac{x^4}{4} + 2x + 1\right).$

Step 6: Find $y(1)$

Substitute $x = 1$:

$y(1) = \frac{2+e^{-2}}{1^3+2}\left(\frac{1}{4} + 2 + 1\right).$

$y(1) = \frac{2+e^{-2}}{3}\left(\frac{1}{4} + 3\right) = \frac{2+e^{-2}}{3} \cdot \frac{13}{4}$

$y(1) = \frac{13}{12}(2 + e^{-2}).$

Given that

$y(1) = \alpha(2 + e^{-2}),$

so

$\alpha = \frac{13}{12}.$

Final Answer

$\boxed{\frac{13}{12}}$

So, the correct option is **D**.

Question19

Let $f : [1, \infty) \to \mathbf{R}$ be a differentiable function defined as $f(x) = \int_1^x f(\mathrm{t})\mathrm{dt} + (1 - x)\left(\log_{\mathrm{e}} x - 1\right) + \mathrm{e}$.

Then the value of $f(f(1))$ is :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

$(1 + \mathrm{e}^{\mathrm{e}})$

B.

$(1 + \mathrm{e})$

C.

$(1 + \mathrm{e} + \mathrm{e}^{\mathrm{e}})$

D.

$1 + 2e$

Answer: A

Solution:

Given

$f(x) = \int_1^x f(t)\,dt + (1-x)(\ln x - 1) + e, \qquad x \in [1, \infty)$

We have to find $f(f(1))$.

First, put $x = 1$ in the given equation:

$f(1) = \int_1^1 f(t)\,dt + (1-1)(\ln 1 - 1) + e$

Now,

$\int_1^1 f(t)\,dt = 0, \qquad (1-1)(\ln 1 - 1) = 0$

So,

$f(1) = e$

Hence,

$f(f(1)) = f(e)$

Now we find $f(x)$.

Differentiate both sides of

$f(x) = \int_1^x f(t)\,dt + (1-x)(\ln x - 1) + e$

Using the Fundamental Theorem of Calculus,

$\frac{d}{dx}\left(\int_1^x f(t)\,dt\right) = f(x)$

So,

$f'(x) = f(x) + \frac{d}{dx}\left[(1-x)(\ln x - 1)\right]$

Now differentiate the second term carefully:

$\frac{d}{dx}\left[(1-x)(\ln x - 1)\right] = -(\ln x - 1) + (1-x) \cdot \frac{1}{x}$

Simplifying,

$= -\ln x + 1 + \frac{1-x}{x}$

$= -\ln x + 1 + \frac{1}{x} - 1$

$= \frac{1}{x} - \ln x$

Therefore,

$f'(x) = f(x) + \frac{1}{x} - \ln x$

So,

$f'(x) - f(x) = \frac{1}{x} - \ln x$

Now let us check whether

$f(x) = \ln x + 1$

satisfies this equation.

Then

$f'(x) = \frac{1}{x}$

So,

$f'(x) - f(x) = \frac{1}{x} - (\ln x + 1)$

This does not match directly. So instead, let us use the original equation to test $f(x) = \ln x + 1$.

Right-hand side:

$\int_1^x (\ln t + 1)\, dt + (1-x)(\ln x - 1) + e$

Now,

$\int (\ln t + 1)\, dt = t \ln t$

Therefore,

$\int_1^x (\ln t + 1)\, dt = x \ln x - 1 \cdot \ln 1 = x \ln x$

since $\ln 1 = 0$.

Thus RHS becomes

$x \ln x + (1-x)(\ln x - 1) + e$

Expand:

$= x \ln x + (1-x) \ln x - (1-x) + e$

$= \ln x - 1 + x + e$

This is not equal to $\ln x + 1$, so this trial is not correct.

Now solve the differential equation properly:

$f'(x) - f(x) = \frac{1}{x} - \ln x$

This is a linear differential equation.

Its integrating factor is

$I.F. = e^{-x}$

Multiplying both sides by e^{-x},

$e^{-x} f'(x) - e^{-x} f(x) = e^{-x} \left(\frac{1}{x} - \ln x\right)$

So,

$\frac{d}{dx}(f(x) e^{-x}) = e^{-x} \left(\frac{1}{x} - \ln x\right)$

This looks difficult, so let us instead use the original functional equation more intelligently.

Write

$g(x) = f(x) - (\ln x + 1)$

Then

$f(x) = g(x) + \ln x + 1$

Substitute into the original equation:

$g(x) + \ln x + 1 = \int_1^x \big(g(t) + \ln t + 1\big)\, dt + (1 - x)(\ln x - 1) + e$

Now,

$\int_1^x (\ln t + 1)\, dt = x \ln x$

Hence,

$g(x) + \ln x + 1 = \int_1^x g(t)\, dt + x \ln x + (1 - x)(\ln x - 1) + e$

Simplify the terms:

$x \ln x + (1 - x)(\ln x - 1) = x \ln x + (1 - x) \ln x - (1 - x)$

$= \ln x - 1 + x$

So,

$g(x) + \ln x + 1 = \int_1^x g(t)\, dt + \ln x - 1 + x + e$

Thus,

$g(x) + 1 = \int_1^x g(t)\, dt - 1 + x + e$

This is not simplifying nicely. Let us now directly guess from options.

Since

$f(1) = e$

we need $f(e)$.

Use the original equation at $x = e$:

$f(e) = \int_1^e f(t)\, dt + (1 - e)(\ln e - 1) + e$

But

$\ln e = 1$

so

$(1 - e)(\ln e - 1) = 0$

Hence

$f(e) = \int_1^e f(t)\, dt + e$

Now from the given equation,

$f'(x) = f(x) + \frac{1}{x} - \ln x$

Try

$f(x) = \ln x + 1 + Ce^x$

Then

$f'(x) = \frac{1}{x} + Ce^x$

So,

$f'(x) - f(x) = \frac{1}{x} + Ce^x - (\ln x + 1 + Ce^x) = \frac{1}{x} - \ln x - 1$

We are short by -1. So modify the trial.

Try

$f(x) = \ln x + A + Ce^x$

Then

$f'(x) = \frac{1}{x} + Ce^x$

Hence

$f'(x) - f(x) = \frac{1}{x} + Ce^x - (\ln x + A + Ce^x) = \frac{1}{x} - \ln x - A$

Comparing with

$\frac{1}{x} - \ln x$

we get

$A = 0$

So,

$f(x) = \ln x + Ce^x$

Now use $f(1) = e$:

$f(1) = \ln 1 + Ce = e$

$0 + Ce = e$

So,

$C = 1$

Therefore,

$f(x) = \ln x + e^x$

Now,

$f(e) = \ln e + e^e = 1 + e^e$

Hence,

$f(f(1)) = 1 + e^e$

So the correct answer is:

$\boxed{1 + e^e}$

Therefore, Option A is correct.

Question20

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in \mathbf{R}$ and $f(0) \neq 0$. Let $g : [1, \infty) \rightarrow \mathbf{R}$ be a differentiable function such that

$$x^2 g(x) = \int_1^x \left(\mathrm{t}^2 f(\mathrm{t}) - \mathrm{tg}(\mathrm{t})\right) dt$$

Then $g(2)$ is equal to :

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

$\frac{13}{8}$

B.

$\frac{11}{16}$

C.

$\frac{15}{32}$

D.

$\frac{17}{64}$

Answer: C

Solution:

We are given

$f(xy) = f(x)f(y) \quad \text{for all } x, y \in \mathbb{R},$

and

$f(0) \neq 0.$

Also,

$x^2 g(x) = \int_1^x \left(t^2 f(t) - t g(t)\right) dt, \qquad x \geq 1.$

We have to find $g(2)$.

Step 1: Use the condition on f

Given

$f(xy) = f(x)f(y)$

for all real x, y.

Now put $x = 0$ and any real y. Then

$f(0 \cdot y) = f(0)f(y)$

that is,

$f(0) = f(0)f(y)$.

Since $f(0) \neq 0$, we can divide by $f(0)$:

$f(y) = 1 \quad \text{for all } y \in \mathbb{R}$.

So,

$f(t) = 1 \quad \text{for all } t$.

Step 2: Substitute into the given integral equation

The equation becomes

$x^2 g(x) = \int_1^x (t^2 - tg(t))\, dt$.

Step 3: Differentiate both sides

Differentiate with respect to x.

Left side:

$\frac{d}{dx}[x^2 g(x)] = x^2 g'(x) + 2xg(x)$.

Right side, by the Fundamental Theorem of Calculus:

$\frac{d}{dx} \int_1^x (t^2 - tg(t))\, dt = x^2 - xg(x)$.

Hence,

$x^2 g'(x) + 2xg(x) = x^2 - xg(x)$.

So,

$x^2 g'(x) + 3xg(x) = x^2$.

Divide by x (since $x \geq 1$):

$xg'(x) + 3g(x) = x$.

Or,

$g'(x) + \frac{3}{x} g(x) = 1$.

Step 4: Solve the differential equation

This is a linear differential equation:

$g'(x) + \frac{3}{x} g(x) = 1$.

Its integrating factor is

$\text{I. F.} = e^{\int \frac{3}{x}\, dx} = e^{3 \ln x} = x^3$.

Multiply both sides by x^3:

$x^3 g'(x) + 3x^2 g(x) = x^3.$

The left side is

$\frac{d}{dx}(x^3 g(x)).$

So,

$\frac{d}{dx}(x^3 g(x)) = x^3.$

Integrate:

$x^3 g(x) = \frac{x^4}{4} + C.$

Thus,

$g(x) = \frac{x}{4} + \frac{C}{x^3}.$

Step 5: Find the constant using the original equation

Put $x = 1$ in

$x^2 g(x) = \int_1^x (t^2 - tg(t))\,dt.$

Then

$1^2 g(1) = \int_1^1 (t^2 - tg(t))\,dt = 0.$

So,

$g(1) = 0.$

Now use

$g(1) = \frac{1}{4} + C = 0.$

Hence,

$C = -\frac{1}{4}.$

Therefore,

$g(x) = \frac{x}{4} - \frac{1}{4x^3}.$

Step 6: Find $g(2)$

$g(2) = \frac{2}{4} - \frac{1}{4(2^3)} = \frac{1}{2} - \frac{1}{32} = \frac{16-1}{32} = \frac{15}{32}.$

So the correct answer is

$\boxed{\frac{15}{32}}.$

Hence, **Option C**.

Question21

Let $y = y(x)$ be the solution of the differential equation $x\sqrt{1-x^2}dy + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right)dx = 0, x \in (0,1), \lim_{x\to 1^-} y(x) = 1$. Then $y\left(\frac{1}{2}\right)$ equals :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$3 - \frac{\pi}{\sqrt{3}}$

B.

$4 - \sqrt{3}\pi$

C.

$4 - \frac{2\pi}{\sqrt{3}}$

D.

$3 - \frac{\pi}{2\sqrt{3}}$

Answer: A

Solution:

$\left(x\sqrt{1-x^2}\right)dy + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right)dx = 0$

$\Rightarrow \left(x\sqrt{1-x^2}\right)\frac{dy}{dx} + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right) = 0$

$\Rightarrow \frac{dy}{dx} + \frac{\left(y\sqrt{1-x^2} - x\cos^{-1}x\right)}{\left(x\sqrt{1-x^2}\right)} = 0$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{\cos^{-1}x}{\sqrt{1-x^2}} \ldots .(\text{i})$

Bernoulli's linear ODE of first order :

$\frac{dy}{dx} + P(x) \cdot y = Q(x)$ having $\boldsymbol{I} \cdot \boldsymbol{F} = \boldsymbol{e}^{\int \boldsymbol{P(x)dx}}$

$\boldsymbol{P(x)}$	$\boldsymbol{Q(x)}$
$\frac{1}{\boldsymbol{x}}$	$\frac{\cos^{-1}\boldsymbol{x}}{\sqrt{1-\boldsymbol{x}^2}}$

$\Rightarrow \boldsymbol{I}.\, \boldsymbol{F} = e^{\int P(x)dx} = e^{\int \frac{dx}{x}} = e^{\ln x} = x \ldots\ldots (ii)$

Apply the I.F from equation (ii) in ODE(i) :

$-x\frac{dy}{dx}+y=\frac{x\cos^{-1}x}{\sqrt{1-x^2}} \Rightarrow xdy+ydx=\left(\frac{x\cos^{-1}x}{\sqrt{1-x^2}}\right)dx$

$\Rightarrow d(xy)=\left(\frac{x\cos^{-1}x}{\sqrt{1-x^2}}\right)dx \Rightarrow \int d(xy)=\int\left(\frac{x\cos^{-1}x}{\sqrt{1-x^2}}\right)dx \ldots\ldots(iii)$

We choose the first and second functions

$\boldsymbol{u}$	$\boldsymbol{v}$
$\cos^{-1}\boldsymbol{x}$	$\frac{\boldsymbol{x}}{\sqrt{1-x^2}}$

for the integrand in equation $(\boldsymbol{iii})$ to proceed further:

$$
\begin{aligned}
xy &= \left(\cos^{-1}x\right)\cdot\int\left(\frac{x}{\sqrt{1-x^2}}\right)dx+\int\left[\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\int\left(\frac{x}{\sqrt{1-x^2}}\right)dx\right]dx \\
&= -\left(\cos^{-1}x\right)\cdot\sqrt{1-x^2}-\int\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\left(\sqrt{1-x^2}\right)dx \\
&= -\left(\cos^{-1}x\right)\cdot\sqrt{1-x^2}-\int\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\left(\sqrt{1-x^2}\right)dx \\
&= -\left(\cos^{-1}x\right)\cdot\sqrt{1-x^2}-\int dx \\
&= -\left(\cos^{-1}x\right)\cdot\sqrt{1-x^2}-x+C
\end{aligned}
$$

$y(x)=\frac{C}{x}-1-\frac{(\cos^{-1}x)\cdot\sqrt{1-x^2}}{x} \ldots.(iv)$ where C is the arbitrary constant

$\lim\limits_{x\to1^-}y(x)=\lim\limits_{x\to1^-}\left[\frac{C}{x}-1-\frac{(\cos^{-1}x)\cdot\sqrt{1-x^2}}{x}\right]=C-1-\frac{(0)\cdot(0)}{1}=C-1\ldots\ldots(v)$

We have the boundary condition

$\lim\limits_{\boldsymbol{x}\to1^-}\boldsymbol{y}(\boldsymbol{x})=1\Rightarrow\boldsymbol{C}=2\ldots..(\boldsymbol{vi})\leftarrow$ From equation $(\boldsymbol{v})$

We have solved the ODE to reach a particular solution $(\boldsymbol{C}=2)$ from $(\boldsymbol{iv})y(x)=\frac{2}{x}-1-\frac{(\cos^{-1}x)\cdot\sqrt{1-x^2}}{x}$ …. (vii)

Substitute $\boldsymbol{x}=\frac{1}{2}$ in equation (vii) to evaluate $\boldsymbol{y}\left(\frac{1}{2}\right)=4-1-(\mathbf{2})\cdot\left(\frac{\pi}{3}\right)\cdot\left(\frac{\sqrt{3}}{\mathbf{2}}\right)$

$=3-\frac{\pi}{\sqrt{3}}$

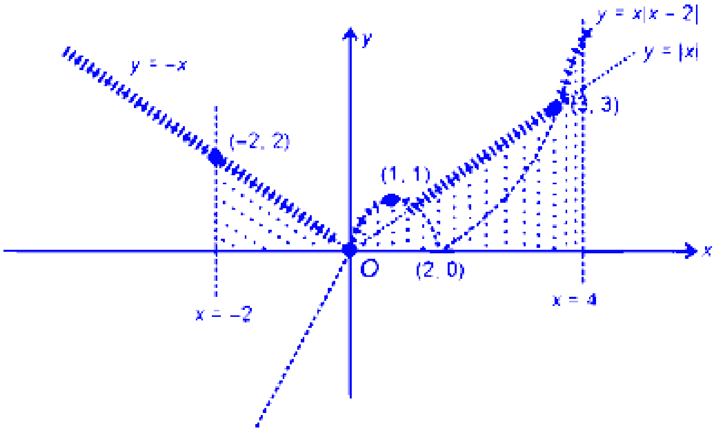
Question22

The area of the region bounded by the curve $y=\max\{|x|,x|x-2|\}$, the x-axis and the lines $x=-2$ and $x=4$ is equal to__________

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 12

Solution:



Area =

$$\frac{1}{2} \times 2 \times 2 + \int_0^1 (2x - x^2) dx + \int_1^3 x dx + \int_3^4 (x^2 - 2x) dx$$

$$= 2 + \left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^1 + \left[\frac{x^2}{2}\right]_1^3 + \left[\frac{x^3}{3} - \frac{2x^2}{2}\right]_3^4$$

= 12 square units

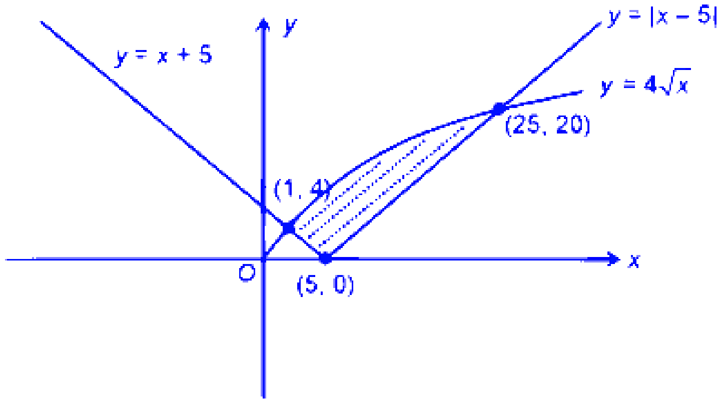
Question23

If the area of the region $\{(x, y) : |x - 5| \leq y \leq 4\sqrt{x}\}$ is A, then $3A$ is equal to __________.

JEE Main 2025 (Online) 4th April Morning Shift

Answer: 368

Solution:



$$\text{Area } = \int_1^{25} 4\sqrt{x}dx - \frac{1}{2} \times (5-1) \cdot 4 - \frac{1}{2} \times (25-5) \times 20$$

$$= 4\left[\frac{2}{3}x^{\frac{3}{2}}\right]_1^{25} - 8 - 200 = \frac{368}{3}$$

$$\Rightarrow 3A = 368$$

Question24

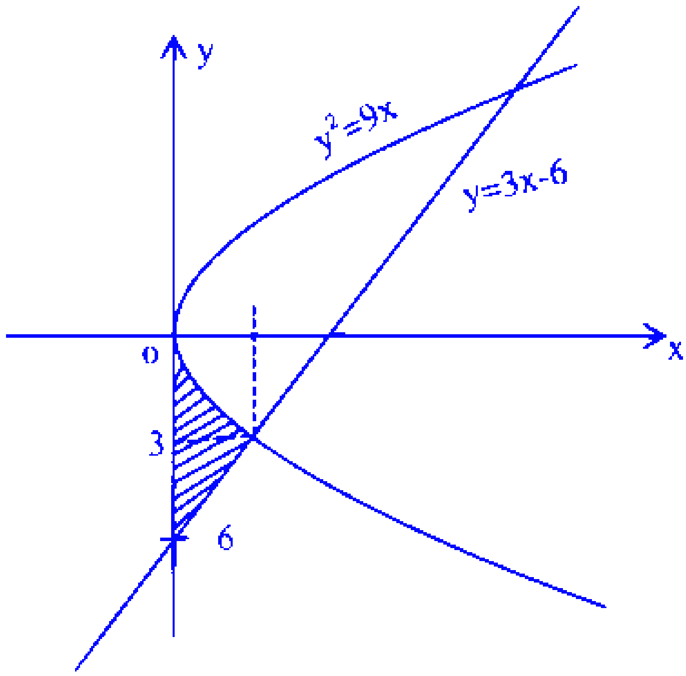
Let the area of the bounded region $\left\{(x, y) : 0 \leq 9x \leq y^2, y \geq 3x - 6\right\}$ be A. Then $6A$ is equal to _________.

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 15

Solution:

$0 \leq 9x \leq y^2$ & $y \geq 3x - 6$



$$A = \text{ Required Area } = \left[\int_0^1 (-3\sqrt{x})dx - \int_0^1 (3x-6)dx\right]$$

$$A = -3\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)\Bigg|_0^1 - \left(\frac{3x^2}{2} - 6x\right)\Bigg|_0^1$$

$$A = -2[1-0]\left[\frac{3}{2} - 6\right]$$

$$A = -2 - \frac{3}{2} + 6 = \frac{5}{2} \text{ Sq. unit}$$

$$\therefore 6A = 6 \times \frac{5}{2} = 15$$

Question25

Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0, x = \frac{3\pi}{2}$, and the x-axis be A . Then, $\mathrm{A} + \mathrm{A}^2$ is equal to ____.

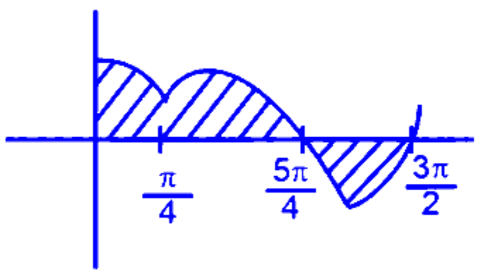
JEE Main 2026 (Online) 23rd January Morning Shift

Answer: 12

Solution:

$f(x) = \max(\sin x, \cos x)$

$= \begin{cases} \cos x, x \in \left(0, \frac{\pi}{4}\right) \\ \sin x, x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \\ \cos x, x \in \left(\frac{5\pi}{4}, \frac{3\pi}{2}\right) \end{cases}$

$\Rightarrow$ 

$\Rightarrow \int_0^{\frac{3\pi}{4}} |f(x)| dx = \int_0^{\frac{\pi}{4}} (\cos x) dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x) dx + \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} (\cos x) dx$

$= 3 \Rightarrow \; \mathrm{A} = 3$

$A + A^2 = 12$

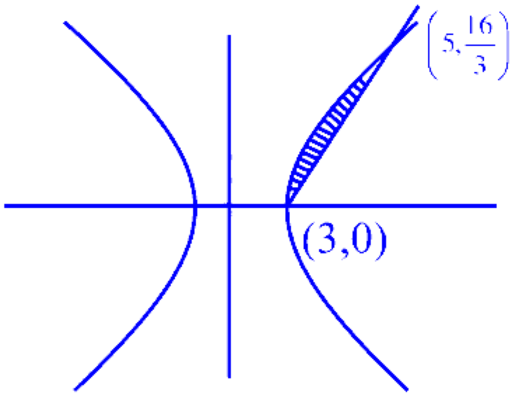
Question26

If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A, then $3(A + 6\log_e(3))$ is equal to ________.

JEE Main 2026 (Online) 2nd April Evening Shift

Answer: 24

Solution:



$16x^2 - 9y^2 = 144$

$16x^2 - (8x - 24)^2 = 144$

$16x^2 - 64(x-3)^2 = 144 \rightarrow x^2 - 4(x-3)^2 = 9$

$3x^2 - 24x + 45 \rightarrow x^2 - 8x + 15 = 0$

$x = 3, 5$

$\text{Area} = \int\limits_3^5 \sqrt{\frac{16x^2-144}{9}} - \frac{1}{2} \cdot 2 \cdot \frac{16}{3}$

$= \frac{4}{3} \int\limits_3^5 \sqrt{x^2-9} - \frac{16}{3}$

$= \frac{4}{3}\left(\frac{x}{2}\sqrt{x^2-9} - \frac{9}{2}\log_e\left(x + \sqrt{x^2-9}\right)\right)_3^5 - \frac{16}{3}$

$= \frac{4}{3}\left(\frac{5}{2} \cdot 4 - \frac{9}{2}\log_e 9 - \frac{3}{2} \cdot 0 + \frac{9}{2}\log_e 3\right) - \frac{16}{3}$

$\text{Area} = 8 - 6\log_e 3 = A$

$\therefore A + 6\ell n 3 = 8$

$\Rightarrow 3(A + 6\ell n 3) = 24$

Question27

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x, x \in \mathbf{R}$. Let the maximum value of f on $\mathbf{R}$ be α. If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3, \beta > 0$, is α^2, then $30\beta^3$ is equal to _____ .

JEE Main 2026 (Online) 5th April Evening Shift

Answer: 16

Solution:

$f(x)+3f\left(\frac{\pi}{2}-x\right)=\sin x \qquad \ldots(1)$

Now replace x by $\frac{\pi}{2}-x$ in (1). This is a standard NCERT step to get a second equation in the same two terms.

$\Rightarrow f\left(\frac{\pi}{2}-x\right)+3f(x)=\cos x \qquad \ldots(2)$

Now we solve the two linear equations (1) and (2) to find $f(x)$.

Multiply (1) by 3:

$3f(x)+9f\left(\frac{\pi}{2}-x\right)=3\sin x \qquad \ldots(3)$

Subtract (2) from (3):

$\left(3f(x)-3f(x)\right)+\left(9-1\right)f\left(\frac{\pi}{2}-x\right)=3\sin x-\cos x$

$8f\left(\frac{\pi}{2}-x\right)=3\sin x-\cos x$

Now put $x\rightarrow\frac{\pi}{2}-x$ in this result to get $f(x)$:

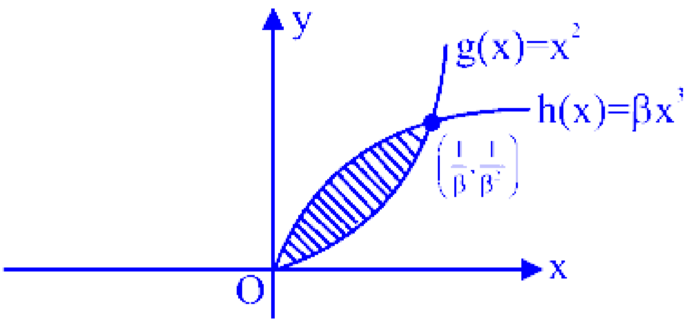
$8f(x)=3\cos x-\sin x$

$f(x)=\frac{1}{8}(3\cos x-\sin x)$

To find the maximum value, write $3\cos x-\sin x$ in the form $R\cos(x+\phi)$, where $R=\sqrt{3^2+(-1)^2}=\sqrt{10}$. So the maximum value of $3\cos x-\sin x$ is $\sqrt{10}$.

$f_{\max}=\frac{\sqrt{10}}{8}$

$\mathrm{y}=\mathrm{g}(\mathrm{x}) \& \mathrm{y}=\mathrm{h}(\mathrm{x})$ intersect as shown in the figure



The curves are $g(x)=x^2$ and $h(x)=\beta x^3$ with $\beta>0$. Their points of intersection come from $x^2=\beta x^3$:

$x^2=\beta x^3 \Rightarrow x^2(1-\beta x)=0 \Rightarrow x=0$ and $x=\frac{1}{\beta}$

For \$0\beta x^3$(since$\beta x<1$), sotheareabetweenthecurvesistheintegralof$(x^2-\beta x^3)$from0to$\frac{1}{\beta}\$.

$$\therefore \text{ Area bounded } = \Delta = \left|\int_0^{\frac{1}{\beta}} (\beta \mathrm{x}^3 - \mathrm{x}^2)\mathrm{dx}\right|$$

$$= \left|\left[\frac{\beta x^4}{4} - \frac{x^3}{3}\right]_0^{1/\beta}\right|$$

$$= \left|\left(\frac{\beta}{4} \cdot \frac{1}{\beta^4} - \frac{1}{3} \cdot \frac{1}{\beta^3}\right)\right|$$

$$= \left|\left(\frac{1}{4\beta^3} - \frac{1}{3\beta^3}\right)\right| = \frac{1}{12\beta^3} = \alpha^2 \text{ (given)}$$

$$\Rightarrow \frac{1}{12\beta^3} = \left(\frac{\sqrt{10}}{8}\right)^2 = \frac{10}{64} = \frac{5}{32}$$

$$\Rightarrow \beta^3 = \frac{1}{12} \cdot \frac{32}{5} = \frac{8}{15}$$

$$\Rightarrow 30\beta^3 = 30 \cdot \frac{8}{15} = 16$$

Question28

If the area of the region $\{(x, y) : 1 - 2x \leq y \leq 4 - x^2,\ x \geq 0,\ y \geq 0\}$ is $\frac{\alpha}{\beta}$, $\alpha, \beta \in \mathbb{N}, \operatorname{gcd}(\alpha, \beta) = 1$, then the value of $(\alpha + \beta)$ is:

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

73

B.

85

C.

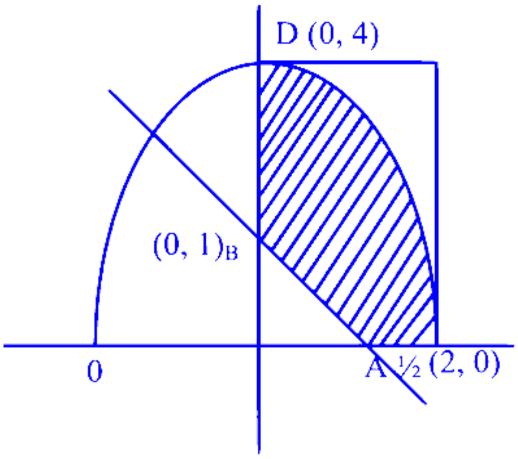
67

D.

91

Answer: A

Solution:



$\int_0^2 (4 - x^2)dx - \frac{1}{2} \times 1 \times \frac{1}{2}$

Question29

Let the line $x = -1$ divide the area of the region $\{(x, y) : 1 + x^2 \leq y \leq 3 - x\}$ in the ratio $m : n, \gcd(m, n) = 1$. Then $m + n$ is equal to

JEE Main 2026 (Online) 22nd January Morning Shift

Options:

A.

27

B.

28

C.

25

D.

26

Answer: A

Solution:

$$m=\int_{-2}^{-1}(3-x)-\left(1+x^2\right)dx$$
$$m=2+\frac{3}{2}-\frac{7}{3}=\frac{7}{6}\quad M=\int_{-1}^{1}(3-x)-\left(1+x^2\right)dx=4-\frac{1}{2}(0)-\frac{1}{3}(1+1)=4-\frac{2}{3}=\frac{10}{3}$$
$$m:M=\frac{7}{6}:\frac{10}{3}=\frac{7}{20}$$
$$m+n=27$$

Question30

The area of the region $\mathrm{A}=\left\{(x,y):4x^2+y^2\leqslant 8\right.$ and $\left.y^2\leqslant 4x\right\}$ is:

JEE Main 2026 (Online) 22nd January Evening Shift

Options:

A.

$\pi+\frac{2}{3}$

B.

$\frac{\pi}{2}+2$

C.

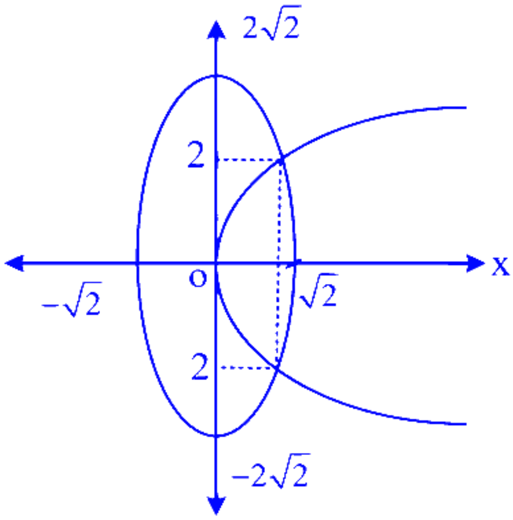
$\pi+4$

D.

$\frac{\pi}{2}+\frac{1}{3}$

Answer: A

Solution:



$A = 2\int_0^1 2\sqrt{x}dx + 2\int_1^{\sqrt{2}} \sqrt{8-4x^2}dx = \frac{8}{3}\left(X^{\frac{3}{2}}\right)_0^1 + 4\int_1^{\sqrt{2}} \sqrt{2-x^2}dx$

$\frac{8}{3} + 4 \times \frac{1}{2}\left[x\sqrt{2-x^2} + 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\right]_1^{\sqrt{2}}$

$\frac{8}{3} + 2\left[2 \times \frac{\pi}{2} - 1 - 2 \times \frac{\pi}{4}\right] = \frac{8}{3} + 2\pi - 2 - \pi = \pi + \frac{2}{3}$ Sq. unit

----

Question31

The area of the region enclosed between the circles $x^2 + y^2 = 4$ and $x^2 + (y-2)^2 = 4$ is:

JEE Main 2026 (Online) 23rd January Evening Shift

Options:

A.

$\frac{2}{3}(4\pi - 3\sqrt{3})$

B.

$\frac{4}{3}(2\pi - \sqrt{3})$

C.

$\frac{4}{3}(2\pi - 3\sqrt{3})$

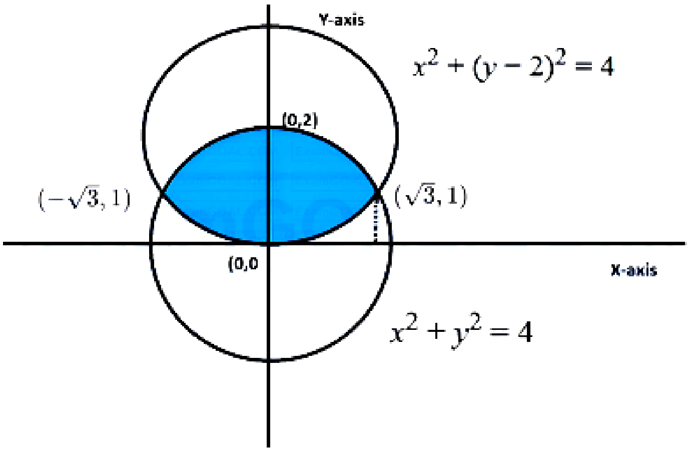
D.

$\frac{2}{3}(2\pi - 3\sqrt{3})$

Answer: A

Solution:

$$x^2 + y^2 = 4 \Longrightarrow C_1(0,0), r = 2$$
$$x^2 + (y-2)^2 = 4 \Longrightarrow C_2(0,2), r = 2$$



Intersection points:

$$x^2 + y^2 = x^2 + (y-2)^2$$
$$y^2 = y^2 - 4y + 4 \Longrightarrow 4y = 4 \Longrightarrow y = 1$$
$$x^2 + 1 = 4 \Longrightarrow x^2 = 3 \Longrightarrow x = \pm\sqrt{3}$$

Points are $(\sqrt{3}, 1)$ and $(-\sqrt{3}, 1)$

For the region between $y = 0$ and $y = 2$:

$$y_{\text{upper}} = \sqrt{4 - x^2} \left(\text{from} x^2 + y^2 = 4 \right)$$
$$y_{\text{lower}} = 2 - \sqrt{4 - x^2} \left(\text{from} x^2 + (y-2)^2 = 4 \right)$$

$$\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} \left(y_{\text{upper}} - y_{\text{lower}} \right) dx$$
$$\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} \left(\sqrt{4 - x^2} - \left(2 - \sqrt{4 - x^2} \right) \right) dx$$
$$\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} \left(2\sqrt{4 - x^2} - 2 \right) dx$$

Property: $\int_{-a}^{a} f(x) dx = 2 \int_0^a f(x) dx$ (for even functions)

Area $= 2\int_0^{\sqrt{3}} \left(2\sqrt{4-x^2}-2\right)dx$

Area $= 4\int_0^{\sqrt{3}} \left(\sqrt{4-x^2}-1\right)dx$

$\int \sqrt{a^2-x^2}dx = \frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$

Area $= 4\left[\frac{x}{2}\sqrt{4-x^2}+2\sin^{-1}\left(\frac{x}{2}\right)-x\right]_0^{\sqrt{3}}$

Area $= 4\left[\frac{\sqrt{3}}{2}(1)+2\left(\frac{\pi}{3}\right)-\sqrt{3}\right]$

Area $= 4\left[\frac{2\pi}{3}-\frac{\sqrt{3}}{2}\right]$

Area $= \frac{8\pi}{3}-2\sqrt{3}=\frac{2}{3}(4\pi-3\sqrt{3})$

Question32

Let A_1 be the bounded area enclosed by the curves $y=x^2+2, x+y=8$ and y-axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y=x^2+2, y^2=x, x=2$, and y-axis that lies in the first quadrant. Then $\mathrm{A}_1-\mathrm{A}_2$ is equal to

JEE Main 2026 (Online) 24th January Morning Shift

Options:

A.

$\frac{2}{3}(2\sqrt{2}+1)$

B.

$\frac{2}{3}(3\sqrt{2}+1)$

C.

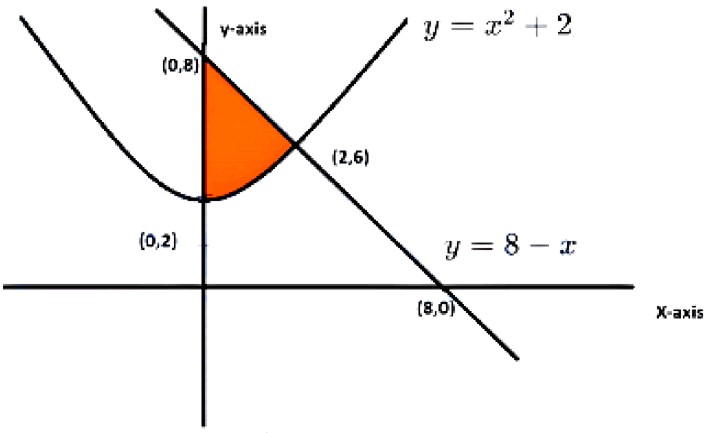
$\frac{2}{3}(\sqrt{2}+1)$

D.

$\frac{2}{3}(4\sqrt{2}+1)$

Answer: A

Solution:



Area A_1

Area A_1 enclosed by curves

$y = x^2 + 2, x + y = 8, y$-axis in first quadrant as shown in graph.

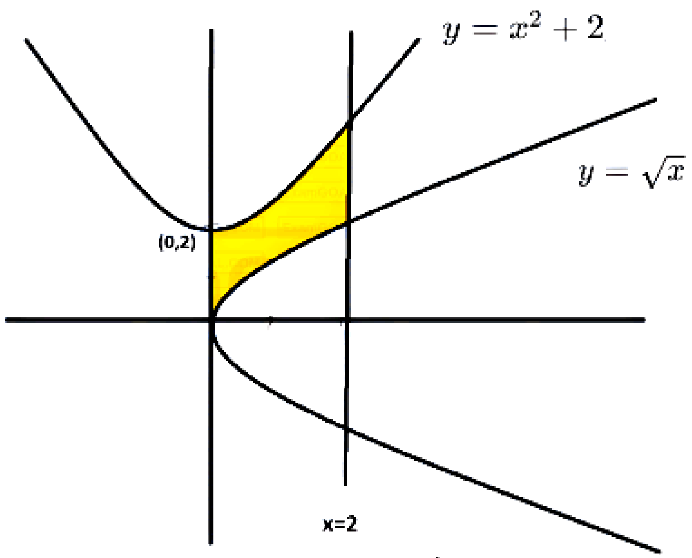
Solve $y = x^2 + 2$ and $x + y = 8$ to find point of intersection.

Substituting $y = 8 - x$ in $y = x^2 + 2$

$$
\begin{aligned}
&8 - x = x^2 + 2 \\
&x^2 + x - 6 = 0 \\
&x^2 + 3x - 2x - 6 = 0 \\
&x(x+3) - 2(x+3) = 0 \\
&(x+3)(x-2) = 0 \\
&x = -3, 2
\end{aligned}
$$

Since area in first quadrant, point of intersection at $x = 2$, $y = 8 - 2 = 6 \Rightarrow (2, 6)$.

$$
\begin{aligned}
A_1 &= \int_0^2 \left[(8 - x) - \left(x^2 + 2\right)\right] dx \\
&= \int_0^2 \left(-x^2 - x + 6\right) dx \\
&= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 6x\right]_0^2 \\
&= \left[-\frac{8}{3} - \frac{4}{2} + 12 - \{0 - 0 - 0\}\right] \\
&= \frac{-16 - 12 + 72}{6} = \frac{72 - 28}{6} = \frac{44}{6} \\
A_1 &= \frac{22}{3}
\end{aligned}
$$



Area A_2

Now, A_2 is area enclosed by curves

$y = x^2 + 2, y^2 = x, x = 2$

Solve $y = x^2 + 2$ and $y = \sqrt{x}$

$y = y^4 + 2$

$y^4 - y + 2 = 0$

$D = (-1)^2 - 4 \times 1 \times 2 = 1 - 8 = -7$

discriminant (D) is negative. This means curves do not intersect.

So A_2 is bounded by curves y-~~axis~~, $2, y = x^2 + 2$ and $y = \sqrt{x}$ in first quadrant is:

$$A_2 = \int_0^2 (x^2 + 2 - \sqrt{x}) dx$$

$$= \left[\frac{x^3}{3} + 2x - \frac{2x^{3/2}}{3}\right]_0^2$$

$$= \left[\frac{8}{3} + 2 \times 2 - \frac{2 \times 2\sqrt{2}}{3} - \{0 + 0 - 0\}\right]$$

$$= \left[\frac{8}{3} + 4 - \frac{4\sqrt{2}}{3}\right] = \left[\frac{20}{3} - \frac{4\sqrt{2}}{3}\right]$$

$$A_1 - A_2 = \frac{22}{3} - \left(\frac{20}{3} - \frac{4\sqrt{2}}{3}\right)$$

$$= \frac{22}{3} - \frac{20}{3} + \frac{4\sqrt{2}}{3} = \frac{2}{3} + \frac{4\sqrt{2}}{3}$$

$$= \frac{2}{3}(2\sqrt{2} + 1)$$

Question33

Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0, x = 1, y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

JEE Main 2026 (Online) 24th January Evening Shift

Options:

A.

12

B.

14

C.

9

D.

7

Answer: D

Solution:

Given Curves $y^2 = x, y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2 x = 0$ and $x = 1$

The area of region bounded by these curve is $f(\alpha)$.

To find $f(0) + f(1)$.

for $\alpha = 0$

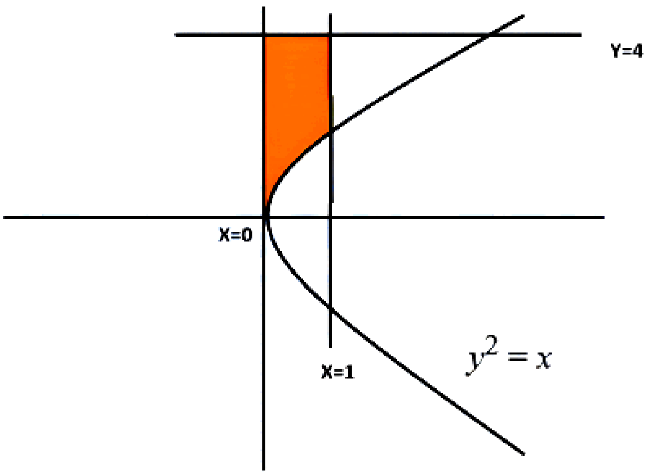
$y = |0 \times x - 5| - |1 - 0 \times x| + 0 \times x^2$
$y = 5 - 1 = 4$

so the region bounded by $x = 0, x = 1, y = \sqrt{x}$

and $y = 4$ is shown in graph.

$$f(0) = \int_0^1 (4 - \sqrt{x})dx$$

$$f(0) = \left[4x - \frac{2}{3}x^{3/2}\right]_0^1 = 4 - \frac{2}{3} = \frac{10}{3}$$



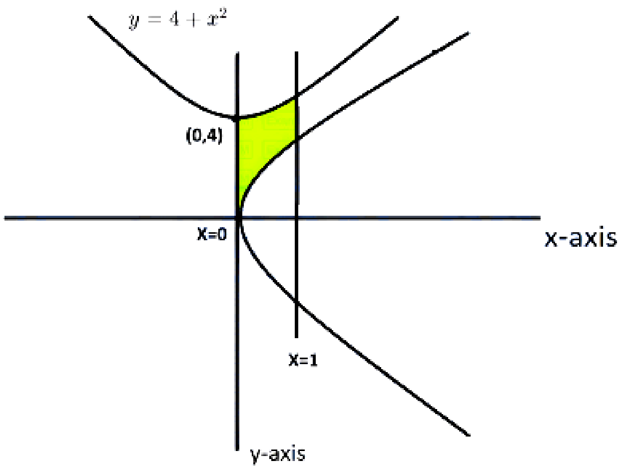
If, When $\alpha = 1$, the second equation becomes:

$y = |x-5| - |1-x| + x^2$

For the interval $0 \leq x \leq 1$:

$|x-5| = -(x-5) = 5-x$
$|1-x| = 1-x$
$y = (5-x) - (1-x) + x^2 = 4 + x^2$

The region is bounded by $x = 0, x = 1, y = \sqrt{x}$, and $y = 4 + x^2$ as shown in graph is.



$$f(1) = \int_0^1 (4 + x^2 - \sqrt{x})dx$$

$$f(1) = \left[4x + \frac{x^3}{3} - \frac{2}{3}x^{3/2}\right]_0^1 = 4 + \frac{1}{3} - \frac{2}{3} = 4 - \frac{1}{3} = \frac{11}{3}$$

So, $(f(0) + f(1)) = \frac{10}{3} + \frac{11}{3} = \frac{21}{3} = 7$

Question34

The area of the region $\mathrm{R}=\left\{(x, y): x y \leq 8,1 \leq y \leq x^{2}, x \geq 0\right\}$ is

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

$\frac{2}{3}(20 \log_e(2)+9)$

B.

$\frac{1}{3}(40 \log_e(2)+27)$

C.

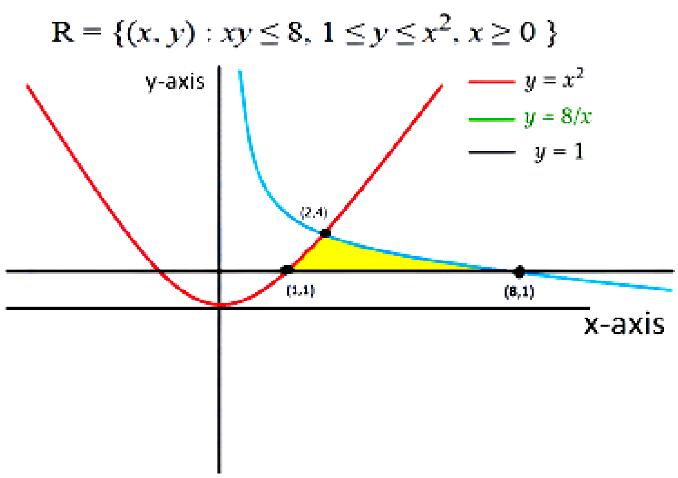
$\frac{1}{3}(49 \log_e(2)-15)$

D.

$\frac{2}{3}(24 \log_e(2)-7)$

Answer: D

Solution:



The region is bounded by three curves in the first quadrant ($x \geq 0$) :

- 1. $y=1$: A horizontal line.
- 2. $y=x^{2}$: A parabola opening upwards.
- 3. $y=\frac{8}{x}$: A rectangular hyperbola.

Find the Intersection Points

We need to know where these curves meet to set our integration limits:

- Parabola and Line $\left(y=x^{2}\right.$ and $\left.y=1\right)$: $x^{2}=1 \Longrightarrow x=1$ (since $x \geq 0$). Point: $(1,1)$.

- Parabola and Hyperbola $\left(y=x^2 \text{ and } y=\frac{8}{x}\right): x^2=\frac{8}{x} \Longrightarrow x^3=8 \Longrightarrow x=2$. Point: $(2,4)$.
- Hyperbola and Line $\left(y=\frac{8}{x} \text{ and } y=1\right): \frac{8}{x}=1 \Longrightarrow x=8$. Point: $(8,1)$.

Understanding the Inequalities

- $x \geq 0$: The region is located in the first and fourth quadrants (but limited to the first quadrant by $1 \leq y$).
- $1 \leq y \leq x^2$: This gives us a lower bound of the horizontal line $y=1$ and an upper bound of the parabola $y=x^2$.
- $xy \leq 8$: This can be rewritten as $y \leq \frac{8}{x}$. This represents the area below the rectangular hyperbola.

Setting up the Integrals

The region starts at $x=1$ and ends at $x=8$. Because the "upper" boundary changes at $x=2$, we must split the area into two parts:

- Part 1 (from $x=1$ to $x=2$): Bounded above by the parabola $y=x^2$ and below by $y=1$.
- Part 2 (from $x=2$ to $x=8$): Bounded above by the hyperbola $y=\frac{8}{x}$ and below by $y=1$.

Area $=\int\limits_1^2\left(x^2-1\right)dx+\int\limits_2^8\left(\frac{8}{x}-1\right)dx$

$=\left[\frac{x^3}{3}-x\right]_1^2+[8\ln x-x]_2^8$

$=\left[\frac{8}{3}-2-\left\{\frac{1}{3}-1\right\}\right]+[(8\ln 8-8)-\{8\ln 2-2\}]$

$=\frac{7}{3}-1+8\ln 2^3-8\ln 2-6$

$=\frac{7}{3}-7+24\ln 2-8\ln 2 \quad \{\ln a^m=m\ln a\}$

$=-\frac{14}{3}+16\ln 2$

$=\frac{2}{3}[24\ln 2-7]$

Question35

Let $P_1: y=4x^2$ and $P_2: y=x^2+27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y=\alpha x$, $\alpha>0$ and P_1, then α is equal to :

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

12

B.

15

C.

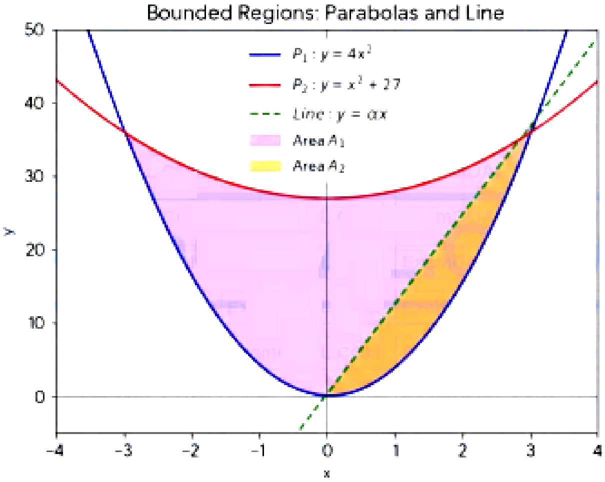
8

D.

6

Answer: A

Solution:



1. Calculate the Area A_1 between P_1 and P_2

The two parabolas are given by:

- $P_1 : y = 4x^2$
- $P_2 : y = x^2 + 27$

Find the intersection points: Setting the equations equal to each other:

$4x^2 = x^2 + 27$

$\Rightarrow 3x^2 = 27$

$\Rightarrow x^2 = 9 \Rightarrow x = \pm 3$

Find the intersection points : Setting the equations equal to each other :

$4x^2 = x^2 + 27$

$3x^2 = 27 \quad , \quad x^2 = 9 \Rightarrow x = \pm 3$

Calculate the area: On the interval $[-3, 3]$, the parabola P_2 lies above P_1. The area A_1 is:

$$A_1 = \int_{-3}^{3} (x^2 + 27 - 4x^2) dx = \int_{-3}^{3} (27 - 3x^2) dx$$
$$A_1 = \left[27x - x^3\right]_{-3}^{3}$$
$$A_1 = \left(27(3) - 3^3\right) - \left(27(-3) - (-3)^3\right)$$
$$A_1 = (81 - 27) - (-81 + 27) = 54 - (-54) = 108$$

Calculate the Area A_2 between P_1 and the line $y = \alpha x$

Find the intersection points:

$4x^2 = \alpha x \implies x(4x - \alpha) = 0$

The points are $x = 0$ and $x = \frac{\alpha}{4}$.

Calculate the area: On the interval $\left[0, \frac{\alpha}{4}\right]$, the line $y = \alpha x$ lies above $y = 4x^2$. The area A_2 is:

$$A_2 = \int_{0}^{\alpha/4} (\alpha x - 4x^2) dx$$
$$A_2 = \left[\frac{\alpha x^2}{2} - \frac{4x^3}{3}\right]_0^{\alpha/4}$$
$$A_2 = \frac{\alpha}{2}\left(\frac{\alpha}{4}\right)^2 - \frac{4}{3}\left(\frac{\alpha}{4}\right)^3$$
$$A_2 = \frac{\alpha^3}{32} - \frac{4\alpha^3}{192} = \frac{\alpha^3}{32} - \frac{\alpha^3}{48} = \frac{3\alpha^3 - 2\alpha^3}{96} = \frac{\alpha^3}{96}$$

The problem states that A_1 is six times A_2 :

$$A_1 = 6 \times A_2$$
$$108 = 6 \times \left(\frac{\alpha^3}{96}\right)$$
$$108 = \frac{\alpha^3}{16}$$
$$\alpha^3 = 108 \times 16 = 1728$$
$$\alpha = \sqrt[3]{1728} = 12$$

Question36

The area of the region $\{(x, y) : y \leq \pi - |x|, y \leq |x \sin x|, y \geq 0\}$ is:

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A. $1 + \frac{\pi^2}{8}$

B.

$2 + \frac{\pi^2}{4}$

C.

$\frac{\pi^2}{8} - 1$

D.

$4 + \frac{\pi^2}{2}$

Answer: B

Solution:

We need the area of the region

$\{(x, y) : y \leq \pi - |x|,\ y \leq |x \sin x|,\ y \geq 0\}$.

This means the required area is the area under the curve

$y = \min(\pi - |x|,\ |x \sin x|)$

and above the x-axis.

So the area is

$A = \int \min(\pi - |x|,\ |x \sin x|)\, dx$

over the values of x where the region exists.

Since $y \geq 0$ and $y \leq \pi - |x|$, we must have

$\pi - |x| \geq 0 \implies |x| \leq \pi$.

Hence the region lies in $[-\pi, \pi]$.

Also, both functions $\pi - |x|$ and $|x \sin x|$ are even functions, so the region is symmetric about the y-axis. Therefore,

$A = 2 \int_0^\pi \min(\pi - x,\ x \sin x)\, dx$.

Now for $0 \leq x \leq \pi$, we have $\sin x \geq 0$, so

$|x \sin x| = x \sin x$.

Thus,

$A = 2 \int_0^\pi \min(\pi - x,\ x \sin x)\, dx$.

Now compare $\pi - x$ and $x \sin x$ on $[0, \pi]$.

At $x = 0$,

$\pi - x = \pi, \qquad x \sin x = 0.$

At $x = \pi$,

$\pi - x = 0, \qquad x \sin x = 0.$

Check where they meet:

$x \sin x = \pi - x$.

It is easy to see that $x = \dfrac{\pi}{2}$ satisfies this, because

$\frac{\pi}{2} \sin \frac{\pi}{2} = \frac{\pi}{2} = \pi - \frac{\pi}{2}$.

Also, on $[0, \pi/2]$, we have $x \sin x \leq \pi - x$, and on $[\pi/2, \pi]$, we have $\pi - x \leq x \sin x$.

So,

$A = 2\left[\int_0^{\pi/2} x\sin x\,dx + \int_{\pi/2}^{\pi}(\pi - x)\,dx\right].$

Now evaluate each integral.

First,

$\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x.$

Therefore,

$\int_0^{\pi/2} x\sin x\,dx = [-x\cos x + \sin x]_0^{\pi/2} = 1.$

Next,

$\int_{\pi/2}^{\pi}(\pi - x)\,dx.$

Let $u = \pi - x$. Then as x goes from $\pi/2$ to π, u goes from $\pi/2$ to 0. Or directly,

$\int_{\pi/2}^{\pi}(\pi - x)\,dx = \left[\pi x - \frac{x^2}{2}\right]_{\pi/2}^{\pi}.$

Compute:

At $x = \pi$,

$\pi^2 - \frac{\pi^2}{2} = \frac{\pi^2}{2}.$

At $x = \frac{\pi}{2}$,

$\pi \cdot \frac{\pi}{2} - \frac{1}{2}\left(\frac{\pi}{2}\right)^2 = \frac{\pi^2}{2} - \frac{\pi^2}{8} = \frac{3\pi^2}{8}.$

So,

$\int_{\pi/2}^{\pi}(\pi - x)\,dx = \frac{\pi^2}{2} - \frac{3\pi^2}{8} = \frac{\pi^2}{8}.$

Hence,

$A = 2\left(1 + \frac{\pi^2}{8}\right) = 2 + \frac{\pi^2}{4}.$

Therefore, the correct answer is

$$\boxed{2 + \frac{\pi^2}{4}}$$

So, Option B is correct.

Question37

The area of the region bounded by the curves $x + 3y^2 = 0$ and $x + 4y^2 = 1$ is equal to :

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$\frac{1}{3}$

B.

$\frac{2}{3}$

C.

$\frac{4}{3}$

D.

$\frac{5}{3}$

Answer: C

Solution:

Given curves are

$x + 3y^2 = 0 \quad \text{and} \quad x + 4y^2 = 1$

First, write them in the form $x =$:

$x = -3y^2$

and

$x = 1 - 4y^2$

To find the bounded area, we first find their points of intersection.

Set the two values of x equal:

$-3y^2 = 1 - 4y^2$

$y^2 = 1$

$y = \pm 1$

So the curves intersect at $y = -1$ and $y = 1$.

Now, for a fixed value of y, the right curve is

$x = 1 - 4y^2$

and the left curve is

$x = -3y^2$

So the horizontal width of the region is

$(1 - 4y^2) - (-3y^2) = 1 - y^2$

Hence, area of the enclosed region is

$A = \int_{-1}^{1} (1 - y^2)\, dy$

Now evaluate:

$A=\left[y-\frac{y^3}{3}\right]_{-1}^{1}$

$A=\left(1-\frac{1}{3}\right)-\left(-1+\frac{1}{3}\right)$

$A=\frac{2}{3}-\left(-\frac{2}{3}\right)$

$A=\frac{4}{3}$

Therefore, the required area is

$$\boxed{\frac{4}{3}}$$

So, the correct option is:

$\boxed{\text{Option C}}$

Question38

The area of the region $\mathrm{R}=\left\{(x, y): xy \leq 27, 1 \leq y \leq x^2\right\}$ is equal to :

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

$78 \log_e 3-\frac{52}{3}$

B.

$54 \log_e 3-\frac{52}{3}$

C.

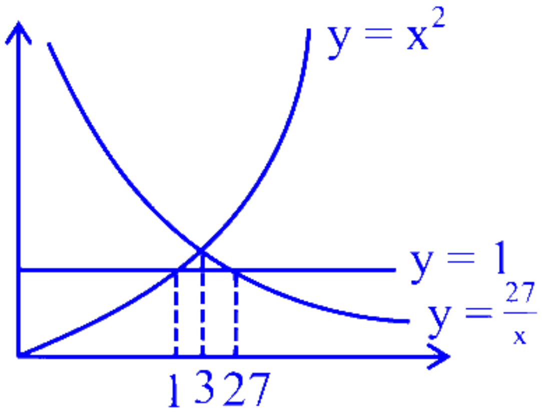
$54 \log_e 3-\frac{26}{3}$

D.

$54 \log_e 3+\frac{26}{3}$

Answer: B

Solution:



$$\text{Area } = \int_1^3 (x^2 - 1)dx + \int_3^{27} \left(\frac{27}{x} - 1\right)dx$$
$$= \left[\frac{x^3}{3} - x\right]_1^3 + [27\ell nx - x]_3^{27}$$
$$= \frac{26}{3} - 2 + 27\ln 9 - 24$$
$$= 27\ln 9 - \frac{26 \times 2}{3}$$
$$= 54\ln 3 - \frac{52}{3}$$

Question39

Let e be the base of natural logarithm and let $f : \{1, 2, 3, 4\} \rightarrow \{1, e, e^2, e^3\}$ and g : $\{1, e, e^2, e^3\} \rightarrow \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x) = \left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region $\mathrm{R} = \{(x, y) : x^2 \leq y \leq \phi(x), 0 \leq x \leq 1\}$ is :

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A. $\frac{3-\log_e(2)}{3\log_e(2)}$

B.

$\frac{1}{3\log_e(2)}$

C.

$3+\log_e(2)$

D.

$\frac{3+\log_e(2)}{2+\log_e(3)}$

Answer: A

Solution:

Since $f:\{1,2,3,4\}\to\{1,e,e^2,e^3\}$ is bijective and strictly decreasing, we match the largest element of the domain to the smallest element of the codomain.

So,

$f(1)=e^3,\quad f(2)=e^2,\quad f(3)=e,\quad f(4)=1$

Hence,

$f^{-1}(1)=4,\quad f^{-1}(e)=3,\quad f^{-1}(e^2)=2,\quad f^{-1}(e^3)=1$

Now $g:\{1,e,e^2,e^3\}\to\left\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4}\right\}$ is bijective and strictly increasing.

Since

$$ 1

and

$\frac{1}{4}<\frac{1}{3}<\frac{1}{2}<1$

for g to be strictly increasing, we must assign:

$g(1)=\frac{1}{4},\quad g(e)=\frac{1}{3},\quad g(e^2)=\frac{1}{2},\quad g(e^3)=1$

Therefore,

$g^{-1}\left(\frac{1}{2}\right)=e^2$

Now,

$f^{-1}\left(g^{-1}\left(\frac{1}{2}\right)\right)=f^{-1}(e^2)=2$

So,

$\phi(x)=\left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x=2^x$

Thus the region is

$R=\{(x,y):x^2\le y\le 2^x,\ 0\le x\le 1\}$

Its area is

$\int_0^1(2^x-x^2)\,dx$

Now,

$\int 2^x\,dx=\frac{2^x}{\ln 2}$

So,

$\int_0^1 2^x\,dx = \left[\frac{2^x}{\ln 2}\right]_0^1 = \frac{2-1}{\ln 2} = \frac{1}{\ln 2}$

Also,

$\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

Hence area

$= \frac{1}{\ln 2} - \frac{1}{3}$

Taking LCM,

$= \frac{3-\ln 2}{3\ln 2}$

Therefore, the required area is

$$\boxed{\frac{3-\log_e 2}{3\log_e 2}}$$

So the correct option is $\boxed{\text{A}}$.

Question40

The area of the region $\left\{(x, y) : 0 \leq y \leq 6 - x, y^2 \geq 4x - 3, x \geq 0\right\}$ is :

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

8

B.

9

C.

12

D.

15

Answer: B

Solution:

We need to find the area of the region

$\{(x, y) : 0 \leq y \leq 6 - x,\ y^2 \geq 4x - 3,\ x \geq 0\}$.

Let us understand each condition carefully.

1. From $0 \leq y \leq 6 - x$

This means:

- $y \geq 0$, so the region lies above the x-axis.
- $y \leq 6 - x$, or

$x + y \leq 6$,

so the region lies below the line

$y = 6 - x$.

2. From $y^2 \geq 4x - 3$

Rewrite it in terms of x:

$4x - 3 \leq y^2$

$4x \leq y^2 + 3$

$x \leq \frac{y^2+3}{4}$.

So the region lies to the left of the parabola

$x = \frac{y^2+3}{4}$.

Also, we are given

$x \geq 0$.

So for a fixed y, x lies between 0 and whichever is smaller:

$x = \frac{y^2+3}{4} \quad \text{and} \quad x = 6 - y$.

So the horizontal width is

$0 \leq x \leq \min\left(\frac{y^2+3}{4}, 6 - y\right)$.

Now we find where these two curves intersect.

3. Point of intersection

Set

$\frac{y^2+3}{4} = 6 - y$.

Multiply by 4:

$y^2 + 3 = 24 - 4y$

$y^2 + 4y - 21 = 0$

$(y + 7)(y - 3) = 0$.

So,

$y = 3 \quad \text{or} \quad y = -7$.

Since $y \geq 0$, we take

$y = 3.$

Now compare the two right boundaries:

- For $0 \leq y \leq 3$, we have

$\frac{y^2+3}{4} \leq 6 - y,$

so width is

$\frac{y^2+3}{4}.$

- For $3 \leq y \leq 6$, we have

$6 - y \leq \frac{y^2+3}{4},$

so width is

$6 - y.$

Hence area is

$A = \int_0^3 \frac{y^2+3}{4}\, dy + \int_3^6 (6-y)\, dy.$

4. Evaluate the integrals

First integral:

$\int_0^3 \frac{y^2+3}{4}\, dy = \frac{1}{4} \int_0^3 (y^2+3)\, dy$

$= \frac{1}{4}\left[\frac{y^3}{3} + 3y\right]_0^3 = \frac{1}{4}\left(\frac{27}{3} + 9\right) = \frac{1}{4}(9+9) = \frac{18}{4} = \frac{9}{2}.$

Second integral:

$\int_3^6 (6-y)\, dy = \left[6y - \frac{y^2}{2}\right]_3^6$

$= (36 - 18) - \left(18 - \frac{9}{2}\right) = 18 - \frac{27}{2} = \frac{9}{2}.$

So total area is

$A = \frac{9}{2} + \frac{9}{2} = 9.$

Therefore, the correct answer is:

$\boxed{9}$

So, **Option B** is correct.

Question41

The area of the region $\left\{(x, y) : x^2 - 8x \leq y \leq -x\right\}$ is :

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

$\frac{343}{6}$

B.

$\frac{637}{6}$

C.

$\frac{437}{6}$

D.

$\frac{523}{6}$

Answer: A

Solution:

For the region

$\{(x, y) : x^2 - 8x \leq y \leq -x\},$

the lower curve is

$y = x^2 - 8x$

and the upper curve is

$y = -x.$

To find the area between the curves, we first find their points of intersection.

Set

$x^2 - 8x = -x$

$x^2 - 7x = 0$

$x(x - 7) = 0$

So, the curves intersect at

$x = 0 \quad \text{and} \quad x = 7.$

Now, on the interval $[0, 7]$, the line $y = -x$ lies above the parabola $y = x^2 - 8x$.

Hence, the required area is

$\text{Area} = \int_0^7 \left[(-x) - (x^2 - 8x)\right] dx.$

Simplify the integrand:

$(-x) - (x^2 - 8x) = -x - x^2 + 8x = 7x - x^2.$

Therefore,

Area $= \int_0^7 (7x - x^2)\,dx.$

Now integrate:

$\int (7x - x^2)\,dx = \frac{7x^2}{2} - \frac{x^3}{3}.$

Applying the limits from 0 to 7:

Area $= \left[\frac{7x^2}{2} - \frac{x^3}{3}\right]_0^7$

$= \left(\frac{7\cdot 49}{2} - \frac{343}{3}\right) - 0$

$= \frac{343}{2} - \frac{343}{3}$

Taking LCM 6:

$= \frac{1029-686}{6} = \frac{343}{6}.$

Hence, the correct answer is

$$\boxed{\frac{343}{6}}$$

So, the correct option is **A**.

Question42

Let $f : (1, \infty) \to \mathbf{R}$ be a function defined as $f(x) = \frac{x-1}{x+1}$. Let $f^{i+1}(x) = f\left(f^i(x)\right), i = 1, 2, \ldots, 25$, where $f^1(x) = f(x)$. If $g(x) + f^{26}(x) = 0, x \in (1, \infty)$, then the area of the region bounded by the curves $y = g(x), 2y = 2x - 3, y = 0$ and $x = 4$ is :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$\frac{1}{8} + \log_e 2$

B.

$\frac{1}{4} + \log_e 2$

C.

$\frac{5}{6} + 3\log_e 2$

D.

$\frac{5}{6} + \log_e 2$

Answer: A

Solution:

We first focus on the orbit periodicity of $\boldsymbol{f^n(x)}$:

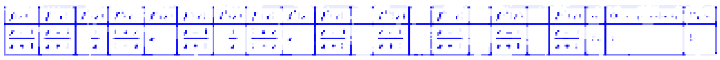
$f^1(x) = f(x) = \frac{x-1}{x+1}; x \in (1, \infty) \ldots\ldots (i)$

$f^2(x) = f\left\{f^1(x)\right\} = f\{f(x)\} = f\left(\frac{x-1}{x+1}\right) = \frac{\left(\frac{x-1}{x+1}\right)-1}{\left(\frac{x-1}{x+1}\right)+1} = -\frac{2}{2x} = -\frac{1}{x}; x \in (1, \infty) \ldots\ldots (ii)$

$\boldsymbol{f^3(x) = f\left\{f^2(x)\right\} = f\left(-\frac{1}{x}\right) = \frac{\left(-\frac{1}{x}\right)-1}{\left(-\frac{1}{x}\right)+1} = \frac{1+x}{1-x}; x \in (1, \infty) \ldots\ldots (iii)}$

$f^4(x) = f\left\{f^3(x)\right\} = f\left(\frac{1+x}{1-x}\right) = \frac{\left(\frac{1+x}{1-x}\right)-1}{\left(\frac{1+x}{1-x}\right)+1} = \frac{2x}{2} = x; x \in (1, \infty) \ldots\ldots (iv)$

$f^5(x) = f\left\{f^4(x)\right\} = f(x) = \frac{x-1}{x+1}; x \in (1, \infty) \ldots\ldots (v)$



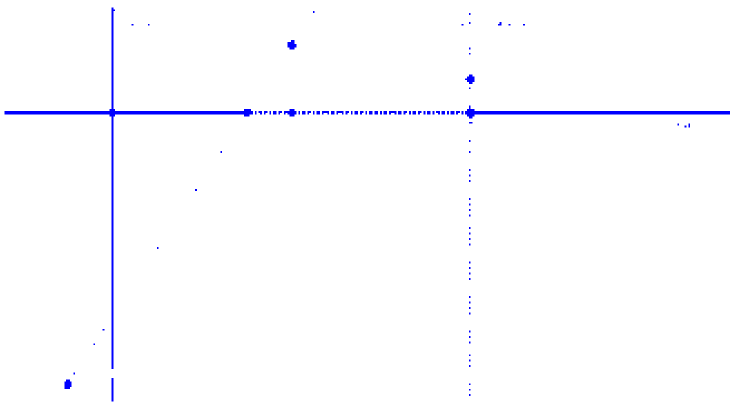
From the nature of table which completes the 25 iterations in 6 cycles with orbit periodicity 4 :

$f^{25}(x) = f(x) = f^1(x) \Rightarrow f^{26}(x) = f^2(x) = -\frac{1}{x} \ldots\ldots (vi)$

Equation (vi) exactly identifies the function $\boldsymbol{g(x)}$ defined under SOW :

$\boldsymbol{g(x) + f^{26}(x) = 0 \Rightarrow g(x) = -f^{26}(x) = \frac{1}{x} \ldots\ldots (vii)}$

Equation (vii) has identified the function $\boldsymbol{g(x)}$ and we can mathematically plot the bounded area between the curves :



$g(x)$	Slant line	Horizontal line	Vertical line
$\frac{1}{x}$	$y = x - \left(\frac{3}{2}\right)$	X-axis	$x = 4$

Pair	$\boldsymbol{g(x)}$	Slant line	Feasible	Intersection Points
1	$\boldsymbol{\frac{1}{x}}$	$\boldsymbol{y = x - \left(\frac{3}{2}\right)}$	Yes	$\boldsymbol{A, K}$

$\Rightarrow \frac{1}{x} = x - \left(\frac{3}{2}\right) \Rightarrow 2x^2 - 3x - 2 = 0 \Rightarrow x = \left(2, -\frac{1}{2}\right) \Rightarrow y = \left(\frac{1}{2}, -2\right)$

Intersections	$\boldsymbol{A}$	$\boldsymbol{K}$	Discarded point	Reason
Coordinates	$\left(2, \frac{1}{2}\right)$	$\left(-\frac{1}{2}, -2\right)$	$\boldsymbol{K}$	Falls outside the scope of bounded area

Pair	$g(x)$	Vertical line	Feasible	Intersection Points
2	$\frac{1}{x}$	$x = 4$	Yes	B

$\Rightarrow x = 4 \Rightarrow y = 0.25$

Intersection	B
Coordinate	$(4, 0.25)$

Pair	Horizontal line	Vertical line	Feasible	Intersection point
3	X-axis	$x = 4$	Yes	C

$\Rightarrow x = 4, y = 0$

Intersection	C
Coordinate	$(4, 0)$

Pair	Slant line	Horizontal line	Feasible	Intersection point
4	$y = x - \left(\frac{3}{2}\right)$	X-axis	Yes	D

$\Rightarrow y = 0 \Rightarrow x = \frac{3}{2}$

Intersection	D
Coordinate	$\left(\frac{3}{2}, 0\right)$

g(x)	X-axis	Feasible	Intersection point	Reason
1/x	y = 0	No	None	X-axis is a horizontal asymptote to g(x)
Slant line	Vertical line	Feasible	Intersection point	Reason
y = x − (3/2)	x = 4	No	(4, 5/2)	Intersects much above B and outside the scope of bounded area

Sides	Left boundary	Equation	Top boundary	Equation	Right boundary	Equation	Bottom boundary	Equation
Curves	Slant line	y = x − (3/2)	Rectangular Hyperbola	y = 1/x	Vertical line	x = 4	X-axis	y = 0

Vertices	Top left	Coordinates	Top right	Coordinates	Bottom right	Coordinates	Bottom left	Coordinates
Points	A	A (2, 1/2)	B	B (4, 1/4)	C	C (4, 0)	D	D (3/2, 0)

We can visualize that both A and D lie on the slant line but A is ahead of D along the X-axis. From A we can drop a $\perp$ on the X-axis at M to form a right angled ΔAMD. From A and D the remaining area upto the vertical $x = 4$ lies under the rectangular hyperbola $g(x) = \frac{1}{x}$:

$$\begin{aligned}\text{Area } &= \left(\frac{1}{2}\right) \cdot (DM) \cdot (AM) + \int_2^4 g(x)dx = \left(\frac{1}{2}\right) \cdot (2 - 1 \cdot 5) \cdot \left(\frac{1}{2}\right) + \int_2^4 \left(\frac{1}{x}\right) dx \\ &= \frac{1}{8} + [\ln|x|]_2^4 \\ &= \frac{1}{8} + \ln 2\end{aligned}$$

Question43

Let $y = f(x)$ be the solution of the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}, -1 < x < 1$ such that $f(0) = 0$. If

$6 \int_{-1/2}^{1/2} f(x) \mathrm{d}x = 2\pi - \alpha$ **then** α^2 **is equal to _________ .**

JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 27

Solution:

I.F. $\mathrm{e}^{-\frac{1}{2}\int \frac{2\mathrm{x}}{1-\mathrm{x}^2}\mathrm{dx}} = \mathrm{e}^{-\frac{1}{2}\ln\left(1-\mathrm{x}^2\right)} = \sqrt{1-\mathrm{x}^2}$

$y \times \sqrt{1-x^2} = \int \left(x^6 + 4x\right) dx = \frac{x^7}{7} + 2x^2 + c$

Given $y(0) = 0 \Rightarrow c = 0$

$y = \frac{\frac{x^7}{7}+2x^2}{\sqrt{1-x^2}}$

Now, $6\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\frac{x^7}{7}+2x^2}{\sqrt{1-x^2}} dx = 6\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{2x^2}{\sqrt{1-x^2}} dx$

$= 24\int_0^{\frac{1}{2}} \frac{\mathrm{x}^2}{\sqrt{1-\mathrm{x}^2}} \mathrm{dx}$

Put $x = \sin\theta$

$dx = \cos\theta d\theta$

$= 24\int_0^{\frac{\pi}{6}} \frac{\sin^2\theta}{\cos\theta} \cos\theta d\theta$

$= 24\int_0^{\frac{\pi}{6}} \left(\frac{1-\cos 2\theta}{2}\right) d\theta = 12\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{6}}$

$= 12\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)$

$= 2\pi - 3\sqrt{3}$

$\alpha^2 = (3\sqrt{3})^2 = 27$

Question44

Let f **be a differentiable function such that** $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2) f(t) dt, x \geq 0$**. Then** $f(2)$ **is equal to _________ .**

JEE Main 2025 (Online) 24th January Morning Shift

Answer: 19

Solution:

Differentiate both sides

$4(x+2)f(x)+2(x+2)^2 f'(x)-6(x+2)=10(x+2)f(x)$

$2(x+2)^2 f'(x)-6(x+2)f(x)=6(x+2)$

$(x+2)\frac{dy}{dx}-3y=3$

$\int \frac{dy}{dx}=3\int \frac{dx}{x+2}$

$\ln(y+1)=3\ln(x+2)+C$

$(y+1)=C(x+2)^3$

$f(0)=\frac{3}{2}$

$f(2)=19$

Question45

Let $y=y(x)$ be the solution of the differential equation

$2\cos x\frac{\mathrm{dy}}{\mathrm{d}x}=\sin 2x-4y\sin x, x\in\left(0,\frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right)=0$, then $y'\left(\frac{\pi}{4}\right)+y\left(\frac{\pi}{4}\right)$ is equal to _________.

JEE Main 2025 (Online) 24th January Evening Shift

Answer: 1

Solution:

$\frac{dy}{dx}+2y\tan x=\sin x$

I.F. $=e^{2\int \tan x dx}=\sec^2 x$

$y\sec^2 x=\int \frac{\sin x}{\cos^2 x}dx$

$=\int \tan x\sec x dx$

$=\sec x+C$

$C=-2$
$y=\cos x-2\cos^2 x$
$y\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}-1$
$y'=-\sin x+4\cos^2 x\sin x$
$y'\left(\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}}+2$
$y'\left(\frac{\pi}{4}\right)+y\left(\frac{\pi}{4}\right)=1$

Question46

If $y=y(x)$ is the solution of the differential equation, $\sqrt{4-x^2}\frac{\mathrm{d}y}{\mathrm{d}x}=\left(\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2-y\right)\sin^{-1}\left(\frac{x}{2}\right), -2\leq x\leq 2, y(2)=\frac{\pi^2-8}{4}$, then $y^2(0)$ is equal to ___________.

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 4

Solution:

$\frac{dy}{dx}+\frac{\left(\sin^{-1}\frac{x}{2}\right)}{\sqrt{4-x^2}}y=\frac{\left(\sin^{-3}\frac{x}{2}\right)^3}{\sqrt{4-x^2}}$
$ye^{\frac{\left(\sin^{-1}\frac{x}{2}\right)^2}{2}}=\int\frac{\left(\sin^{-3}\frac{x}{2}\right)^3}{4-x^2}e^{\frac{\left(\sin^{-1}\frac{x}{2}\right)^2}{2}}dx$
$y=\left(\sin^{-1}\frac{x}{2}\right)^2-2+c\cdot e^{\frac{-\left(\sin^{-1}\frac{x}{2}\right)^2}{2}}$
$y(2)=\frac{\pi^2}{4}-2\Rightarrow c=0$
$y(0)=-2$

Question47

Let $y = y(x)$ be the solution of the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y\sec^2 x = 2\sec^2 x + 3\tan x \cdot \sec^2 x$ such that $y(0) = \frac{5}{4}$. Then $12\left(y\left(\frac{\pi}{4}\right) - \mathrm{e}^{-2}\right)$ is equal to____________________

JEE Main 2025 (Online) 2nd April Evening Shift

Answer: 21

Solution:

$$\frac{dy}{dx} + 2y\sec^2 x = 2\sec^2 x + 3\tan x\sec^2 x$$
$$\text{I.F. } = e^{\int 2\sec^2 x dx}$$
$$\text{I.F. } = e^{2\tan x}$$
$$y \cdot e^{2\tan x} = \int e^{2\tan x}(2 + 3\tan x)\sec^2 x dx$$
$$\text{Put } \tan x = u$$
$$\sec^2 x dx = du$$
$$y \cdot e^{2u} = \int e^{2u}(2+3u)du$$
$$y \cdot e^{2u} \Rightarrow \frac{2e^{2u}}{2} + 3\int e^{2u} \cdot udu$$
$$y \cdot e^{2u} = e^{2u} + 3\left[\frac{ue^{2u}}{2} - \int \frac{e^{2u}}{2}\right]$$
$$ye^{2u} = e^{2u} + 3\left[\frac{ue^{2u}}{2} - \frac{e^{2u}}{4}\right] + C$$
$$ye^{2\tan x} = e^{2\tan x} + 3\left[\frac{\tan xe^{2\tan x}}{2} - \frac{e^{2\tan x}}{4}\right] + C$$

$$F(0) = \frac{5}{4}$$
$$\frac{5}{4} = 1 - \frac{3}{4} + C$$
$$\frac{5}{4} - \frac{1}{4} = C$$
$$1 = C$$
$$y = 1 + 3\left(\frac{\tan x}{2} - \frac{1}{4}\right) + 1 \cdot e^{-2\tan x}$$
$$y\left(\frac{\pi}{4}\right) = 1 + 3\left(\frac{1}{2} - \frac{1}{4}\right) + \frac{1}{e^2}$$
$$y\left(\frac{\pi}{4}\right) = \frac{7}{4} + \frac{1}{e^2}$$
$$12\left(y\left(\frac{x}{4}\right) - \frac{1}{e^2}\right) = 12\left(\frac{7}{4} + \frac{1}{e^2} - \frac{1}{e^2}\right) = 21$$

Question48

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a twice differentiable function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbf{R}$. If $f'(0) = 4\mathrm{a}$ and f satisfies $f''(x) - 3\mathrm{a}f'(x) - f(x) = 0, \mathrm{a} > 0$, then the area of the region $\mathrm{R} = \{(x, y) \mid 0 \leq y \leq f(ax), 0 \leq x \leq 2\}$ is :

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $\mathrm{e}^2 - 1$

B. $e^4 + 1$

C. $\mathrm{e}^2 + 1$

D. $e^4 - 1$

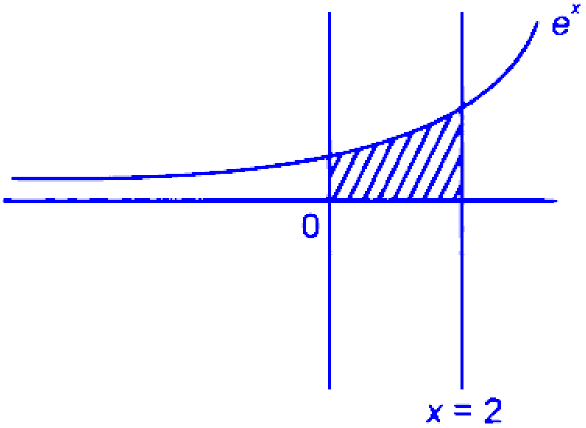
Answer: A

Solution:

$\mathrm{f(x+y) = f(x)f(y)}$
$\Rightarrow \mathrm{f(x) = e^{\lambda x} f'(0) = 4a}$
$\Rightarrow \mathrm{f'(x) = \lambda e^{\lambda x}} \Rightarrow \lambda = 4\mathrm{a}$

So, $\mathrm{f(x) = e^{4ax}}$

$\mathrm{f''(x) - 3af'(x) - f(x) = 0}$
$\Rightarrow \lambda^2 - 3\mathrm{a}\lambda - 1 = 0$
$\Rightarrow 16\mathrm{a}^2 - 12\mathrm{a}^2 - 1 = 0 \Rightarrow 4\mathrm{a}^2 = 1 \Rightarrow \mathrm{a} = \frac{1}{2}$



$\mathrm{F(x) = e^{2x}}$

$$\text{Area } = \int_0^2 \mathrm{e^x dx = e^2 - 1}$$

Question49

Let $f(x)$ be a real differentiable function such that $f(0)=1$ and $f(x+y)=f(x)f'(y)+f'(x)f(y)$ for all $x, y \in \mathbf{R}$. Then $\sum\limits_{n=1}^{100} \log_e f(n)$ is equal to :

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. 2406

B. 5220

C. 2525

D. 2384

Answer: C

Solution:

$$
\begin{aligned}
& f(x+y)=f(x)f'(y)+f'(x)f(x) \\
& \text{ Put } =x=y=0 \\
& f(0)=f(0)f'(0)+f'(0)f(0) \\
& f'(0)=\frac{1}{2} \\
& \text{ Put } y=0 \\
& f(x)=f(x)f'(0)+f'(x)f(0) \\
& f(x)=\frac{1}{2}f(x)+f'(x) \\
& f'(x)=\frac{f(x)}{2}
\end{aligned}
$$

$\frac{dy}{dx}=\frac{y}{2} \Rightarrow \int \frac{dy}{y}=\int \frac{dx}{2}$

$\Rightarrow \ln y=\frac{x}{2}+c$

$$
\begin{aligned}
& \because \mathrm{f}(0)=1 \Rightarrow \mathrm{C}=0 \\
& \ell \mathrm{ny}=\frac{\pi}{2} \Rightarrow \mathrm{f}(\mathrm{x})=\mathrm{e}^{\mathrm{x}/2} \\
& \ln \mathrm{f}(\mathrm{n})=\frac{\mathrm{n}}{2} \\
& \sum_{\mathrm{n}=1}^{100} \ell \mathrm{f}(\mathrm{n})=\frac{1}{2} \sum_{\mathrm{n}=1}^{100} \mathrm{n}=\frac{5050}{2} \\
& =2525
\end{aligned}
$$

Question50

Let $x = x(y)$ be the solution of the differential equation $y^2 \, \mathrm{d}x + \left(x - \frac{1}{y}\right)\mathrm{d}y = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is :

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $\frac{3}{2} + \mathrm{e}$

B. $\frac{1}{2} + \mathrm{e}$

C. $3 + e$

D. $3 - e$

Answer: D

Solution:

$$y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$$
$$y^2 dx = \left(\frac{1}{y} - x\right) dy$$
$$\Rightarrow y^2 \frac{dx}{dy} = \frac{1}{y} - x$$
$$\Rightarrow \frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$$
$$\text{I.F. } = e^{\int \frac{1}{y^2} dy} = e^{\frac{-1}{y}}$$

$\therefore$ Solution is

$$xe^{\frac{-1}{y}} = \int e^{-\frac{1}{y}} \times \frac{1}{y^3} dy + C$$

Let $\frac{-1}{y} = t$

$\Rightarrow \frac{1}{y^2} dy = dt$

$$\Rightarrow xe^{-\frac{1}{y}} = -\int e^t t dt + C$$
$$\Rightarrow xe^{-\frac{1}{y}} = -e^t(t-1) + C$$
$$\Rightarrow xe^{-\frac{1}{y}} = -e^{\frac{-1}{y}}\left(\frac{-1}{y} - 1\right) + C$$

$x(1)=1$
$\Rightarrow e^{-1}=-e^{-1}(-2)+C$
$\Rightarrow C=-e^{-1}$
$\Rightarrow x=\frac{1}{y}+1-e^{-1+\frac{1}{y}}$
$x\left(\frac{1}{2}\right)=3-e$

Question51

If $x=f(y)$ is the solution of the differential equation $\left(1+y^2\right)+\left(x-2\mathrm{e}^{\tan^{-1} y}\right) \frac{\mathrm{d} y}{\mathrm{~d} x}=0, y \in\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $f(0)=1$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. $\mathrm{e}^{\pi/4}$

B. $e^{\pi/12}$

C. $\mathrm{e}^{\pi/6}$

D. $e^{\pi/3}$

Answer: C

Solution:

$\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{2e^{\tan^{-1} y}}{1+y^2}$
I.F. $=e^{\tan^{-1} y}$
$x^{\tan^{-1} y}=\int \frac{2\left(e^{\tan^{-1} y}\right)^2 dy}{1+y^2}$
Put $\tan^{-1} y=t, \frac{dy}{1+y^2}=dt$

$$x e^{\tan^{-1} y}=\int 2 e^{2 t} d t$$
$$x e^{\tan^{-1} y}=e^{2 \tan^{-1} y}+c$$
$$x=e^{\tan^{-1} y}+c e^{-\tan^{-1} y}$$
$$\because y=0, x=1$$
$$1=1+c \Rightarrow c=0$$
$$y=\frac{1}{\sqrt{3}}, x=e^{\pi / 6}$$

Question52

Let a curve $y=f(x)$ pass through the points $(0,5)$ and $(\log_e 2, k)$. If the curve satisfies the differential equation $2(3+y) e^{2 x} d x-\left(7+e^{2 x}\right) d y=0$, then k is equal to

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. 32

B. 8

C. 4

D. 16

Answer: B

Solution:

$$\frac{d y}{d x}=\frac{2(3+y) \cdot e^{2 x}}{7+e^{2 x}}$$
$$\frac{d y}{d x}-\frac{2 y \cdot e^{2 x}}{7+e^{2 x}}=\frac{6 \cdot e^{2 x}}{7+e^{2 x}}$$
$$\text{I.F. }=e^{-\int \frac{2 e^{2 x}}{7+e^{2 x}}}=\frac{1}{7+e^{2 x}}$$
$$\therefore y \cdot \frac{1}{7+\mathrm{e}^{2 \mathrm{x}}}=\int \frac{6 \mathrm{e}^{2 \mathrm{x}}}{\left(7+3^{2 x}\right)^2} \mathrm{dx}$$

$$\frac{\mathrm{y}}{7+\mathrm{e}^{2\mathrm{x}}}=\frac{-3}{7+\mathrm{e}^{2\mathrm{x}}}+C$$
$$(0,5) \Rightarrow \frac{5}{8}=\frac{-3}{8}+C \Rightarrow C=1$$
$$\therefore y=-3+7+\mathrm{e}^{2\mathrm{x}}$$
$$y=\mathrm{e}^{2\mathrm{x}}+4$$
$$\therefore \mathrm{k}=8$$

Question53

Let $x=x(y)$ be the solution of the differential equation $y=\left(x-y\frac{\mathrm{d}x}{\mathrm{d}y}\right)\sin\left(\frac{x}{y}\right), y>0$ and $x(1)=\frac{\pi}{2}$. Then $\cos(x(2))$ is equal to :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

A. $2\left(\log_e 2\right)-1$

B. $1-2(\log_e 2)^2$

C. $1-2\left(\log_{\mathrm{e}} 2\right)$

D. $2(\log_e 2)^2-1$

Answer: D

Solution:

$$y d y=(x d y-y d x) \sin \left(\frac{x}{y}\right)$$
$$\frac{d y}{y}=\left(\frac{x d y-y d x}{y^2}\right) \sin \left(\frac{x}{y}\right)$$
$$\frac{d y}{y}=\sin \left(\frac{x}{y}\right) d\left(-\frac{x}{y}\right)$$
$$\Rightarrow \quad \ell n y=\cos \frac{x}{y}+C$$

$$x(1)=\frac{\pi}{2} \Rightarrow 0=\cos \frac{\pi}{2}+C \Rightarrow C=0$$
$$\text{ચ}\mathrm{ny}=\cos \frac{x}{y}$$

$$\text{but } y = 2 \Rightarrow \cos\frac{x}{2} = \ln 2$$
$$\begin{aligned}\cos x &= 2\cos^2\frac{x}{2} - 1 \\ &= 2(\ln 2)^2 - 1\end{aligned}$$

Question54

Let $\mathrm{y} = \mathrm{y}(\mathrm{x})$ be the solution of the differential equation $\left(xy - 5x^2\sqrt{1+x^2}\right)dx + \left(1+x^2\right)dy = 0, y(0) = 0$. Then $y(\sqrt{3})$ is equal to

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. $\frac{5\sqrt{3}}{2}$

B. $\sqrt{\frac{15}{2}}$

C. $\sqrt{\frac{14}{3}}$

D. $2\sqrt{2}$

Answer: A

Solution:

$$\left(1+x^2\right)\frac{dy}{dx} + xy = 5x^1\sqrt{1+x^2}$$
$$\frac{dy}{dx} + \frac{xy}{1+x^2} = \frac{5x^2}{\sqrt{1+x^2}}$$
$$\therefore \text{ I.F. } = e^{\int \frac{x}{1+x^2}dx} = e^{\frac{\ln\left(1+x^2\right)}{2}} = \sqrt{1+x^2}$$
$$\therefore y\sqrt{1+x^2} = \int \frac{5x^2}{\sqrt{1+x^2}} \cdot \sqrt{1+x^2}dx$$
$$\therefore y\sqrt{1+x^2} = \int \frac{5x^2}{\sqrt{1+x^2}} \cdot \sqrt{1+x^2}dx$$
$$y\sqrt{1+x^2} = \frac{5x^3}{3} + C$$

$\because y(0) = 0 \Rightarrow 0 = 0 + C \Rightarrow C = 0$
$\therefore y = \dfrac{5x^3}{3\sqrt{1+x^2}}$
$y(\sqrt{3}) = \dfrac{15\sqrt{3}}{32} = \dfrac{5\sqrt{3}}{2}$

Question55

Let for some function y $= f(x), \int_0^x tf(t)dt = x^2 f(x), x > 0$ and $f(2) = 3$. Then $f(6)$ is equal to

JEE Main 2025 (Online) 28th January Morning Shift

Options:

A. 1

B. 6

C. 2

D. 3

Answer: A

Solution:

$\int_0^x \mathrm{tf(t)dt} = \mathrm{x}^2 + (\mathrm{x}), \mathrm{x} > 0$
Diff. both side w.r. to x
$xf(x) = x^2 f'(x) + 2xf(x)$
$-xf(x) = x^2 f'(x)$
$\int \dfrac{f'(x)}{f(x)} dx = \int \dfrac{-1}{x} dx$
$\mathrm{logdf}(x) = -\log x + \log c$
$f(x) = \dfrac{c}{x}$
$\mathrm{f}(2) = 3 \Rightarrow 3 = \dfrac{\mathrm{c}}{2} \Rightarrow \mathrm{c} = 6$

$f(x)=\dfrac{6}{x}$
$f(6)=1 \quad \therefore (1)$

Question56

Let *y* = *y(x)* be the solution of the differential equation :

$\cos x(\log_e(\cos x))^2 dy+(\sin x-3y\sin x\log_e(\cos x))dx=0$, x ∈ (0, $\frac{\pi}{2}$). If $y(\frac{\pi}{4})=-\frac{1}{\log_e 2}$, then $y(\frac{\pi}{6})$ is equal to :

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

$\frac{2}{\log_e(3)-\log_e(4)}$

B.

$-\frac{1}{\log_e(4)}$

C.

$\frac{1}{\log_e(4)-\log_e(3)}$

D.

$\frac{1}{\log_e(3)-\log_e(4)}$

Answer: D

Solution:

$\cos x(\ln(\cos x))^2 dy+(\sin x-3y(\sin x)\ln(\cos x))dx=0$
$\cos x(\ln(\cos x))^2\dfrac{dy}{dx}-3\sin x\cdot\ln(\cos x)y=-\sin x$

$$\frac{dy}{dx}-\frac{3\tan x}{\ln(\cos x)}y=\frac{-\tan x}{(\ln(\cos x))^2}$$
$$\frac{dy}{dx}+\frac{3\tan x}{\ln(\sec x)}y=\frac{-\tan x}{(\ln(\sec x))^2}$$
$$\text{I.F. }=e^{\int\frac{3\tan x}{\ln(\sec x)}dx}=(\ln(\sec x))^3$$
$$y\times(\ln(\sec x))^3=-\int\frac{\tan x}{(\ln(\sec x))^2}(\ln(\sec x))^3dx$$
$$y\times(\ln(\sec x))^3=-\frac{1}{2}(\ln(\sec x))^2+C$$
$$\text{Given }: x=\frac{\pi}{4}, y=-\frac{1}{\ln 2}$$
$$\frac{-1}{\ln 2}\times(\ln\sqrt{2})^3=-\frac{1}{2}\times(\ln\sqrt{2})^2+C$$
$$\Rightarrow\frac{-1}{8\ln 2}\times(\ln 2)^3=\frac{-1}{2}\times\frac{1}{4}(\ln 2)^2+C$$
$$-\frac{1}{8}(\ln 2)^2=\frac{-1}{8}(\ln 2)^2+C$$
$$\Rightarrow C=0$$
$$\therefore y(\ln(\sec x))^3=\frac{-1}{2}(\ln(\sec x))^2+0$$
$$y=\frac{-1}{2\ln(\sec x)}$$
$$y=\frac{1}{2\ln(\cos x)}$$
$$\therefore y\left(\frac{\pi}{6}\right)=\frac{1}{2\ln\left(\cos\frac{\pi}{6}\right)}$$
$$=\frac{1}{2\ln\left(\frac{\sqrt{3}}{2}\right)}$$
$$=\frac{1}{2\left(\frac{1}{2}\ln 3-\ln 2\right)}$$
$$=\frac{1}{\ln 3-\ln 4}$$

Question57

If for the solution curve $y=f(x)$ of the differential equation $\frac{dy}{dx}+(\tan x)y=\frac{2+\sec x}{(1+2\sec x)^2}$, $x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$, $f\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A. $\frac{5-\sqrt{3}}{2\sqrt{2}}$

B.

$\frac{4-\sqrt{2}}{14}$

C.

$\frac{9\sqrt{3}+3}{10(4+\sqrt{3})}$

D.

$\frac{\sqrt{3}+1}{10(4+\sqrt{3})}$

Answer: B

Solution:

$$\text{If } \mathrm{e}^{\int \tan x dx} = \mathrm{e}^{\ln(\sec x)} = \sec x$$
$$\therefore y \cdot \sec x = \int \left\{ \frac{2+\sec x}{(1+2\sec x)^2} \right\} \sec x dx$$
$$= \int \frac{2\cos x + 1}{(\cos x + 2)^2} dx \text{ Let } \cos x = \frac{1-t^2}{1+t^2}$$
$$= \int \frac{2\left(\frac{1-t^2}{1+t^2}\right) + 1}{\left(\frac{1-t^2}{1+t^2} + 2\right)^2} 2dt$$
$$= \int \frac{2-2t^2+1+t^2}{(1-t^2+2+2t^2)^2} \times 2dt$$
$$= 2\int \frac{3-t^2}{(t^2+3)^2} dt$$
$$\text{Let } \mathrm{t} + \frac{3}{\mathrm{t}} = \mathrm{u}$$
$$\left(1 - \frac{3}{\mathrm{t}^2}\right)\mathrm{dt} = \mathrm{du}$$

$= -2\int \frac{\mathrm{du}}{\mathrm{u}^2}$

$$y \cdot (\sec x) = \frac{2}{u} + c$$
$$y \cdot \sec x = \frac{2}{t + \frac{3}{t}} + c \quad \text{..... (I)}$$

$$\text{At } \mathrm{x} = \frac{\pi}{3}, \mathrm{t} = \tan\frac{\mathrm{x}}{2} = \frac{1}{\sqrt{3}}$$
$$2. \frac{\sqrt{3}}{10} = \frac{2}{\frac{1}{\sqrt{3}} + 3\sqrt{3}} + \mathrm{c}$$
$$2. \frac{\sqrt{3}}{10} = \frac{2\sqrt{3}}{10} + \mathrm{c} \Rightarrow \mathrm{C} = 0$$
$$\text{At } \mathrm{x} = \frac{\pi}{4}, \mathrm{t} = \tan\frac{\mathrm{x}}{2} = \sqrt{2} - 1$$

$$\therefore y \cdot \sqrt{2} = \frac{2}{\sqrt{2} - 1 + \frac{3}{\sqrt{2}-1}}$$

$$y \cdot \sqrt{2} = \frac{2(\sqrt{2}-1)}{6 - 2\sqrt{2}}$$

$$y = \frac{\sqrt{2}(\sqrt{2}-1)}{2(3-\sqrt{2})} = \frac{1}{\sqrt{2}} \times \frac{2\sqrt{2}-1}{7}$$

$$= \frac{4-\sqrt{2}}{14}$$

Question58

Let g be a differentiable function such that $\int_0^x g(t)dt = x - \int_0^x \mathrm{t}g(t)dt, x \geq 0$ and let $y = y(x)$ satisfy the differential equation $\frac{dy}{dx} - y\tan x = 2(x+1)\sec x g(x), x \in \left[0, \frac{\pi}{2}\right)$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. $\frac{4\pi}{3}$

B. $\frac{2\pi}{3}$

C. $\frac{2\pi}{3\sqrt{3}}$

D. $\frac{4\pi}{3\sqrt{3}}$

Answer: A

Solution:

To solve the given problem, let's start by considering the equation:

$\int_0^x g(t)\, dt = x - \int_0^x \tan g(t)\, dt$

Differentiate both sides with respect to x:

$g(x) = 1 - xg(x)$

Rearranging gives:

$g(x)(1+x) = 1 \implies g(x) = \frac{1}{1+x}$

Now, consider the differential equation:

$\frac{dy}{dx} - y\tan x = 2(x+1)\sec x g(x)$

Substitute $g(x) = \frac{1}{1+x}$:

$\frac{dy}{dx} - y \tan x = 2(x+1) \sec x \cdot \frac{1}{1+x}$

This simplifies to:

$\frac{dy}{dx} - y \tan x = 2 \sec x$

To solve this, we use an Integrating Factor (I.F):

$\text{I.F} = e^{\int \tan x \, dx} = e^{-\ln \cos x} = \cos x$

Multiply the entire differential equation by the Integrating Factor:

$\cos x \cdot \frac{dy}{dx} - y \cdot \sin x = 2$

This implies:

$\frac{d}{dx}(y \cos x) = 2$

Integrate both sides with respect to x:

$y \cos x = \int 2 \, dx = 2x + c$

Given the initial condition $y(0) = 0$:

$0 \cdot 1 = 2 \cdot 0 + c \implies c = 0$

Thus:

$y \cos x = 2x$

Evaluate at $x = \frac{\pi}{3}$:

$y \cdot \frac{1}{2} = 2 \cdot \frac{\pi}{3}$

Solving for y:

$y = \frac{4\pi}{3}$

Therefore, $y\left(\frac{\pi}{3}\right) = \frac{4\pi}{3}$.

Question59

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} + 3\left(\tan^2 x\right)y + 3y = \sec^2 x, y(0) = \frac{1}{3} + e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. $\frac{4}{3}$

B. $\frac{2}{3} + e^3$

C. $\frac{4}{3}+e^3$

D. $\frac{2}{3}$

Answer: A

Solution:

$$\frac{dy}{dx}+3\left(\tan^2 x\right)y+3y=\sec^2 x$$
$$\Rightarrow \frac{dy}{dx}+3\sec^2 xy=\sec^2 x$$
$$\text{I.F } =e^{\int 3\sec^2 x dx}$$
$$=e^{3\tan x}$$

$$y\cdot e^{\tan x}=\int e^{3\tan x}\cdot\sec^2 x dx+c$$
$$y\cdot e^{3\tan x}=\frac{e^{3\tan x}}{3}+c$$
$$\text{Also } f(0)=\frac{1}{3}+e^3$$
$$\Rightarrow\left(\frac{1}{3}+e^3\right)=\frac{1}{3}+c$$
$$\Rightarrow c=e^3$$
$$\therefore y\cdot e^{3\tan x}=\frac{e^{3\tan x}}{3}+e^3$$
$$\text{Put } x=\frac{\pi}{4}$$

$ye^3=\frac{e^3}{3}+e^3\Rightarrow y=\frac{4}{3}$

Question60

If a curve $y=y(x)$ passes through the point $\left(1,\frac{\pi}{2}\right)$ and satisfies the differential equation $\left(7x^4\cot y-\mathrm{e}^x\operatorname{cosec} y\right)\frac{\mathrm{d}x}{\mathrm{d}y}=x^5, x\geq 1$, then at $x=2$, the value of $\cos y$ is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. $\frac{2\mathrm{e}^2+\mathrm{e}}{64}$

B. $\frac{2\mathrm{e}^2-\mathrm{e}}{64}$

C. $\frac{2\mathrm{e}^2-\mathrm{e}}{128}$

D. $\frac{2e^2+e}{128}$

Answer: C

Solution:

$$\left(7x^4 \cot y - e^x \operatorname{cosec} y\right)\frac{dx}{dy} = x^5$$
$$x^5 \frac{dy}{dx} - 7x^4 \cot y = -e^x \operatorname{cosec} y$$
$$\frac{dy}{dx} - \frac{7}{x}\cot y = -\frac{e^x}{x^5}\operatorname{cosec} y$$
$$\sin y \frac{dy}{dx} - \frac{7}{x}\cos y = -\frac{e^x}{x^5}$$
$$\text{Let } -\cos y = t$$
$$\sin y \frac{dy}{dx} = \frac{dt}{dx}$$
$$\therefore \frac{dt}{dx} + \frac{7}{x}t = -\frac{e^x}{x^5}$$
$$\therefore \text{ I.F. } = e^{\int \frac{7}{x}dx} = x^7$$
$$t \cdot x^7 = \int \frac{-e^x}{x^5} \cdot x^7 dx$$
$$-\cos y \cdot x^7 = -\int e^x x^2 dx$$
$$\cos y x^7 = e^x\left(x^2 - 2x + 2\right) + c$$
$$\because \quad x = 1 \text{ then } y = \frac{\pi}{2} \Rightarrow c = -e$$
$$\therefore \quad \cos y \cdot x^7 = e^x\left(x^2 - 2x + 2\right) - e$$
$$\text{When } x = 2 \text{ then } \cos y = \frac{2e^2 - e}{128}$$

Question61

Let $y = y(x)$ be the solution curve of the differential equation

$x\left(x^2 + e^x\right)dy + \left(\mathrm{e}^x(x-2)y - x^3\right)\mathrm{d}x = 0, x > 0$, passing through the point $(1, 0)$. Then $y(2)$ is equal to :

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. $\frac{2}{2+e^2}$

B. $\frac{4}{4-e^2}$

C. $\frac{4}{4+e^2}$

D. $\frac{2}{2-e^2}$

Answer: C

Solution:

$$x\left(x^2+e^x\right)dy+\left(e^x(x-2)y-x^3\right)dx=0$$
$$\frac{dy}{dx}+\frac{e^x(x-2)}{x\left(x^2+e^x\right)}y=\frac{x^3}{x\left(x^2+e^x\right)}$$
$$\text{I.F. }=e^{\iint\frac{e^x(x-2)}{x\left(x^2+e^x\right)}dx}$$
$$=e^{\int\frac{e^x+2x}{e^x+x^2}dx-\int\frac{2}{x}dx}$$
$$=e^{\ln e^{-x}+x^2-2\ln x}$$

$$=\frac{e^x+x^2}{x^2}$$
$$\therefore\quad y\left(\frac{e^2+x^2}{x^2}\right)=\int dx+c$$
$$\Rightarrow\quad y\left(\frac{e^x+x^2}{x^2}\right)=x+c$$

Also, $y(1)=0$

$$\Rightarrow\quad c=-1$$
$$\therefore\quad y\left(\frac{e^2+x^2}{x^2}\right)=x-1$$

Hence, $y(2)=\frac{4}{e^2+4}$

Question62

Let y = y(x) be the solution of the differential equation $(x^2+1)y'-2xy=(x^4+2x^2+1)\cos x$,

$y(0)=1$. Then $\int\limits_{-3}^{3}y(x)\,dx$ is :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

36

B.

24

C.

18

D.

30

Answer: B

Solution:

$$\left(x^2+1\right)\frac{dy}{dx}-2xy=\left(x^4+2x^2+1\right)\cos x$$

$$\frac{dy}{dx}-\left(\frac{2x}{x^2+1}\right)y=\frac{\left(x^2+1\right)^2\cos x}{cx^2+1}=\left(x^2+1\right)\cos x$$

(Linear D.E)

$$\mathrm{P}=\frac{-2\mathrm{x}}{\mathrm{x}^2+1},\mathrm{Q}=\left(\mathrm{x}^2+1\right)\cos\mathrm{x}$$

$$\text{I.F }=\mathrm{e}^{\int Pdx}=\mathrm{e}^{\int\frac{-2x}{x^2+1}dx}=\frac{1}{x^2+1}$$

$$y\cdot\frac{1}{x^2+1}=\int\left(x^2+1\right)\cos x\cdot\frac{1}{x^2+1}dx$$

$$\frac{y}{x^2+1}=\sin x+c\Rightarrow y\cos=1\Rightarrow c=1$$

$$y=\left(x^2+1\right)(\sin x+1)$$

$$\int_{-3}^{3}ydx=\int_{-3}^{3}\left(x^2+1\right)(\sin x+1)$$

$$dx=\int_{-3}^{3}x^2\sin x+x^2\sin x+1dx$$

$$\Rightarrow\int_{-3}^{3}x^2\sin xdx+\int_{-3}^{3}x^2dx+\int_{-3}^{3}\sin xdx+\int_{-3}^{3}1dx$$

$$=0+18+0+6=24$$

Question63

Let $f(x) = x - 1$ and $g(x) = e^x$ for $x \in \mathbb{R}$. If $\frac{dy}{dx} = \left(e^{-2\sqrt{x}} g\left(f(f(x))\right) - \frac{y}{\sqrt{x}}\right)$, $y(0) = 0$, then $y(1)$ is

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

$\frac{1-e^3}{e^4}$

B.

$\frac{e-1}{e^4}$

C.

$\frac{1-e^2}{e^4}$

D.

$\frac{2e-1}{e^3}$

Answer: B

Solution:

$f(x) = x - 1$
$f(f(x)) = f(x) - 1 = x - 1 - 1 = x - 2$
$g(f(f(x))) = e^{x-2}$
$\therefore \frac{dy}{dx} = e^{-2\sqrt{x}} \times e^{x-2} - \frac{1}{\sqrt{x}} y$
$\frac{dy}{dx} + \frac{1}{\sqrt{x}} y = e^{x-2\sqrt{x}-2}$ which is L.D.E
I.F. $= e^{\int \frac{dy}{\sqrt{x}}} = e^{2\sqrt{x}}$

Its solution is

$$y \times e^{2\sqrt{x}} = \int e^{2\sqrt{x}} \times e^{x-2\sqrt{x}-2} dx + c$$

$$y \times e^{2\sqrt{x}} = \int e^{x-2} dx + c$$

$$y \times e^{2\sqrt{x}} = e^{x-2} + c$$

Given $\mathrm{x} = 0, \mathrm{y} = 0 \Rightarrow 0 = \mathrm{e}^{-2} + \mathrm{c} \quad ; \mathrm{c} = -\mathrm{e}^{-2}$

$\therefore \mathrm{y} \times \mathrm{e}^{2\sqrt{x}} = \mathrm{e}^{\mathrm{x}-2} - \mathrm{e}^{-2}$

when $\mathrm{x} = 1, \mathrm{y} \times \mathrm{e}^{2} = \mathrm{e}^{-1} - \mathrm{e}^{-2}$

$$y = \frac{e^{-1}-e^{-2}}{e^{2}} = \frac{\frac{1}{e}-\frac{1}{e^{2}}}{e^{2}} = \frac{e^{2}-e}{e^{5}} = \frac{e-1}{e^{4}}$$

Option (1) is correct

Question64

Let x = x(t) and y = y(t) be solutions of the differential equations $\frac{dx}{dt} + ax = 0$ and $\frac{dy}{dt} + by = 0$ respectively, a, b ∈ R. Given that

x(0) = 2; y(0) = 1 and 3y(1) = 2x(1), the value of t, for which x(t) = y(t), is : [27-Jan-2024 Shift 1]

Options:

A. $\log_{\frac{2}{3}} 2$

B. $\log_4 3$

C. $\log_3 4$

D. $\log_{\frac{4}{3}} 2$

Answer: D

Solution:

$$\frac{dx}{dt} + ax = 0$$

$$\frac{dx}{x} = -a\,\text{dt}$$

$$\int \frac{dx}{x} = -a\int \text{dt}$$

$$\ln|x| = -at + c$$

$$att = 0, x = 2$$

$$\ln 2 = 0 + c$$

$$\ln x = -at + \ln 2$$

$$\frac{x}{2} = e^{-at}$$

$x = 2e^{-at}$ ....(I)

$$\frac{dy}{dt} + by = 0$$

$$\frac{dy}{y} = -b\,\text{dt}$$

$$\ln|y| = -bt + \lambda$$

$$\text{t} = 0, \text{y} = 1$$

$$0 = 0 + \lambda$$

$\text{y} = \text{e}^{-\text{bt}}$ ....(II)

According to question

$$3\text{y}(1) = 2\text{x}(1)$$

$$3\text{e}^{-\text{b}} = 2(2\text{e}^{-\text{a}})$$

$$\text{e}^{\text{a}-\text{b}} = \frac{4}{3}$$

For $x(t) = y(t)$

$$\Rightarrow 2\text{e}^{-\text{at}} = \text{e}^{-\text{bt}}$$

$$2 = \text{e}^{(\text{a}-\text{b})\text{t}}$$

$$2 = \left(\frac{4}{3}\right)^t$$

$$\log_{\frac{4}{3}} 2 = \text{t}$$

Question65

If the solution of the differential equation
$(2x+3y-2)\,dx+(4x+6y-7)dy=0$, $y(0)=3$, is
$\alpha x+\beta y+3\log_e 2x+3y-\gamma\,|\ =6$, then $\alpha+2\beta+3\gamma$ is equal to____
[27-Jan-2024 Shift 1]

Answer: 29

Solution:

$2x+3y-2=t$

$4x+6y-4=2t$

$2+3\frac{dy}{dx}=\frac{dt}{dx}$

$4x+6y-7=2t-3$

$\frac{dy}{dx}=\frac{-(2x+3y-2)}{4x+6y-7}$

$\frac{dt}{dx}=\frac{-3t+4t-6}{2t-3}=\frac{t-6}{2t-3}$

$\int\frac{2t-3}{t-6}\,\mathrm{dt}=\int\mathrm{dx}$

$\int\left(\frac{2t-12}{t-6}+\frac{9}{t-6}\right)\cdot\mathrm{dt}=x$

$2t+9\ln(t-6)=x+c$

$2(2x+3y-2)+9\ln(2x+3y-8)\ =x+c$

$x=0, y=3$

$c=14$

$4x+6y-4+9\ln(2x+3y-8)\ =x+14$

$x+2y+3\ln(2x+3y-8)=6$

$\alpha=1, \beta=2, \gamma=8$

$\alpha+2\beta+3\gamma=1+4+24=29$

Question66

If y = y(x) is the solution curve of the differential equation
$(x^2-4)dy-(y^2-3y)dx=0$,
$x>2$, $y(4)=\frac{3}{2}$ and the slope of the curve is never zero, then the value of y(10) equals :
[27-Jan-2024 Shift 2]

Options:

A. $\frac{3}{1+(8)^{1/4}}$

B. $\frac{3}{1+2\sqrt{2}}$

C. $\frac{3}{1-2\sqrt{2}}$

D. $\frac{3}{1-(8)^{1/4}}$

Answer: A

Solution:

$(x^2-4)dy-(y^2-3y)dx=0$

$\Rightarrow \int \frac{dy}{y^2-3y}=\int \frac{dx}{x^2-4}$

$\Rightarrow \frac{1}{3}\int \frac{y-(y-3)}{y(y-3)}dy=\int \frac{dx}{x^2-4}$

$\Rightarrow \frac{1}{3}(\ln|y-3|-\ln|y|)=\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+C$

$\Rightarrow \frac{1}{3}\ln\left|\frac{y-3}{y}\right|=\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+C$

At $x=4, y=\frac{3}{2}$

$\therefore C = \frac{1}{4} \ln 3$

$\therefore \frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + \frac{1}{4} \ln(3)$

At $x = 10$

$\frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{2}{3} \right| + \frac{1}{4} \ln(3)$

$\ln \left| \frac{y-3}{y} \right| = \ln 2^{3/4}, \forall x > 2, \frac{dy}{dx} < 0$

as $y(4) = \frac{3}{2} \Rightarrow y \in (0, 3)$

$-y + 3 = 8^{1/4} \cdot y$

$y = \frac{3}{1 + 8^{1/4}}$

Question67

If the solution curve, of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passing through the point (2, 1) is
$\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{\beta}\log_e\left(\alpha + \left(\frac{y-1}{x-1}\right)^2\right) = \left|\log_e x - 1\right|$
then $5\beta + \alpha$ is equal to__
[27-Jan-2024 Shift 2]

Answer: 11

Solution:

$\frac{dy}{dx} = \frac{x+y-2}{x-y}$

$x = X + h, y = Y + k$

$\frac{dY}{dX} = \frac{X+Y}{X-Y}$

$\left.\begin{array}{r} h+k-2=0 \\ h-k=0 \end{array}\right\} h = k = 1$

$Y = vX$

$v + \frac{dv}{dX} = \frac{1+v}{1-v} \Rightarrow X - \frac{dv}{dX} = \frac{1+v^2}{1-v}$

$\frac{1-v}{1+v^2} dv = \frac{dX}{X}$

$\tan^{-1} v - \frac{1}{2} \ln(1+v^2) = \ln|X| + C$

As curve is passing through (2, 1)

$\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{2}\ln\left(1 + \left(\frac{y-1}{x-1}\right)^2\right) = \ln|x-1|$

$\therefore \alpha = 1$ and $\beta = 2$

$\Rightarrow 5\beta + \alpha = 11$

Question68

A function $y = f(x)$ satisfies $f(x) \sin 2x + \sin x - (1 + \cos^2 x) f'(x) = 0$ with condition $f(0) = 0$. Then $f\left(\frac{\pi}{2}\right)$ is equal to

[29-Jan-2024 Shift 1]

Options:

A. 1

B. 0

C. -1

D. 2

Answer: A

Solution:

$$\frac{dy}{dx}-\left(\frac{\sin 2x}{1+\cos^2 x}\right)y=\sin x$$

$$\text{I.F. } =1+\cos^2 x$$

$$y\cdot(1+\cos^2 x)=\int(\sin x)\,dx$$

$$=-\cos x+C$$

$$x=0,\ C=1$$

$$y\left(\frac{\pi}{2}\right)=1$$

Question69

If the solution curve y = y(x) of the differential equation $(1+y^2)(1+\log_e x)\,dx+x\,dy=0$, x > 0 passes through the point (1, 1) and $y(e)=\dfrac{\alpha-\tan\left(\frac{3}{2}\right)}{\beta+\tan\left(\frac{3}{2}\right)}$, then $\alpha+2\beta$ is

[29-Jan-2024 Shift 1]

Answer: 3

Solution:

$$\int\left(\frac{1}{x}+\frac{\ln x}{x}\right)dx+\int\frac{dy}{1+y^2}=0$$

$$\ln x+\frac{(\ln x)^2}{2}+\tan^{-1}y=C$$

Put $x=y=1$

$$\therefore C=\frac{\pi}{4}$$

$$\Rightarrow \ln x + \frac{(\ln x)^2}{2} + \tan^{-1} y = \frac{\pi}{4}$$

Put $x = e$

$$\Rightarrow y = \tan\left(\frac{\pi}{4} - \frac{3}{2}\right) = \frac{1 - \tan\frac{3}{2}}{1 + \tan\frac{3}{2}}$$

$\therefore \alpha = 1, \beta = 1$

$\Rightarrow \alpha + 2\beta = 3$

Question70

If $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ and $y = \frac{\pi}{3}$, then α^2 is equal to

[29-Jan-2024 Shift 2]

Options:

A. 3

B. 12

C. 4

D. 9

Answer: A

Solution:

Differential equation :-

$$x\cos\frac{y}{x}\frac{dy}{dx}=y\cos\frac{y}{x}+x$$

$$\cos\frac{y}{x}\left[x\frac{dy}{dx}-y\right]=x$$

Divide both sides by x^2

$$\cos\frac{y}{x}\left(\frac{x\frac{dy}{dx}-y}{x^2}\right)=\frac{1}{x}$$

Let $\frac{y}{x}=t$

$$\cos t\left(\frac{dt}{dx}\right)=\frac{1}{x}$$

$$\cos tt=\frac{1}{x}dx$$

Integrating both sides

$$sin\ t=\ln|x|+c$$

$$sin\ \frac{y}{x}=\ln|x|+c$$

Using $y(1)=\frac{\pi}{3}$, we get $c=\frac{\sqrt{3}}{2}$

So, $\alpha=\sqrt{3}\Rightarrow\alpha^2=3$

Question71

Let y = y(x) be the solution of the differential equation $(1-x^2)dy=\left[xy+(x^3+2)\sqrt{3(1-x^2)}\right]dx\ -1<x<1,\ y(0)=0$. If $y\left(\frac{1}{2}\right)=\frac{m}{n}$, m and n are coprime numbers, then m + n is equal to
[30-Jan-2024 Shift 1]

Answer: 97

Solution:

$$\frac{dy}{dx}-\frac{xy}{1-x^2}=\frac{(x^3+2)\sqrt{3(1-x^2)}}{1-x^2}$$

$$\text{IF} \ = e^{-\int \frac{x}{1-x^2}dx} = e^{+\frac{1}{2}\ln(1-x^2)} = \sqrt{1-x^2}$$

$$y\sqrt{1-x^2}=\sqrt{3}\int(x^3+2)\,dx$$

$$y\sqrt{1-x^2}=\sqrt{3}\left(\frac{x^4}{4}+2x\right)+c$$

$$\Rightarrow y(0)=0 \qquad \therefore c=0$$

$$y\left(\frac{1}{2}\right)=\frac{65}{32}=\frac{m}{n}$$

$$m+n=97$$

Question72

Let y = y(x) be the solution of the differential equation sec xd y + {2(1 − x) tan x + x(2 − x)} dx = 0 such that y(0) = 2. Then y(2) is equal to :
[30-Jan-2024 Shift 1]

Options:

A. 2

B. $2\{1-\sin(2)\}$

C. $2\{\sin(2)+1\}$

D. 1

Answer: A

Solution:

$$\frac{dy}{dx} = 2(x-1)\sin x + (x^2 - 2x)\cos x$$

Now both side integrate

$$y(x) = \int 2(x-1)\sin x \, dx + [(x^2 - 2x)(\sin x) - \int (2x-2)\sin x \, dx]$$

$$y(x) = (x^2 - 2x)\sin x + \lambda$$

$$y(0) = 0 + \lambda \Rightarrow 2 = \lambda$$

$$y(x) = (x^2 - 2x)\sin x + 2$$

$$y(2) = 2$$

Question73

The solution curve of the differential equation $y\frac{dx}{dy} = x(\log_e x - \log_e y + 1), x > 0, y > 0$ passing through the point (e, 1) is
[31-Jan-2024 Shift 1]

Options:

A. $\left|\log_e \frac{y}{x}\right| = x$

B. $\left|\log_e \frac{y}{x}\right| = y^2$

C. $\left|\log_e \frac{x}{y}\right| = y$

D. $2\left|\log_e \frac{x}{y}\right| = y + 1$

Answer: C

Solution:

$$\frac{dx}{dy}=\frac{x}{y}\left(\ln\left(\frac{x}{y}\right)+1\right)$$

Let $\frac{x}{y}=t \Rightarrow x=ty$

$$\frac{dx}{dy}=t+y\frac{dt}{dy}$$

$$t+y\frac{dt}{dy}=t(\ln(t)+1)$$

$$y\frac{dt}{dy}=t\ln(t) \Rightarrow \frac{dt}{t\ln(t)}=\frac{dy}{y}$$

$$\Rightarrow \int\frac{dt}{t\cdot\ln(t)}=\int\frac{dy}{y}$$

$$\Rightarrow \int\frac{dp}{p}=\int\frac{dy}{y}$$

let $\ln t=p$

$$\frac{1}{t}dt=dp$$

$$\Rightarrow \ln p=\ln y+c$$

$$\ln(\ln t)=\ln y+c$$

$$\ln\left(\ln\left(\frac{x}{y}\right)\right)=\ln y+c$$

at $x=e, y=1$

$$\ln\left(\ln\left(\frac{e}{1}\right)\right)=\ln(1)+c \Rightarrow c=0$$

$$\ln\left|\ln\left(\frac{x}{y}\right)\right|=\ln y$$

$$\left|\ln\left(\frac{x}{y}\right)\right|=e^{\ln y}$$

$$\left|\ln\left(\frac{x}{y}\right)\right|=y$$

Question74

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx}=\frac{(\tan x)+y}{\sin x(\sec x-\sin x\tan x)}$, **$x\in\left(0,\frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right)=2$.**

Then, $y\left(\frac{\pi}{3}\right)$ is

[31-Jan-2024 Shift 1]

Options:

A. $\sqrt{3}(2+\log_e\sqrt{3})$

B. $\frac{\sqrt{3}}{2}(2+\log_e 3)$

C. $\sqrt{3}(1+2\log_e 3)$

D. $\sqrt{3}(2+\log_e 3)$

Answer: A

Solution:

$$\frac{dy}{dx}=\frac{\sin x+y\cos x}{\sin x\cdot\cos x\left(\frac{1}{\cos x}-\sin x\cdot\frac{\sin x}{\cos x}\right)}$$

$$=\frac{\sin x+y\cos x}{\sin x(1-\sin^2 x)}$$

$$\frac{dy}{dx}=\sec^2 x+y\cdot 2(\operatorname{cosec} 2x)$$

$$\frac{dy}{dx}-2\operatorname{cosec}(2x)\cdot y=\sec^2 x$$

$$\frac{dy}{dx}+p\cdot y=Q$$

$$\text{I.F.}\ =e^{\int p\,dx}=e^{\int -2\operatorname{cosec}(2x)\,dx}$$

Let $2x=t$

$$2\frac{dx}{dt}=1$$

$dx = \frac{dt}{2}$

$= e^{-\int \operatorname{cosec}(t)\,dt}$

$= e^{-\ln\left|\tan\frac{t}{2}\right|}$

$= e^{-\ln|\tan x|} = \frac{1}{|\tan x|}$

$y(\mathrm{IF}) = \int Q(\mathrm{IF})\,dx + c$

$\Rightarrow y\,\frac{1}{|\tan x|} = \int sec^2 x \cdot \frac{1}{|\tan x|} + c$

$y \cdot \frac{1}{|\tan x|} = \int \frac{dt}{|t|} + c \quad$ for $\tan x = t$

$y \cdot \frac{1}{|\tan x|} = \ln|t| + c$

$y = |\tan x|\,(\ln|\tan x| + c)$

Put $x = \frac{\pi}{4}, y = 2$

$2 = \ln 1 + c \Rightarrow c = 2$

$y = |\tan x|\,(\ln|\tan x| + 2)$

$y\left(\frac{\pi}{3}\right) = \sqrt{3}(\ln\sqrt{3} + 2)$

Question75

The temperature T(t) of a body at time t = 0 is 160° F and it decreases continuously as per the differential equation $\frac{dT}{dt} = -K(T-80)$, where K is positive constant. If T(15) = 120°F, then T(45) is equal to

[31-Jan-2024 Shift 2]

Options:

A. $85^{\circ}F$

B. $95^{\circ}F$

C. $90^{\circ}F$

D. $80^{\circ}F$

Answer: C

Solution:

$$\frac{dT}{dt} = -k(T-80)$$

$$\int_{160}^{T} \frac{dT}{(T-80)} = \int_{0}^{t} -Kdt$$

$$[\ln|T-80|]_{160}^{T} = -kt$$

$$\ln|T-80| - \ln 80 = -kt$$

$$\ln\left|\frac{T-80}{80}\right| = -kt$$

$$T = 80 + 80e^{-kt}$$

$$120 = 80 + 80e^{-k \cdot 15}$$

$$\frac{40}{80} = e^{-k15} = \frac{1}{2}$$

$$\therefore T(45) = 80 + 80e^{-k \cdot 45}$$

$$= 80 + 80(e^{-k.15})^{3}$$

$$= 80 + 80 \times \frac{1}{8}$$

$$= 90$$

Question76

Let y = y(x) be the solution of the differential equation
$\sec^2 x\,dx + (e^{2y}\tan^2 x + \tan x)dy = 0$
$0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = 0$. If $y\left(\frac{\pi}{6}\right) = \alpha$
Then $e^{8\alpha}$ is equal to
[31-Jan-2024 Shift 2]

Answer: 9

Solution:

$$sec^2 x\frac{dx}{dy} + e^{2y}\tan^2 x + \tan x = 0$$

$$\left(\text{ Put } \tan x = t \Rightarrow sec^2 x\frac{dx}{dy} = \frac{dt}{dy}\right)$$

$$\frac{dt}{dy} + e^{2y} \times t^2 + t = 0$$

$$\frac{dt}{dy} + t = -t^2 \cdot e^{2y}$$

$$\frac{1}{t^2}\frac{dt}{dy} + \frac{1}{t} = -e^{2y}$$

$$\left(\text{ Put } \frac{1}{t} = u\ \frac{-1}{t^2}\frac{dt}{dy} = \frac{du}{dy}\right)$$

$$\frac{-du}{dy} + u = -e^{2y}$$

$$\frac{du}{dy} - u = e^{2y}$$

$$\text{I.F.} = e^{-\int dy} = e^{-y}$$

$$ue^{-y} = \int e^{-y} \times e^{2y} dy$$

$$\frac{1}{\tan x} \times e^{-y} = e^{y} + c$$

$x = \frac{\pi}{4}, y = 0, c = 0$

$x = \frac{\pi}{6}, y = \alpha$

$\sqrt{3}e^{-\alpha} = e^{\alpha} + 0$

$e^{2\alpha} = \sqrt{3}$

$e^{8\alpha} = 9$

Question77

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1, y(0) = 1.$

Then, $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals :

[1-Feb-2024 Shift 1]

Options:

A. $\frac{4}{4+\sqrt{e}}$

B. $\frac{3}{3-\sqrt{e}}$

C. $\frac{2}{1+\sqrt{e}}$

D. $\frac{1}{2-\sqrt{e}}$

Answer: D

Solution:

$$\frac{dy}{dx}=2x(x+y)^3-x(x+y)-1$$

$$x+y=t$$

$$\frac{dt}{dx}-1=2xt^3-xt-1$$

$$\frac{dt}{2t^3-t}=x\,dx$$

$$\frac{tdt}{2t^4-t^2}=xdx$$

Let $t^2=z$

$$\int\frac{dz}{2(2z^2-z)}=\int x\,dx$$

$$\int\frac{dz}{4z\left(z-\frac{1}{2}\right)}=\int x\,dx$$

$$\ln\left|\frac{z-\frac{1}{2}}{z}\right|=x^2+k$$

$$z=\frac{1}{2-\sqrt{e}}$$

Question78

If x = x(t) is the solution of the differential equation $(t+1)\,dx=(2x+(t+1)^4)\,dt$, x(0) = 2, then, x(1) equals ________

[1-Feb-2024 Shift 1]

Answer: 14

Solution:

$(t+1)\,dx = (2x + (t+1)^4)\,dt$

$\frac{dx}{dt} = \frac{2x + (t+1)^4}{t+1}$

$\frac{dx}{dt} - \frac{2x}{t+1} = (t+1)^3$

$I \cdot F = e^{-\int \frac{2}{t+1} dt} = e^{-2\ln(t+1)} = \frac{1}{(t+1)^2}$

$\frac{x}{(t+1)^2} = \int \frac{1}{(t+1)^2}(t+1)^3\,dt + c$

$\frac{x}{(t+1)^2} = \frac{(t+1)^2}{2} + c$

$\Rightarrow c = \frac{3}{2}$

$x = \frac{(t+1)^4}{2} + \frac{3}{2}(t+1)^2$

put, $t = 1$

$x = 2^3 + 6 = 14$

Question79

Let α be a non-zero real number. Suppose f : R⟶ R is a differentiable function such that f (0) = 2 and $\lim_{x \to -\infty} f(x) = 1$. If $f'(x) = \alpha f(x) + 3$, for all x ∈ R, then f $(-\log_e 2)$ is equal to
[1-Feb-2024 Shift 2]

Options:

A. 3

B. 5

C. 9

D. 7

Answer: A

Solution:

$f(0) = 2, \lim_{x \to -\infty} f(x) = 1$

$f'(x) - \alpha \cdot f(x) = 3$

$I \cdot F = e^{-\alpha x}$

$y(e^{-\alpha x}) = \int 3 \cdot e^{-\alpha x} dx$

$f(x) \cdot (e^{-\alpha x}) = \frac{3e^{-\alpha x}}{-\alpha} + c$

$x = 0 \Rightarrow 2 = \frac{-3}{\alpha} + c \Rightarrow \frac{3}{\alpha} = c - 2$

$f(x) = \frac{-3}{\alpha} + c \cdot e^{\alpha x}$

$x \to -\infty \Rightarrow 1 = \frac{-3}{\alpha} + c(0)$

$\alpha = -3 \therefore c = 1$

$f(-\ln 2) = \frac{-3}{\alpha} + c \cdot e^{\alpha x}$

$= 1 + e^{3 \ln 2} = 9$

(But α should be greater than 0 for finite value of c)

Question80

If $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, x(1) = 1, then 5x(2) is equal to : [1-Feb-2024 Shift 2]

Answer: 5

Solution:

$$\frac{dx}{dy} - \frac{x}{y} = \frac{1 - y^2}{y}$$

$$\text{Integrating factor } = e^{\int -\frac{1}{y}} = \frac{1}{y}$$

$$x \cdot \frac{1}{y} = \int \frac{1 - y^2}{y^2} dy$$

$$\frac{x}{y} = \frac{-1}{y} - y + c$$

$$x = -1 - y^2 + cy$$

$$x(1) = 1$$

$$1 = -1 - 1 + c \Rightarrow c = 3$$

$$x = -1 - y^2 + 3y$$

$$5x(2) = 5(-1 - 4 + 6)$$

$$= 5$$

Question81

Let y = y(x) be the solution of the differential equation $x^3 dy + (xy - 1)dx = 0$, $x > 0$, $y\left(\frac{1}{2}\right) = 3 - e\cdot$ Then y(1) is equal to [24-Jan-2023 Shift 1]

Options:

A. 1

B. e

C. $2 - e$

D. 3

Answer: A

Solution:

$$\frac{dy}{dx} = \frac{1 - xy}{x^3} = \frac{1}{x^3} - \frac{y}{x^2}$$

$$\frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^3}$$

$$\text{If } = e^{\int \frac{1}{x^2} dx} = e^{-\frac{1}{x}}$$

$$y \cdot e^{-\frac{1}{x}} = \int e^{-\frac{1}{x}} \cdot \frac{1}{x^3} dx \left(\text{put } -\frac{1}{x} = t \right)$$

$$y \cdot e^{-\frac{1}{x}} = -\int e^t \cdot t\, dt$$

$$y = \frac{1}{x} + 1 + Ce^{\frac{1}{x}}$$

Where C is constant

Put $x = \frac{1}{2}$

$$3 - e = 2 + 1 + Ce^2$$

$$C = -\frac{1}{e}$$

$$y(1) = 1$$

Question82

Let y = y(x) be the solution of the differential equation left$(x^2 - 3y^2$ right) dx + 3xyd y = 0, y(1) = 1. Then $6y^2(e)$ is equal to [24-Jan-2023 Shift 2]

Options:

A. $3e^2$

B. e^2

C. $2e^2$

D. $\frac{3e^2}{2}$

Answer: C

Solution:

$$(x^2 - 3y^2)\, dx + 3xy\, dy = 0$$

$$\frac{dy}{dx} = \frac{3y^2 - x^2}{3xy} \Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{3} \frac{x}{y}$$

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$(1) \Rightarrow v + x \frac{dv}{dx} = v - \frac{1}{3} \frac{1}{v}$$

$$\Rightarrow v\, dv = \frac{-1}{3x}$$

Integrating both side

$\frac{v^2}{2} = \frac{-1}{3}\ln x + c$

$\Rightarrow \frac{y^2}{2x^2} = \frac{-1}{3}\ln x + c$

$y(1) = 1$

$\Rightarrow \frac{1}{2} = c$

$\Rightarrow \frac{y^2}{2x^2} = \frac{-1}{3}\ln x + \frac{1}{2}$

$\Rightarrow y^2 = -\frac{2}{3}x^2 \ln x + x^2$

$y^2(e) = -\frac{2}{3}e^2 + e^2 = \frac{e^2}{3}$

$\Rightarrow 6y^2(e) = 2e^2$

Question83

Let y = y(x) be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}(1 + xy^2(1 + \log_e x))$, x > 0, y(1) = 3. Then $\frac{y^2(x)}{9}$ is equal to : [25-Jan-2023 Shift 1]

Options:

A. $\frac{x^2}{5 - 2x^3(2 + \log_e x^3)}$

B. $\frac{x^2}{2x^3(2 + \log_e x^3) - 3}$

C. $\frac{x^2}{3x^3(1 + \log_e x^2) - 2}$

D. $\frac{x^2}{7 - 3x^3(2 + \log_e x^2)}$

Answer: A

Solution:

$\frac{dy}{dx} - \frac{y}{x} = y^3(1 + \log_e x)$

$\frac{1}{y^3}\frac{dy}{dx} - \frac{1}{xyy^2} = 1 + \log_e x$

Let $-\frac{1}{y^2} = t \Rightarrow \frac{2}{y^3}\frac{dy}{dx} = \frac{dt}{dx}$

$\therefore \frac{dt}{dx} + \frac{2t}{x} = 2(1 + \log_e x)$

I.F. $= e^{\int \frac{2}{x} dx} = x^2$

$$\frac{-x^2}{y^2} = \frac{2}{3}\left((1+\log_e x)x^3 - \frac{x^3}{3}\right) + C$$

$$y(1) = 3$$

$$\frac{y^2}{9} = \frac{x^2}{5 - 2x^3(2+\log_e x^3)}$$

$$x\,dy = y\,dx + xy^3(1+\log_e x)\,dx$$

$$\frac{x\,dy - y\,dx}{y^3} = x(1+\log_e x)\,dx$$

$$-\frac{x}{y}d\left(\frac{x}{y}\right) = x^2(1+\log_e x)\,dx$$

$$-\left(\frac{x}{y}\right)^2 = 2\int x^2(1+\log_e x)\,dx$$

Question84

Let y = y(t) be a solution of the differential equation $\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$ Where, $\alpha > 0$, $\beta > 0$ and $\gamma > 0$. Then $\lim_{t \to \infty} y(t)$

[25-Jan-2023 Shift 2]

Options:

A. is 0

B. does not exist

C. is 1

D. is - 1

Answer: A

Solution:

$$\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$$

$$\text{I.F.} \quad = e^{\int \alpha\,dt} = e^{\alpha t}$$

$$\text{Solution} \quad \Rightarrow y \cdot e^{\alpha t} = \int \gamma e^{-\beta T} \cdot e^{\alpha t}\,dt$$

$$\Rightarrow ye^{\alpha t} = \gamma\,\frac{e^{(\alpha-\beta)t}}{(\alpha-\beta)} + c$$

$$\Rightarrow y = \frac{\gamma}{e^{\beta t}(\alpha-\beta)} + \frac{c}{e^{\alpha t}}$$

So, $\lim_{t \to \infty} y(t) = \frac{\gamma}{\infty} + \frac{c}{\infty} = 0$

Question85

Let $y = f(x)$ be the solution of the differential equation $y(x+1)\,dx - x^2 d\,y = 0$, $y(1) = e$. Then $\lim_{x \to 0^+} f(x)$ is equal to
[29-Jan-2023 Shift 1]

Options:

A. 0

B. $\frac{1}{e}$

C. e^2

D. $\frac{1}{e^2}$

Answer: A

Solution:

$$\frac{x+1}{x^2}dx = \frac{dy}{y}$$
$$\ln x - \frac{1}{x} = \ln y + c$$
$$(1, e)$$
$$c = -2$$
$$\ln x - \frac{1}{x} = \ln y - 2$$
$$y = e^{\ln x} - \frac{1}{x} + 2$$
$$\lim_{x \to 0^+} e^{\ln x - 1} - \frac{1}{x} + 2$$
$$= e^{-\infty}$$
$$= 0$$

Question86

Let $y = y(x)$ be the solution of the differential equation $x\log_e x\,\frac{dy}{dx} + y = x^2\log_e x$, $(x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to
[29-Jan-2023 Shift 2]

Options:

A. $\frac{4+e^2}{4}$

B. $\frac{1+e^2}{4}$

C. $\frac{2+e^2}{2}$

D. $\frac{1+e^2}{2}$

Answer: A

Solution:

$x\log_e x \frac{dy}{dx} + y = x^2\log_e x, (x > 1)$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x\ln x} = x$

Linear differential equation

$\text{I.F.} = e^{\int \frac{1}{x\ln x}dx} = \left|\ln x\right|$

∴ Solution of differential equation

$y|\ln x| = \int x|\ln x|dx$

$= |\ln x| \frac{x^2}{2} - \int \frac{1}{x}\cdot\frac{x^2}{2}dx$

$\Rightarrow y|\ln x| = |\ln x|\left(\frac{x^2}{2}\right) - \frac{x^2}{4} + c$

For constant

$y(2) = 2 \Rightarrow c = 1$

So, $y(x) = \frac{x^2}{2} - \frac{x^2}{4|\ln x|} + \frac{1}{|\ln x|}$

Hence, $y(e) = \frac{e^2}{2} - \frac{e^2}{4} + 1 = 1 + \frac{e^2}{4}$

Question87

Let the solution curve y = y(x) of the differential equation

$\frac{dy}{dx} - \frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{\frac{3}{2}}}y = 2x$

exp $\frac{x^3 - \tan^{-1}x^3}{\sqrt{(1+x)^6}}$ **pass through the origin.**

Then y(1) is equal to :

[30-Jan-2023 Shift 1]

Options:

A. $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

B. $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$

C. $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$

D. $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$

Answer: A

Solution:

$$\frac{dy}{dx}+\left(\frac{-3x^5\tan^{-1}x^3}{(1+x^6)^{3/2}}\right)y=2e^{\left\{\frac{x-\tan x}{\sqrt{1+x^6}}\right\}}$$

$$\text{I.F.}\ =e^{\int\frac{-3x^5\tan^{-1}x^3}{(1+x^6)^{3/2}}dx}$$

$$=e^{\frac{\tan^{-1}x^3-x^3}{\sqrt{1+x^6}}}$$

Solution of differential equation

$$y\cdot e^{\frac{\tan^{-1}x^3-x^3}{\sqrt{1+x^6}}}=\int 2xe^{\left(\frac{x^3-\tan^{-1}x^3}{\sqrt{1+x^6}}\right)}\cdot e^{\left(\frac{\tan^{-1}(x^3)-x^3}{\sqrt{1+x^6}}\right)}dx$$

$$=\int 2x\,dx+c$$

$$y\cdot e^{\frac{\tan^{-1}x^3-x^3}{\sqrt{1+x^6}}}=x^2+c$$

Also it passes through origin

$c=0$

$$y(1)\cdot e^{\frac{\tan^{-1}(1)-1}{\sqrt{2}}}=1$$

$$y(1)\cdot e^{\frac{\frac{\pi}{4}-1}{\sqrt{2}}}=1$$

$$y(1)\cdot e^{\frac{\pi-4}{4\sqrt{2}}}=1$$

$$y(1)=\frac{1}{e^{\frac{\pi-4}{4\sqrt{2}}}}=e^{\frac{4-\pi}{4\sqrt{2}}}$$

Question88

The solution of the differential equation

$\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right)$, **$y(1)=0$ is**

[30-Jan-2023 Shift 2]

Options:

A. $\log_e|x+y| - \frac{xy}{(x+y)^2} = 0$

B. $\log_e|x+y| + \frac{xy}{(x+y)^2} = 0$

C. $\log_e|x+y| + \frac{2xy}{(x+y)^2} = 0$

D. $\log_e|x+y| - \frac{2xy}{(x+y)^2} = 0$

Answer: C

Solution:

Put $y = vx$

$$v + x\frac{dv}{dx} = -\left(\frac{1+3v^2}{3+v^2}\right)$$

$$x\frac{dv}{dx} = -\frac{(v+1)^3}{3+v^2}$$

$$\frac{(3+v^2)dv}{(v+1)^3} + \frac{dx}{x} = 0$$

$$\int\frac{4dv}{(v+1)^3} + \int\frac{dv}{v+1} - \int\frac{2dv}{(v+1)^2} + \int\frac{dx}{x} = 0$$

$$\frac{-2}{(v+1)^2} + \ln(v+1) + \frac{2}{v+1} + \ln x = c$$

$$\frac{-2x^2}{(x+y)^2} + \ln\left(\frac{x+y}{x}\right) + \frac{2x}{x+y} + \ln x = c$$

$$\frac{2xy}{(x+y)^2} + \ln(x+y) = c$$

$\therefore c = 0$, as $x = 1, y = 0$

$$\therefore \frac{2xy}{(x+y)^2} + \ln(x+y) = 0$$

Question89

Let $y = y(x)$ be the solution of the differential equation $(3y^2 - 5x^2)y\,dx + 2x(x^2 - y^2)dy = 0$ such that $y(1) = 1$. then $|(y(2))^3 - 12y(2)|$ is equal to:
[31-Jan-2023 Shift 2]

Options:

A. $32\sqrt{2}$

B. 64

C. $16\sqrt{2}$

D. 32

Answer: A

Solution:

$$(3y^2-5x^2)y\cdot dx+2x(x^2-y^2)dy=0$$
$$\Rightarrow \frac{dy}{dx}=\frac{y(5x^2-3y^2)}{2x(x^2-y^2)}$$
Put $y=mx$
$$\Rightarrow m+x\cdot\frac{dm}{dx}=\frac{m(5-3m^2)}{2(1-m^2)}$$
$$x\cdot\frac{dm}{dx}=\frac{(5-3m^2)m-2m(1-m^2)}{2(1-m^2)}$$
$$\Rightarrow \frac{dx}{x}=\frac{2(m^2-1)}{m(m^2-3)}dm$$
$$\Rightarrow \frac{dx}{x}=\left(\frac{2}{m}-\frac{\frac{4}{3}}{m}+\frac{\frac{4m}{3}}{m^2-3}\right)dm$$
$$\Rightarrow \int\frac{dx}{x}=\int\frac{\left(\frac{2}{3}\right)}{m}+\int\frac{2}{3}\left(\frac{2m}{m^2-3}\right)dm$$
$$\Rightarrow \ln|x|=\frac{2}{3}\ln|m|+\frac{2}{3}\ln|m^2-3|+C$$
$$\text{Or, } \ln|x|=\frac{2}{3}\ln\left|\frac{y}{x}\right|+\frac{2}{3}\ln\left|\left(\frac{y}{x}\right)^2-3\right|+C$$
Put $(x=1, y=1)$: we get $c=-\frac{2}{3}\ln(2)$
$$\Rightarrow \ln|x|=\frac{2}{3}\ln\left|\frac{y}{x}\right|+\frac{2}{3}\ln\left|\left(\frac{y}{x}\right)^2-3\right|-\frac{2}{3}\ln(2)$$
$$\Rightarrow \left(\frac{y}{x}\right)\left[\left(\frac{y}{x}\right)^2-3\right]=2.(x^{3/2})$$
Put $x=2$ to get $y(2)$
$$\Rightarrow y(y^2-12)=4\times2\times2\times2\sqrt{2}$$
$$\Rightarrow y^3-12y=32\sqrt{2}$$
$$\Rightarrow |y^3(2)-12y(2)|=32\sqrt{2}$$

Question90

If $y=y(x)$ is the solution curve of the differential equation

$\frac{dy}{dx}+y\tan x=x\sec x,\quad 0\le x\le\frac{\pi}{3}$, $y(0)=1$, then $y\left(\frac{\pi}{6}\right)$ is equal to

[1-Feb-2023 Shift 1]

Options:

A. $\frac{\pi}{12}-\frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

B. $\frac{\pi}{12}+\frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

C. $\frac{\pi}{12}-\frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

D. $\frac{\pi}{12}+\frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

Answer: A

Solution:

Here I.F. $=\sec x$

Then solution of D.E :

$y(\sec x)=x\tan x-\ln(\sec x)+c$

Given $y(0)=1\Rightarrow c=1$

$\therefore y(\sec x)=x\tan x-\ln(\sec x)+1$

At $x=\frac{\pi}{6}, y=\frac{\pi}{12}+\frac{\sqrt{3}}{2}\ln\frac{\sqrt{3}}{2}+\frac{\sqrt{3}}{2}$

Question91

Let $\alpha x=\exp(x^{\beta}y^{\gamma})$ be the solution of the differential equation $2x^2y\,dy-(1-xy^2)\,dx=0$, $x>0$, $y(2)=\sqrt{\log_e 2}$. Then $\alpha+\beta-\gamma$ equals : [1-Feb-2023 Shift 2]

Options:

A. 1

B. −1

C. 0

D. 3

Answer: A

Solution:

$\alpha x = e^{x^{8} \cdot y^{y}}$

$2x^2 y \frac{dy}{dx} = 1 - x \cdot y^2 \quad y^2 = t$

$x^2 \frac{dt}{dx} = 1 - xt$

$\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2} \quad \text{I.F.} \quad = e^{tnx} = x$

$t(x) = \int \frac{1}{x^2} \cdot x \, dx$

$y^2 \cdot x = \ell nx + C$

$\therefore 2 . \ln 2 = \ln 2 + C$

$\therefore C = \ell n2$

Hence, $xy^2 = \ell n2x$

$\therefore 2x = e^{x \cdot y^2}$

Hence $\alpha = 2, \beta = 1, \gamma = 2$

Question92

Let y = y(x) be a solution of the differential

$(x \cos x) dy + (xy \sin x + y \cos x - 1) dx = 0$, $0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3} y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then

$\left| \frac{\pi}{6} y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right) \right|$ is equal to ________.

[6-Apr-2023 shift 1]

Answer: 2

Solution:

$(x \cos x) dy + (xy \sin x + y \cos x - 1) dx = 0, 0 < x < \frac{\pi}{2}$

$\frac{dy}{dx} + \left(\frac{x \sin x + \cos x}{x \cos x} \right) y = \frac{1}{x \cos x}$

$\text{I F} = x \sec x$

$y \cdot x \sec x = \int \frac{x \sec x}{x \cos x} dx = \tan x + c$

Since $y\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{\pi}$ Hence $c = \sqrt{3}$

Hence $\left| \frac{\pi}{6} y''\left(\frac{\pi}{6}\right) + y'\left(\frac{\pi}{6}\right) \right| = |-2| = 2$

Question93

If the solution curve $f(x, y) = 0$ of the differential equation $(1 + \log_e x)\frac{dx}{dy} - x\log_e x = e^y$, $x > 0$, passes through the points $(1, 0)$ and $(\alpha, 2)$, then α^α is equal to :
[6-Apr-2023 shift 2]

Options:

A. $e^{\sqrt{2c^2}}$

B. e^{c^2}

C. $e^{2e^{\sqrt{2}}}$

D. e^{2c^2}

Answer: D

Solution:

$(1 + \ell \ln x)\frac{dx}{dy} - x\ell \ln x = e^y$

Let $x\ell nx = t$

$(1 + \ln x)\frac{dx}{dy} = \frac{dt}{dy}$

$\frac{dt}{dy} - t = e^y \quad P = -1, Q = e^y$

$I \cdot F = e^{\int -dy} = e^{-y}$

Solution -

$(t)(e^{-y}) = \int (e^{-y})(e^y)dy$

$t(e^{-y}) = y + c$

$(x\ell \ln x)e^{-y} = y + c \quad$ pass $(1, 0) \Rightarrow c = 0$

pass $(\alpha, 2)$

$\alpha^\alpha = e^{2e^2}$

Ans. Option 4

Question94

Let the solution curve $x = x(y)$, $0 < y < \frac{\pi}{2}$, of the differential equation $(\log_e(\cos y))^2 \cos y \, dx - (1 + 3x\log_e.(\cos y))\sin y dy = 0$ satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_c 2}$. If $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_c n}$, where m and n are co-prime, then mn is

equal to
[8-Apr-2023 shift 2]

Answer: 12

Solution:

$\operatorname{Cos} y \ln^2 \cos y\, dx = (1 + 3x \ln \cos y) \sin y d y$

$\frac{dx}{dy} = \tan y \left(\frac{3x}{\ln \cos y} + \frac{1}{\ln^2 \cos y} \right)$

$\frac{dx}{dy} - \left(\frac{3 \tan y}{\ln \cos y} \right) x = \frac{\tan y}{\ln^2 \cos y}$

If $= e^{-\frac{3}{\ln \cos y \cos y} dy}$

$\ln \cos y = t$

$\frac{1}{\cos y} - \sin y d y = d t$

If $= e^{\frac{3}{t} dt} = e^{3 \ln t} = t^3 = \ln^3 \cos y$

solution is $x \cdot \ln^3 \cos y = \int \frac{\sin y}{\cos y} \cdot \ln \cos y \, d\, y + C.$

$x \ln^3 \cos y = \frac{-\ln^2 \cos y}{2} + C$

$x\left(\frac{\pi}{3} \right) = \frac{1}{2 \ln 2}$ so $\frac{1}{2 \ln^2} \times \ln^3 \left(\frac{1}{2} \right) = -\frac{\ln^3 \left(\frac{1}{2} \right)}{2} + C$

$C = 0 \quad x \quad \ln^3 \frac{\sqrt{3}}{2} = -\frac{1}{2} \ln^2 \frac{\sqrt{3}}{2} + 0$

$y = \frac{\pi}{6}$

$x = -\frac{1}{2 \ln \left(\frac{\sqrt{3}}{2} \right)}$

$x = \frac{1}{\ln \frac{4}{3}} = \frac{1}{\ln 4 - \ln 3}$

$mn = 12$

Question95

Let f be a differentiable function such that $x^2 f(x) - x = 4 \int_0^x t f(t)\, dt$, $f(1) = \frac{2}{3}$..
Then 18f(3) is equal to :
[10-Apr-2023 shift 1]

Options:

A. 180

B. 150

C. 210

D. 160

Answer: D

Solution:

$$x^2 f(x) - x = 4\int_0^x t f(t)\,dt$$

Differentiate w.r.t. x

$$x^2 f'(x) + 2x f(x) - 1 = 4x f(x)$$

Let $y = f(x)$

$$\Rightarrow x^2 \frac{dy}{dx} - 2xy - 1 = 0$$

$$\frac{dy}{dx} - \frac{2}{x}y = \frac{1}{x^2}$$

$$\text{I.F.} = e^{\int \frac{-2}{x} dx} = \frac{1}{x^2}$$

Its solution is

$$\frac{y}{x^2} = \int \frac{1}{x^4} dx + C$$

$$\frac{y}{x^2} = \frac{-1}{3x^3} + C$$

$$\because f(1) = \frac{2}{3} \Rightarrow y(1) = \frac{2}{3}$$

$$\Rightarrow \frac{2}{3} = -\frac{1}{3} + C$$

$$\Rightarrow C = 1$$

$$\because y = -\frac{1}{3x} + x^2$$

$$f(x) = -\frac{1}{3x} + x^2$$

$$f(3) = -\frac{1}{9} + 9 = \frac{80}{9} \Rightarrow 18f(3) = 160$$

Question96

The slope of tangent at any point (x, y) on a curve y = y(x) is $\frac{x^2+y^2}{2xy}$, x > 0. If y(2) = 0, then a value of y(8) is:
[10-Apr-2023 shift 1]

Options:

A. $4\sqrt{3}$

B. $-4\sqrt{2}$

C. $-2\sqrt{3}$

D. $2\sqrt{3}$

Answer: A

Solution:

$$\frac{dy}{dx}=\frac{x^2+y^2}{2xy}$$
$y = vx$
$y(2)=0$
$y(8)=?$
$$\frac{dv}{dx}=v+x\frac{dv}{dx}$$
$$v+\frac{xdv}{dx}=\frac{x^2+v^2x^2}{2vx^2}$$
$$x\cdot\frac{dv}{dx}=\left(\frac{v^2+1}{2v}-v\right)$$
$$\frac{2vdv}{(1-v^2)}=\frac{dx}{x}$$
$-\ln(1-v^2)=\ln x+C$
$\ln x+\ln(1-v^2)=C$
$$\ln\left[x\left(1-\frac{y^2}{x^2}\right)\right]=C$$
$$\ln\left[\left(\frac{x^2-y^2}{x}\right)\right]=C$$
$x^2-y^2=cx$
$y(2)=0$ at $x=2, y=0$
$4=2C\Rightarrow C=2$
$x^2-y^2=2x$
Hence, at $x=8$
$64-y^2=16$
$y=\sqrt{48}=4\sqrt{3}$
$y(8)=4\sqrt{3}$
Option (1)

Question97

Let the tangent at any point P on a curve passing through the points (1, 1) and $\left(\frac{1}{10}, 100\right)$, intersect positive x axis and y-axis at the points A and B respectively. If PA : PB = 1 : k and y = y(x) is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}$, y(0) = k, then 4y then $4y(1) - 5\log e^3$ is equal to

________.

[10-Apr-2023 shift 2]

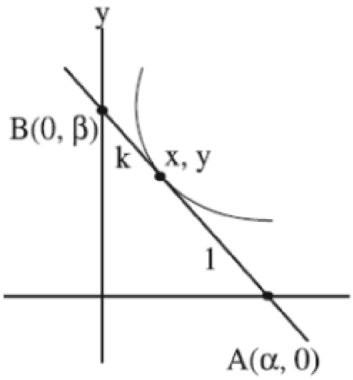
Answer: 5

Solution:

$Y - y = \frac{dy}{dx}(X - x)$

$Y = 0$

$X = \frac{-y\,dx}{dy} + x$



$\frac{k\alpha + 0}{k+1} = x, \alpha = \frac{k+1}{k}x$

$\frac{k+1}{k}x = -y\frac{dx}{dy} + x$

$x + \frac{x}{k} = -y\frac{dx}{dy} + x$

$x\frac{dy}{dx} + ky = 0$

$\frac{dy}{dx} + \frac{k}{x}y = 0$

$y \cdot x^k = C$

$C = 1$

$100 \cdot \left(\frac{1}{10}\right)^k = 1$

$K = 2$

$\frac{dy}{dx} = \ln(2x+1)$

$y = \frac{(2x+1)}{2}(\ln(2x+1) - 1) + c$

$2 = \frac{1}{2}(0 - 1) + C$

$C = 2 + \frac{1}{2} = \frac{5}{2}$

$y(1) = \frac{3}{2}(\ell\ln 3 - 1) + \frac{5}{2}$

$= \frac{3}{2}\ln 3 + 1$

$4y(1) = 6\ln 3 + 4$
$4y(1) - 5\ln 3 = 4 + \ln 3$

Question98

Let y = y(x) be a solution curve of the differential equation. $(1 - x^2y^2)\,dx = y\,dx + x\,dy$. If the line x = 1 intersects the curve y = y(x) at y = 2 and the line x = 2 intersects the curve y = y(x) at y = α, then a value of α is : [11-Apr-2023 shift 1]

Options:

A. $\dfrac{1+3e^2}{2(3e^2-1)}$

B. $\dfrac{1-3e^2}{2(3e^2+1)}$

C. $\dfrac{3e^2}{2(3e^2-1)}$

D. $\dfrac{3e^2}{2(3e^2+1)}$

Answer: A

Solution:

$(1 - x^2y^2)\,dx = y\,dx + x\,dy,\ y(1) = 2$
$y(2) = \infty$

$$dx = \frac{d(xy)}{1-(xy)^2}$$

$$\int dx = \int \frac{d(xy)}{1-(xy)^2}$$

$$x = \frac{1}{2}\ln\left|\frac{1+xy}{1-xy}\right| + C$$

Put $x = 1$ and $y = 2$:

$$1 = \frac{1}{2}\ln\left|\frac{1+2}{1-2}\right| + C$$

$$C = 1 - \frac{1}{2}\ln 3$$

Now put $x = 2$:

$$2 = \frac{1}{2}\ln\left|\frac{1+2\alpha}{1-2\alpha}\right| + 1 - \frac{1}{2}\ln 3$$

$$1 + \frac{1}{2}\ln 3 = \frac{1}{2}\left|\frac{1+2\alpha}{1-2\alpha}\right|$$

$$2 + \ln 3 = \left|\frac{1+2\alpha}{1-2\alpha}\right|$$

$\left|\frac{1+2\alpha}{1-2\alpha}\right| = 3e^2$

$\frac{1+2\alpha}{1-2\alpha} = 3e^2, -3e^2$

$\frac{1+2\alpha}{1-2\alpha} = 3e^2 \Rightarrow \alpha = \frac{3e^2-1}{2(3e^2+1)}$

And $\frac{1+2\alpha}{1-2\alpha} = -3e^2 \Rightarrow \alpha = \frac{3e^2+1}{2(3e^2-1)}$

Question99

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x(x^5+1)}y = \frac{(x^5+1)^2}{x^7}$, x > 0. If y(1) = 2, then y(2) is equal to [11-Apr-2023 shift 2]

Options:

A. $\frac{693}{128}$

B. $\frac{637}{128}$

C. $\frac{697}{128}$

D. $\frac{679}{128}$

Answer: A

Solution:

$\text{I.F} = = e^{\int \frac{5\,dx}{x(x^5+1)}} = e^{e\frac{5x^{-6}-6x}{(x^{-5}+1)}}$

Put, $1+x^{-5} = t \Rightarrow -5x^{-6}dx = dt$

$\Rightarrow e^{\int \frac{-dt}{1}} = \frac{1}{t} = \frac{x^5}{1+x^5}$

$y\cdot\frac{x^5}{1+x^5} = \int \frac{x^5}{(1+x^5)} \times \frac{(1+x^5)^2}{x^7}dx$

$= \int x^3 dx + \int x^{-2}dx$

$y\cdot\frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + c$

Given than: $x = 1 \Rightarrow y = 2$

$2\cdot\frac{1}{2} = \frac{1}{4} - 1 + c$

$c = \frac{7}{4}$

$$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + \frac{7}{4}$$

Now put, $x = 2$

$$y \cdot \left(\frac{32}{33}\right) = \frac{21}{4}$$

$$y = \frac{693}{128}$$

Question100

Let $y = y(x)$, $y > 0$, be a solution curve of the differential equation $(1 + x^2) dy = y(x - y) dx$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then [12-Apr-2023 shift 1]

Options:

A. $e^{3\beta^{-1}} = e(5 + \sqrt{2})$

B. $e^{3\beta^{-1}} = e(3 + 2\sqrt{2})$

C. $e^{\beta^{-1}} = e^{-2}(3 + 2\sqrt{2})$

D. $e^{\beta^{-1}} = e^{-2}(5 + \sqrt{2})$

Answer: B

Solution:

$$(1 + x^2) dy = y(x - y) dx$$

$$Y(0) = 1 \cdot y(2\sqrt{2}) = \beta$$

$$\frac{dy}{dx} = \frac{yx - y^2}{1 + x^2}$$

$$\frac{dy}{dx} + y\left(\frac{-x}{1+x^2}\right) = \left(\frac{-1}{1+x^2}\right)y^2$$

$$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\left(\frac{-x}{1+x^2}\right) = \frac{-1}{1+x^2}$$

$$\text{Put } \frac{1}{y} = t \text{ then } \frac{-1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + t\frac{x}{1+x^2} = \frac{1}{1+x^2}$$

$$\text{If } = e^{\int \frac{x}{1+x^2} dx} = e^{\frac{1}{2}\ln(1+x^2)} \sqrt{1+x^2}$$

$$t\sqrt{1+x^2} = \int \frac{1}{\sqrt{1+x^2}} dx$$

$$\frac{\sqrt{1+x^2}}{y} = \ln\left(x + \sqrt{x^2+1}\right) + c$$

$$y(0) = 1 \Rightarrow c = 1$$

$$\Rightarrow \sqrt{1+x^2} = y\ln\left(e\left(x+\sqrt{x^2+1}\right)\right)$$
$$\beta = \frac{3}{\ln(e(3+2\sqrt{2}))} \Rightarrow \frac{3}{\beta} = \ln(e(3+2\sqrt{2}))$$
$$e^{\frac{3}{\beta}} = e(3+2\sqrt{2})$$

Question101

Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves of the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at
[13-Apr-2023 shift 1]

Options:

A. no point

B. infinite number of points

C. one point

D. two points

Answer: A

Solution:

$$\frac{dy}{dx} = y + 7 \Rightarrow \frac{dy}{dx} - y = 7$$
$$\text{I.F.} = e^{-x}$$
$$ye^{-x} = \int 7e^{-x}\,dx$$
$$\Rightarrow ye^{-x} = -7e^{-x} + c$$
$$\Rightarrow y = -7 + ce^{x}$$
$$-7 + 7e^{x} = -7 + 8e^{x} \Rightarrow e^{x} = 0$$
No solution

Question102

If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ such that $y(2) = \frac{2}{9}\log_e(2+\sqrt{3})$ and $y(\sqrt{2}) = \alpha\log_e(\sqrt{\alpha}+\beta) + \beta - \sqrt{\gamma}$, $\alpha, \beta, \gamma, \in N$, then $\alpha\beta\gamma$ is equal to ________
[13-Apr-2023 shift 2]

Answer: 6

Solution:

given differential equation $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{5/2}}$ is linear D.E.

$$\text{I.F.} = e^{\int \frac{4x}{x^2-1} dx} = e^{2\ln(x^2-1)} = e^{\ln(x^2-1)^2} = (x^2-1)^2$$

$$y(x^2-1)^2 = \int \frac{x+2}{(x^2-1)^{5/2}}(x^2-1)^2\,dx$$

$$= \int \frac{x}{\sqrt{x^2-1}}dx + \int \frac{2\,dx}{\sqrt{x^2-1}}$$

$$= \sqrt{x^2-1} + 2\ln\left[x + \sqrt{x^2-1}\right] + C$$

put $y(2) = \frac{2}{9}\ln(2+\sqrt{3})$

$$\frac{2}{9}\ln(2+\sqrt{3})(9) = \sqrt{3} + 2\ln[2+\sqrt{3}] + C$$

$$= C = -\sqrt{3}$$

put $x = \sqrt{2}$

$$y = 1 + 2\ln[\sqrt{2}+1] - \sqrt{3}$$

$$\alpha = 2, \beta = 1 = \gamma = 3$$

$$\alpha\beta\gamma = 2(1)(3) = 6$$

Question103

Let $x = x(y)$ be the solution of the differential equation $2(y+2)\log_e(y+2)\,dx + (x + 4 - 2\log_e(y+2))\,dy = 0$, $y > -1$ with $x(e^4 - 2) = 1$. Then $x(e^9 - 2)$ is equal to
[15-Apr-2023 shift 1]

Options:

A. $\frac{4}{9}$

B. $\frac{32}{9}$

C. $\frac{10}{3}$

D. 3

Answer: B

Solution:

$$2(y+2)\ln(y+2)\,dx+(x+4-2\ln(y+2))\,dy=0$$
$$2\ln(y+2)+(x+4-2\ln(y+2))\,\frac{1}{y+2}\cdot\frac{dy}{dx}=0$$
let, $\ln(y+2)=t$
$$\frac{1}{y+2}\cdot\frac{dy}{dx}=\frac{dt}{dx}$$
$$2t+(x+4-2t)\cdot\frac{dt}{dx}=0$$
$$(x+4-2t)\frac{dt}{dx}=-2t$$
$$\frac{dx}{dt}=\frac{2t-4-x}{2t}$$
$$\frac{dx}{dt}+\frac{x}{2t}=\frac{2t-4}{2t}$$
$$x\cdot t^{1/2}=\int\frac{2t-4}{2t}\cdot t^{1/2}\cdot dt$$
$$x\cdot t^{1/2}=\int\left(t^{1/2}-\frac{2}{t^{1/2}}\right)\cdot dt$$
$$=\frac{t^{\frac{3}{2}}}{\frac{3}{2}}-2\cdot\frac{t^{\frac{1}{2}}}{\frac{1}{2}}+C$$
$$x\cdot t^{\frac{1}{2}}=\frac{2t^{\frac{3}{2}}}{3}-4t^{\frac{1}{2}}+C$$
$$x=\frac{2}{3}\cdot t-4+C\cdot t^{\frac{-1}{2}}$$
$$x=\frac{2}{3}\ln(y+2)-4+C\cdot(\ln(y+2))^{\frac{-1}{2}}$$
Put $y=e^4-2,\ x=1$
$$1=\frac{2}{3}\times4-4+C\times\frac{1}{2}$$
$$\frac{C}{2}=5-\frac{8}{3}=\frac{7}{3}$$
$$\Rightarrow C=\frac{14}{3}$$
$$x=\frac{2}{3}\times9-4+\frac{14}{3}\times\frac{1}{3}$$
$$=2+\frac{14}{9}$$
$$=3$$

Question104

The slope of normal at any point (x, y), x > 0, y > 0 on the curve y = y(x) is given by $\frac{x^2}{xy-x^2y^2-1}$. If the curve passes through the point (1, 1), then e. y(e) is equal to
[24-Jun-2022-Shift-2]

Options:

A. $\frac{1-\tan(1)}{1+\tan(1)}$

B. $\tan(1)$

C. 1

D. $\frac{1+\tan(1)}{1-\tan(1)}$

Answer: D

Solution:

$\because -\frac{dx}{dy} = \frac{x^2}{xy - x^2y^2 - 1}$

$\therefore \frac{dy}{dx} = \frac{x^2y^2 - xy + 1}{x^2}$

Let $xy = v \Rightarrow y + x\frac{dy}{dx} = \frac{dv}{dx}$

$\therefore \frac{dv}{dx} - y = \frac{(v^2 - v + 1)y}{v}$

$\therefore \frac{dv}{dx} = \frac{v^2 + 1}{x}$

$\because y(1) = 1 \Rightarrow \tan^{-1}(xy) = \ln x + \tan^{-1}(1)$

Put $x = e$ and $y = y(e)$ we get

$\tan^{-1}(e \cdot y(e)) = 1 + \tan^{-1}1$

$\tan^{-1}(e \cdot y(e)) - \tan^{-1}1 = 1$

$\therefore e(y(e)) = \frac{1+\tan(1)}{1-\tan(1)}$

Question105

Let y = y(x) be the solution of the differential equation $(x+1)y' - y = e^{3x}(x+1)^2$, with $y(0) = \frac{1}{3}$. Then, the point $x = -\frac{4}{3}$ for the curve y = y(x) is :
[25-Jun-2022-Shift-1]

Options:

A. not a critical point

B. a point of local minima

C. a point of local maxima

D. a point of inflection

Answer: B

Solution:

$(x+1)\dfrac{dy}{dx}-y=e^{3x}(x+1)^2$

$\dfrac{dy}{dx}-\dfrac{y}{x+1}=e^{3x}(x+1)$

If $e^{-\int\frac{1}{x+1}x}=e^{-\log(x+1)}=\dfrac{1}{x+1}$

$\therefore y\left(\dfrac{1}{x+1}\right)=\int\dfrac{e^{3x}(x+1)}{x+1}dx$

$\dfrac{y}{x+1}=\int e^{3x}dx$

$\dfrac{y}{x+1}=\dfrac{e^{3x}}{3}+c$

$\because y(0)=\dfrac{1}{3}$

$\dfrac{1}{3}=\dfrac{1}{3}+c$

$\therefore c=0$

So : $y=\dfrac{e^{3x}}{3}(x+1)$

$y'=e^{3x}(x+1)+\dfrac{e^{3x}}{3}=e^{3x}\left(x+\dfrac{4}{3}\right)$

$y''=3e^{3x}\left(x+\dfrac{4}{3}\right)+e^{3x}=e^{3x}(3x+5)$

$y'=0$ at $x=\dfrac{-4}{3}$ & $y''=e^{-4}(1)>0$ at $x=\dfrac{-4}{3}$

$\Rightarrow x=\dfrac{-4}{3}$ is point of local minima

Question106

If the solution curve y = y(x) of the differential equation $y^2dx+(x^2-xy+y^2)dy=0$, which passes through the point (1, 1) and intersects the line $y=\sqrt{3}x$ at the point $(\alpha, \sqrt{3}\alpha)$, then value of $\log_e(\sqrt{3}\alpha)$ is

equal to :
[25-Jun-2022-Shift-1]

Options:

A. $\frac{\pi}{3}$

B. $\frac{\pi}{2}$

C. $\frac{\pi}{12}$

D. $\frac{\pi}{6}$

Answer: C

Solution:

$$\frac{dy}{dx} = \frac{y^2}{xy - x^2 - y^2}$$

Put $y = vx$ we get

$$v + x\frac{dv}{dx} = \frac{v^2}{v - 1 - v^2}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v^2 - v^2 + v + v^3}{v - 1 - v^2}$$

$$\Rightarrow \int \frac{v - 1 - v^2}{v(1 + v^2)} dv = \int \frac{dx}{x}$$

$$\tan^{-1}\left(\frac{y}{x}\right) - \ln\left(\frac{y}{x}\right) = \ln x + c$$

As it passes through $(1, 1)$

$$c = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) \ln\left(\frac{y}{x}\right) = \ln x + \frac{\pi}{4}$$

Put $y = \sqrt{3}x$ we get

$$\Rightarrow \frac{\pi}{3} - \ln\sqrt{3} = \ln x + \frac{\pi}{4}$$

$$\Rightarrow \ln x = \frac{\pi}{12} - \ln\sqrt{3} = \ln\alpha$$

$$\therefore \ln(\sqrt{3}\alpha) = \ln\sqrt{3} + \ln\alpha$$

$$= \ln\sqrt{3} + \frac{\pi}{12} - \ln\sqrt{3} = \frac{\pi}{12}$$

Question107

If $y = y(x)$ is the solution of the differential equation $2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$ such that $y(e) = \frac{e}{3}$, then $y(1)$ is equal to
[25-Jun-2022-Shift-2]

Options:

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{3}{2}$

D. 3

Answer: B

Solution:

$2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$

$\Rightarrow 2x(xdy - ydx) + 3y^3dx = 0$

$\Rightarrow 2\left(\frac{xdy - ydx}{y^2}\right) + 3\frac{dx}{x} = 0$

$\Rightarrow -\frac{2x}{y} + 3\ln x = C$

$\because y(e) = \frac{e}{3} \Rightarrow -6 + 3 = C \Rightarrow C = -3$

Now, at $x = 1, -\frac{2}{y} + 0 = -3$

$y = \frac{2}{3}$

Question108

Let the solution curve $y = y(x)$ of the differential equation $(4 + x^2)dy - 2x(x^2 + 3y + 4)dx = 0$ pass through the origin. Then $y(2)$ is

equal to____

[26-Jun-2022-Shift-1]

Answer: 12

Solution:

$(4+x^2)dy - 2x(x^2+3y+4)dx = 0$

$\Rightarrow \frac{dy}{dx} = \left(\frac{6x}{x^2+4}\right)y + 2x$

$\Rightarrow \frac{dy}{dx} - \left(\frac{6x}{x^2+4}\right)y = 2x$

I. F. $= e^{-3\ln(x^2+4)} = \frac{1}{(x^2+4)^3}$

So $\frac{y}{(x^2+4)^3} = \int \frac{2x}{(x^2+4)^3}dx + c$

$\Rightarrow y = -\frac{1}{2}(x^2+4) + c(x^2+4)^3$

When $x = 0$, $y = 0$ gives $c = \frac{1}{32}$,

So, for $x = 2$, $y = 12$

Question109

Let $S = (0, 2\pi) - \left\{\frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}\right\}$. Let $y = y(x)$, $x \in S$, be the solution curve of the differential equation $\frac{dy}{dx} = \frac{1}{1+\sin 2x}$, $y\left(\frac{\pi}{4}\right) = \frac{1}{2}$. If the sum of abscissas of all the points of intersection of the curve $y = y(x)$ with the curve $y = \sqrt{2}\sin x$ is $\frac{k\pi}{12}$, then k is equal to____

[26-Jun-2022-Shift-1]

Answer: 42

Solution:

$\frac{dy}{dx} = \frac{1}{1+\sin 2x}$

$\Rightarrow dy = \frac{\sec^2 x dx}{(1+\tan x)^2}$

$\Rightarrow y = -\frac{1}{1+\tan x} + c$

When $x = \frac{\pi}{4}, y = \frac{1}{2}$ gives c = 1

So $y = \frac{\tan x}{1+\tan x} \Rightarrow y = \frac{\sin x}{\sin x + \cos x}$

Now, $y = \sqrt{2}\sin x \Rightarrow \sin x = 0$

or $\sin x + \cos x = \frac{1}{\sqrt{2}}$

$\sin x = 0$ gives $x = \pi$ only.

and $\sin x + \cos x = \frac{1}{\sqrt{2}} \Rightarrow \sin\left(x + \frac{\pi}{4}\right) = \frac{1}{2}$

So $x + \frac{\pi}{4} = \frac{5\pi}{6}$ or $\frac{13\pi}{6} \Rightarrow x = \frac{7\pi}{12}$ or $\frac{23\pi}{12}$

Sum of all solutions $= \pi + \frac{7\pi}{12} + \frac{23\pi}{12} = \frac{42\pi}{12}$

Hence, $k = 42$.

Question110

If the solution of the differential equation $\frac{dy}{dx} + e^x(x^2-2)y = (x^2-2x)(x^2-2)e^{2x}$ satisfies y(0) = 0, then the value of y(2) is__

[26-Jun-2022-Shift-2]

Options:

A. -1

B. 1

C. 0

D. e

Answer: C

Solution:

$\because \frac{dy}{dx} + e^x(x^2-2)y = (x^2-2x)(x^2-2)e^{2x}$

Here, $I.F. = e^{\int e^x(x^2-2)dx}$

$= e^{(x^2-2x)e^x}$

$\therefore$ Solution of the differential equation is

$y \cdot e^{(x^2-2x)e^x} = \int (x^2-2x)(x^2-2)e^{2x} \cdot e^{(x^2-2x)e^x} dx$

$= \int (x^2-2x)e^x \cdot (x^2-2)e^x \cdot e^{(x^2-2x)e^x} dx$

Let $(x^2-2x)e^x = t$

$\therefore (x^2-2)e^x dx = dt$

$y \cdot e^{(x^2-2x)e^x} = \int t \cdot e^t dt$

$y \cdot e^{(x^2-2x)e^x} = (x^2-2x-1)e^{(x^2-2x)e^x} + c$

$\therefore y(0) = 0$

$\therefore c = 1$

$\therefore y = (x^2-2x-1) + e^{(2x-x^2)e^x}$

$\therefore y(2) = -1 + 1 = 0$

Question111

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, where a, b, c are constants, represent a circle passing through the point (2, 5). Then the shortest distance of the point (11, 6) from this circle is :
[27-Jun-2022-Shift-1]

Options:

A. 10

B. 8

C. 7

D. 5

Answer: B

Solution:

$\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$
$= bxdy + cydy + ady = axdx - bydx + adx$
$= cydy + ady - axdx - adx + b(xdy + ydx) = 0$

$= c\int y\,dy + a\int x\,dx - a\int dx + b\int d(xy) = 0$

$= \frac{cy^2}{2} + ay - \frac{ax^2}{2} - ax + bxy = k$

$= ax^2 - cy^2 + 2ax - 2ay - 2bxy = k$

Above equation is circle

$\Rightarrow a = -c$ and $b = 0$

$ax^2 + ay^2 + 2ax - 2ay = k$

$\Rightarrow x^2 + y^2 + 2x - 2y = \lambda \quad \left[\lambda = \frac{k}{a}\right]$

Passes through $(2, 5)$

$4 + 25 + 4 - 10 = \lambda \Rightarrow \lambda = 23$

Circle $\equiv x^2 + y^2 + 2x - 2y - 23 = 0$

Centre $(-1, 1) r = \sqrt{(-1)^2 + 1^2 + 23} = 5$

Shortest distance of $(11, 6) = \sqrt{12^2 + 5^2} - 5$

$= 13 - 5$

$= 8$

Question112

If $\frac{dy}{dx} + \frac{2^{x-y}(2^y - 1)}{2^x - 1} = 0$, x, y > 0, y(1) = 1, then y(2) is equal to : [27-Jun-2022-Shift-1]

Options:

A. $2 + \log_2 3$

B. $2 + \log_3 2$

C. $2 - \log_3 2$

D. $2 - \log_2 3$

Answer: D

Solution:

$\frac{dy}{dx} + \frac{2^{x-y}(2^y - 1)}{2^x - 1} = 0, x, y > 0, y(1) = 1$

$\frac{dy}{dx} = -\frac{2^x(2^y - 1)}{2^y(2^z - 1)}$

$\int \frac{2^y}{2^y - 1} dy = -\int \frac{2^x}{2^x - 1} dx$

$= \frac{\log_e(2^y - 1)}{\log_{e^2}} = -\frac{\log_e(2^x - 1)}{\log_e 2} + \frac{\log_e c}{\log_e 2}$

$= |(2^y - 1)(2^x - 1)| = c$

$\because y(1) = 1$

$\therefore c = 1$

$= |(2^y - 1)(2^x - 1)| = 1$
For $x = 2$
$|(2^y - 1)3| = 1$
$2^y - 1 = \frac{1}{3} \Rightarrow 2y = \frac{4}{3}$
Taking log to base 2.
$\therefore y = 2 - \log_2 3$

Question113

If the solution curve of the differential equation $((\tan^{-1}y) - x)d\,y = (1 + y^2)d\,x$ passes through the point (1, 0), then the abscissa of the point on the curve whose ordinate is tan(1), is [27-Jun-2022-Shift-2]

Options:

A. 2e

B. $\frac{2}{e}$

C. 2

D. $\frac{1}{e}$

Answer: B

Solution:

$((\tan^{-1}y) - x)d\,y = (1 + y^2)\,dx$
$\frac{d\,x}{d\,y} + \frac{x}{1 + y^2} = \frac{\tan^{-1}y}{1 + y^2}$
I. F. $= e^{\int \frac{1}{1+y^2}d\,y} = e^{\tan^{-1}y}$
$\therefore$ Solution
$x.\ e^{\tan^{-1}y} = \int \frac{e^{\tan^{-1}y}\tan^{-1}y}{1 + y^2}d\,y$
Let $e^{\tan^{-1}y} = t$
$\frac{e^{\tan - 1}y}{1 + y^2} = d\,t$
$= xe^{\tan^{-1}y} = \int \ln t\,dt = t\ln t - t + c$
$\therefore = xe^{\tan^{-1}y} = e^{\tan^{-1}y}\tan^{-1}y - e^{\tan^{-1}y} + c.....$ (i)
$\because$ It passes through $(1, 0) \Rightarrow c = 2$
Now put $y = \tan 1$, then
$ex = e - e + 2$

$\Rightarrow x = \frac{2}{e}$

Question114

Let y = y(x) be the solution of the differential equation $(1-x^2)dy = \left(xy + (x^3+2)\sqrt{1-x^2}\right)dx$, $-1 < x < 1$, and $y(0) = 0$. If $\int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}\, y(x)\,dx = k$, then k^{-1} is equal to____

[27-Jun-2022-Shift-2]

Answer: 320

Solution:

$(1-x^2)dy = \left(xy + (x^3+2)\sqrt{1-x^2}\right)dx$

$\therefore \frac{dy}{dx} - \frac{x}{1-x^2}y = \frac{x^3+3}{\sqrt{1-x^2}}$

$\therefore$ I. F. $= e^{\int -\frac{x}{1-x^2}dx} = \sqrt{1-x^2}$

Solution is

y. $\sqrt{1-x^2} = \int (x^3+3)\,dx$

$y \cdot \sqrt{1-x^2} = \frac{x^4}{4} + 3x + c$

$\because y(0) = 0 \Rightarrow c = 0$

$\therefore y(x) = \frac{x^4+12x}{4\sqrt{1-x^2}}$

$\therefore \int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}\, y(x)\,dx = \int_{\frac{-1}{2}}^{\frac{1}{2}} \left(\frac{x^4+12x}{4}\right)dx = \int_0^{\frac{1}{2}} \frac{x^4}{2}dx$

$\therefore k = \frac{1}{320}$

$\therefore = k^{-1} = 320$

Question115

Let the solution curve y = y(x) of the differential equation

$$\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right] x\frac{dy}{dx} = x + \left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right] y$$

pass through the points (1, 0) and (2α, α), α > 0. Then α is equal to [28-Jun-2022-Shift-1]

Options:

A. $\frac{1}{2}\exp\left(\frac{\pi}{6}+\sqrt{e}-1\right)$

B. $\frac{1}{2}\exp\left(\frac{\pi}{6}+e-1\right)$

C. $\exp\left(\frac{\pi}{6}+\sqrt{e}+1\right)$

D. $2\exp\left(\frac{\pi}{3}+\sqrt{e}-1\right)$

Answer: A

Solution:

$$\left(\frac{1}{\sqrt{1-\frac{y^2}{x^2}}}+e^{\frac{y}{x}}\right)\frac{dy}{dx} = 1 + \left(\frac{1}{\sqrt{1-\frac{y^2}{x^2}}}+e^{\frac{y}{x}}\right)\frac{y}{x}$$

Putting $y = tx$

$$\left(\frac{1}{\sqrt{1-t^2}}+e^t\right)\left(t+x\frac{dt}{dx}\right) = 1+\left(\frac{1}{\sqrt{1-t^2}}+e^t\right)t$$

$$\Rightarrow x\left(\frac{1}{\sqrt{1-t^2}}+e^t\right)\frac{dt}{dx} = 1$$

$$\Rightarrow \sin^{-1}t + e^t = \ln x + C$$

$$\Rightarrow \sin^{-1}\left(\frac{y}{x}\right) + e^{y/x} = \ln x + C$$

at $x = 1, y = 0$

So, $0+e^0 = 0+C \Rightarrow C = 1$

at $(2\alpha, \alpha)$

$$\sin -1\left(\frac{y}{x}\right) + e^{y/x} = \ln x + 1$$

$$\Rightarrow \frac{\pi}{6}+e^{\frac{1}{2}}-1 = \ln(2\alpha)$$

$$\Rightarrow \alpha = \frac{1}{2}e^{\left(\frac{\pi}{6}+e^{\frac{1}{2}}-1\right)}$$

Question116

Let y = y(x) be the solution of the differential equation $x(1-x^2)\frac{dy}{dx}+(3x^2y-y-4x^3)=0$, $x>1$, with $y(2)=-2$. Then y(3) is equal to [28-Jun-2022-Shift-1]

Options:

A. −18

B. −12

C. −6

D. −3

Answer: A

Solution:

$$\frac{dy}{dx}+\frac{y(3x^2-1)}{x(1-x^2)}=\frac{4x^3}{x(1-x^2)}$$

$$IF=e^{\int\frac{3x^2-1}{x-x^3}dx}=e^{-\ln\left|x^3-x\right|}=e^{-\ln(x^3-x)}=\frac{1}{x^3-x}$$

Solution of D.E. can be given by

$$y.\ \frac{1}{x^3-x}=\int\frac{4x^3}{x(1-x^2)}\cdot\frac{1}{x(x^2-1)}dx$$

$$\Rightarrow\frac{y}{x^3-x}=\int\frac{-4x}{(x^2-1)^2}dx$$

$$\Rightarrow\frac{y}{x^3-x}=\frac{2}{(x^2-1)}+c$$

at $x=2$, $y=-2$

$$\frac{-2}{6}=\frac{2}{3}+c\Rightarrow c=-1$$

at $x=3\Rightarrow\ \frac{y}{24}=\frac{2}{8}-1\Rightarrow y=-18$

Question117

Let x = x(y) be the solution of the differential equation $2ye^{x/y^2}dx+\left(y^2-4xe^{x/y^2}\right)dy=0$ such that $x(1)=0$. Then, x(e) is equal to : [28-Jun-2022-Shift-2]

Options:

A. $e\log_e(2)$

B. $-\log_e(2)$

C. $e^2\log_e(2)$

D. $-e^2\log_e(2)$

Answer: D

Solution:

Given differential equation

$2ye^{\frac{x}{y^2}}dx + \left(y^2 - 4xe^{\frac{x}{y^2}}\right)dy = 0,\ x(1) = 0$

$\Rightarrow e^{\frac{x}{y^2}}[2y\,dx - 4x\,dy] = -y^2dy$

$\Rightarrow e^{\frac{x}{y^2}}\left[\frac{2y^2dx - 4xy\,dy}{y^4}\right] = \frac{-1}{y}dy$

$\Rightarrow 2e^{\frac{x}{y^2}}d\left(\frac{x}{y^2}\right) = -\frac{1}{y}dy$

$\Rightarrow 2e^{\frac{x}{y^2}} = -\ln y + c.....\ \text{(i)}$

Now, using $x(1) = 0,\ c = 2$

So, for $x(e)$, Put $y = e$ in (i)

$2e^{\frac{x}{e^2}} = -1 + 2$

$\Rightarrow \frac{\ }{e^2} = \ln\left(\frac{1}{2}\right) \Rightarrow x(e) = -e^2\ln 2$

Question118

Let the solution curve of the differential equation $x\frac{dy}{dx} - y = \sqrt{y^2 + 16x^2}$, $y(1) = 3$ be $y = y(x)$. Then $y(2)$ is equal to: [29-Jun-2022-Shift-1]

Options:

A. 15

B. 11

C. 13

D. 17

Answer: A

Solution:

Given,

$x\frac{dy}{dx} - y = \sqrt{y^2 + 16x}$

$\Rightarrow x\frac{dy}{dx} = y + \sqrt{y^2 + 16x}$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x} + \sqrt{\left(\frac{y}{x}\right)^2 + 16}$

This is a homogenous different equation.

Let $\frac{y}{x} = v$

$\Rightarrow y = vx$

$\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore v + x\frac{dv}{dx} = v + \sqrt{v^2 + 16}$

$\Rightarrow x\frac{dv}{dx} = \sqrt{v^2 + 16}$

$\Rightarrow \frac{dv}{\sqrt{v^2 + 16}} = \frac{dx}{x}$

Integrating both sides, we get

$\int \frac{dv}{\sqrt{v^2 + 16}} = \int \frac{dx}{x}$

$\Rightarrow \ln\left|v + \sqrt{v^2 + 16}\right| = \ln x + \ln c$

$\Rightarrow v + \sqrt{v^2 + 16} = cx$

Now putting, $v = \frac{y}{x}$, we get

$\frac{y}{x} + \sqrt{\frac{y^2}{x^2} + 16} = cx$

$\Rightarrow \frac{y}{x} + \sqrt{\frac{y^2 + 16x^2}{x^2}} = cx$

$\Rightarrow y + \sqrt{y^2 + 16x^2} = cx^2 \ldots\ldots$

Given, $y(1) = 3$

$\therefore$ When $x = 1$ then $y = 3$.

Putting in equation (1) we get,

$3 + \sqrt{9 + 16} = c.1$

$\Rightarrow c = 8$

$\therefore$ Solution of equation,

$y + \sqrt{y^2 + 16x^2} = 8x^2$

Now, $y(2)$ means when $x = 2$ then $y =$?

$\therefore y + \sqrt{y^2 + 16 \times 4} = 8 \times 4$

$\Rightarrow y = 15$

Question119

Let y = y(x) be the solution of the differential equation

$$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos^2 x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)},\ 0 < x < \frac{\pi}{2} \text{ with } y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32} \text{ If}$$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}, \text{ then the value of } 3\alpha^2 \text{ is equal to\_\_\_\_}$$

[29-Jun-2022-Shift-1]

Answer: 2

Solution:

$$\frac{dy}{dx} + \frac{\sqrt{2}}{2\cos^4 x - \cos 2x}y = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}$$

$$\int \frac{dx}{2\cos^4 x - \cos 2x}$$

$$= \int \frac{dx}{\cos^4 x + \sin^4 x} = \int \frac{\operatorname{cosec}^4 x\, dx}{1 + \cot^4 x}$$

$$= -\int \frac{t^2+1}{t^4+1}dt = -\int \frac{1}{\left(t - \frac{1}{t}\right)^2 + 2}dt = \frac{-1}{\sqrt{2}}\tan^{-1}\left(\frac{t - \frac{1}{t}}{\sqrt{2}}\right)$$

$\text{Cot}\, x = t$

$$= -\frac{1}{\sqrt{2}}\tan^{-1}(\sqrt{2}\cot 2x)$$

$$\therefore \text{IF} = e^{-\tan^{-1}(\sqrt{2}\cot 2x)}$$

$$ye^{-\tan^{-1}(\sqrt{2}\cot 2x)} = \int x dx$$

$$ye^{-\tan^{-1}(\sqrt{2}\cot 2x)} = \frac{x^2}{2} + c$$

$$y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32} + c \Rightarrow c = 0$$

$$y = \frac{x^2}{2}e^{\tan^{-1}(\sqrt{2}\cot 2x)}$$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{\tan^{-1}\left(\sqrt{2}\cot\frac{2\pi}{3}\right)}$$

$$= \frac{\pi^2}{18}e^{-\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)}$$

$$\alpha = \sqrt{\frac{2}{3}} \Rightarrow 3\alpha^2 = 2$$

Question120

If $y = y(x)$ is the solution of the differential equation $(1+e^{2x})\frac{dy}{dx} + 2(1+y^2)e^x = 0$ and $y(0) = 0$ then $6(y'(0) + (y(\log_e\sqrt{3}))^2)$ is equal to
[29-Jun-2022-Shift-2]

Options:

A. 2

B. −2

C. −4

D. −1

Answer: C

Solution:

$$\frac{dy}{1+y^2} + \frac{2e^x}{1+e^{2x}}dx = 0$$

on integration

$\tan^{-1}y + 2\tan^{-1}e^x = c$

$\because y(0) = 0$

so, $C = \frac{\pi}{2} \Rightarrow \tan^{-1}y + 2\tan^{-1}e^x = \frac{\pi}{4}$ from eq.(i), $\left(\frac{dy}{dx}\right)_{x=0} = -1$ arg $y(\ln\sqrt{3}) = -\frac{1}{\sqrt{3}}$

$6\left[y'(0) + (y(\ln\sqrt{3})^2\right] = 6\left[-1 + \frac{1}{3}\right] = -4.$

Question121

Let $y = y(x)$, $x > 1$, be the solution of the differential equation $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with $y(2) = \frac{1+e^4}{2e^4}$. If $y(3) = \frac{e^\alpha+1}{\beta e^\alpha}$, then the value of $\alpha+\beta$ is equal to____
[29-Jun-2022-Shift-2]

Answer: 14

Solution:

$$\frac{dy}{dx}+\frac{2x}{x-1}\cdot y=\frac{1}{(x-1)^2}$$

$$y=\frac{1}{(x-1)^2}\left[\frac{e^{2x}+1}{2e^{2x}}\right]$$

$$y(3)=\frac{e^6+1}{8e^6}$$

$\alpha+\beta=14$

Question122

If $x=x(y)$ is the solution of the differential equation $y\frac{dx}{dy}=2x+y^3(y+1)e^y$, $x(1)=0$; then $x(e)$ is equal to: [24-Jun-2022-Shift-1]

Options:

A. $e^3(e^e-1)$

B. $e^e(e^3-1)$

C. $e^2(e^e+1)$

D. $e^e(e^2-1)$

Answer: A

Solution:

$$\frac{dx}{dy}-\frac{2x}{y}=y^2(y+1)e^y$$

$$\text{If }=e^{\int-\frac{2}{y}dy}=e^{-2\ln y}=\frac{1}{y^2}$$

Solution is given by

$$x\cdot\frac{1}{y^2}=\int y^2(y+1)e^y\cdot\frac{1}{y^2}dy$$

$$\Rightarrow\frac{x}{y^2}=\int(y+1)e^ydy$$

$$\Rightarrow\frac{x}{y^2}=ye^y+c$$

$\Rightarrow x=y^2(ye^y+c)$ at, $y=1, x=0$

$\Rightarrow 0=1(1\cdot e^1+c)\Rightarrow c=-e$ at $y=e$

$x\ =e^2(e\cdot e^e-e)$

Question123

The slope of the tangent to a curve C : y = y(x) at any point (x, y) on it is $\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}$. If C passes through the points $\left(0, \frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to :
[25-Jul-2022-Shift-1]

Options:

A. $\frac{3+\sqrt{2}}{3-\sqrt{2}}$

B. $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

C. $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

D. $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Answer: B

Solution:

$$\frac{dy}{dx}=\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}=e^{2x}-\frac{6e^{-x}}{2+9e^{-2x}}$$

$$\int dy=\int e^{2x}dx-3\int 1+\left(\frac{3e^{-x}}{\sqrt{2}}\right)^2 dx(\text{ put } e^{-x}=t)$$

$$=\frac{e^{2x}}{2}+3\int\frac{dt}{1+\left(\frac{3t}{\sqrt{2}}\right)^2}$$

$$=\frac{e^{2x}}{2}+\sqrt{2}\tan^{-1}\frac{3t}{\sqrt{2}}+C$$

$$y=\frac{e^{2x}}{2}+\sqrt{2}\tan^{-1}\left(\frac{3e^{-x}}{\sqrt{2}}\right)+C$$

It is given that the curve passes through

$$\left(0, \frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$$

$$\frac{1}{2}+\frac{\pi}{2\sqrt{2}}=\frac{1}{2}+\sqrt{2}\tan^{-1}\left(\frac{3}{\sqrt{2}}\right)+C$$

$$\Rightarrow C=\frac{\pi}{2\sqrt{2}}-\sqrt{2}\tan^{-1}\left(\frac{3}{\sqrt{2}}\right)$$

Now if $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$ satisfies the curve, then

$$\frac{1}{2}e^{2\alpha}=\frac{e^{2\alpha}}{2}+\sqrt{2}\tan^{-1}\left(\frac{3e^{-\alpha}}{\sqrt{2}}\right)+\frac{\pi}{2\sqrt{2}}-\sqrt{2}\tan^{-1}\left(\frac{3}{\sqrt{2}}\right)$$

$$\tan^{-1}\left(\frac{3}{\sqrt{2}}\right)-\tan^{-1}\left(\frac{3e^{-\alpha}}{\sqrt{2}}\right)=\frac{\pi}{2\sqrt{2}}\times\frac{1}{\sqrt{2}}=\frac{\pi}{4}$$

$$\frac{\frac{3}{\sqrt{2}}-\frac{3e^{-\alpha}}{\sqrt{2}}}{1+\frac{9}{2}e^{-\alpha}}=1$$

$$\frac{3}{\sqrt{2}}e^{\alpha}-\frac{3}{\sqrt{2}}=e^{\alpha}+\frac{9}{2}$$

$$e^{\alpha}=\frac{\frac{9}{2}+\frac{3}{\sqrt{2}}}{\frac{3}{\sqrt{2}}-1}=\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$$

Question124

The general solution of the differential equation $(x-y^2)dx+y(5x+y^2)dy=0$ is:
[25-Jul-2022-Shift-1]

Options:

A. $(y^2+x)^4=C\,|(y^2+2x)^3|$

B. $(y^2+2x)^4=C\,|(y^2+x)^3|$

C. $|(y^2+x)^3| = C(2y^2+x)^4$

D. $|(y^2+2x)^3| = C(2y^2+x)^4$

Answer: A

Solution:

$$(x-y^2)dx + y(5x+y^2)dy = 0$$

$$y\frac{dy}{dx} = \frac{y^2-x}{5x+y^2}$$

Let $y^2 = t$ $\frac{1}{2}\cdot\frac{dt}{dx} = \frac{t-x}{5x+t}$

Now substitute, $t = vx$

$$\frac{dt}{dx} = v + x\frac{dv}{dx}$$

$$\frac{1}{2}\left\{v + x\frac{dv}{dx}\right\} = \frac{v-1}{5+v}$$

$$x\frac{dv}{dx} = \frac{2v-2}{5+v} - v = \frac{-3v-v^2-2}{5+v}$$

$$\int \frac{5+v}{v^2+3v+2}dv = \int -\frac{dx}{x}$$

$$\int \frac{4}{v+1}dv - \int \frac{3}{v+2}dv = -\int \frac{dx}{x}$$

$$4\ln|v+1| - 3\ln|v+2| = -\ln x + \ln C$$

$$\left|\frac{(v+1)^4}{(v+2)^3}\right| = \frac{c}{x}$$

$$\left|\frac{\left(\frac{y^2}{x}+1\right)^4}{\left(\frac{y^2}{x}+2\right)^3} = \frac{c}{x}\right|$$

$$|(y^2+x)^4| = C\,|(y^2+2x)^3|$$

Question125

Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points $(1, 2)$ and $(8, 1)$, then $y\left(\frac{1}{8}\right)\Big|$ is equal to
[25-Jul-2022-Shift-2]

Options:

A. $2\log_e 2$

B. 4

C. 1

D. $4\log_e 2$

Answer: B

Solution:

$\frac{dy}{dx} \propto \frac{-y}{x}$
$\frac{dy}{dx} = \frac{-ky}{x} \Rightarrow \int \frac{dy}{y} = -K \int \frac{dx}{x}$
$\ln|y| = -K \ln|x| + C$
If the above equation satisfy $(1, 2)$ and $(8, 1)$
$\ln 2 = -K \times 0 + C \Rightarrow C = \ln 2$
$\ln 1 = -K \ln 8 + \ln 2 \Rightarrow K = \frac{1}{3}$
So, at $x = \frac{1}{8}$
$\ln|y\Big| = -\frac{1}{3}\ln\left(\frac{1}{8}\right) + \ln 2 = 2\ln 2$
$|y| = 4$

Question126

Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points $(1, 2)$ and $(8, 1)$, then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to
[25-Jul-2022-Shift-2]

Options:

A. $2\log_e 2$

B. 4

C. 1

D. $4\log_e 2$

Answer: B

Solution:

$\frac{dy}{dx} \propto \frac{-y}{x}$

$\frac{dy}{dx} = \frac{-ky}{x} \Rightarrow \int \frac{dy}{y} = -K \int \frac{dx}{x}$

$\ln|y| = -K \ln|x| + C$

If the above equation satisfy $(1, 2)$ and $(8, 1)$

$\ln 2 = -K \times 0 + C \Rightarrow C = \ln 2$

$\ln 1 = -K \ln 8 + \ln 2 \Rightarrow K = \frac{1}{3}$

So, at $x = \frac{1}{8}$

$\ln|y| = -\frac{1}{3}\ln\left(\frac{1}{8}\right) + \ln 2 = 2\ln 2$

$|y| = 4$

Question127

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{4y^3 + 2yx^2}{3xy^2 + x^3}$, $y(1) = 1$. If for some $n \in \mathbb{N}$, $y(2) \in [n-1, n)$, then n is equal to __ [25-Jul-2022-Shift-2]

Answer: 3

Solution:

$\frac{dy}{dx} = \frac{y}{x}\frac{(4y^2 + 2x^2)}{(3y^2 + x^2)}$

Put $y = vx$

$\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{v(4v^2+2)}{(3v^2+1)}$

$\Rightarrow x\frac{dv}{dx} = v\left(\frac{(4v^2+2-3v^2-1)}{3v^2+1}\right)$

$\Rightarrow \int(3v^2+1)\frac{dv}{v^3+v} = \int\frac{dx}{x}$

$\Rightarrow \ln|v^3+v| = \ln x + c$

$\Rightarrow \ln\left|\left(\frac{y}{x}\right)^3 + \left(\frac{y}{x}\right)\right| = \ln x + C$

$\downarrow y(1) = 1$

$\Rightarrow C = \ln 2$

$\therefore$ for $y(2)$

$\ln\left(\frac{y^3}{8} + \frac{y}{2}\right) = 2\ln 2 \Rightarrow \frac{y^3}{8} + \frac{y}{2} = 4$

$\Rightarrow [y(2)] = 2$

$\Rightarrow n = 3$

Question128

If $\frac{dy}{dx} + 2y\tan x = \sin x$, $0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of $y(x)$ is :
[26-Jul-2022-Shift-1]

Options:

A. $\frac{1}{8}$

B. $\frac{3}{4}$

C. $\frac{1}{4}$

D. $\frac{3}{8}$

Answer: A

Solution:

$\frac{dy}{dx} + 2y\tan x = \sin x$

which is a first order linear differential equation.

Integrating factor (I. F.) $= e^{\int 2\tan x\,dx}$

$= e^{2\ln|\sec x|} = \sec^2 x$

Solution of differential equation can be written as

$y \cdot \sec^2 x = \int \sin x \cdot \sec^2 x\,dx = \int \sec x \cdot \tan x\,dx\ y\sec^2 x = \sec x + C$

$y\left(\frac{\pi}{3}\right) = 0, 0 = \sec\frac{\pi}{3} + C \Rightarrow C = -2$

$$y = \frac{\sec x - 2}{\sec^2 x} = \cos x - 2\cos^2 x$$

$$= \frac{1}{8} - 2\left(\cos x - \frac{1}{4}\right)^2$$

$$y_{max} = \frac{1}{8}$$

Question129

Let a curve y = y(x) pass through the point (3, 3) and the area of the region under this curve, above the x-axis and between the abscissae 3 and x(>3) be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $(\alpha, 6\sqrt{10})$ in the first quadrant, then α is equal to __________.
[26-Jul-2022-Shift-1]

Answer: 6

Solution:

$$\int_3^x f(x)dx = \left(\frac{f(x)}{x}\right)^3$$

$$x^3 \cdot \int_3^x f(x)dx = f^3(x)$$

Differentiate w.r.t. x

$$x^3 f(x) + 3x^2 \cdot \frac{f^3(x)}{x^3} = 3f^2(x)f'(x)$$

$$\Rightarrow 3y^2 \frac{dy}{dx} = x^3 y + \frac{3y^3}{x}$$

$$3xy \frac{dy}{dx} = x^4 + 3y^2$$

Let $y^2 = t$

$$\frac{3}{2}\frac{dt}{dx} = x^3 + \frac{3t}{x}$$

$$\frac{dt}{dx} - \frac{2t}{x} = \frac{2x^3}{3}$$

$$\text{I. F.} = e^{\int -\frac{2}{x}dx} = \frac{1}{x^2}$$

Solution of differential equation

$$t.\ \frac{1}{x^2} = \int \frac{2}{3}x\,dx$$

$$\frac{y^2}{x^2} = \frac{x^2}{3} + C$$

$y^2 = \frac{x^4}{3} + Cx^2$

Curve passes through $(3, 3) \Rightarrow C = -2$

$y^2 = \frac{x^4}{3} - 2x^2$

Which passes through $(\alpha, 6\sqrt{10})$

$\frac{\alpha^4 - 6\alpha^2}{3} = 360$

$\alpha^4 - 6\alpha^2 - 1080 = 0$

$\alpha = 6$

Question130

Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$, $x \in (-1, 1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is equal to

[26-Jul-2022-Shift-2]

Options:

A. $\frac{\pi}{3} - \frac{1}{4}$

B. $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

C. $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

D. $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Answer: B

Solution:

$\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$

which is first order linear differential equation.

Integrating factor $(I.F) = .e^{\int \frac{x}{x^2-1}dx}$

$= e^{\frac{1}{2}\ln|x^2-1|} = \sqrt{|x^2-1|}$

$= \sqrt{1-x^2}$

$\because x \in (-1, 1)$

Solution of differential equation

$$y\sqrt{1-x^2} = \int (x^4 + 2x)dx = \frac{x^5}{5} + x^2 + c$$

Curve is passing through origin, $c = 0$

$$y = \frac{x^5 + 5x^2}{5\sqrt{1-x^2}}$$

$$\int_{\frac{-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^5 + 5x^2}{5\sqrt{1-x^2}} dx = 0 + 2\int_0^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$$

put $x = \sin\theta$

$dx = \cos\theta \, d\theta$

$$I = 2\int_0^{\frac{\pi}{3}} \frac{\sin^2\theta \cdot \cos\theta \, d\theta}{\cos\theta}$$

$$= \int_0^{\frac{\pi}{3}} (1 - \cos 2\theta) d\theta$$

$$= \left(\theta - \frac{\sin 2\theta}{2}\right)\Bigg|_0^{\frac{\pi}{3}}$$

Question131

Suppose $y = y(x)$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim_{x \to \infty} y(x)$ is finite. If a and b are respectively the x - and y-intercepts of the tangent to the curve at $x = 0$, then the value of $a - 4b$ is equal to _________.
[26-Jul-2022-Shift-2]

Answer: 3

Solution:

IF $= e^{-x}$

$$y \cdot e^{-x} = -2e^{-x} + \frac{e^{-2x}}{2} + C$$

$\Rightarrow y = -2 + e^{-x} + Ce^{x}$

$\lim_{x \to \infty} y(x)$ is finite so $C = 0$

$y = -2 + e^{-x}$

$$\Rightarrow \frac{dy}{dx} = -.e^{-x} \Rightarrow \frac{dy}{dx}\Big|_{x=0} = -1$$

Equation of tangent

$y + 1 = -1(x - 0)$

or $y + x = -1$

So $a=-1, b=-1$
$\Rightarrow a-4b=3$

Question132

Let $y=y_1(x)$ and $y=y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx}=x+y$, with $y_1(0)=0$ and $y_2(0)=1$ respectively. Then, the number of points of intersection of $y=y_1(x)$ and $y=y_2(x)$ is [27-Jul-2022-Shift-1]

Options:

A. 0

B. 1

C. 2

D. 3

Answer: A

Solution:

$\frac{dy}{dx}=x+y$
Let $x+y=t$
$1+\frac{dy}{dx}=\frac{dt}{dx}$
$\frac{dt}{dx}-1=t \Rightarrow \int \frac{dt}{t+1}=\int dx$
$\ln|t+1|=x+C'$
$|t+1|=Ce^x$
$|x+y+1|=Ce^x$
For $y_1(x)$, $y_1(0)=0 \Rightarrow C=1$
For $y_2(x)$, $y_2(0)=1 \Rightarrow C=2$
$y_1(x)$ is given by $|x+y+1|=e^x$
$y_2(x)$ is given by $|x+y+1|=2e^x$
At point of intersection
$e^x=2e^x$
No solution
So, there is no point of intersection of $y_1(x)$ and $y_2(x)$.

Question133

Let $y = y(x)$ be the solution curve of the differential equation $\sin(2x^2)\log_e(\tan x^2)\,dy + \left(4xy - 4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)\right)dx = 0$, $0 < x < \sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}}, 1\right)$. Then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to ________.

[27-Jul-2022-Shift-1]

Answer: 1

Solution:

$$\frac{dy}{dx} + y\left(\frac{4x}{\sin(2x^2)\ln(\tan x^2)}\right) = \frac{4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)}{\sin(2x^2)\ln(\tan x^2)}$$

$$I.F = e^{\int \frac{4x}{\sin(2x^2)\ln(\tan x^2)}dx}$$

$$= e^{\ln|\ln(\tan x^2)|} = \ln(\tan x^2)$$

$$\therefore \int d(y\cdot\ln(\tan x^2)) = \int \frac{4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)}{\sin(2x^2)}dx$$

$$\Rightarrow y\ln(\tan x^2) = \ln\left|\frac{\sec x^2 + \tan x^2}{\operatorname{cosec} x^2 - \cot x^2}\right| + C$$

$$\ln\left(\frac{1}{\sqrt{3}}\right) = \ln\left(\frac{\frac{3}{\sqrt{3}}}{2 - \sqrt{3}}\right) + C$$

$$e = \ln\left(\frac{1}{\sqrt{3}}\right) - \ln\left(\frac{\sqrt{3}}{2 - \sqrt{3}}\right)$$

For $y\left(\sqrt{\frac{\pi}{3}}\right)$

$$y\ln(\sqrt{3}) = \ln\left|\frac{2 + \sqrt{3}}{\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}}\right| + \ln\left(\frac{1}{\sqrt{3}}\right) - \ln\left(\frac{\sqrt{3}}{2\sqrt{3}}\right)$$

$$= \ln(2 + \sqrt{3}) + \ln\left(\frac{1}{\sqrt{3}}\right) + \ln\left(\frac{1}{\sqrt{3}}\right) - \ln\left(\frac{\sqrt{3}}{2 - \sqrt{3}}\right)$$

$$\Rightarrow y\ln\sqrt{3} = \ln\left(\frac{1}{\sqrt{3}}\right)$$

$$\Rightarrow \frac{y}{2}\ln 3 = -\frac{1}{2}\ln 3$$

$$\Rightarrow y = 1$$

$$\therefore \left|y\left(\sqrt{\frac{\pi}{3}}\right)\right| = 1$$

Question134

Let the solution curve of the differential equation $xdy = \left(\sqrt{x^2 + y^2} + y\right)dx$, $x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is :
[28-Jul-2022-Shift-1]

Options:

A. $\frac{1}{2}$

B. $\frac{3}{2}$

C. $-\frac{3}{2}$

D. $\frac{5}{2}$

Answer: B

Solution:

$\frac{xdy - ydx}{\sqrt{x^2 + y^2}} = dx$

$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{x^2 + y^2}}{x} + \frac{y}{x}$

$\Rightarrow \frac{dy}{dx} = \sqrt{1 + \frac{y^2}{x^2}} + \frac{y}{x}$

Let $\frac{y}{x} = v$

$\Rightarrow v + x\frac{dv}{dx} = \sqrt{1 + v^2} + v$

$\Rightarrow \frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$

OR $\ln\left(v + \sqrt{1 + v^2}\right) = \ln x + C$

at $x = 1, y = 0$

$\Rightarrow C = 0$

$\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} = x$

At $x = 2$

$\frac{y}{2} + \sqrt{1 + \frac{y^2}{4}} = 2$

$\Rightarrow 1 + \frac{y^2}{4} = 4 + \frac{y^2}{4} - 2y$

OR $y = \frac{3}{2}$

Question135

If $y = y(x)$, $x \in (0, \pi/2)$ be the solution curve of the differential equation $(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y - 2e^{-4x}(2\sin 2x + \cos 2x)$,

with $y\left(\frac{\pi}{4}\right) = e^{-\pi}$, then $y\left(\frac{\pi}{6}\right)$ is equal to:

[28-Jul-2022-Shift-1]

Options:

A. $\frac{2}{\sqrt{3}}e^{-2\pi/3}$

B. $\frac{2}{\sqrt{3}}e^{2\pi/3}$

C. $\frac{1}{\sqrt{3}}e^{-2\pi/3}$

D. $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Answer: A

Solution:

$(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y$

$= 2e^{-4x}(2\sin 2x + \cos 2x)$

$\frac{dy}{dx} + (8 + 4\cot 2x)y = 2e^{-4x}\left(\frac{2\sin 2x + \cos 2x}{\sin^2 2x}\right)$

Integrating factor

$(I.F.) = e^{\int(8 + 4\cot 2x)dx}$

$= e^{8x + 2\ln \sin 2x}$

Solution of differential equation

$y.\, e^{8x + 2\ln \sin 2x}$

$= \int 2e^{(4x + 2\ln \sin 2x)}\frac{(2\sin 2x + \cos 2x)}{\sin^2 2x}dx$

$= 2\int e^{4x}(2\sin 2x + \cos 2x)dx$

$y.\, e^{8x + 2\ln \sin 2x} = e^{4x}\sin 2x + c$

$y\left(\frac{\pi}{4}\right) = e^{-\pi}$

$e^{-\pi} \cdot e^{2\pi} = e^{\pi} + c \Rightarrow c = 0$

$y\left(\frac{\pi}{6}\right) = \frac{e^{\frac{2\pi}{3}}\frac{\sqrt{3}}{2}}{e^{\left(\frac{4\pi}{3} + 2\ln\frac{\sqrt{3}}{2}\right)}}$

$= e^{\frac{-2\pi}{3}} \cdot \frac{2}{\sqrt{3}}$

Question136

Let y = y(x) be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{1/2}$, x > 1 passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}y(8)$ is equal to:

[28-Jul-2022-Shift-2]

Options:

A. $11 + 6\log_e 3$

B. 19

C. $12 - 2\log_e 3$

D. $19 - 6\log_e 3$

Answer: B

Solution:

$\frac{dy}{dx} + \frac{1}{x^2-1}y = \sqrt{\frac{x-1}{x+1}}, x > 1$

Integrating factor I.F. $= e^{\int \frac{1}{x^2-1}dx} = e^{\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right|}$

$= \sqrt{\frac{x-1}{x+1}}$

Solution of differential equation

$y\sqrt{\frac{x-1}{x+1}} = \int \frac{x-1}{x+1}dx = \int\left(1 - \frac{2}{x+1}\right)dx$

$y\sqrt{\frac{x-1}{x+1}} = x - 2\ln\left|x+1\right| + C$

Curve passes through $\left(2, \sqrt{\frac{1}{3}}\right)$

$\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = 2 - 2\ln 3 + C$

$C = 2\ln 3 - \frac{5}{3}$

$y(8) \times \frac{\sqrt{7}}{3} = 8 - 2\ln 9 + 2\ln 3 - \frac{5}{3}$

$\sqrt{7} \cdot y(8) = 19 - 6\ln 3$

Question137

The differential equation of the family of circles passing through the points (0, 2) and (0, −2) is :
[28-Jul-2022-Shift-2]

Options:

A. $2xy\frac{dy}{dx} + (x^2 - y^2 + 4) = 0$

B. $2xy\frac{dy}{dx} + (x^2 + y^2 - 4) = 0$

C. $2xy\frac{dy}{dx} + (y^2 - x^2 + 4) = 0$

D. $2xy\frac{dy}{dx} - (x^2 - y^2 + 4) = 0$

Answer: A

Solution:

Family of circles passing through the points $(0, 2)$ and $(0, -2)$
$x^2 + (y-2)(y+2) + \lambda x = 0, \lambda \in R$
$x^2 + y^2 + \lambda x - 4 = 0 \ldots\ldots \ (1)$
Differentiate w.r.t x
$2x + 2y\frac{dy}{dx} + \lambda = 0 \ldots\ldots \ (2)$
Using (1) and (2), eliminate λ
$x^2 + y^2 - \left(2x + 2y\frac{dy}{dx}\right)x - 4 = 0$
$2xy\frac{dy}{dx} + x^2 - y^2 + 4 = 0$

Question138

Let the solution curve $y = y(x)$ of the differential equation $(1 + e^{2x})\left(\frac{dy}{dx} + y\right) = 1$ pass through the point $\left(0, \frac{\pi}{2}\right)$. Then, $\lim\limits_{x \to \infty} e^x y(x)$ is equal to :
[29-Jul-2022-Shift-1]

Options:

A. $\frac{\pi}{4}$

B. $\frac{3\pi}{4}$

C. $\frac{\pi}{2}$

D. $\frac{3\pi}{2}$

Answer: B

Solution:

D.E. $(1+e^{2x})\left(\frac{dy}{dx}+y\right)=1$

$\Rightarrow \frac{dy}{dx}+y=\frac{1}{1+e^{2x}}$

I.F. $=e^{\int 1.dx}=e^{x}$

$\therefore$ Solution

$e^{x}y(x)=\int \frac{e^{x}}{1+e^{2x}}dx$

$\Rightarrow e^{x}y(x)=\tan^{-1}(e^{x})+C$

$\because$ It passes through $\left(0,\ \frac{\pi}{2}\right)$, $C=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{4}$

$\therefore \lim\limits_{x\to\infty} e^{x}y(x)=\lim\limits_{x\to\infty}\tan^{-1}(e^{x})+\frac{\pi}{4}$

$=\frac{3\pi}{4}$

Question139

If the solution curve of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passes through the points (2, 1) and (k + 1, 2), k > 0, then
[29-Jul-2022-Shift-2]

Options:

A. $2\tan^{-1}\left(\frac{1}{k}\right)=\log_{e}(k^{2}+1)$

B. $\tan^{-1}\left(\frac{1}{k}\right)-\log_{e}(k^{2}+1)$

C. $2\tan^{-1}\left(\frac{1}{k+1}\right)=\log_{e}(k^{2}+2k+2)$

D. $2\tan^{-1}\left(\frac{1}{k}\right)=\log_{e}\left(\frac{k^{2}+1}{k^{2}}\right)$

Answer: A

Solution:

$$\frac{dy}{dx}=\frac{x+y-2}{x-y}=\frac{(x-1)+(y-1)}{(x-1)-(y-1)}$$

Let $x-1=X,\ y-1=Y$

$$\frac{dY}{dX}=\frac{X+Y}{X-Y}$$

Let $Y=tX \Rightarrow \frac{dY}{dX}=t+X\frac{dt}{dX}$

$$t+X\frac{dt}{dX}=\frac{1+t}{1-t}$$

$$X\frac{dt}{dX}=\frac{1+t}{1-t}-t=\frac{1+t^2}{1-t}$$

$$\int\frac{1-t}{1+t^2}dt=\int\frac{dX}{X}$$

$$\tan^{-1}t-\frac{1}{2}\ln(1+t^2)=\ln|X|+c$$

$$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ln|x-1|+c$$

Curve passes through $(2, 1)$

$0-0=0+c \Rightarrow c=0$

If $(k+1, 2)$ also satisfies the curve

$$\tan^{-1}\left(\frac{1}{k}\right)-\frac{1}{2}\ln\left(\frac{1+k^2}{k^2}\right)=\ln k$$

$$2\tan^{-1}\left(\frac{1}{k}\right)=\ln(1+k^2)$$

Question140

Let y = y(x) be the solution curve of the differential equation $\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{(x+3)}{x+1}$, $x>-1$, which passes through the point (0, 1). Then y(1) is equal to: [29-Jul-2022-Shift-2]

Options:

A. $\frac{1}{2}$

B. $\frac{3}{2}$

C. $\frac{5}{2}$

D. $\frac{7}{2}$

Answer: B

Solution:

$$\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{(x+3)}{x+1}, x>-1$$

Integrating factor I.F. $=e^{\int \frac{2x^2+11x+13}{x^3+6x^2+11x+6}dt}$

Let $\frac{2x^2+11x+13}{(x+1)(x+2)(x+3)}=\frac{A}{x+1}+\frac{B}{x+2}+\frac{C}{x+3}$ $A=2, B=1, C=-1$

I.F. $=e^{(2\ln|x+1|+\ln|x+2|-\ln|x+3|)}$

$=\frac{(x+1)^2(x+2)}{x+3}$

Solution of differential equation

$y\cdot\frac{(x+1)^2(x+2)}{x+3}=\int(x+1)(x+2)\,dt$

$y\frac{(x+1)^2(x+2)}{x+3}=\frac{x^3}{3}+\frac{3x^2}{2}+2x+c$

Curve passes through $(0,1)$

$1\times\frac{1\times 2}{3}=0+c\Rightarrow c=\frac{2}{3}$

So, $y(1)=\frac{\frac{1}{3}+\frac{3}{2}+2+\frac{2}{3}}{\frac{(2^2\times 3)}{4}}=\frac{3}{2}$

Question141

The population $P=P(t)$ at time ' t ' of a certain species follows the differential equation $\frac{dP}{dt}=0.5P-450$. If $P(0)=850$, then the time at which population becomes zero is :
24 Feb 2021 Shift 1

Options:

A. $\log_e 18$

B. $\log_e 9$

C. $1/2\log_e 18$

D. $2\log_e 18$

Answer: D

Solution:

Given that $\frac{dP}{dt} = 0.5P - 450$

$\Rightarrow \int_0^t \frac{dp}{P-900} = \int_0^t \frac{dt}{2}$

$\Rightarrow [\ell n\,|P(t) - 900|]_0^t = \left[\frac{t}{2}\right]_0^t$

$\Rightarrow \ell n\,|P(t) - 900| - \ell n\,|P(0) - 900| = \frac{t}{2}$

$\Rightarrow \ell n\,|P(t) - 900| - \ell n\,|50| = \frac{t}{2}$

For $P(t) = 0$

$\Rightarrow \ell n\left|\frac{900}{50}\right| = \frac{t}{2}$

$\Rightarrow t = 2\ell\,\ln 18$

Question142

Let f be a twice differentiable function defined on R, such that

$f(0) = 1, f'(0) = 2$ and $f''(x) \neq 0$ for all $x \in R$. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$, for all

$x \in R$, then the value of $f(1)$ lies in the interval [24 Feb 2021 Shift 2]

Options:

A. (9, 12)

B. (6, 9)

C. (0, 3)

D. (3, 6)

Answer: B

Solution:

Given, $f(0) = 1$,

$f'(0) = 2$,

$f''(x) \neq 0$

$$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$$

$\Rightarrow f(x)f''(x) - f'(x)f'(x) = 0$

$\Rightarrow \frac{f''(x)}{f'(x)} = \frac{f'(x)}{f(x)}$

$\Rightarrow \int \frac{f''(x)}{f'(x)} dx = \int \frac{f'(x)}{f(x)} dx$
$\Rightarrow \log f'(x) = \log f(x) + \log c$
$\Rightarrow f'(x) = cf(x)$
Now, put $x = 0$, we get
$f'(0) = cf(0)$
$\Rightarrow 2 = c \times 1$
$\Rightarrow c = 2$
Putting the value of $c = 2$ in Eq. (i), we get
$\log f'(x) = \log f(x) + \log 2$
$\Rightarrow f'(x) = 2f(x) \Rightarrow \int \frac{f'(x)}{f(x)} dx = \int 2 dx$
$\Rightarrow \log f(x) = 2x + D \Rightarrow f(x) = e^{2x+D}$
$\Rightarrow f(x) = e^D \cdot e^{2x}$
$\Rightarrow f(x) = k \cdot e^{2x}$ [Let $k = e^D$]
Put $x = 0$, we get
$f(0) = k \cdot e^0$
$\Rightarrow 1 = k \Rightarrow f(x) = k \cdot e^{2x}$
$\therefore f(x) = e^{2x}$
Put $x = 1$, we get
$f(1) = e^2$
Clearly, e^2 lies in $(6, 9)$.

Question143

The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a\left(x + \frac{\sqrt{a}}{2}\right)$, a > 0 is .... [26 Feb 2021 Shift 1]

Answer: 2

Solution:

Given, $y^2 = a\left[x + \frac{\sqrt{a}}{2}\right], a > 0$ ...(i)
Differentiating both sides w.r.t. ' x ',
$2y \frac{dy}{dx} = a[1 + 0] = a$ . . . (ii)
Use Eq. (ii) in Eq. (i) to eliminate the constant ' a '.
$y^2 = 2y \frac{dy}{dx}\left(x + \sqrt{2y} \sqrt{\frac{dy}{dx}}\right)$
$y^2 - 2xy \frac{dy}{dx} = 2\sqrt{2} \cdot y\sqrt{y} \cdot \frac{dy}{dx} \sqrt{\frac{dy}{dx}}$
Squaring on both sides,

$y^4 + 4x^2y^2\left(\frac{dy}{dx}\right)^2 - 4xy^3\frac{dy}{dx} = 8y^3\left(\frac{dy}{dx}\right)^3$
Thus, degree of above differential equation is 3 and its order is 1 .
Difference between degree and order $= 3 - 1 = 2$

Question144

Let slope of the tangent line to a curve at any point P(x, y) be given by $\frac{xy^2 + y}{x}$. If the curve intersects the line $x + 2y = 4$ at $x = -2$, then the value of y, for which the point (3, y) lies on the curve, is
[26 Feb 2021 Shift 2]

Options:

A. $\frac{18}{35}$

B. $-\frac{4}{3}$

C. $-\frac{18}{19}$

D. $-\frac{18}{11}$

Answer: C

Solution:

Given, slope of tangent line to curve at (x, y) is $\frac{xy^2 + y}{x}$

i.e., $\frac{dy}{dx} = \frac{xy^2 + y}{x}$

$\Rightarrow \frac{dy}{dx} = y^2 + \frac{y}{x} \Rightarrow x\,dy = xy^2dx + y\,dx$

$\Rightarrow x\,dy - y\,dx = xy^2dx \Rightarrow \frac{x\,dy - y\,dx}{y^2} = x\,dx$

Integrating both sides, we get

$\frac{-x}{y} = \frac{x^2}{2} + C \ldots$ (i)

The curve intersect line at $x = -2$
Then, $x = -2$, is satisfied by $x + 2y = 4$
Hence, $(-2) + 2y = 4$ Gives, $y = 3$
∴ Curve passes through $(2, -3)$.
Use $(2, -3)$ is Eq. (i), we get

$\frac{-2}{-3} = \frac{(-2)^2}{2} + C \Rightarrow C = \frac{-4}{3}$

∴ The curve is

$\frac{-x}{y} = \frac{x^2}{2} = \frac{-4}{3} \ldots$ (ii)

It also passes through $(3, y)$.
Put $(3, 4)$ in Eq. (ii), we get

$$\Rightarrow \frac{-3}{y} = \frac{(3)^2}{2} - \frac{4}{3} \Rightarrow y = -\frac{18}{19}$$

Question145

Let $f(x) = \int_0^x e^t f(t) dt + e^x$ be a differentiable function for all $x \in R$. Then, $f(x)$ equals
[26 Feb 2021 Shift 2]

Options:

A. $2e^{(e^x - 1)} - 1$

B. $e^{e^x} - 1$

C. $2e^{e^x} - 1$

D. $e^{(e^x - 1)}$

Answer: A

Solution:

Given, $f(x) = \int_0^x e^t f(t) dt + e^x \ldots$ (i)
Since, $f(x)$ is differentiable function, differentiate Eq. (i)
$f'(x) = e^x f(x) + e^x$[Using Newton Leibnitz theorem]

$f'(x) = e^x (f(x) + 1) \Rightarrow \frac{f'(x)}{f(x) + 1} = e^x$

Integrating it, $\int \frac{f'(x)}{f(x) + 1} dx = \int e^x dx + C$

Let $f(x) + 1 = u$, then $f'(x) dx = du$

$\int \frac{du}{u} = e^x + C \Rightarrow \log u = e^x + C$

$\log(f(x) + 1) = e^x + C \quad [\because u = f(x) + 1] \ldots$ (ii)
Now

$f(x) = \int_0^x e^t f(t) dt + e^x \Rightarrow f(0) = e^0 = 1$

Put $x = 0$, in Eq. (ii), we get
$\log(2) = e^{\circ} + C \Rightarrow C = \log(2) - 1$
From Eq. (ii), we get

$$\begin{aligned} \log(f(x) + 1) &= e^x + \log 2 - 1 \\ f(x) + 1 &= e^{e^x + \log 2 - 1} = e^{e^x} \cdot e^{\log 2} \cdot e^{-1} \\ f(x) + 1 &= 2e^{e^x} \cdot e^{-1} = 2e^{e^x - 1} \\ \therefore \ f(x) &= 2e^{e^x - 1} - 1 \end{aligned}$$

Question146

The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time t = 0. The number of bacteria is increased by 20% in 2h. If the population of bacteria is 2000 after $\frac{k}{\log_e(6/5)}$h, then $\left(\frac{k}{\log_e 2}\right)^2$ is equal to

[26 Feb 2021 Shift 1]

Options:

A. 4

B. 8

C. 2

D. 16

Answer: A

Solution:

Let x be the number of bacteria at any time t.

Given that, $\frac{dx}{dt} \propto x$ $\left(\because \text{Rate of growth } = \frac{dx}{dt}\right)$

$\Rightarrow \frac{dx}{dt} = \lambda x \Rightarrow \frac{dx}{x} = \lambda dt$

After integrating it, we get

$\log x = \lambda t + C \ldots$ (i)

$= 0, x = 1000$ which gives

$\log 1000 = 0 + C \Rightarrow C = \log 1000$

Given, when $t = 0, x = \log 1000$

$\log x - \log 1000 = \lambda t$ or $\log\left(\frac{x}{1000}\right) = \lambda t \ldots$ (ii)

Given that in 2h, number of bacteria increased by 20% i.e. when $t = 2h, x = 1200$ Put, $t = 2$ and $x = 1200$ in Eq. (ii), $\log\left(\frac{1200}{1000}\right) = 2\lambda$ gives, $\lambda = \frac{1}{2}\log\left(\frac{6}{5}\right)$

Again, from Eq. (ii),

$\log\left(\frac{x}{1000}\right) = \frac{1}{2}\log\left(\frac{6}{5}\right)$

or $\frac{x}{1000} = e^{\frac{t}{2}\log\left(\frac{6}{5}\right)} \ldots$ (iii)

Given, $x = 2000$ at $t = k/\log_e(6/5)$,

put in Eq. (iii),

$\frac{2000}{1000} = e^{\frac{k}{2}\log\left(\frac{6}{5}\right)/\log\left(\frac{6}{5}\right)}$

$2 = e^{k/2}$ or $\log 2 = k/2$

$k/\log 2 \quad = 2$
$\therefore \ (k/\log_e 2)^2 \quad = (2)^2 = 4$

Question147

If y = y(x) is the solution of the equation $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x$, y(0) = 0, then $1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2} y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}} y\left(\frac{\pi}{4}\right)$ is ....
[26 Feb 2021 Shift 1]

Answer: 1

Solution:

Given $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x$ . . . (i)
Let $e^{\sin y} = t$, then $e^{\sin y} \cdot \cos y \cdot \frac{dy}{dx} = \frac{dt}{dx}$,
Putting in Eq. (i),
$\cos x$ ... (ii) (Linear form)
Then, IF $= e^{\int \cos x \, dx} = e^{\sin x}$
Solution of differential Eq. (ii) is,
$t \cdot IF = \int \cos x \cdot IF \, dx + C$
$t \cdot e^{\sin x} = \int \cos x \cdot e^{\sin x} dx + C$
$= e^u$ i.e. let $\sin x = u$ then $\cos x \, dx = du$
$\Rightarrow \ t \cdot e^{\sin x} = \int e^u du + C = e^u + C$
Put $u = \sin x$ and $t = e^{\sin y}$
$\Rightarrow \ e^{\sin y} \cdot e^{\sin x} = e^{\sin x} + C$
Given, $y(0) = 0$, this gives $C = 0$
$\Rightarrow \ e^{\sin y} \cdot e^{\sin x} = e^{\sin x}$
$\Rightarrow \ e^{\sin y + \sin x} = e^{\sin x}$
$\Rightarrow \ \sin y + \sin x = \sin x$
$\Rightarrow \ \sin y = 0$
$\Rightarrow \ y = 0$
$\therefore y(\pi/6) = y(\pi/3) = y(\pi/4) = 0$
Hence, $1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2} y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}} y\left(\frac{\pi}{4}\right)$
$= 1 + 0 + 0 + 0 = 1$

Question148

If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2-4x+y+8}{x-2}$, then this curve also passes through the point [25 Feb 2021 Shift 1]

Options:

A. (5, 4)

B. (4, 5)

C. (4, 4)

D. (5, 5)

Answer: D

Solution:

Given, slope $=\frac{x^2-4x+y+8}{x-2}$

$\Rightarrow \frac{dy}{dx}=\frac{x^2-4x+y+8}{x-2}=\frac{(x-2)^2+(y+4)}{(x-2)}$

$=(x-2)+\frac{y+4}{x-2}$

Let $(x-2)=t \Rightarrow dx=dt$

and $(y+4)=u \Rightarrow dy=du$

$\therefore \frac{dy}{dx}=\frac{du}{dt}$

Now, $\frac{dy}{dx}=(x-2)+\frac{(y+4)}{(x-2)}$

$\Rightarrow \frac{du}{dt}=t+\frac{u}{t} \Rightarrow \frac{du}{dt}-\frac{u}{t}=t$

Here, integrating factor (IF) $=1/t$

$\Rightarrow u\cdot\left(\frac{1}{t}\right)=\int t\left(\frac{1}{t}\right)dt \Rightarrow u/t=t+c$

$\Rightarrow \frac{(y+4)}{(x-2)}=(x-2)+c$

$\because$ It passes through origin [i.e. $(0,0)$], then

$\therefore \frac{4}{-2}=-2+c$

$\Rightarrow -2+2=c \Rightarrow c=0$

Hence, $\frac{(y+4)}{(x-2)}=(x-2)+0$

$\Rightarrow y+4=(x-2)^2$

Clearly, this curve passes through $(5,5)$ as it satisfies the equation.

Question149

If the curve y = y(x) represented by the solution of the differential equation $(2xy^2-y)dx+xdy=0$, passes through the intersection of the lines

$2x-3y=1$ and$3x+2y=8$, then $|y(1)|$ is equal to ....
[25 Feb 2021 Shift 2]

Answer: 1

Solution:

Given, $(2xy^2-y)dx+xdy=0$
$\Rightarrow \frac{dy}{dx}-\frac{y}{x}=-2y^2$
$\Rightarrow \frac{-1}{y^2}\frac{dy}{dx}+\frac{1}{xy}=2$ ...(i) [divide by y^2]
Let $\frac{1}{y}=v$, then $-\frac{1}{y^2}\cdot\frac{dy}{dx}=\frac{dv}{dx}$, putting in Eq. (i)
$\frac{dv}{dx}+v\left(\frac{1}{x}\right)=2$ (this is a linear form)
Now, integrating factor (IF) $=e^{\int\frac{1}{x}dx}=e^{\log x}=x$
$\therefore\ (IF)v=\int 2.(IF)dx=\int 2xdx=2\frac{x^2}{2}+C$
$\therefore\ (IA)v=x^2+C$
Put $v=\frac{1}{y}$, this gives
$x^2+c=\frac{x}{y}$
Now, first find point of intersection of lines
$2x-3y=1$ and $3x=-2y+8$ by elimination method, we get $x=2,\ y=1$
$\therefore$ The curve $x^2+c=\frac{1}{y}$ passes through $(2,1)$.
Put $x=2,\ y=1$, we get $c=-2$
$\frac{x}{y}=x^2-2$
or $y=\frac{x}{x^2-2}$
Put $x=1$, we get $y(1)=\frac{1}{1-2}=-1$
$\therefore |y(1)|=1$

Question150

If a curve $y=f(x)$ passes through the point (1, 2) and satisfies $x\frac{dy}{dx}+y=bx^4$, then for what value of b, $\int_1^2 f(x)dx=\frac{62}{5}$?
[24 Feb 2021 Shift 2]

Options:

A. 5

B. 10

C. $\frac{62}{5}$

D. $\frac{31}{5}$

Answer: B

Solution:

Given, curve $y = f(x)$ passes through $(1, 2)$ and satisfies

$x\frac{dy}{dx} + y = bx^4$

$\Rightarrow x\frac{dy}{dx} + y = bx^4$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = bx^3$

$\text{IF} = e^{\int \frac{1}{x} \cdot dx} = x$

$yx = \int bx^4 dx = \frac{bx^5}{5} + C$

$\therefore y = \frac{bx^4}{5} + \frac{C}{x} = f(x)$. . . (i)

$\because$ This curve passes through $(1, 2)$.

$\therefore 2 \times 1 = \frac{b \times (1)^5}{5} + C$

$\Rightarrow 2 = \frac{b}{5} + C$ . . . (ii)

Also, $\int_1^2 f(x) dx = \frac{62}{5}$

$\Rightarrow \int_1^2 \left(\frac{bx^4}{5} + \frac{C}{x}\right) dx = \frac{62}{5}$

$\Rightarrow \left[b \times \frac{x^5}{25} + C \log x\right]_1^2 = \frac{62}{5}$

$\Rightarrow \left[\left(\frac{b \times 32}{25} + C \log 2\right) - \left(\frac{b}{25} + c \log 1\right)\right] = \frac{62}{5}$

$\Rightarrow \frac{b \times 32}{25} + C \log 2 - \frac{b}{25} = \frac{62}{5} + 0 \log 2$

[from Eq. (i)] On comparing, we get

$\frac{b}{25} \times 31 = \frac{62}{5}$ and $c = 0$

$b = \frac{62 \times 25}{31 \times 5}$

$b = 10$

Hence, the required value of $b = 10$.

Question151

The differential equation satisfied by the system of parabolas $y^2 = 4a(x + a)$ is
[18 Mar 2021 Shift 1]

Options:

A. $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) - y = 0$

B. $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) + y = 0$

C. $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

D. $y\left(\frac{dy}{dx}\right) + 2x\left(\frac{dy}{dx}\right) - y = 0$

Answer: C

Solution:

Given, equation of curve is $y^2 = 4a(x + a)$
$\Rightarrow\ y^2 = 4ax + 4a^2$...(i)
Differentiating Eq. (i) w.r.t. x, we get
$2y\frac{dy}{dx} = 4a$
$\Rightarrow\ a = \left(\frac{y}{2}\right)\cdot\frac{dy}{dx}$....(ii)
∴ Required differential equation is
$y^2 = 4\times\frac{y}{2}\times\frac{dy}{dx}x + 4\left(\frac{y}{2}\cdot\frac{dy}{dx}\right)^2$ [From Eqs. (i) and (ii)]
$\Rightarrow\ y^2\left(\frac{dy}{dx}\right)^2 + 2xy\left(\frac{dy}{dx}\right) - y^2 = 0$
$\Rightarrow\ y\left[y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y\right] = 0$
As, $y \neq 0$
$\therefore\ y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

Question152

Let $y = y(x)$ be the solution of the differential equation $xdy - ydx = \sqrt{(x^2 - y^2)}dx$, $x \geq 1$, with $y(1) = 0$. If the area bounded by the line $x = 1$, $x = e^\pi$, $y = 0$ and $y = y(x)$ is $\alpha e^{2\pi} + \beta$, then the value of $10(\alpha + \beta)$ is equal to
[18 Mar 2021 Shift 2]

Answer: 4

Solution:

$x\,dy - y\,dx = \sqrt{x^2 - y^2}\,dx$

$\Rightarrow \dfrac{x\,dy - y\,dx}{x^2} = \dfrac{1}{x}\sqrt{1 - \dfrac{y^2}{x^2}}\,dx \Rightarrow \dfrac{d\left(\frac{y}{x}\right)}{\sqrt{1-\left(\frac{y}{x}\right)^2}} = \dfrac{1}{x}dx$

On integrating. $\int \dfrac{1}{\sqrt{1-\left(\frac{y}{x}\right)^2}} \cdot d(y/x) = \int \dfrac{1}{x}dx$

$\Rightarrow \sin^{-1}(y/x) = \log|x| + C$

Now, at $x = 1, y = 0$

$\therefore C = 0$

Hence, $y = x\sin(\log x)$

$\therefore A = \int_1^{e^x} x\sin(\log x)\,dx$

Put $x = e^t \Rightarrow dx = e^t dt$

$\therefore A = \int_0^{\pi} e^{2t}\sin(t)\,dt$

$\left(\text{using} \int e^{ax} \cdot \sin bx = \dfrac{e^{2x}}{a^2+b^2}(a\sin bx - b\cos bx)\right)$

$\alpha e^{2\pi} + \beta = \left(\dfrac{e^{2t}}{5}(2\sin t - \cos t)\right)_0^{\pi} = \dfrac{1+e^{2\pi}}{5}$

$\therefore \alpha = \dfrac{1}{5}, \beta = \dfrac{1}{5}$

$\therefore 10(\alpha + \beta) = 4$

Question153

Which of the following is true for y(x) that satisfies the differential equation $\frac{dy}{dx} = xy - 1 + x - y$; $y(0) = 0$ [17 Mar 2021 Shift 1]

Options:

A. $y(1) = e^{-\frac{1}{2}} - 1$

B. $y(1) = e^{\frac{1}{2}} - e^{-\frac{1}{2}}$

C. $y(1) = 1$

D. $y(1) = e^{\frac{1}{2}} - 1$

Answer: A

Solution:

$$\frac{dy}{dx} = xy - 1 + x - y$$
$$\frac{dy}{dx} = x(y+1) - 1(y+1) \Rightarrow \frac{dy}{dx} = (x-1)(y+1)$$
$$\int \frac{dy}{y+1} = \int (x-1)dx$$
$$\log_e(y+1) = \frac{x^2}{2} - x + c \quad [\because \log_e 1 = 0]$$
$$\log_e 1 = 0 + C \Rightarrow C = 0$$
$$y(1) \Rightarrow \log_e(y+1) = \frac{1}{2} - 1$$
$$y + 1 = e^{-1/2}$$
$$y(1) = -1 + e^{-1/2}$$

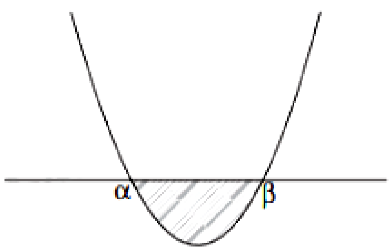
Question154

Let the curve y = y(x) be the solution of the differential equation $\frac{dy}{dx} = 2(x+1)$.
If the numerical value of area bounded by the curve y = y(x) and x-axis is $\frac{4\sqrt{8}}{3}$, then the value of y(1) is equal to.........
[16 Mar 2021 Shift 1]

Answer: 2

Solution:

We have, $dy/dx = 2(x+1)$
$y = \int 2(x+1)dx = 2(x^2/2 + x) + C$
$y = x^2 + 2x + C$...(i)



Now, $\int_{\alpha}^{\beta} y\,dx = \frac{4\sqrt{8}}{3}$

$\int_{\alpha}^{\beta}(x^2 + 2x + c)\,dx = \frac{4\sqrt{8}}{3}$

$\Rightarrow \left[\frac{x^3}{3} + x^2 + cx\right]_{\alpha}^{\beta} = \frac{4\sqrt{8}}{3}$

$\Rightarrow \left(\frac{\beta^3 - \alpha^3}{3}\right) + (\beta^2 - \alpha^2) + c(\beta - \alpha) = \frac{4\sqrt{8}}{3}$ ....(ii)

From Eq. (i),

$\alpha + \beta = -2$

$\alpha\beta = c$

$\therefore\ \beta - \alpha = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$

$= \sqrt{4 - 4c} = 2\sqrt{1 - c}$

From Eq. (ii), $\frac{1}{3}(\beta - \alpha)(\alpha^2 + \beta^2 + \alpha\beta) + (\beta - \alpha)(\beta + \alpha) + c(\beta - \alpha) = \frac{4\sqrt{8}}{3}$

$\Rightarrow \frac{1}{3}(2\sqrt{1 - c})(4 - c) + (2\sqrt{1 - c})(-2 + c) = \frac{4\sqrt{8}}{3}$

$\Rightarrow\ (2\sqrt{1 - c})[4 - c + (-6 + 3c)] = 4\sqrt{8}$

$\Rightarrow (2\sqrt{1 - c})[2(c - 1)] = 4\sqrt{8}$

$\Rightarrow\ (1 - c)^{3/2} = -\sqrt{8}$

$\Rightarrow\ (1 - c)^3 = 8$

$\Rightarrow\ 1 - c = 2$

$\therefore\ c = -1$

Now, $y = x^2 + 2x - 1$

$\therefore\ y(1) = 1^2 + 2 \cdot 1 - 1 = 2$

Question155

Let C_1 be the curve obtained by the solution of differential equation $2xy\frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the solution of $\frac{2xy}{x^2 - y^2} = \frac{dy}{dx}$. If both the curves pass through (1, 1), then the area enclosed by the curves C_1 and C_2 is equal to

[16 Mar 2021 Shift 2]

Options:

A. $\pi - 1$

B. $\frac{\pi}{2} - 1$

C. $\pi + 1$

D. $\frac{\pi}{4} + 1$

Answer: B

Solution:

Given, $2xy\frac{dy}{dx} = y^2 - x^2, x > 0$

$\frac{dy}{dx} = \left(\frac{y^2 - x^2}{2xy}\right), x > 0$

Let $y = vx$

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

$v + x\frac{dv}{dx} = \frac{v^2x^2 - x^2}{2x \cdot vx} = \frac{v^2 - 1}{2v}$

$\Rightarrow \ x\frac{dv}{dx} = \frac{v^2 - 1}{2v} - v \Rightarrow x\frac{dv}{dx} = \frac{v^2 - 1 - 2v^2}{2v}$

$\Rightarrow x\frac{dv}{dx} = \frac{-(v^2 + 1)}{2v} \Rightarrow \int -\left(\frac{2v}{v^2 + 1}\right)dv = \int \frac{dx}{x}$

$\Rightarrow \ -\log|v^2 + 1| = \log x + c$

$\Rightarrow \ \log|x| + \log|v^2 + 1| = c$

$\Rightarrow \ \log|v^2 + 1)x| = c \Rightarrow \left(\frac{y^2}{x^2} + 1\right)x = c$

$\Rightarrow \ \frac{(y^2 + x^2)}{x} = c \Rightarrow y^2 + x^2 = cx$

Similarly, for second curve, $x^2 + y^2 = cy$

Both passes through $(1, 1)$,

$C_1 \Rightarrow 1 + 1 = C_1$

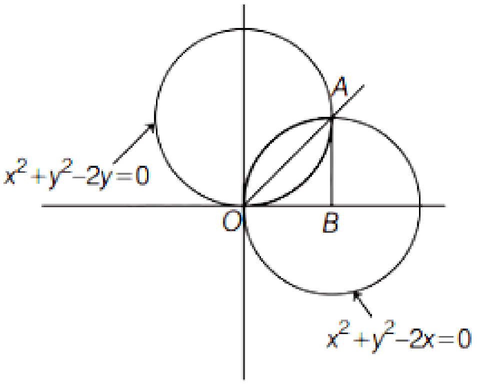
$\Rightarrow \ C_1 = 2$

$C_1 \Rightarrow x^2 + y^2 = 2x$

$C_2 \Rightarrow x^2 + y^2 = C_2y$

$\Rightarrow \ 1 + 1 = C_2 \Rightarrow C_2 = 2$

$C_2 \Rightarrow x^2 + y^2 = 2y$



$\therefore$ Required area $= 2\left[\frac{\pi \cdot 1^2}{4} - \frac{1}{2} \cdot 1 \cdot 1\right]$

$= 2\left[\left(\frac{\pi - 2}{4}\right)\right] = \frac{\pi}{2} - 1$

Question156

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx} = (y + 1)\left[(y + 1)e^{x^2/2} - x\right]$ **0 < x < 2. 1, with y(2) = 0. Then the value of** $\frac{dy}{dx}$

at x = 1 is equal to
[18 Mar 2021 Shift 2]

Options:

A. $\frac{-e^{3/2}}{(e^2+1)^2}$

B. $-\frac{2e^2}{(1+e^2)^2}$

C. $\frac{e^{5/2}}{(1+e^2)^2}$

D. $\frac{5e^{1/2}}{(e^2+1)^2}$

Answer: A

Solution:

Given, $\frac{dy}{dx} = (y+1)\left[(y+1)e^{x^2/2} - x\right]$...(i)

$\Rightarrow \frac{dy}{dx} = (y+1)^2 e^{x^2/2} - x(y+1)$

$\Rightarrow \frac{dy}{dx} + x(y+1) = (y+1)^2 e^{x^2/2}$

$\Rightarrow \frac{1}{(y+1)^2} \cdot \frac{dy}{dx} + \frac{1}{y+1} \cdot x = e^{x^2/2}$...(ii)

Let $\frac{1}{y+1} = t \Rightarrow \frac{-1}{(y+1)^2} \cdot \frac{dy}{dx} = \frac{dt}{dx}$

$\Rightarrow -\frac{dt}{dx} + tx = e^{x^2/2}$

$\Rightarrow \frac{dt}{dx} + (-x)t = -e^{x^2/2}$

which is linear differential equation

$IF = e^{\int -x\,dx} = e^{-x^2/2}$

Now, solution of differential equation is,

$t \cdot e^{-x^2/2} = -\int e^{-x^2/2} \cdot e^{x^2/2} dx$

$\Rightarrow \left(\frac{1}{y+1}\right) e^{-x^2/2} = -x + c$, where c is constant of integration....(iii)

Given, $y(2) = 0$ i.e. when $x = 2$, then $y = 0$.

From Eq. (iii),

$e^{-2} = -2 + c \Rightarrow c = 2 + e^{-2}$

Now, at $x = 1$

$\frac{1}{y+1} e^{-1/2} = -1 + e^{-2} + 2$

$\Rightarrow (y+1) = \frac{e^{-1/2}}{1+e^{-2}}$

Now, putting the value of $(y+1)$ in Eq. (i), we get

$y'(1) = \frac{e^{-1/2}}{1+e^{-2}}\left(\frac{e^{-1/2}}{1+e^{-2}} \cdot e^{1/2} - 1\right)$

$$= \frac{e^{-1/2}}{1+e^{-2}}\left(\frac{1}{1+e^{-2}} - 1\right)$$
$$= \frac{e^{-1/2}}{1+e^{-2}}\left(\frac{1-1-e^{-2}}{1+e^{-2}}\right)$$
$$= \frac{-e^{-5/2}}{(1+e^{-2})^2} = \frac{-e^{-5/2}}{\frac{(e^2+1)^2}{e^4}} = \frac{-e^{3/2}}{(e^2+1)^2}$$

Question157

Let $y = y(x)$ be the solution of the differential equation $\cos x(3\sin x + \cos x + 3)\,dy = [1 + y\sin x(3\sin x + \cos x + 3)]\,dx$ $0 \le x \le \frac{\pi}{2}$, $y(0) = 0$. Then, $y\left(\frac{\pi}{3}\right)$ is equal to

[17 Mar 2021 Shift 2]

Options:

A. $2\log_e\left(\frac{2\sqrt{3}+9}{6}\right)$

B. $2\log_e\left(\frac{2\sqrt{3}+10}{11}\right)$

C. $2\log_e\left(\frac{\sqrt{3}+7}{2}\right)$

D. $2\log_e\left(\frac{3\sqrt{3}-8}{4}\right)$

Answer: B

Solution:

Given, $\cos x(3\sin x + \cos x + 3)\,dy$
$= [1 + y\sin x(3\sin x + \cos x + 3)]\,dx$

$\Rightarrow \cos x\,dy = \left(\frac{1}{3\sin x + \cos x + 3} + y\sin x\right)dx$

$\Rightarrow \cos x \cdot \frac{dy}{dx} = y\sin x + \frac{1}{3\sin x + \cos x + 3}$

$\Rightarrow \frac{dy}{dx} = y\frac{\sin x}{\cos x} + \frac{1}{\cos x(3\sin x + \cos x + 3)}$

$\Rightarrow \frac{dy}{dx} - (\tan x)y = \frac{1}{\cos x(3\sin x + \cos x + 3)}$....(i)

Which is linear differential equation.

$\therefore$ Integrating factor $(IF) = e^{\int(-\tan x)dx}$
$= e^{\log|\cos x|} = |\cos X|$

$\because |\cos x| > 0, \forall x \in [0, \pi/2)$
$\therefore |\cos x| = \cos x$
Hence, solution of Eq. (i) is

$$y(\cos x) = \int (\cos x) \cdot \frac{1}{\cos x(3 \sin x + \cos x + 3)} d x$$

$$\Rightarrow \ y \cos x = \int \frac{1}{3 \sin x + \cos x + 3} d x$$

$$\Rightarrow y \cos x = \int \frac{\sec^2 \frac{x}{2} d x}{2\tan^2 \frac{x}{2} + 6 \tan \frac{x}{2} + 4} d x = \int \frac{\sec^2 \frac{x}{2} d x}{2\left(\tan^2 \frac{x}{2} + 3 \tan \frac{x}{2} + 2\right)} d x$$

Putting $\tan \frac{x}{2} = z$

$$\Rightarrow \frac{1}{2} \sec^2 \frac{x}{2} d x = d z$$

$$\therefore y \cos x = \int \frac{d z}{z^2 + 3z + 2} = \int \frac{d z}{(z+1)(z+2)}$$

$$= \int \frac{1}{z+1} d z - \int \frac{1}{z+2} d z = \log(z+1) - \log(z+2) + c$$

$$\Rightarrow y \cos x = \log\left| \frac{z+1}{z+2} \right| + c = \log \left| \frac{\tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right| + c \ldots \text{(i)}$$

Since, $y(0) = 0$

$$\therefore \ C = -\log\left(\frac{1}{2} \right) = \log 2$$

From Eq. (i), $y \cos x = \log \left| \frac{\tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right| + \log 2$

$$\therefore \ y\left(\frac{\pi}{3} \right) = 2 \left[\log \left| \frac{\frac{1}{\sqrt{3}} + 1}{\frac{1}{\sqrt{3}} + 2} \right| + \log 2 \right] = 2 \log \left| 2\left(\frac{\sqrt{3}+1}{2\sqrt{3}+1} \right) \right|$$

$$= 2 \log\left|2\left(\frac{\sqrt{3}+1}{2\sqrt{3}+1} \times \frac{2\sqrt{3}-1}{2\sqrt{3}-1} \right) \right| = 2 \log\left|\frac{2\sqrt{3}+10}{11}\right|$$

Question158

If the curve $y = y(x)$ is the solution of the differential equation $2(x^2 + x^{5/4}) d y - y(x + x^{1/4}) d x = 2x^{9/4} d x$, $x > 0$ which passes through the point $\left(1, 1 - \frac{4}{3}\log_e 2\right)$, then the value of $y(16)$ is equal to
[17 Mar 2021 Shift 2]

Options:

A. $4\left(\frac{31}{3} + \frac{8}{3}\log_e 3 \right)$

B. $\left(\frac{31}{3}+\frac{8}{3}\log_e 3\right)$

C. $4\left(\frac{31}{3}-\frac{8}{3}\log_e 3\right)$

D. $\left(\frac{31}{3}-\frac{8}{3}\log_e 3\right)$

Answer: C

Solution:

Given, $2(x^2+x^{5/4})dy-y\cdot(x+x^{1/4})dx =$
$2x^{9/4}dx$, where $x>0$
After rearranging, we get
$\frac{dy}{dx}-\frac{y}{2x}=\frac{x^{9/4}}{x^{5/4}(x^{3/4}+1)}$
This is of the form $\frac{dy}{dx}+Py=Q$, where P and Q are constants or function of x.
$\therefore$ Integrating factor (IF) $=e^{\int P\,dx}$
$=e^{\int -\frac{1}{2x}dx}=e^{\frac{-1}{2}\int x\,dx}$
$=e^{-\frac{1}{2}\log x}=e^{\log(x)^{-1/2}}=\frac{1}{x^{1/2}}$
Its solution is
$y\times(IF)=\int Q\times(IF)dx$

$\Rightarrow\ y\times\frac{1}{x^{1/2}}=\int\frac{x^{9/4}}{x^{5/4}(x^{3/4}+1)}\times x^{-1/2}dx=\int\frac{(x)^{\frac{9}{4}-\frac{5}{4}-\frac{1}{2}}}{(x^{3/4}+1)}dx$

$\Rightarrow\ y\times\frac{1}{x^{1/2}}=\int\frac{x^{1/2}dx}{(x^{3/4}+1)}$...(i)
Putting $x=z^4$
$\Rightarrow\ dx=4z^3\cdot dz$
RHS of Eq. (i) becomes,
$\int\frac{z^2\cdot 4z^3}{(z^3+1)}\cdot dz=4\int\frac{z^2(z^3+1-1)}{(z^3+1)}dz$

$=4\left[\int\frac{z^2(z^3+1)}{(z^3+1)}dz-\int\frac{z^2}{(z^3+1)}dz\right]$

$=4\left[\frac{z^3}{3}-\frac{1}{3}\cdot\int\frac{3z^2}{z^3+1}dz\right]$

$=4\left[\frac{z^3}{3}-\frac{1}{3}\cdot\log|z^3+1|\right]$

$\left(\because\int\frac{f'(x)}{f(x)}dx=\log|f(x)|+C\right)$

$=\frac{4z^3}{3}-\frac{4}{3}\log|z^3+1|+C$

$=\frac{4x^{3/4}}{3}-\frac{4}{3}\log|x^{3/4}+1|+C\quad\left(\begin{array}{c}\because x=z^4\\ \therefore x^{3/4}=z^3\end{array}\right)$

$\therefore$ Eq. (i) becomes,

$y \times x^{-1/2} = \frac{4x^{3/4}}{3} - \frac{4}{3}\log|x^{3/4}+1| + C$

Since, this passes through $\left(1, 1 - \frac{4}{3}\log_e 2\right)$

Then,

$\left(1 - \frac{4}{3}\log_e 2\right) \times 1 = \frac{4 \times 1}{3} - \frac{4}{3}\log\left|1+1\right| + C$

$\Rightarrow C = 1 - \frac{4}{3} \Rightarrow C = \frac{-1}{3}$

Hence,

$y = \frac{4}{3}x^{5/4} - \frac{4}{3}\sqrt{x}\log x^{3/4} + 1\Big| - \frac{\sqrt{x}}{3}$

$\because x > 0$

$\therefore x^{3/4} > 0 \Rightarrow x^{3/4} + 1 > 0$ i.e., $x^{3/4} + 1\,| = x^{3/4} + 1$

$\therefore\ y = \frac{4}{3}x^{5/4} - \frac{4}{3}\sqrt{x}\log(x^{3/4}+1) - \frac{\sqrt{x}}{3}$

Now, putting $x = 16$, we get

$y(16) = \frac{4}{3} \times 32 - \frac{4}{3} \times 4\log 9 - \frac{4}{3}$

$= \frac{124}{3} - \frac{32}{3}\log 3$

$= 4\left(\frac{31}{3} - \frac{8}{3}\log 3\right)$

Question159

If y = y(x) is the solution of the differential equation, $\frac{dy}{dx} + 2y\tan x = \sin x$, $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of the function y(x) over R is equal to [16 Mar 2021 Shift 1]

Options:

A. 8

B. $\frac{1}{2}$

C. $-\frac{15}{4}$

D. $\frac{1}{8}$

Answer: D

Solution:

Given, $\frac{dy}{dx} + 2y\tan x = \sin x$

This differential equation is of the form $\frac{dy}{dx} + Py = Q$ where P and Q is function of x.

which is a linear differential equation.

Here, $P = 2\tan x$ and $Q = \sin x$

The integrating factor of linear differential equation is $e^{\int P dx}$.

Here, $e^{\int 2\tan x\, dx} = e^{\int \frac{2\sin x}{\cos x} dx} = e^{-2\log(\cos x)} = \sec^2 x$

Now, $\frac{dy}{dx} + 2y\tan x = \sin x$

On multiplying $\sec^2\times$ both the sides,

$\sec^2 x \frac{dy}{dx} + 2y\sec^2 x\tan x = \sin x\sec^2 x$

$\Rightarrow \frac{d}{dx}(y\sec^2 x) = \sin x\sec^2 x$

$\Rightarrow y\sec^2 x = \int \sin x\sec^2 x\, dx$

$\Rightarrow y\sec^2 x = \int \frac{\sin x}{\cos^2 x} dx$

Let $\cos x = t$

$(-\sin x)dx = dt$

$\int \frac{-dt}{t^2} = \frac{1}{t} + c$

So, $y\sec^2 x = \frac{1}{\cos x} + c$

$y\sec^2 x = \sec x + c$

Now, $x = \pi/3, y = 0$

$0 = 2 + c$

$\Rightarrow c = -2$

So, $y\sec^2 x = \sec x - 2$

$y = \cos^2 x\left(\frac{1}{\cos x} - 2\right) = -2\cos^2 x + \cos x$

$\Rightarrow y = -2\left(\cos^2 x - \frac{\cos x}{2}\right)$

$\Rightarrow y = -2\left[\left(\cos^2 x - \frac{\cos x}{2} + \frac{1}{16}\right) - \frac{1}{16}\right]$

$\Rightarrow y = -2\left[(\cos x - 1/4)^2 - \frac{1}{16}\right]$

$\Rightarrow y = \frac{1}{8} - 2(\cos x - 1/4)^2$

So, $y_{min} = \frac{1}{8}$

Question160

If y = y(x) is the solution of the differential equation $\frac{dy}{dx} + (\tan x)y = \sin x$, $0 \le x \le \frac{\pi}{3}$, with $y(0) = 0$, then $y\left(\frac{\pi}{4}\right)$ is equal to [16 Mar 2021 Shift 2]

Options:

A. $\frac{1}{4}\log_e 2$

B. $\left(\frac{1}{2\sqrt{2}}\right)\log_e 2$

C. $\log_e 2$

D. $\frac{1}{2}\log_e 2$

Answer: B

Solution:

Given, $\frac{dy}{dx}+(\tan x)y=\sin x,\ x\in\left[0,\frac{\pi}{3}\right]$

which is a linear differential equation of the form of $\frac{dy}{dx}+Py=Q$

Here, $P=\tan x$

$\therefore\quad IF=e^{\int P\,dx}$

$\Rightarrow e^{\int \tan x\,dx}=e^{\log(\sec x)}=\sec x$

Multiplying by $\sec x$ on both sides

$\frac{dy}{dx}+(\tan x)y=\sin x$

$\sec x\frac{dy}{dx}+(\tan x\sec x)y=\sin x\sec x$

$\Rightarrow\quad \frac{d}{dx}(y\sec x)=\tan x\Rightarrow y\sec x=\int\tan x\,dx$

$\Rightarrow\quad y\sec x=\log(\sec x)+c$

$y=\cos x\log(\sec x)+c\cdot\cos x$

$y(0)=0$

$\Rightarrow\ 0=1\cdot 0+c\cdot 1\Rightarrow c=0$

$\therefore\ y=\cos x\cdot\log(\sec x)$

$\Rightarrow\ y\left(\frac{\pi}{4}\right)=\cos\left(\frac{\pi}{4}\right)\cdot\log\left(\sec\frac{\pi}{4}\right)$

$=\frac{1}{\sqrt{2}}\log(\sqrt{2})$

$=\frac{1}{2\sqrt{2}}\log 2$

Question161

In a triangle ABC, if $|\overrightarrow{BC}|=8, |\overrightarrow{CA}|=7, |\overrightarrow{AB}|=10$, then the projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to [18 Mar 2021 Shift 2]

Options:

A. $\frac{25}{4}$

B. $\frac{85}{14}$

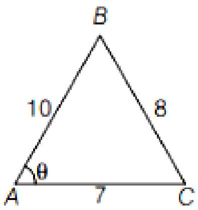
C. $\frac{127}{20}$

D. $\frac{115}{16}$

Answer: B

Solution:

Projection of AB or AC = AB cos theta = 10 cdot cos theta



$= 10 . \left(\frac{10^2 + 7^2 - 8^2}{2 \times 7 \times 10} \right) = \frac{85}{14}$ $\left(\text{using } \cos\theta = \frac{c^2 + b^2 - a^2}{2bc} \right)$

Question162

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 1 + xe^{y-x}$, $-\sqrt{2} < x < \sqrt{2}$, $y(0) = 0$then, the minimum value of $y(x)$, $x \in (-\sqrt{2}, \sqrt{2})$ is equal to:
[25 Jul 2021 Shift 1]

Options:

A. $(2 - \sqrt{3}) - \log_e 2$

B. $(2 + \sqrt{3}) + \log_e 2$

C. $(1 + \sqrt{3}) - \log_e(\sqrt{3} - 1)$

D. $(1 - \sqrt{3}) - \log_e(\sqrt{3} - 1)$

Answer: D

Solution:

$\frac{dy-dx}{e^{y-x}} = xdx$

$\Rightarrow \frac{dy-dx}{e^{y-x}} = xdx$

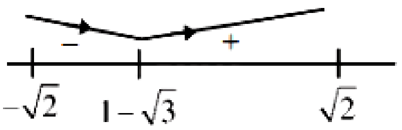
$\Rightarrow -e^{x-y} = \frac{x^2}{2} + c$

At $x = 0, y = 0 \Rightarrow c = -1$

$\Rightarrow e^{x-y} = \frac{2-x^2}{2}$

$\Rightarrow y = x - \ln\left(\frac{2-x^2}{2}\right)$

$\Rightarrow \frac{dy}{dx} = 1 + \frac{2x}{2-x^2} = \frac{2+2x-x^2}{2-x^2}$



So minimum value occurs at $x = 1-\sqrt{3}$

$y(1-\sqrt{3}) = (1-\sqrt{3}) - \ln\left(\frac{2-(4-2\sqrt{3})}{2}\right)$

$= (1-\sqrt{3}) - \ln(\sqrt{3}-1)$

Question163

Let $y = y(x)$ be solution of the differential equation $\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, with $y(0) = 0$ If $y\left(-\frac{2}{3}\log_e 2\right) = \alpha\log_e 2$, then the value of α is equal to:

[27 Jul 2021 Shift 1]

Options:

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. 2

D. $-\frac{1}{2}$

Answer: A

Solution:

$$\frac{dy}{dx} = e^{3x} . e^{4y} \Rightarrow \int e^{-4y} dy = \int e^{3x} dx$$

$$\frac{e^{-4y}}{-4} = e^{3x}3 + C \Rightarrow -\frac{1}{4} - \frac{1}{3} = C \Rightarrow C = -\frac{7}{12}$$

$$\frac{e^{-4y}}{-4} = \frac{e^{3x}}{3} - \frac{7}{12} \Rightarrow e^{-4y} = \frac{4e^{3x} - 7}{-3}$$

$$e^{4y} = \frac{3}{7 - 4e^{3x}} \Rightarrow 4y = \ln\left(\frac{3}{7 - 4e^{3x}}\right)$$

$$4y = \ln\left(\frac{3}{6}\right) \text{ when } x = -\frac{2}{3}\ln 2$$

$$y = \frac{1}{4}\ln\left(\frac{1}{2}\right) = -\frac{1}{4}\ln 2$$

Question164

If $y = y(x)$, $y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation

$\sec y \frac{dy}{dx} - \sin(x + y) - \sin(x - y) = 0$, with $y(0) = 0$ then $5y'\left(\frac{\pi}{2}\right)$ is equal to

__________.

[27 Jul 2021 Shift 1]

Answer: 2

Solution:

$$\sec y \frac{dy}{dx} = 2 \sin x \cos y$$

$$\sec^2 y\, dy = 2 \sin x\, dx$$

$$\tan y = -2 \cos x + c$$

$$c = 2$$

$$\tan y = -2 \cos x + 2 \Rightarrow \text{at } x = \frac{\pi}{2}$$

$$\tan y = 2$$

$$\sec^2 y \frac{dy}{dx} = 2 \sin x$$

$$5\frac{dy}{dx} = 2$$

Question165

Let $y = y(x)$ be the solution of the differential equation
$(x - x^3) dy = (y + yx^2 - 3x^4) dx$, $x > 2$If $y(3) = 3$, then $y(4)$ is equal to :

[27 Jul 2021 Shift 2]

Options:

A. 4

B. 12

C. 8

D. 16

Answer: B

Solution:

$(x-x^3)dy=(y+yx^2-3x^4)dx$
$\Rightarrow xdy-ydx=(yx^2-3x^4)dx+x^3dy$
$\Rightarrow \dfrac{xdy-ydx}{x^2}=(ydx+xdy)-3x^2dx$
$\Rightarrow d\left(\dfrac{y}{x}\right)=d(xy)-d(x^3)$
Integrate $\Rightarrow \dfrac{y}{x}=xy-x^3+c$
given $f(3)=3$
$\Rightarrow \dfrac{3}{3}=3\times 3-3^3+c$
$\Rightarrow c=19$
$\therefore \dfrac{y}{x}=xy-x^3+19$
at $x=4,\ \dfrac{y}{4}=4y-64+19$
$15y=4\times 45$
$\Rightarrow y=12$

Question166

Let y = y(x) be the solution of the differential equation $dy=e^{\alpha x+y}dx$; $\alpha\in N$. If $y(\log_e 2)=\log_e 2$ and $y(0)=\log_e\left(\frac{1}{2}\right)$, then the value of α is equal to ____.
[27 Jul 2021 Shift 2]

Answer: 2

Solution:

$\int e^{-y} d y=\int e^{ex} d x$
$\Rightarrow e^{-y}=\frac{e^{\alpha x}}{\alpha}+c$ ....(i)
Put $(x, y)=(l n 2, \ln 2)$
$\frac{-1}{2}=\frac{2^{\alpha}}{\alpha}+C$ ....(ii)
Put $(x, y) \equiv(0,-\ln 2)$ in (i)
$-2=\frac{1}{\alpha}+C$ ......(iii)
(ii) - (iii)
$\frac{2^{\alpha}-1}{\alpha}=\frac{3}{2}$
$\Rightarrow \alpha=2$ (as $\alpha \in N$)

Question167

Let y = y(x) be the solution of the differential equation $x d y=\left(y+x^{3} \cos x\right) d x$ with $y(\pi)=0$, then $y\left(\frac{\pi}{2}\right)$ is equal to: [25 Jul 2021 Shift 2]

Options:

A. $\frac{\pi^{2}}{4}+\frac{\pi}{2}$

B. $\frac{\pi^{2}}{2}+\frac{\pi}{4}$

C. $\frac{\pi^{2}}{2}-\frac{\pi}{4}$

D. $\frac{\pi^{2}}{4}-\frac{\pi}{2}$

Answer: A

Solution:

$x d y=\left(y+x^{3} \cos x\right) d x$
$x d y=y d x+x^{3} \cos x d x$
$\frac{x d y-y d x}{x^{2}}=\frac{x^{3} \cos x d x}{x^{2}}$
$\frac{d}{d x}\left(\frac{y}{x}\right)=\int x \cos x d x$
$\Rightarrow \frac{y}{x}=x \sin x-\int 1 . \sin x d x$

$$\frac{y}{x} = x\sin x + \cos x + C$$
$$\Rightarrow 0 = -1 + C \Rightarrow C = 1, x = \pi, y = 0$$
so $\frac{y}{x} = x\sin x + \cos x + 1$
$$y = x^2 \sin x + x\cos x + x \quad x = \frac{\pi}{2}$$
$$y\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4} + \frac{\pi}{2}$$

Question168

Let a curve $y = f(x)$ pass through the point $(2, (\log_e 2)^2)$ and have slope $\frac{2y}{x\log_c x}$ for all positive real value of x. Then the value of $f(e)$ is equal to ________.
[25 Jul 2021 Shift 2]

Answer: 1

Solution:

$$y' = \frac{2y}{x\ln x}$$
$$\Rightarrow \frac{dy}{y} = \frac{2dx}{x\ln x}$$
$$\Rightarrow \ln|y| = 2\ln|\ln x| + C$$
put $x = 2, y = (\ln 2)^2$
$$\Rightarrow c = 0$$
$$\Rightarrow y = (\ln x)^2$$
$$\Rightarrow f(e) = 1$$

Question169

Let $y = y(x)$ be the solution of the differential equation $\left((x+2)e^{\left(\frac{y+1}{x+2}\right)} + (y+1)\right)dx = (x+2)dy, y(1) = 1$. If the domain of $y = y(x)$ is an open interval (α, β), then $|\alpha + \beta|$ is equal to ________.
[22 Jul 2021 Shift 2]

Answer: 4

Solution:

$y+1=Y \Rightarrow dy=dY$
$x+2=X \Rightarrow dx=dX$
$\Rightarrow \left(Xe^{\frac{Y}{X}}+Y\right)dX = XdY$
$\Rightarrow XdY-YdX = Xe^{Y/x}dX$
$\Rightarrow d\left(\frac{Y}{X}\right)e^{-\frac{Y}{x}}=\frac{dX}{X}$
$-e^{-Y/x}=1|X|+c$
$(3,2)\rightarrow -e^{-\frac{2}{3}}=1\left|3\right|+c$
$-e^{-\frac{Y}{x}}=1n\left|X\right|-e^{-\frac{2}{3}}-1n3$
$e^{-\frac{Y}{x}}=e^{2/3}+1n3-1n\left|X\right|>0$
$\ln|X|<(e^{2/3}+1n3)$
Let $\lambda=(e^{2/3}+1n3)$
$|x+2|<e^{\lambda}$
$-e^{\lambda}<x+2<e^{\lambda}$
$-e^{\lambda}-2<x<e^{\lambda}-2$
$\alpha+\beta=-4 \Rightarrow |\alpha+\beta|=4$
Although $x=-2$ should be excluded from domain but according to the given problem it will the most appropriate solution.

Question170

Let y = y(x) be the solution of the differential equation $x\tan\left(\frac{y}{x}\right)dy=\left(y\tan\left(\frac{y}{x}\right)-x\right)dx - 1\le x\le 1, y\left(\frac{1}{2}\right)=\frac{\pi}{6}$. Then the area of the region bounded by the curves $x=0$, $x=\frac{1}{\sqrt{2}}$ and $y=y(x)$ in the upper half plane is: [20 Jul 2021 Shift 1]

Options:

A. $\frac{1}{8}(\pi-1)$

B. $\frac{1}{12}(\pi-3)$

C. $\frac{1}{4}(\pi-2)$

D. $\frac{1}{6}(\pi-1)$

Answer: A

Solution:

We have

$$\frac{dy}{dx}=\frac{x\left(\frac{y}{x}.\tan\frac{y}{x}-1\right)}{x\tan\frac{y}{x}}$$

$$\therefore\frac{dy}{dx}=\frac{y}{x}-\cot\left(\frac{y}{x}\right)$$

Put $\frac{y}{x}=v$

$\Rightarrow y=vn$

$$\therefore\frac{dy}{dx}=v+\frac{ndv}{dx}$$

Now, we get $v+n\frac{dv}{dx}=v-\cot(v)$

$$\Rightarrow\int(\tan)dv=-\int\frac{dx}{x}$$

$$\therefore\ln\left|\sec\left(\frac{y}{x}\right)\right|=-1n|x|+c$$

$$\text{As}\left(\frac{1}{2}\right)=\left(\frac{y}{x}\right)\Rightarrow C=0$$

$$\therefore\sec\left(\frac{y}{x}\right)=\frac{1}{x}$$

$$\Rightarrow\cos\left(\frac{y}{x}\right)=x$$

$\therefore y=x\cos^{-1}(x)$

So, required bounded area

$$=\int_0^{1/\sqrt{2}}x(\cos^{-1}x)dx=\left(\frac{\pi-1}{8}\right)$$

$\therefore$ option (1) is correct.

Question171

Let $y=y(x)$ be the solution of the differential equation $e^x\sqrt{1-y^2}dx+\left(\frac{y}{x}\right)dy=0$, $y(1)=-1$. Then the value of $(y(3))^2$ is equal to: [20 Jul 2021 Shift 1]

Options:

A. $1-4e^3$

B. $1-4e^6$

C. $1+4e^3$

D. $1+4e^6$

Answer: B

Solution:

$e^x\sqrt{1-y^2}\,dx+\frac{y}{x}dy=0$

$\Rightarrow e^x\sqrt{1-y^2}\,dx+\frac{-y}{x}dy$

$\Rightarrow\int\frac{-y}{\sqrt{1-y^2}}dy=\int e^x x\,dx$

$\Rightarrow\sqrt{1-y^2}=e^x(x-1)+c$

Given : At $x=1,\ y=-1$

$\Rightarrow 0=0+c\Rightarrow c=0$

$\therefore\sqrt{1-y^2}=e^x(x-1)$

At $x=3\ 1-y^2=(e^3 2)^2\Rightarrow y^2=1-4e^6$

Question172

Let a curve $y=y(x)$ be given by the solution of the differential equation

$$\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx=\sqrt{e^{2x}-1}\,dy$$

If it intersects y-axis at $y=-1$, and the intersection point of the curve with x-axis is $(\alpha,0)$, then e^{α} is equal to ________.
[20 Jul 2021 Shift 2]

Answer: 2

Solution:

$\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx=\sqrt{e^{2x}-1}\,dy$

Put $\cos^{-1}(e^{-x})\theta,\ \theta\in[0,\pi]$

$\cos\theta=e^{-x}\Rightarrow 2\cos^2\theta 2-1=e^{-x}$

$\cos\frac{\theta}{2}=\sqrt{\frac{e^{-x}+1}{2}}=\sqrt{\frac{e^x+1}{2c^x}}$

$\sqrt{\frac{e^x+1}{2c^x}}dx=\sqrt{e^{2x}-1}\,dy$

$\frac{1}{\sqrt{2}}\int\frac{dx}{\sqrt{e^x}\sqrt{e^x-1}}=\int dy$

Put $e^x=t,\ \frac{dt}{dx}=e^x$

$\frac{1}{\sqrt{2}}\int\frac{dt}{e^x\sqrt{e^x}\sqrt{e^x-1}}=\int dy$

$\int\frac{dt}{t\sqrt{t^2-t}}=\sqrt{2}y$

Put $t=\frac{1}{z},\frac{dt}{dz}=-\frac{1}{z^2}$

$\int\frac{-\frac{dz}{z^2}}{\frac{1}{z}\sqrt{\frac{1}{z^2}-\frac{1}{z}}}=\frac{\sqrt{2}}{y}$

$-\int\frac{dz}{\sqrt{1-z}}=\sqrt{2}y$

$\frac{-2(1-z)^{1/2}}{-1}=\sqrt{2}y+c$

$2\left(1-\frac{1}{t}\right)^{1/2}=\sqrt{2}y+c$

$2(1-e^{-x})^{1/2}=\sqrt{2y}+c\xrightarrow{(0,-1)}\Rightarrow c=\sqrt{2}$

$2(1-e^{-x})^{1/2}=\sqrt{2}(y+1)$, passes through $(\alpha,0)$

$2(1-e^{-\alpha})^{1/2}=\sqrt{2}$

$\sqrt{1-e^{-\alpha}}=\frac{1}{\sqrt{2}}\Rightarrow 1-e^{-\alpha}=\frac{1}{2}$

$e^{-\alpha}=\frac{1}{2}\Rightarrow e^{\alpha}=2$

Question173

Let $y = y(x)$ be solution of the following differential equation $e^y\frac{dy}{dx}-2e^y\sin x+\sin x\cos^2 x=0$, $y\left(\frac{\pi}{2}\right)=0$ If $y(0)=\log_e(\alpha+\beta e^{-2})$, then $4(\alpha+\beta)$ is equal to ________.

[25 Jul 2021 Shift 1]

Answer: 4

Solution:

Let $e^y=t$

$\Rightarrow\frac{dt}{dx}-(2\sin x)t=-\sin x\cos^2 x$

I.F. $=e^{2\cos x}$

$\Rightarrow t\,.\,e^{2\cos x}=\int e^{2\cos x}\,.\,(-\sin x\cos^2 x)dx$

$\Rightarrow e^y\,.\,e^{2\cos x}=\int e^{2z}\,.\,z^2dz,\ z=e^{2\cos x}$

at $x=\frac{\pi}{2},\ y=0\Rightarrow C=\frac{3}{4}$

$\Rightarrow e^{y}=\frac{1}{2}\cos^{2}x-\frac{1}{2}\cos x+\frac{1}{4}+\frac{3}{4}\cdot e^{-2\cos x}$
$\Rightarrow y=\log\left[\frac{\cos^{2}x}{2}-\frac{\cos x}{2}+\frac{1}{4}+\frac{3}{4}e^{-2\cos x}\right]$
Put $x=0$
$\Rightarrow y=\log\left[\frac{1}{4}+\frac{3}{4}e^{-2}\right]\Rightarrow\alpha=\frac{1}{4},\beta=\frac{3}{4}$

Question174

Let $y=y(x)$ be the solution of the differential equation $\operatorname{cosec}^{2}x\,dy+2dx=(1+y\cos 2x)\operatorname{cosec}^{2}x\,dx$, with $y\left(\frac{\pi}{4}\right)=0$. Then, the value of $(y(0)+1)^{2}$ is equal to:
[22 Jul 2021 Shift 2]

Options:

A. $e^{1/2}$

B. $e^{-1/2}$

C. e^{-1}

D. e

Answer: C

Solution:

$\frac{dy}{dx}+2\sin^{2}x=1+y\cos 2x$
$\Rightarrow\frac{dy}{dx}+(-\cos 2x)y=\cos 2x$
I.F. $=e^{\int-\cos 2x\,dx}=e^{-\frac{\sin 2x}{2}}$
Solution of D.E.
$y\left(e^{-\frac{\sin 2x}{2}}\right)=\int(\cos 2x)\left(e^{-\frac{\sin 2x}{2}}\right)dx+c$
$\Rightarrow y\left(e^{-\frac{\sin 2x}{2}}\right)=-e^{-\frac{\sin 2x}{2}}+c$
Given
$y\left(\frac{\pi}{4}\right)=0$
$\Rightarrow 0=-e^{-\frac{1}{2}}+c\Rightarrow c=e^{-\frac{1}{2}}$
$\Rightarrow y\left(e^{-\frac{\sin 2x}{2}}\right)=-e^{-\frac{\sin 2x}{2}}+e^{-1/2}$
at $x=0$

$y=-1+e^{-\frac{1}{2}}$
$\Rightarrow y(0)=-1+e^{-\frac{1}{2}} \Rightarrow (y(0)+1)^2=e^{-1}$

Question175

Let y = y(x) satisfies the equation $\frac{dy}{dx}-|A|=0$,for all x > 0, where

$A=\left[\begin{array}{ccc} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{array}\right]$. If $y(\pi)=\pi+2$, then the value of $y\left(\frac{\pi}{2}\right)$ is :

[20 Jul 2021 Shift 2]

Options:

A. $\frac{\pi}{2}+\frac{4}{\pi}$

B. $\frac{\pi}{2}-\frac{1}{\pi}$

C. $\frac{3\pi}{2}-\frac{1}{\pi}$

D. $\frac{\pi}{2}-\frac{4}{\pi}$

Answer: A

Solution:

$|A|=-\frac{y}{x}+2\sin x+2$
$\frac{dy}{dx}=|A|$
$\frac{dy}{dx}=-\frac{y}{x}+2\sin x+2$
$\frac{dy}{dx}+\frac{y}{x}=2\sin x+2$
I.F. $=e^{\int\frac{1}{x}dx}=x$
$\Rightarrow yx=\int x(2\sin x+2)dx$
$xy=x^2-2x\cos x+2\sin x+c$ ....(i)
Now $x=\pi,\ y=\pi+2$
Use in (i) $c=0$
Now (i) be comes
$xy=x^2-2x\cos x+2\sin x$
put $x=\pi/2$

$\frac{\pi}{2}y = \left(\frac{\pi}{2}\right)^2 - 2 . \frac{\pi}{2}\cos\frac{\pi}{2} + 2\sin\frac{\pi}{2}$
$\frac{\pi}{2}y = \frac{\pi^2}{4} + 2$

Question176

Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers.
If $\vec{a} . \vec{b} = -1$ and $\vec{b} . \vec{c} = 10$, then $\left(\vec{a} \times \vec{b}\right) . \vec{c}$ is equal to ____ .
[27 Jul 2021 Shift 2]

Answer: 9

Solution:

$\vec{a} = (1, -\alpha, \beta)$
$\vec{b} = (3, \beta, -\alpha)$
$\vec{c} = (-\alpha, -2, 1)$; $\alpha, \beta \in I$
$\vec{a} . \vec{b} = -1 \Rightarrow 3 - \alpha\beta - \alpha\beta = -1$
$\Rightarrow \alpha\beta = 2$

1	2
2	1
−1	−2
−2	−1

$\vec{b} . \vec{c} = 10$
$\Rightarrow -3\alpha - 2\beta - \alpha = 10$
$\Rightarrow 2\alpha + \beta + 5 = 0$
$\therefore \alpha = -2; \beta = -1$

$$\left[\begin{array}{ccc} \vec{a} & \vec{b} & \vec{c} \end{array}\right] = \left|\begin{array}{ccc} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 1 \end{array}\right|$$
$= 1(-1 + 4) - 2(3 - 4) - 1(-6 + 2)$
$= 3 + 2 + 4 = 9$

Question177

A differential equation representing the family of parabolas with axis parallel to Y-axis and whose length of latus rectum is the distance of the

point (2, −3) from the line 3x + 4y = 5, is given by
[27 Aug 2021 Shift 2]

Options:

A. $10\frac{d^2y}{dx^2} = 11$

B. $11\frac{d^2x}{dy^2} + 10$

C. $10\frac{d^2x}{dy^2} = 11$

D. $11\frac{d^2y}{dx^2} = 10$

Answer: D

Solution:

Let (h, k) be the vertex of parabola. Then, equation of parabola parallel to Y-axis is
$(x-h)^2 = 4a(y-k)$ ...(i)
Also,
Length of latusrectum = Distance of point (2, − 3) from the line
3x + 4y = 5
$\Rightarrow 4a = \frac{|6-12-5|}{\sqrt{3^2+4^2}}$
$\Rightarrow 4a = \frac{11}{5}$
∴ From Eq. (i),
$(x-h)^2 = \frac{11}{5}(y-k)$
Differentiating w.r.t. x, we get
$2(x-h) = \frac{11}{5}\frac{dy}{dx}$
Again, differentiating w.r.t. x
$2 = \frac{11}{5}\frac{d^2y}{dx^2}$
$\Rightarrow 11\frac{d^2y}{dx^2} = 10$

Question178

Let f be a non-negative function in [0, 1] and twice differentiable in (0, 1). If $\int_0^x \sqrt{1-(f'(t))^2}\,dt = \int_0^x f(t)\,dt$, $0 \le x \le 1$ and f(0) = 0, then $\lim_{x \to 0} \frac{1}{x^2}\int_0^x f(t)\,dt$

[31 Aug 2021 Shift 1]

Options:

A. equals 0

B. equals 1

C. does not exist

D. equals $\frac{1}{2}$

Answer: D

Solution:

$\int_0^x [1-(f'(t))^2]^{\frac{1}{2}} = \int_0^x f(t)$

Differentiating on both sides,

$\sqrt{1-[f'(x)]^2} = [f(x)]$

$\Rightarrow 1-[f'(x)]^2 = [f(x)]^2$

$\Rightarrow 1-[f(x)]^2 = [f'(x)]^2$

$\Rightarrow f'(x) = \sqrt{1-[f(x)]^2}$

$\Rightarrow \int \frac{f'(x)dx}{\sqrt{1-[f(x)]^2}} = \int 1 dx$

$\Rightarrow \sin^{-1} f(x) = x + C$

$\because f(0) = 0$

C = 0

$f(x) = \sin x$

$\Rightarrow \lim\limits_{x\to 0} \frac{\int_0^x f(t)dt}{x^2} = \lim\limits_{x\to 0} \frac{\int_0^x \sin x\, dt}{x^2}$ [applying L'Hopital Rule]

$= \lim\limits_{x\to 0} \left(\frac{\sin x}{2x}\right) = \frac{1}{2}$

Question179

If $\frac{dy}{dx} = \frac{2^{x+y}-2^x}{2^y}$, y(0) = 1, then y(1) is equal to
[31 Aug 2021 Shift 1]

Options:

A. $\log_2(2+e)$

B. $\log_2(1+e)$

C. $\log_2(2e)$

D. $\log_2(1+e^2)$

Answer: B

Solution:

$\frac{dy}{dx}=\frac{2^{x+y}-2^{x}}{2^{y}}=\frac{2^{x}(2^{y}-1)}{2^{y}}$

$\int\frac{2^{y}}{2^{y}-1}dy=\int 2^{x}dx$

$\frac{\ln(2^{y}-1)}{\ln 2}=\frac{2^{x}}{\ln 2}+C$

As, $y(0)=1$

$\Rightarrow 0=\frac{1}{\log 2}+C$

For $y(1)$, $\ln_{2}(2^{y}-1)=2^{1}-1$

$\Rightarrow 2^{y}-1=e$

$y=\log_{2}(e+1)$

Question180

If $\frac{dy}{dx}=\frac{2^{x}y+2^{y}.2^{x}}{2^{x}+2^{x+y}\log_{e}2}$, $y(0)=0$, then for $y=1$, the value of x lies in the interval [31 Aug 2021 Shift 2]

Options:

A. $(1, 2)$

B. $\left(\frac{1}{2}, 1\right]$

C. $(2, 3)$

D. $\left(0, \frac{1}{2}\right]$

Answer: A

Solution:

$\frac{dy}{dx}=\frac{2^{x}y+2^{y}.2^{x}}{2^{x}+2^{x+y}\log_{e}2}$, $y(0)=0$

$\Rightarrow\frac{dy}{dx}=\frac{y+2^{y}}{1+2^{y}\log_{e}2}$

$\Rightarrow\int\frac{1+2^{y}.\log_{e}2dy}{y+2^{y}}=\int dx$

$\Rightarrow\log|y+2^{y}|=x+C$

$\because y(0)=0$

$\Rightarrow C = 0$
$\therefore \ln|y+2^y| = x$
For y = 1
$x = \ln|1+2| = \ln 3$
$\Rightarrow x \in (1, 2)$

Question181

If $y\frac{dy}{dx} = x\left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$, $x > 0, \phi > 0$, and $y(1) = -1$, then $\phi\left(\frac{y^2}{4}\right)$ is equal to
[31 Aug 2021 Shift 2]

Options:

A. $4\phi(2)$

B. $4\phi(1)$

C. $2\phi(1)$

D. $\phi(1)$

Answer: B

Solution:

Given,

$$y\frac{dy}{dx} = x\left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right] \quad ...(i)$$

Let $t = \frac{y}{x}$
$\Rightarrow y = xt$
$\Rightarrow \frac{dy}{dx} = t + x\frac{dt}{dx}$
$\therefore$ Eq. (i) becomes

$$t\left(t + x\frac{dt}{dx}\right) = \left(t^2 + \frac{\phi(t^2)}{\phi'(t^2)}\right)$$

$$\Rightarrow xt\frac{dt}{dx} = \frac{\phi(t^2)}{\phi'(t^2)}$$

$$\Rightarrow \frac{t\phi'(t^2)}{\phi(t^2)}dt = \frac{dx}{x}$$

Integrating both sides

$\int \frac{t\phi'(t^2)}{\phi t^{2)}} dt = \int \frac{dx}{x}$
Let $\phi(t^2) = u$
$\Rightarrow t\phi'(t^2) dt = \frac{du}{2}$
$\therefore \frac{1}{2}\int \frac{du}{u} = \int \frac{dx}{x}$
$\Rightarrow \frac{1}{2}\ln u = \ln x + C$
$\Rightarrow \frac{1}{2}\ln \phi(t^2) = \ln x + C$
$\Rightarrow \frac{1}{2}\ln\left(\phi\left(\frac{y^2}{x^2}\right)\right) = \ln x + C$
If $x = 1, y = -1$, then $C = \frac{1}{2}\ln(\phi(1))$
$\therefore \frac{1}{2}\ln\left(\phi\left(\frac{y^2}{x^2}\right)\right) = \ln x + 2\ln(\phi(1))$
If $x = 2$, then $\ln\left(\phi\left(\frac{y^2}{4}\right)\right) = \ln 4 + \ln[\phi(1)]$
Or $\phi\left(\frac{y^2}{4}\right) = 4\phi(1)$

Question182

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = 2(y + 2\sin x - 5)x - 2\cos x$ such that, y(0) = 7. Then y(π) is equal to [27 Aug 2021 Shift 1]

Options:

A. $2e^{\pi^2} + 5$

B. $e^{\pi^2} + 5$

C. $3e^{\pi^2} + 5$

D. $7e^{\pi^2} + 5$

Answer: A

Solution:

Given, $\frac{dy}{dx} = 2(y + 2\sin x - 5)x - 2\cos x,\ y(0) = 7$
$\Rightarrow \frac{dy}{dx} + 2\cos x = 2(y + 2\sin x - 5)x$ ...(i)
Let $y + 2\sin x - 5 = t$

$\Rightarrow \frac{dy}{dx} + 2\cos x = \frac{dt}{dx}$
Then, Eq. (i) becomes
$\frac{dt}{dx} = 2tx$
$\Rightarrow \frac{dt}{t} = 2x dx$
On integrating
$\ln t = x^2 + C$
$\Rightarrow \ln(y + 2\sin x - 5) = x^2 + C$ ...(ii)
$\therefore y(0) = 7$
$\Rightarrow \ln(7 + 0 - 5) = 0 + C$
$\Rightarrow C = \ln 2$
$\therefore$ From Eq. (ii),
$\ln(y + 2\sin x - 5) = x^2 + \ln 2$
Now, at $x = \pi$
$\ln(y(\pi) + 2\sin\pi - 5) = \pi^2 + \ln 2$
$\Rightarrow \ln(y(\pi) - 5) = \pi^2 + \ln 2$
$\Rightarrow y(\pi) - 5 = e^{\pi^2 + \ln 2}$
$\Rightarrow y(\pi) = 2e^{\pi^2} + 5$

Question183

Let y(x) be the solution of the differential equation $2x^2 dy + (e^y - 2x)dx = 0$, $x > 0$. If $y(e) = 1$, then $y(1)$ is equal to [26 Aug 2021 Shift 2]

Options:

A. 0

B. 2

C. $\log_e 2$

D. $\log_e(2e)$

Answer: C

Solution:

We have, $2x^2 dy + (e^y - 2x)dx = 0$
$\frac{dy}{dx} + \frac{e^y - 2x}{2x^2} = 0$
$\frac{dy}{dx} + \frac{e^y}{2x^2} - \frac{1}{x} = 0$
$e^{-y}\frac{dy}{dx} - \frac{e^{-y}}{x} = -\frac{1}{2x^2}$ ...(i)
$e^{-y} = t$ ...(ii)

$-e^{-y} d y = d t$

$d y = -\left(\frac{d t}{t}\right)$...(iii)

$\frac{-d t}{d x} - \frac{t}{x} = -\frac{1}{2x^2}$ [From Eq. (i)]

$x d t + t d x = \frac{d x}{2x}$

$\int d(xt) = \int \frac{d x}{2x}$

$xt = \frac{1}{2}\log(x) + \frac{C}{2}$

$2xe^{-y} = \log x + c$

When $x = e, y = 1$

$2e \cdot e^{-1} = \log e + c$

c = 1

$\therefore 2xe^{-y} = \log x + 1$

When x = 1,

$e^{-y} = 0 + 1$

$e^{y} = 2$

$\Rightarrow y = \log_e 2$

Question184

Let us consider a curve, $y = f(x)$ passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point $(x, f(x))$ is given by $f(x) + xf'(x) = x^2$ Then [27 Aug 2021 Shift 1]

Options:

A. $x^2 + 2xf(x) - 12 = 0$

B. $x^3 + 2xf(x) + 12 = 0$

C. $x^3 - 3xf(x) - 4 = 0$

D. $x^2 + 2xf(x) + 4 = 0$

Answer: C

Solution:

Given, $f(x) + xf'(x) = x^2$

$\Rightarrow f'(x) + \frac{f(x)}{x} = x$

$\Rightarrow \frac{d y}{d x} + \frac{1}{x} y = x \left[\because y = f(x) \Rightarrow \frac{d y}{d x} = f'(x)\right]$

This is linear differential equation.

∴ Integrating factor IF $= e^{\int \frac{1}{x} dx} = x$
Solution, $y \cdot x = \int x \cdot x dx + C$
or $xy = \frac{x^3}{3} + C$
∵ It passes through $(-2, 2)$.
$\therefore -2.2 = \frac{(-2)^3}{3} + C$
$C = -\frac{4}{3}$
Hence, $xf(x) = \frac{x^3}{3} - \frac{4}{3}$
or $x^3 - 3f(x) \cdot x - 4 = 0$

Question185

If the solution curve of the differential equation $(2x - 10y^3)dy + ydx = 0$, passes through the points (0, 1) and (2, β), then β is a root of the equation [27 Aug 2021 Shift 2]

Options:

A. $y^5 - 2y - 2 = 0$

B. $2y^5 - 2y - 1 = 0$

C. $2y^5 - y^2 - 2 = 0$

D. $y^5 - y^2 - 1 = 0$

Answer: D

Solution:

Given, differential equation
$(2x - 10y^3)dy + ydx = 0$
$\Rightarrow \frac{dx}{dy} + \frac{2x}{y} = 10y^2$ ...(i)
This is Linear differential equation
Integrating factor $\mathrm{IF} = e^{2\int \frac{1}{y} dy} = y^2$
Solution of differential Eq. (i),
$x \cdot y^2 = \int 10y^2 \cdot y^2 dy + C$
$\Rightarrow xy^2 = 2y^5 + C$ ...(ii)
Solution Eq. (ii) passes through (0, 1)
$\Rightarrow 0.1^2 = 21^5 + C$
⇒ C = − 2
∴ Solution of Eq. (i) is
$xy^2 = 2y^5 - 2$
Now, this equation passes through (2, β).

$\therefore 2 . \beta^2 = 2\beta^5 - 2$
$\Rightarrow \beta^5 - \beta^2 - 1 = 0$
$\Rightarrow$ β is root of the equation $y^5 - y^2 - 1 = 0$

Question186

Let y = y(x) be a solution curve of the differential equation $(y+1)\tan^2 x\,dx + \tan x\,dy + y\,dx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$. If $\lim_{x \to 0^+} xy(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is

[26 Aug 2021 Shift 1]

Options:

A. $-\frac{\pi}{4}$

B. $\frac{\pi}{4} - 1$

C. $\frac{\pi}{4} + 1$

D. $\frac{\pi}{4}$

Answer: D

Solution:

We have, $(y+1)\tan^2 x\,dx + \tan x\,dy + y\,dx = 0$
$\Rightarrow [(y+1)\tan^2 x + y]dx + \tan x\,dy = 0$
$\Rightarrow \frac{dy}{dx} + (y+1)\tan x + \frac{y}{\tan x} = 0$
$\Rightarrow \frac{dy}{dx} + \frac{y\tan^2 x + \tan^2 x + y}{\tan x} = 0$
$\Rightarrow \frac{dy}{dx} + \frac{y\sec^2 x}{\tan x} + \tan^2 x = 0$
$\Rightarrow \frac{dy}{dx} + \left(\frac{\sec^2 x}{\tan x}\right)y = -\tan x$
This is a linear differential equation
$\therefore IF = e^{\int \frac{\sec^2 x}{\tan x}dx} = e^{\ln(\tan x)} = \tan x$
So, solution is given by
$(y\tan x) = \int -\tan^2 x\,dx = \int (1 - \sec^2 x)dx = x - \tan x + C$
$y = x\cot x - 1 + C\cot x$
Now, $\lim_{x \to 0^+} x . y = 1$

$\Rightarrow \lim\limits_{x \to 0^+} (x^2 \cot x - x + Cx \cot x) = 1$

$\Rightarrow \lim\limits_{x \to 0} + \left(x \cdot \frac{x}{\tan x} - x + \frac{Cx}{\tan x} \right) = 1$

$\Rightarrow 0 - 0 + C = 1$

$\Rightarrow C = 1$

$\therefore y = x \cot x - 1 + \cot x$

Now, $x = \frac{\pi}{4}$

$y = \frac{\pi}{4} - 1 + 1 = \frac{\pi}{4}$

Question187

Let a, b and c be three vectors mutually perpendicular to each other and have same magnitude. If a vector r satisfies. a × {(r - b) × a} + b × {(r - c) × b} + c × {(r - a) × c} = 0, then r is equal to [31 Aug 2021 Shift 2]

Options:

A. $\frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$

B. $\frac{1}{3}(2\mathbf{a} + \mathbf{b} - \mathbf{c})$

C. $\frac{1}{2}(\mathbf{a} + \mathbf{b} + \mathbf{c})$

D. $\frac{1}{2}(\mathbf{a} + \mathbf{b} + 2\mathbf{c})$

Answer: C

Solution:

a × [(r - b) × a] + b × [(r - c) × b] + c × [(r - a) × c] = 0

⇒ a . a (r - b) - (a . (r - b)) a + b . b (r - c) - (b . (r - c)) b + c . c (r - a) - (c . (r - a)) c = 0

$\Rightarrow |a|^2 (r - b) - (r \cdot a)a + |b|^2 (r - c) - (r \cdot b)b + |c|^2 (r - a) - (r \cdot c)c = 0$ [∵ a b, c are mutually perpendicular; ∴ a . b = b . c = c . a = 0]

$\Rightarrow |a|^2 [3r - (a + b + c)] - [(r \cdot a)a + (r \cdot b)b + (r \cdot c)c] = 0$ [∵ |a| = |b| = |c|]

$\Rightarrow |a|^2 [3r - (a + b + c) - r] = 0$

∴ 3r - (a + b + c) - r = 0

$\Rightarrow r = \frac{a + b + c}{2}$

Question188

If y = y(x) is the solution curve of the differential equation

$x^2 dy + \left(y - \frac{1}{x}\right) dx = 0$; $x > 0$

and y(1) = 1, then $y\left(\frac{1}{2}\right)$ is equal to

[1 Sep 2021 Shift 2]

Options:

A. $\frac{3}{2} - \frac{1}{\sqrt{e}}$

B. $3 + \frac{1}{\sqrt{e}}$

C. $3 + e$

D. $3 - e$

Answer: D

Solution:

$x^2 dy + \left(y - \frac{1}{x}\right) dx = 0$

$\Rightarrow x^2 dy + y dx = \frac{dx}{x}$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^3}$

This in linear differential equations

$IF = e^{\int \frac{1}{x^2} dx} = e^{-\frac{1}{x}}$

$\Rightarrow y \cdot e^{-1/x} = \int e^{-1/x} \cdot \frac{1}{x^3} dx + c$

Let $\frac{-1}{x} = t$

$\Rightarrow \frac{1}{x^2} dx = dt$

$\Rightarrow y \cdot e^{-1/x} = \int -te^t dt + c = -(te^t - e^t) + c$

$\Rightarrow ye^{-1/x} = \frac{1}{x} e^{-\frac{1}{x}} + e^{-\frac{1}{x}} + c$

Put $x = 1, y = 1$

(1) $e^{-1} = \frac{e^{-1}}{1} + e^{-1} + c$

$\Rightarrow c = -e^{-1}$

Solution $y \cdot e^{-1/x} = \frac{1}{x} e^{-\frac{1}{x}} + e^{-\frac{1}{x}} - e^{-1}$

$\Rightarrow y = \frac{1}{x} + 1 - \frac{e^{1/x}}{e}$

Put $x = \frac{1}{2}$

$y=2+1-\frac{e^2}{e}$
$\Rightarrow y=3-e$

Question189

If for $x \geq 0$, $y=y(x)$ is the solution of the differential equation, $(x+1)dy=((x+1)^2+y-3)dx$,$y(2)=0$ then $y(3)$ is equal to _______.
[NA Jan. 09,2020 (I)]

Answer: 3

Solution:

$(x+1)dy=((x+1)^2+(y-3))\,dx=0$

$\Rightarrow \frac{dy}{dx}=(1+x)+\left(\frac{y-3}{1+x}\right)$

$\frac{dy}{dx}-\frac{1}{(1+x)}y \ =(1+x)-\frac{3}{(1+x)}$

I.F. $=e^{-\int\frac{1}{(1+x)}dx}=\frac{1}{(1+x)}$

$\therefore \frac{d}{dx}\left(\frac{y}{1+x}\right)=1-\frac{3}{(1+x)^2}$

$y=(1+x)\left[x+\frac{3}{(1+x)}+C\right]$

$\because$ At $x=2, y=0$

$\therefore 0=3(2+1+C) \Rightarrow C=-3$

Then, $y=(1+x)\left[x+\frac{3}{1+x}-3\right]$

Now, at $x=3, y=(1+3)\ \left[3+\frac{3}{1+3}-3\right]=3$

Question190

Let $y=y(x)$ be the solution curve of the differential equation, $(y^2-x)\frac{dy}{dx}=1$, satisfying $y(0)=1$. This curve intersects the x -axis at a point whose abscissa is:
[Jan. 7,2020 (II)]

Options:

A. $2-e$

B. $-e$

C. 2

D. $2+e$

Answer: A

Solution:

The given differential equation is $\frac{dx}{dy}+x=y^2$
Comparing with $\frac{dx}{dy}+Px=Q$, where $P=1, Q=y^2$
Now, I.F. $=e^{\int 1 . dy}=e^y$
$x . e^y=\int (y^2)e^y . dy=y^2 . e^y-\int 2y . e^y . dy$
$=y^2e^y-2(y . e^y-e^y)+C$
$\Rightarrow x . e^y=y^2e^y-2ye^y+2e^y+C$
$\Rightarrow x=y^2-2y+2+C . e^{-y}$ ....(i)
As $y(0)=1$, satisfying the given differential eqn,
$\therefore$ put $x=0, y=1$ in eqn. (i)
$0=1-2+2+\frac{C}{e}$
$C=-e$
$y=0, x=0-0+2+(-e)(e^{-0})$
$x=2-e$

Question191

The differential equation of the family of curves,$x^2=4b(y+b)$, b ∈ R, is: [Jan. 8, 2020 (II)]

Options:

A. $x(y')^2=x+2yy'$

B. $x(y')^2=2yy'-x$

C. $xy''=y'$

D. $x(y')^2=x-2yy'$

Answer: A

Solution:

Since, $x^2 = 4b(y+b)$
$x^2 = 4by + 4b^2$
$2x = 4by'$
$\Rightarrow b = \frac{x}{2y'}$
So, differential equation is
$x^2 = \frac{2x}{y'} . y + \left(\frac{x}{y'}\right)^2$
$x(y')^2 = 2yy' + x$

Question192

If f 2(x) = $\tan^{-1}(\sec x + \tan x)$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $f(0) = 0$, then $f(1)$ is equal to: [Jan. 9, 2020 (I)]

Options:

A. $\frac{\pi+1}{4}$

B. $\frac{1}{4}$

C. $\frac{\pi-1}{4}$

D. $\frac{\pi+2}{4}$

Answer: A

Solution:

$f'(x) = \tan^{-1}(\sec x + \tan x)$

$= \tan^{-1}\left(\frac{1+\sin x}{\cos x}\right) = \tan^{-1}\left(\frac{1-\cos\left(\frac{\pi}{2}+x\right)}{\sin\left(\frac{\pi}{2}+x\right)}\right)$

$= \tan^{-1}\left(\frac{2\sin^2\left(\frac{\pi}{4}+\frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4}+\frac{x}{2}\right)\cos\left(\frac{\pi}{4}+\frac{x}{2}\right)}\right)$

$= \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right) = \frac{\pi}{4}+\frac{x}{2}$

Integrate both sides, we get

$\int (f'(x))dx = \int \left(\frac{\pi}{4}+\frac{x}{2}\right)dx$

$f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + C$

$\because f(0) = 0$

$C=0 \Rightarrow f(x)=\frac{\pi}{4}x+\frac{x^2}{4}$

So, $f(1)=\frac{\pi+1}{4}$

Question193

If $\frac{dy}{dx}=\frac{xy}{x^2+y^2}$; y(1) = 1; then a value of x satisfying y(x) = e is: [Jan. 9, 2020 (II)]

Options:

A. $\frac{1}{2}\sqrt{3}e$

B. $\frac{e}{\sqrt{2}}$

C. $\sqrt{2}e$

D. $\sqrt{3}e$

Answer: D

Solution:

The given differential equation,

$\frac{dy}{dx}=\frac{xy}{x^2+y^2}$

Put $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$

Then, $v+x\frac{dv}{dx}=\frac{vx^2}{x^2+v^2x^2}=\frac{v}{1+v^2}$

$\Rightarrow \frac{1+v^2}{v^3}dv=-\frac{1}{x}dx$

$\Rightarrow \int\left(\frac{1}{v^3}+\frac{1}{v}\right)dv=\int\frac{-1}{x}dx$

$\Rightarrow \frac{-1}{2}\left(\frac{1}{v^2}\right)+\ln v=-\ln x+c$

$\Rightarrow -\frac{x^2}{2y^2}=-\ln y+c \quad \left[\because v=\frac{y}{x}\right]$

When $x=1, y=1$, then $-\frac{1}{2}=c$

$\Rightarrow x^2=y^2(1+2\ln y)$

At $y=e, x^2=e^2(3)$

$\Rightarrow x=\pm\sqrt{3}e$

So, $x=\sqrt{3}e$

Question194

Let $f(x) = (\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1, |x| > 1$. If $\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}(\sin^{-1}(f(x)))$ and $y(\sqrt{3}) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to:
[Jan. 8, 2020 (I)]

Options:

A. $\frac{2\pi}{3}$

B. $-\frac{\pi}{6}$

C. $\frac{5\pi}{6}$

D. $\frac{\pi}{3}$

Answer: B

Solution:

$\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}(\sin^{-1}f(x))$

$2y = \sin^{-1}f(x) + C \ = \sin^{-1}(\sin(2\tan^{-1}x)) + C$

$\Rightarrow 2\left(\frac{\pi}{6}\right) = \sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right) + C$

$\frac{\pi}{3} = \frac{\pi}{3} + C \quad \therefore C = 0$

for $x = -\sqrt{3}, 2y = \sin^{-1}\left(\sin\left(\frac{-2\pi}{6}\right)\right) + 0$

$\Rightarrow 2y = \frac{-\pi}{3} \Rightarrow y = \frac{-\pi}{6}$

Question195

Let $y = y(x)$ be a solution of the differential equation,
$\sqrt{1-x^2}\frac{dy}{dx} + \sqrt{1-y^2} = 0, |x| < 1$

If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to:

[Jan. 8, 2020 (I)]

Options:

A. $\frac{\sqrt{3}}{2}$

B. $-\frac{1}{\sqrt{2}}$

C. $\frac{1}{\sqrt{2}}$

D. $-\frac{\sqrt{3}}{2}$

Answer: C

Solution:

The given differential eqn. is

$\frac{dy}{\sqrt{1-y^2}}+\frac{dx}{\sqrt{1-x^2}}=0 \Rightarrow \sin^{-1}y+\sin^{-1}x=c$

At $x=\frac{1}{2}, y=\frac{\sqrt{3}}{2} \Rightarrow c=\frac{\pi}{2}$

$\Rightarrow \sin^{-1}y=\cos^{-1}x$

Hence, $y\left(-\frac{1}{\sqrt{2}}\right)=\sin\left(\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)\right)$

$=\sin\left(\pi-\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)\right)=\frac{1}{\sqrt{2}}$

Question196

If y = y(x) is the solution of the differential equation, $e^y = e^x$ such that y(0) = 0, then y(l) is equal to: [Jan. 7, 2020 (I)]

Options:

A. $1+\log_e 2$

B. $2+\log_e 2$

C. 2e

D. $\log_e 2$

Answer: A

Solution:

Let $e^y = t$

$e^y\frac{dy}{dx} = \frac{dt}{dx}$

$\therefore dt\,dx - t = e^x \left[\because e^y\frac{dy}{dx} - e^y = e^x\right]$

$I.F. = e^{\int -1.dx} = e^{-x}$

$t(e^{-x}) = \int e^x . e^{-x} dx \Rightarrow e^{y-x} = x + c$

Put $x = 0, y = 0$, then we get $c = 1$

$e^{y-x} = x + 1$

$y = x + \log_e(x+1)$

Put $x = 1 \quad \therefore y = 1 + \log_e 2$

Question197

The general solution of the differential equation $\sqrt{1 + x^2 + y^2 + x^2y^2} + xy\frac{dy}{dx} = 0$ is:
(where C is a constant of integration)
:[Sep. 06, 2020 (I)]

Options:

A. $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right) + C$

B. $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right) + C$

C. $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right) + C$

D. $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right) + C$

Answer: A

Solution:

$\sqrt{1+x^2} . \sqrt{1+y^2} = -xy\frac{dy}{dx}$

$\int \frac{\sqrt{1+x^2}}{x} dx = -\int \frac{y}{\sqrt{1+y^2}} dy$

Let $x = \tan\theta \Rightarrow dx = \sec^2\theta d\theta$

$$\Rightarrow \int \frac{\sec^3\theta d\theta}{\tan\theta} = -\int \frac{2y}{2\sqrt{1+y^2}} dy$$
$$\Rightarrow \int \frac{\sin^2\theta + \cos^2\theta}{\sin\theta . \cos^2\theta} d\theta = -\sqrt{1+y^2}$$
$$\Rightarrow \int (\tan\theta . \sec\theta + \operatorname{cosec}\theta) d\theta = -\sqrt{1+y^2}$$
$$\Rightarrow \sec\theta + \log_e|\operatorname{cosec}\theta - \cot\theta| = -\sqrt{1+y^2} + C$$
$$\therefore \sqrt{1+x^2} + \log_e\left|\frac{\sqrt{1+x^2}-1}{x}\right| = -\sqrt{1+y^2} + C$$
$$\Rightarrow \sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right) + C$$

Question198

If $y = \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x$ is the solution of the differentialequation, $d\, yd\, x + p(x)y = \frac{2}{\pi} \operatorname{cosec} x$, $0 < x < \frac{\pi}{2}$, then the function $p(x)$ is equal to: [Sep. 06, 2020 (II)]

Options:

A. $\cot x$

B. $\operatorname{cosec} x$

C. $\sec x$

D. $\tan x$

Answer: A

Solution:

$$\because y = \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x$$
$$\frac{dy}{dx} = \frac{2}{\pi}\operatorname{cosec} x - \left(\frac{2}{\pi}x - 1\right)\operatorname{cosec} x . \cot x$$
$$= \operatorname{cosec} x\left[\frac{2}{\pi} - \left(\frac{2}{\pi}x - 1\right)\cot x\right]$$
$$\Rightarrow \frac{dy}{dx} - \frac{2}{\pi}\operatorname{cosec} x = y\cot x \text{ ....(i)}$$
It is given that,
$$\Rightarrow \frac{dy}{dx} - \frac{2}{\pi}\operatorname{cosec} x = -yp(x) \text{ ....(ii)}$$
By comparision of (i) and (ii), we get
$p(x) = \cot x$

Question199

If y = y(x) is the solution of the differential equation $\frac{5+e^x}{2+y}\cdot\frac{dy}{dx}+e^x=0$ satisfying y(0) = 1, then a value of $y(\log_e 13)$ is :
[Sep. 05, 2020 (I)]

Options:

A. 1

B. -1

C. 0

D. 2

Answer: B

Solution:

$\frac{5+e^x}{2+y}\cdot\frac{dy}{dx}=-e^x$
$\int\frac{dy}{2+y}=-\int\frac{e^x}{5+e^x}dx$
$\Rightarrow \log_e|2+y|.\log_e|5+e^x| = \log_e C$
$\Rightarrow |(2+y)(5+e^x)| = C \because y(0)=1$
$C=18$
$\therefore (2+y).(5+e^x)=18$
When $x=\log_e 13$ then $(2+y).18=18$
$\Rightarrow 2+y=\pm 1$
$\therefore y=-1,-3$
$\therefore y(\ln 13)=-1$

Question200

The solution of the differential equation $\frac{dy}{dx}-\frac{y+3x}{\log_e(y+3x)}+3=0$ is
(where C is a constant of integration.)
[Sep. 04, 2020 (II)]

Options:

A. $x-\frac{1}{2}(\log_e(y+3x))^2=C$

B. $x-\log_e(y+3x)=C$

C. $y + 3x - \frac{1}{2}(\log_e x)^2 = C$

D. $x - 2\log_e(y + 3x) = C$

Answer: A

Solution:

Let $y + 3x = t$
$\Rightarrow d y d x + 3 = \frac{d t}{d x}$
Putting these value in given differential equation
$\frac{d t}{d x} = \frac{t}{\log_e t}$
$\Rightarrow \int \frac{\log_e t}{t} d t = \int d x$
$\Rightarrow \frac{(\log_e t)^2}{2} = x - C$
$\Rightarrow x - \frac{1}{2}(\ln(y + 3x))^2 = C$

Question201

Let $f : (0, \infty) \to (0, \infty)$ be a differentiable function such that $f(1) = e$ and $\lim_{t \to x} \frac{t^2 f^2(x) - x^2 f^2(t)}{t - x} = 0$

If $f(x) = 1$, then x is equal to :
[Sep. 04, 2020 (II)]

Options:

A. $\frac{1}{e}$

B. 2e

C. $\frac{1}{2e}$

D. e

Answer: A

Solution:

$$\lim_{t \to x} \frac{t^2 f^2(x) - x^2 f^2(t)}{t - x} = 0$$
$$\Rightarrow \lim_{t \to x} \frac{2tf^2(x) - 2x^2 f(t) . f(t)}{1} = 0$$
Using L'Hospital's rule
$\Rightarrow f(x) = xf'(x)$
$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{x} dx$
$\log_e f(x) = \log_e x + \log_e C$
$\Rightarrow f(x) = Cx \quad \because f(1) = e$
$\Rightarrow C = e$; so $f(x) = ex$
When $f(x) = 1 = ex \Rightarrow x = \frac{1}{e}$

Question202

The solution curve of the differential equation, $(1 + e^{-x})(1 + y^2)\frac{dy}{dx} = y^2$, which passes through the point (0, 1), is : [Sep. 03, 2020 (I)]

Options:

A. $y^2 + 1 = y\left(\log_e\left(\frac{1 + e^{-x}}{2}\right) + 2\right)$

B. $y^2 + 1 = y\left(\log_e\left(\frac{1 + e^{x}}{2}\right) + 2\right)$

C. $y^2 = 1 + y\log_e\left(\frac{1 + e^{x}}{2}\right)$

D. $y^2 = 1 + y\log_e\left(\frac{1 + e^{-x}}{2}\right)$

Answer: C

Solution:

$\int \left(\frac{y^2 + 1}{y^2}\right) dy = \int \frac{e^x dx}{e^x + 1}$
$\Rightarrow y - \frac{1}{y} = \log_e\left|e^x + 1\right| + c$
$\because$ Passes through (0,1)
$\therefore c = -\log_e 2$

$$\Rightarrow y^2 - 1 = y\log_e\left(\frac{e^x+1}{2}\right)$$

$$\Rightarrow y^2 = 1 + y\log_e\left(\frac{e^x+1}{2}\right)$$

Question203

If $x^3 dy + xy dx = x^2 dy + 2y dx$; $y(2) = e$ and $x > 1$, then $y(4)$ is equal to : [Sep. 03,2020 (II)]

Options:

A. $\frac{3}{2} + \sqrt{e}$

B. $\frac{3}{2}\sqrt{e}$

C. $\frac{1}{2} + \sqrt{e}$

D. $\frac{\sqrt{e}}{2}$

Answer: B

Solution:

$x^3 dy + xyd yx \ = 2yd x + x^2 dy$
$\Rightarrow (x^3 - x^2)dy = (2-x)yd x$
$\Rightarrow \frac{dy}{y} = \frac{2-x}{x^2(x-1)}dx$
$\Rightarrow \int \frac{dy}{y} = \int \frac{2-x}{x^2(x-1)}dx$
Let $\frac{2-x}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$
$\Rightarrow 2 - x = A(x-1) + B(x-1) + Cx^2$
Compare the coefficients of x, x^2 and constant term.
$C = 1$, $B = -2$ and $A = -1$
$\therefore \int \frac{dy}{y} = \int \left\{ \frac{-1}{x} - \frac{2}{x^2} + \frac{1}{x-1} \right\} dx$
$\Rightarrow \ln y = -\ln x + \frac{2}{x} + \ln\left|x-1\right| + C$
$\therefore y(2) = e$
$\Rightarrow 1 = -\ln 2 + 1 + 0 + C$ [$\because$ log e = 1]
$\Rightarrow C = \ln 2$
$\Rightarrow \ln y = -\ln x + \frac{2}{x} + \ln\left|x-1\right| + \ln 2$
At x = 4,
$\Rightarrow \ln y(4) = -\ln 4\frac{1}{2} + \ln 3 + \ln 2$

$\Rightarrow \ln y(4) = \ln\left(\frac{3}{2}\right) + \frac{1}{2} = \ln\left(\frac{3}{2}e^{1/2}\right)$ $[\because \log m + \log n = \log(mn)]$

$\Rightarrow y(4) = \frac{3}{2}e^{1/2}$

Question204

Let y = y(x) be the solution of the differential equation, $\mathbf{2 + \sin x y + 1 \,.\, \frac{dy}{dx} = -\cos x,\ y > 0,\ y(0) = 1}$**. If** $\mathbf{y(\pi) = a}$ **and** $\mathbf{\frac{dy}{dx}}$ **at** $\mathbf{x = \pi}$ **is b, then the ordered pair (a, b) is equal to :**
[Sep. 02, 2020 (I)]

Options:

A. $\left(2, \frac{3}{2}\right)$

B. (1,-1)

C. (1,1)

D. (2,1)

Answer: C

Solution:

The given differential equation is

$\frac{2 + \sin x}{y + 1}\frac{dy}{dx} = -\cos x,\ y > 0$

$\Rightarrow \frac{dy}{y + 1} = -\frac{\cos x}{2 + \sin x}dx$

Integrate both sides,

$\int \frac{dy}{y + 1} = \int \frac{(-\cos x)dx}{2 + \sin x}$

$\ln|y + 1| = -\ln|2 + \sin x| + \ln C$

$\Rightarrow \ln|y + 1| + \ln|2 + \sin x| = \ln C$

$\Rightarrow \ln|(y + 1)(2 + \sin x)| = \ln C$

$\because y(0) = 1 \Rightarrow \ln 4 = \ln C \Rightarrow C = 4$

$\therefore (y + 1)(2 + \sin x) = 4$

$\Rightarrow y = \frac{4}{2 + \sin x} - 1$

$\therefore y = \frac{2 - \sin x}{2 + \sin x} \Rightarrow y(\pi) = \frac{2 - \sin \pi}{2 + \sin \pi} = 1$

$\Rightarrow a = 1$

Now, $\frac{dy}{dx} = \frac{(2 + \sin x)(-\cos x) - (2 - \sin x)\,.\,\cos x}{(2 + \sin x)^2}$

$\left.\frac{dy}{dx}\right|_{x = \pi} = 1 \Rightarrow b = 1$

Ordered pair $(a, b) = (1, 1)$

Question205

If a curve $y = f(x)$, passing through the point (1,2) , is the solution of the differential equation, $2x^2 dy = (2xy + y^2)dx$, then $f\left(\frac{1}{2}\right)$ is equal to

[Sep. 02, 2020 (II)]

Options:

A. $\frac{1}{1+\log_e 2}$

B. $\frac{1}{1-\log_e 2}$

C. $1+\log_e 2$

D. $\frac{-1}{1+\log_e 2}$

Answer: A

Solution:

$\frac{dy}{dx} = \frac{2xy+y^2}{2x^2}$

It is homogeneous differential equation.

$\therefore$ Put $y = vx$

$\Rightarrow v + x\frac{dv}{dx} = v + \frac{v^2}{2} \Rightarrow \int 2\frac{dv}{v^2} = \int \frac{dx}{x}$

$\Rightarrow \frac{-2}{v} = \log_e x + c \Rightarrow \frac{-2x}{y} = \log_e x + c$

Put $x = 1, y = 2$, we get $c = -1$

$\Rightarrow \frac{-2x}{y} = \log_e x - 1$

Hence, put $x = \frac{1}{2} \Rightarrow y = \frac{1}{1+\log_e 2}$

Question206

Let $y = y(x)$ be the solution of the differential equation

$\cos x \frac{dy}{dx} + 2y \sin x = \sin 2x$, $x \in \left(0, \frac{\pi}{2}\right)$

If $y(\pi/3) = 0$, then $y(\pi/4)$ is equal to :

[Sep. 05, 2020 (II)]

Options:

A. $2-\sqrt{2}$

B. $2+\sqrt{2}$

C. $\sqrt{2}-2$

D. $\frac{1}{\sqrt{2}}-1$

Answer: C

Solution:

$\frac{dy}{dx}+2y\tan x=2\sin x$
I.F. $=e^{\int 2\tan x\,dx}=\sec^2 x$
The solution of the differential equation is
$y\times I.F=\int I.F\times 2\sin x\,dx+C$
$\Rightarrow y.\sec^2 x=\int 2\sin x.\sec^2 x dx+C$
$\Rightarrow y\sec^2 x=2\sec x+C$ ......(i)
When $x=\frac{\pi}{3}$, $y=0$; then $C=-4$
$\therefore$ From(i), $y\sec^2 x=2\sec x-4$
$\Rightarrow y=\frac{2\sec x-4}{\sec^2 x}\Rightarrow y\left(\frac{\pi}{4}\right)=\sqrt{2}-2$

Question207

Let y = y(x) be the solution of the differential equation,$xy'-y=x^2(x\cos x+\sin x)$, $x>0$. If $y(\pi)=\pi$, then$y''\left(\frac{\pi}{2}\right)+y\left(\frac{\pi}{2}\right)$ is equal to : [Sep. 04, 2020 (I)]

Options:

A. $2+\frac{\pi}{2}$

B. $2+\frac{\pi}{2}+\frac{\pi^2}{4}$

C. $2+\frac{\pi}{2}+\frac{\pi^2}{4}$

D. $1+\frac{\pi}{2}$

Answer: A

Solution:

$$\frac{dy}{dx}-\frac{y}{x}=x(x\cos x+\sin x)$$
$$\text{I.F.}=e^{-\int\frac{1}{x}dx}=\frac{1}{x}$$
$$\therefore\int d\left(\frac{y}{x}\right)=\int(x\cos x+\sin x)dx$$
$$\Rightarrow\frac{y}{x}=x\sin x+C\because y(\pi)=\pi\Rightarrow C=1$$
$$y=x^2\sin x+x\Rightarrow y\left(\frac{\pi}{2}\right)=\frac{\pi^2}{4}+\frac{\pi}{2}$$
$$y'=2x\sin x+x^2\cos x+1$$
$$y''=2\sin x-x^2\sin x\Rightarrow y''\left(\frac{\pi}{2}\right)=2-\frac{\pi^2}{4}$$
$$\therefore y''\left(\frac{\pi}{2}\right)+y\left(\frac{\pi}{2}\right)=2-\frac{\pi^2}{4}+\frac{\pi^2}{4}+\frac{\pi}{2}=2+\frac{\pi}{2}$$

Question208

If $y=y(x)$ is the solution of the differential equation, $x\frac{dy}{dx}+2y=x^2$ satisfying $y(1)=1$, then $y\left(\frac{1}{2}\right)$ is equal to:

[Jan. 09, 2019 (I)]

Options:

A. $\frac{7}{64}$

B. $\frac{1}{4}$

C. $\frac{49}{16}$

D. $\frac{13}{16}$

Answer: C

Solution:

Since, $x\frac{dy}{dx}+2y=x^2$

$\Rightarrow\frac{dy}{dx}+\frac{2}{x}y=x$

$I.F. = e^{\int \frac{2}{x} dx} = e^{2\ln x} = e^{\ln x^2} = x^2$
Solution of differential equation is:
$y.x^2 = \int x.x^2 dx$
$y.x^2 = \frac{x^4}{4} + C$ ....(1)
$\because y(1) = 1$
$\therefore C = \frac{3}{4}$
Then, from equation (1)
$y.x^2 = \frac{x^4}{4} + \frac{3}{4}$
$\therefore y = \frac{x^2}{4} + \frac{3}{4x^2}$
$\therefore y\left(\frac{1}{2}\right) = \frac{1}{16} + 3 = \frac{49}{16}$

Question209

Let f : [0, 1] → R be such that f (xy) = f (x), f (y), for all x, y ∈ [0, 1], and f (0) ≠ 0. If y = y(x) satisfies the differential equation, $\frac{dy}{dx}$ = f (x) with y(0) = 1, then $y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is[Jan. 09,2019 equal to:
[Jan. 09, 2019 (II)]

Options:

A. 3

B. 4

C. 2

D. 5

Answer: A

Solution:

$f(xy) = f(x)f(y)$ ......(1)
Put $x = y = 0$ in (1) to get $f(0) = 1$
Put $x = y = 1$ in (1) to get $f(1) = 0$ or $f(1) = 1$
$f(1) = 0$ is rejected else $y = 1$ in (1) gives $f(x) = 0$
imply $f(0) = 0$
Hence, $f(0) = 1$ and $f(1) = 1$
By first principle derivative formula,
$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$$\lim_{h \to 0} f(x)\left(\frac{f\left(1+\frac{h}{x}\right)-f(1)}{h}\right)$$
$$\Rightarrow f'(x)=\frac{f(x)}{x}f'(1)$$
$$\Rightarrow \frac{f'(x)}{f(x)}=\frac{k}{x} \Rightarrow \ln f(x)=k\ln x+c$$
$$f(1)=1 \Rightarrow \ln 1=k\ln 1+c \Rightarrow c=0$$
$\Rightarrow \ln f(x)=k\ln x \Rightarrow f(x)=x^k$ but $f(0)=1$
$$\Rightarrow k=0$$
$$\therefore f(x)=1$$
$$\frac{dy}{dx}=f(x)=1 \Rightarrow y=x+c,\ y(0)=1 \Rightarrow c=1$$
$$\Rightarrow y=x+1$$
$$\therefore y\left(\frac{1}{4}\right)+y\left(\frac{3}{4}\right)=\frac{1}{4}+1+\frac{3}{4}+1=3$$

Question210

If $\frac{dy}{dx}+\frac{3}{\cos^2 x}y=\frac{1}{\cos^2 x}$, $x \in \left(\frac{-\pi}{3},\frac{\pi}{3}\right)$, and $y\left(\frac{\pi}{4}\right)=\frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals: [10 Jan 2019 I]

Options:

A. $\frac{1}{3}+e^6$

B. $\frac{1}{3}$

C. $-\frac{4}{3}$

D. $\frac{1}{3}+e^3$

Answer: A

Solution:

Given, $\frac{dy}{dx}+\frac{3}{\cos^2 x}y=\frac{1}{\cos^2 x}$
$$\frac{dy}{dx}=\sec^2 x(1-3y)$$
$$\Rightarrow \int\frac{dy}{(1-3y)}=\int \sec^2 x\,dx$$
$\Rightarrow -\frac{1}{3}\ln\left|1-3y\right|=\tan x+C$ ...... (i)

$\because y\left(\frac{\pi}{4}\right)=\frac{4}{3}$ (Given)

$\Rightarrow -\frac{1}{3}\ln\left|1-4\right| = \tan\frac{\pi}{4} + C$

$\Rightarrow -\frac{1}{3}\ln 3 = C + 1 \Rightarrow C = -1 - \frac{1}{3}\ln 3$

∴ in eq. (i), we get

$\frac{-1}{3}\ln\left|1-3y\right| = \tan x - 1 - \frac{1}{3}\ln 3$

Put, $x = -\frac{\pi}{4}$

$\Rightarrow -\frac{1}{3}\ln\left|1-3y\right| = \tan\left(-\frac{\pi}{4}\right) - 1 - \frac{1}{3}\ln 3$

$= -1 - 1 - \frac{1}{3}\ln 3$

$\Rightarrow \ln|1-3y| = 6 + \ln 3$

$\Rightarrow \ln|13 - y| = 6 \Rightarrow \left|\frac{1}{3} - y\right| = e^6 \Rightarrow y = \frac{1}{3} \pm e^6$

Question211

The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2)dx + 2xydy = 0$ which passes through (1, 1), is: [Jan. 10, 2019 (II)]

Options:

A. a circle with centre on the x-axis.

B. an ellipse with ma□or axis along the y-axis.

C. a circle with centre on the y-axis.

D. a hyperbola with transverse axis along the x-axis

Answer: A

Solution:

$(x^2 - y^2)dx + 2xydy = 0$

$y^2dx - 2xydy = x^2dx$

$2xydy - y^2dx = -x^2dx$

$d(xy^2) = -x^2dx$

$\frac{xd(y^2) - y^2d(x)}{x^2} = -dx$

$d\left(\frac{y^2}{x}\right) = -dx$

$\int d\left(\frac{y^2}{x}\right) = -\int dx$

$\frac{y^2}{x} = -x + C$

Since, the above curve passes through the point (1,1)

Then, $\frac{1^2}{1} = -1 + C \Rightarrow C = 2$
Now, the curve (1) becomes
$y^2 = -x^2 + 2x$
$\Rightarrow y^2 = -(x-1)^2 + 1$
$(x-1)^2 + y^2 = 1$
The above equation represents a circle with centre (1,0) and centre lies on x -axis.

Question212

Let f be a differentiable function such that $f'(x) = 7 - \frac{3}{4}\frac{f(x)}{x}$, $(x > 0)$ and $f(1) \neq 4$. Then $\lim_{x \to 0^+} xf\left(\frac{1}{x}\right)$

[Jan. 10, 2019 (II)]

Options:

A. exists and equals $\frac{4}{7}$.

B. exists and equals 4 .

C. does not exist.

D. exists and equals 0 .

Answer: B

Solution:

Let $y = f(x)$
$\frac{dy}{dx} + \left(\frac{3}{4x}\right)y = 7$
I.F. $= e^{\int \frac{3}{4x} dx} = e^{\frac{3}{4} \ln x} = x^{\left(\frac{3}{4}\right)}$
Solution of differential equation
$y \cdot x^{\frac{3}{4}} = \int 7 \cdot x^{\frac{3}{4}} dx + C$
$y \cdot x^{\frac{3}{4}} = 7 \cdot \frac{x^{\frac{7}{4}}}{\left(\frac{7}{4}\right)} + C = 4x^{\frac{7}{4}} + C$
$y = 4x + Cx^{-\frac{3}{4}}$
$\Rightarrow f\left(\frac{1}{x}\right) = \frac{4}{x} + Cx^{\frac{3}{4}}$
$\Rightarrow \lim_{x \to 0^-} x \cdot f\left(\frac{1}{x}\right) = \lim_{x \to 0^+}\left(4 + Cx^{\frac{7}{4}}\right) = 4$

Question213

The solution of the differential equation, $\frac{dy}{dx} = (x-y)^2$, when y(1) = 1, is: [Jan. 11, 2019(II)]

Options:

A. $\log_e\left|\frac{2-x}{2-y}\right| = x-y$

B. $-\log_e\left|\frac{1-x+y}{1+x-y}\right| = 2(x-1)$

C. $-\log_e\left|\frac{1+x-y}{1-x+y}\right| = x+y-2$

D. $\log_e\left|\frac{2-y}{2-x}\right| = 2(y-1)$

Answer: B

Solution:

The given differential equation
$\frac{dy}{dx} = (x-y)^2$ ....(1)
Let $x-y=t \Rightarrow 1-\frac{dy}{dx} = \frac{dt}{dx}$
$\Rightarrow \frac{dy}{dx} = 1-\frac{dt}{dx}$
Now, from equation (1)
$\left(1-\frac{dt}{dx}\right) = (t)^2$
$\Rightarrow 1-t^2 = \frac{dt}{dx} \Rightarrow \int dx = \int \frac{dt}{1-t^2}$
$\Rightarrow -x = \frac{1}{2\times 1}\ln\left|\frac{t-1}{t+1}\right| + c$
$\Rightarrow -x = \frac{1}{2}\ln\left|\frac{x-y-1}{x-y+1}\right| + c$
$\because$ The given condition $y(1)=1$
$-1 = \frac{1}{2}\ln\left|\frac{1-1-1}{1-1+1}\right| + c \Rightarrow c = -1$
Hence, $2(x-1) = -\ln\left|\frac{1-x+y}{1-y+x}\right|$

Question214

If y(x) is the solution of the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$, x > 0, where $y(1) = \frac{1}{2}e^{-2}$, then:
[Jan. 11, 2019 (I)]

Options:

A. $y(\log_e 2) = \log_e 4$

B. $y(\log_e 2) = \frac{\log_e 2}{4}$

C. y(x) is decreasing in $\left(\frac{1}{2}, 1\right)$

D. y(x) is decreasing in (0,1)

Answer: C

Solution:

Given differential equation is,

$\frac{dy}{dx} + \left(2 + \frac{1}{x}\right)y = e^{-2x}, x > 0$

$IF = e^{\int\left(2+\frac{1}{x}\right)dx} = e^{2x+\ln x} = xe^{2x}$

Complete solution is given by

$y(x) \cdot xe^{2x} = \int xe^{2x} \cdot e^{-2x} dx + c$

$= \int x dx + c$

$y(x) \cdot e^{2x} \cdot x = \frac{x^2}{2} + c$

Given, $y(1) = \frac{1}{2}e^{-2}$

$\therefore \frac{1}{2}e^{-2} \cdot e^2 \cdot 1 = \frac{1}{2} + c \Rightarrow c = 0$

$\therefore y(x) = \frac{x^2}{2} \cdot \frac{e^{-2x}}{x}$

$y(x) = \frac{x}{2} \cdot e^{-2x}$

Differentiate both sides with respect to x,

$y'(x) = \frac{e^{-2x}}{2}(1-2x) < 0 \ \forall x \in \left(\frac{1}{2}, 1\right)$

Hence, $y(x)$ is decreasing in $\left(\frac{1}{2}, 1\right)$

Question215

Let y = y(x) be the solution of the differential equation,
$x\frac{dy}{dx}+y=x\log_e x$, (x > 1). If $2y(2)=\log_e 4-1$, then y(e) is equal to :
[Jan. 12, 2019 (I)]

Options:

A. $-\frac{e}{2}$

B. $-\frac{e^2}{2}$

C. $\frac{e}{4}$

D. $\frac{e^2}{4}$

Answer: C

Solution:

Consider the differential equation,
$\frac{dy}{dx}+\frac{y}{x}=\log_e x$
$\because IF=e^{\int\frac{1}{x}dx}=x$
$\therefore yx=\int x\ln x\,dx$
$\Rightarrow xy=\ln x\cdot\frac{x^2}{2}-\int\frac{1}{x}\cdot\frac{x^2}{2}dx$
$\Rightarrow xy=\frac{x^2}{2}\cdot\ln x-\frac{x^2}{4}+c$
Given, $2y(2)=\log_e 4-1$
$\therefore 2y=2\ln 2-1+c$
$\Rightarrow \ln 4-1=\ln 4-1+c$
i.e. $c=0$
$\Rightarrow xy=\frac{x^2}{2}\ln x-\frac{x^2}{4}$
$\Rightarrow y=\frac{x}{2}\ln x-\frac{x}{4}$
$\Rightarrow y(e)=\frac{e}{2}-\frac{e}{4}=\frac{e}{4}$

Question216

If a curve passes through the point (1,-2) and has slope of the tangent at any point (x, y) on it as $\frac{x^2-2y}{x}$, then the curve also passes through the point :
[Jan. 12, 2019 (II)]

Options:

A. (3,0)

B. $(\sqrt{3}, 0)$

C. (-1,2)

D. $(-\sqrt{2}, 1)$

Answer: B

Solution:

$\because$ Slope of the tangent $= \frac{x^2 - 2y}{x}$

$\because \frac{dy}{dx} = \frac{x^2 - 2y}{x}$

$\frac{dy}{dx} + \frac{2}{x} y = x$

$I.F = e^{\int \frac{2}{x}} = e^{2 \ln x} = x^2$

Solution of equation

$y . x^2 = \int x . x^2 dx$

$x^2 y = \frac{x^4}{4} + C$

$\therefore$ curve passes through point (1,-2)

$(1)^2(-2) = \frac{1^4}{4} + C$

$\Rightarrow C = \frac{-9}{4}$

Then, equation of curve

$y = \frac{x^2}{4} - \frac{9}{4x^2}$

Since, above curve satisfies the point.

Hence, the curve passes through $(\sqrt{3}, 0)$

Question217

The general solution of the differential equation $(y^2 - x^3)$ $dx - xy dy = 0 (x \neq 0)$ is: (where c is a constant of integration) [April 12, 2019 (II)]

Options:

A. $y^2 - 2x^2 + cx^3 = 0$

B. $y^2 + 2x^3 + cx^2 = 0$

C. $y^2 + 2x^2 + cx^3 = 0$

D. $y^2 - 2x^3 + cx^2 = 0$

Answer: B

Solution:

Given differential equation can be written as,
$y^2 d x - xyd y = x^3 d x$
$\Rightarrow \frac{(yd x - xd y)y}{x^2} = xd x \Rightarrow -yd\left(\frac{y}{x}\right) = xd x$
$\Rightarrow -\frac{y}{x} . d\left(\frac{y}{x}\right) = d x \Rightarrow -\frac{1}{2}\left(\frac{y}{x}\right)^2 = x + c_1$
$\Rightarrow 2x^3 + cx^2 + y^2 = 0$ [Here, $c = 2c_1$]

Question218

If $\cos x \frac{dy}{dx} - y \sin x = 6x$, $\left(0 < x < \frac{\pi}{2}\right)$ and $y\left(\frac{\pi}{3}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to: [April. 09, 2019 (II)]

Options:

A. $\frac{\pi^2}{2\sqrt{3}}$

B. $-\frac{\pi^2}{2}$

C. $-\frac{\pi^2}{2\sqrt{3}}$

D. $-\frac{\pi^2}{4\sqrt{3}}$

Answer: C

Solution:

$\cos x\, d y - (\sin x)yd x = 6xd x$
$\Rightarrow \int d(y \cos x) = \int 6xd x \Rightarrow y \cos x = 3x^2 + C$ .....(1)

Given, $y\left(\frac{\pi}{3}\right) = 0$

Putting $x = \frac{\pi}{3}$ and $y = 0$ in eq. (1), we get

$(10) \times \left(\frac{1}{2}\right) = \frac{3\pi^2}{9} + C \Rightarrow C\frac{-\pi^2}{3}$

So, from(1) $y \cos x = 3x^2 - \frac{\pi^2}{3}$

Now, put $x = \frac{\pi}{6}$ in the above equation,

$y\frac{\sqrt{3}}{2} = \frac{3\pi^2}{36} - \frac{\pi^2}{3} \Rightarrow \frac{\sqrt{3}y}{2} = \frac{-3\pi^2}{12} \Rightarrow y = \frac{-\pi^2}{2\sqrt{3}}$

Question219

Given that the slope of the tangent to a curve $y = y(x)$ at any point (x, y) is $\frac{2y}{x^2}$. If the curve passes through the centre of the circle $x^2 + y^2 - 2x - 2y = 0$, then its equation is:
[April. 08,2019 (II)]

Options:

A. $x\log_e|y| = 2(x-1)$

B. $x\log_e|y| = -2(x-1)$

C. $x^2\log_e|y| = -2(x-1)$

D. $x\log_e|y| = x-1$

Answer: A

Solution:

Given $\frac{dy}{dx} = \frac{2y}{x^2}$

Integrating both sides, $\int\frac{dy}{y} = 2\int\frac{dx}{x^2}$

$\Rightarrow \ln|y| = -\frac{2}{x} + C$

Equation (i) passes through the centre of the circle $x^2 + y^2 - 2x - 2y = 0$, i.e. ,$(1, 1)$

$\therefore C = 2$

Now, $\ln|y| = -\frac{2}{x} + 2$

$x\ln|y| = -2(1-x) \Rightarrow x\ln|y| = 2(x-1)$

Question220

Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$If value of y is 1 when x = 1, then the value of x for which y = 2, is :
[April 12, 2019 (I)]

Options:

A. $\frac{5}{2} + \frac{1}{\sqrt{e}}$

B. $\frac{3}{2} - \frac{1}{\sqrt{e}}$

C. $\frac{1}{2} + \frac{1}{\sqrt{e}}$

D. $\frac{3}{2} - \sqrt{e}$

Answer: B

Solution:

Consider the differential equation,

$y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$

$\Rightarrow \frac{dx}{dy} + \left(\frac{1}{y^2}\right)x = \frac{1}{y^3}$

$I.F. = e^{\int \frac{1}{y^2} dy} = e^{-\frac{1}{y}}$

$\therefore x . e^{-\frac{1}{y}} = \int e^{-\frac{1}{y}} \frac{1}{y^3} dy + c$

Put $-\frac{1}{y} = u \Rightarrow \frac{1}{y^2} dy = du$

$\Rightarrow x . e^{-\frac{1}{y}} = -\int ue^u du + c = -ue^u + e^u + c$

$\Rightarrow x . e^{-\frac{1}{y}} = e^{-\frac{1}{y}}\left(\frac{1}{y} + 1\right) + c$

At $y = 1, x = 1$

$1 = 2 + ce \Rightarrow c = -\frac{1}{e} \Rightarrow x = \left(1 + \frac{1}{y}\right) - \frac{1}{e}e^{\frac{1}{y}}$

On putting $y = 2$, we get $x = \frac{3}{2} - \frac{1}{\sqrt{e}}$

Question221

If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} = (\tan x - y)\sec^2 x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that $y(0) = 0$then $y\left(-\frac{\pi}{4}\right)$ is equal to:
[April 10, 2019 (I)]

Options:

A. $e - 2$

B. $\frac{1}{2} - e$

C. $2 + \frac{1}{e}$

D. $\frac{1}{e} - 2$

Answer: A

Solution:

$\frac{dy}{dx} + y\sec^2 x = \sec^2 \times \tan x$
Given equation is linear differential equation.
$IF = e^{\int \sec^2 x dx} = e^{\tan x}$
$\Rightarrow y \cdot e^{\tan x} = \int e^{\tan x}\sec^2 \times \tan x \, d\, x$
Put $\tan x = u = \sec^2 x d x = d u$
$ye^{\tan x} = \int e^u u d u \Rightarrow ye^{\tan x} = ue^u - e^u + c$
$\Rightarrow ye^{\tan x} = (\tan x - 1)e^{\tan x} + c$
$\Rightarrow y = (\tan x - 1) + c \cdot e^{-\tan x}$
$\therefore y(0) = 0$ (given) $\Rightarrow 0 = -1 + c \Rightarrow c = 1$
Hence, solution of differential equation,
$y\left(-\frac{\pi}{4}\right) = -1 - 1 + e = -2 + e$

Question222

Let $y = y(x)$ be the solution of the differential equation,
$\frac{dy}{dx} + y\tan x = 2x + x^2\tan x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that $y = 1$. Then :
[April 10, 2019 (II)]

Options:

A. $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$

B. $y'\left(\frac{\pi}{4}\right)+y'\left(-\frac{\pi}{4}\right)=-\sqrt{2}$

C. $y\left(\frac{\pi}{4}\right)-y\left(-\frac{\pi}{4}\right)=\sqrt{2}$

D. $y'\left(\frac{\pi}{4}\right)-y'\left(-\frac{\pi}{4}\right)=\pi-\sqrt{2}$

Answer: D

Solution:

Given differential equation is,
$\frac{dy}{dx}+y\tan x=2x+x^2\tan x$
Here, $P=\tan x$, $Q=2x+x^2\tan x$
I.F. $=e^{\int \tan x\,dx}=e^{\ln|\sec x|}=|\sec x|$
$\therefore y(\sec x)=\int(2x+x^2\tan x)\sec x\,dx$
$=\int x^2\tan x\sec x\,dx+\int 2x\sec x\,dx=x^2\sec x+c$
Given $y(0)=1\Rightarrow c=1$
$\therefore y=x^2+\cos x$
Now put $x=\frac{\pi}{4}$ and $x=\frac{-\pi}{4}$ in equation (i),
$y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{16}+\frac{1}{\sqrt{2}}$ and $y\left(-\frac{\pi}{4}\right)=\frac{\pi^2}{16}+\frac{1}{\sqrt{2}}$
$\Rightarrow y\left(\frac{\pi}{4}\right)-y\left(-\frac{\pi}{4}\right)=0$
$\frac{dy}{dx}=2x-\sin x$
$\therefore y'\left(\frac{\pi}{4}\right)=\frac{\pi}{2}-\frac{1}{\sqrt{2}}$ and $y'\left(-\frac{\pi}{4}\right)=-\frac{\pi}{2}+\frac{1}{\sqrt{2}}$
$\Rightarrow y'\left(\frac{\pi}{4}\right)-y'\left(-\frac{\pi}{4}\right)=\pi-\sqrt{2}$

Question223

The solution of the differential equation $x\frac{dy}{dx}+2y=x^2$ $(x\neq 0)$ with $y(1)=1$, is:
[April 09, 2019 (I)]

Options:

A. $y=\frac{4}{5}x^3+\frac{1}{5x^2}$

B. $y=\frac{x^3}{5}+\frac{1}{5x^2}$

C. $y=\frac{x^2}{4}+\frac{3}{4x^2}$

D. $y=\frac{3}{4}x^2+\frac{1}{4x^2}$

Answer: C

Solution:

$\frac{dy}{dx}+\frac{2}{x}y=xy(1)=1(\text{ given })$

Since, the above differential equation is the linear differential equation, then $I.F=e^{\int\frac{2}{x}dx}=x^2$
Now, the solution of the linear differential equation
$y\times x^2=\int x^3dx$
$\Rightarrow yx^2=\frac{x^4}{4}+C\because y(1)=1$
$\therefore 1\times 1=\frac{1}{4}+C\Rightarrow C=\frac{3}{4}$
$\therefore$ Solution becomes.
$y=\frac{x^2}{4}+\frac{3}{4x^2}$

Question224

Let y = y(x) be the solution of the differential equation, $(x^2+1)^2\frac{dy}{dx}+2x(x^2+1)y=1$ such that $y(0)=0$. If $\sqrt{a}y(1)=\frac{\pi}{32}$, then the value of 'a ' is: quad [April [April 08, 2019 (I)]

Options:

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 1

D. $\frac{1}{16}$

Answer: D

Solution:

$$(x^2+1)^2\frac{dy}{dx}+2x(x^2+1)y=1$$
$$\Rightarrow\frac{dy}{dx}+\left(\frac{2x}{1+x^2}\right)y\ =\frac{1}{(1+x^2)^2}$$
Since, the above differential equation is a linear differential equation
$\therefore$ I.F. $=\int e^{\int 2x\,.\,1+x^2 dx}=e^{\log(1+x^2)}\ =1+x^2$
Then, the solution of the differential equation
$$\Rightarrow y(1+x^2)=\int\frac{dx}{1+x^2}+c$$
$\Rightarrow y(1+x^2)=\tan^{-1}x+c$ ....(1)
If $x=0$ then $y=0$ (given)
$\Rightarrow 0=0+c$
$\Rightarrow c=0$
Then, equation (1) becomes,
$\Rightarrow y(1+x^2)=\tan^{-1}x$
Now put $x=1$ in above equation, then
$$2y=\frac{\pi}{4}$$
$$\Rightarrow 2\left(\frac{\pi}{32\sqrt{a}}\right)=\frac{\pi}{4}\left[\sqrt{a}y(1)=\frac{\pi}{32}\right]$$
$$\Rightarrow\sqrt{a}=\frac{1}{4}$$
$$\Rightarrow a=\frac{1}{16}$$

Question225

The differential equation representing the family of ellipse having foci either on the x -axis or on the y-axis centre at the origin and passing through the point (0,3) is:
[Online April 16, 2018]

Options:

A. $xyy'+y^2-9=0$

B. $x+yy''=0$

C. $xyy''+x(y')^2-yy'=0$

D. $xyy'-y^2+9=0$

Answer: C

Solution:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$
Since, it passes through (0,3)

$\therefore \frac{0}{a^2}+\frac{9}{b^2}=1$
$\Rightarrow b^2=9$
$\therefore$ eq. of ellipse becomes:
$\frac{x^2}{a^2}+\frac{y^2}{9}=1$
differential w.r.t. x, we get;
$\frac{2x}{a^2}+\frac{2y}{9}\frac{dy}{dx}=0$
$\Rightarrow \frac{y}{x}\left(\frac{dy}{dx}\right)=\frac{-9}{a^2}$
Again differentiating w.r.t. x, we get;
$\frac{y}{x}\frac{d^2y}{dx^2}+\frac{x\frac{dy}{dx}-y}{x^2}\frac{dy}{dx}=0$
$\Rightarrow xyy''+x(y')^2-yy'=0$

Question226

The curve satisfying the differential equation, $(x^2-y^2)dx+2xydy=0$ and passing through the point (1,1) is
[Online April 15, 2018]

Options:

A. a circle of radius two

B. a circle of radius one

C. a hyprbola

D. an ellipse

Answer: B

Solution:

$(x^2-y^2)dx+2xydy=0$
$\Rightarrow \frac{dy}{dx}=\frac{y^2-x^2}{2xy}$
Let $y=vx$
$\frac{dy}{dx}=v+x\frac{dv}{dx}$
$\Rightarrow v+x\frac{dv}{dx}=\frac{v^2x^2-x^2}{2vx^2} \Rightarrow v+x\frac{dv}{dx}=\frac{v^2-1}{2v}$
$\Rightarrow x\frac{dv}{dx}=\frac{-v^2-1}{2v}$
$\Rightarrow \frac{2vdv}{v^2+1}=-\frac{dx}{x}$
After integrating, we get

$\ln|v^2+1| = -\ln|x| + \ln c$
$\frac{y^2}{x^2}+1=\frac{c}{x}$
As curve passes through the point $(1, 1)$, so $1+1=c$
$\Rightarrow c=2$
$x^2+y^2-2x=0$, which is a circle of radius one.

Question227

Let $y-y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx}+y\cos x=4x$, $x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right)=0$, then $y\left(\frac{\pi}{6}\right)$ is equal to: [2018]

Options:

A. $\frac{-8}{9\sqrt{3}}\pi^2$

B. $-\frac{8}{9}\pi^2$

C. $-\frac{4}{9}\pi^2$

D. $\frac{4}{9\sqrt{3}}\pi^2$

Answer: B

Solution:

Consider the given differential equation the
$\sin x\, dy + y\cos x\, dx = 4x\,dx$
$\Rightarrow d(y \cdot \sin x) = 4x\,dx$
Integrate both sides
$\Rightarrow y \cdot \sin x = 2x^2 + C$ ....(1)
$\Rightarrow y(x) = \frac{2x^2}{\sin x} + c$ ....(2)
$\because$ eq. (2) passes through $\left(\frac{\pi}{2}, 0\right)$
$\Rightarrow 0 = \frac{\pi^2}{2} + C$
$\Rightarrow C = -\frac{\pi^2}{2}$
Now, put the value of C in (1)
Then, $y\sin x = 2x^2 - \frac{\pi^2}{2}$ is the solution
$\therefore y\left(\frac{\pi}{6}\right) = \left(2 \cdot \frac{\pi^2}{36} - \frac{\pi^2}{2}\right)2 = -\frac{8\pi^2}{9}$

Question228

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + 2y = f(x)$, where

$f(x) =$ $\begin{cases} 1, & x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$

If y(0) = 0, then $y\left(\frac{3}{2}\right)$ is

[Online April 15, 2018]

Options:

A. $\frac{e^2 - 1}{2e^3}$

B. $\frac{e^2 - 1}{e^3}$

C. $\frac{1}{2e}$

D. $\frac{e^2 + 1}{2e^4}$

Answer: A

Solution:

When $x \in [0, 1]$, then $\frac{dy}{dx} + 2y = 1 \Rightarrow y = \frac{1}{2} + C_1 e^{-2x}$

$\because y(0) = 0 \Rightarrow y(x) = \frac{1}{2} - \frac{1}{2}e^{-2x}$

Here ,$y(1) = \frac{1}{2} - \frac{1}{2}e^{-2} = \frac{e^2 - 1}{2e^2}$

When $x \notin [0, 1]$, then $\frac{dy}{dx} + 2y = 0 \Rightarrow y = c_2 e^{-2x}$

$\because y(1) = \frac{e^2 - 1}{2} \Rightarrow \frac{e^2 - 1}{2} = c^2 e^{-2} \Rightarrow C_2 = \frac{e^2 - 1}{2}$

$\therefore y(x)\left(\frac{e^2 - 1}{2}\right)e^{-2x} \Rightarrow y\left(\frac{3}{2}\right) = \frac{e^2 - 1}{2e^3}$

Question229

If $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0$ and $y(0)=1$, then $y\left(\frac{\pi}{2}\right)$ is equal to : [2017]

Options:

A. $\frac{4}{3}$

B. $\frac{1}{3}$

C. $-\frac{2}{3}$

D. $-\frac{1}{3}$

Answer: B

Solution:

We have $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0$
$\Rightarrow \frac{d}{dx}(2+\sin x)(y+1)=0$
On integrating, we get
$(2+\sin x)(y+1)=C$
At $x=0,\ y=1$ we have
$(2+\sin 0)(1+1)=C$
$\Rightarrow C=4$
$\Rightarrow y+1=\frac{4}{2+\sin x}$
$y=\frac{4}{2+\sin x}-1$
Now $y\left(\frac{\pi}{2}\right)=\frac{4}{2+\sin\frac{\pi}{2}}-1=\frac{4}{3}-1=\frac{1}{3}$

Question230

The curve satisfying the differential equation, $y\,dx-(x+3y^2)\,dy=0$ and passing through the point (1, 1), also passes through the point : [Online April 8, 2017]

Options:

A. $\left(\frac{1}{4},-\frac{1}{2}\right)$

B. $\left(-\frac{1}{3}, \frac{1}{3}\right)$

C. $\left(\frac{1}{3}, -\frac{1}{3}\right)$

D. $\left(\frac{1}{4}, \frac{1}{2}\right)$

Answer: B

Solution:

$y\,dx - x\,dy - 3y^2 dy = 0$

$\Rightarrow \frac{dx}{dy} = \frac{x}{y} + 3y \Rightarrow \frac{dx}{dy} - \frac{x}{y} = 3y$

if $= e^{-\int \frac{1}{y} dy} = e^{-\ln y} = \frac{1}{y}$

$\therefore$ solution is $\frac{x}{y} = \int 3y \cdot \frac{1}{y} dy$

$\Rightarrow \frac{x}{y} = 3y + c$

which passes through (1,1)

$\therefore 1 = 3 + c \Rightarrow c = -2$

$\therefore$ solution becomes $\Rightarrow x = 3y^2 - 2y$

which also passes through $\left(-\frac{1}{3}, \frac{1}{3}\right)$

Question231

If a curve y = f (x) passes through the point (1,-1) and satisfies the differential equation, $y(1 + xy)dx = x\,dy$, then $f\left(-\frac{1}{2}\right)$ is equal to: [2016]

Options:

A. $\frac{2}{5}$

B. $\frac{4}{5}$

C. $-\frac{2}{5}$

D. $-\frac{4}{5}$

Answer: B

Solution:

$$y(1+xy)dx = xdy$$
$$\frac{xdy - ydx}{y^2} = xdx$$
$$\Rightarrow \int -d\left(\frac{x}{y}\right) = \int xdx$$
$-\frac{x}{y} = \frac{x^2}{2} + C$ as $y(1) = -1 \Rightarrow C = \frac{1}{2}$

Hence, $y = \frac{-2x}{x^2+1} \Rightarrow f\left(\frac{-1}{2}\right) = \frac{4}{5}$

Question232

If f (x) is a differentiable function in the interval ((0, ∞) such that f (a) = 1 and $\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x}$ = 1, for each x > 0,then f $\left(\frac{3}{2}\right)$ is equal to:
[Online April 9, 2016]

Options:

A. $\frac{23}{18}$

B. $\frac{13}{6}$

C. $\frac{25}{9}$

D. $\frac{31}{18}$

Answer: D

Solution:

Let $L = \lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

Applying L.H. rule

$$L = \lim_{t \to x} \frac{2tf(x) - x^2 f'(t)}{1} = 1$$
$$2xf(x) - x^2 f'(x) = 1$$
solving above differential equation, we get
$$f(x) = \frac{2}{3}x^2 + \frac{1}{3x}$$
Put $x = \frac{3}{2}$
$$f\left(\frac{3}{2}\right) = \frac{2}{3} \times \left(\frac{3}{2}\right)^2 + \frac{1}{3} \times \frac{2}{3} = \frac{3}{2} + \frac{2}{9} = \frac{27+4}{18} = \frac{31}{18}$$

Question233

The solution of the differential equaiton $\frac{dy}{dx} + \frac{y}{2} \sec x = \frac{\tan x}{2y}$, where $0 \leq x < \frac{\pi}{2}$, and $y(0) = 1$, is given by :
[Online April 10, 2016]

Options:

A. $y^2 = 1 + \frac{x}{\sec x + \tan x}$

B. $y = 1 + \frac{x}{\sec x + \tan x}$

C. $y = 1 - \frac{x}{\sec x + \tan x}$

D. $y^2 = 1 - \frac{x}{\sec x + \tan x}$

Answer: D

Solution:

$\frac{dy}{dx} + \frac{y}{2} \sec x = \frac{\tan x}{2y}$

$2y\frac{dy}{dx} + y^2 \sec x = \tan x$

Put $y^2 = t \Rightarrow 2y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} + t \sec x = \tan x$

$If = e^{\int \sec x \, dx} = e^{In(\sec x + \tan x)} = \sec x + \tan x$

$\frac{dt}{dx}(\sec x + \tan x) + t \sec x(\sec x + \tan x)$

$= \tan x(\sec x + \tan x)$

$\int d(t(\sec x + \tan x)) = \int \tan x(\sec x + \tan x) \, dx$

$t(\sec x + \tan x) = \sec x + \tan x - x$

$t = 1 - \frac{x}{\sec x + \tan x} \Rightarrow y^2 = 1 - \frac{x}{\sec x + \tan x}$

Question234

The solution of the differential equation $yd x - (x + 2y^2) dy = 0$ is $x = f(y)$. If $f(-1) = 1$, then $f(a)$ is equal to :
[Online April 11, 2015]

Options:

A. 4

B. 3

C. 1

D. 2

Answer: B

Solution:

Given differential equation is
$y\,dx - (x + 2y^2)\,dy = 0$
$\Rightarrow y\,dx - x\,dy - 2y^2\,dy = 0$
$\Rightarrow \frac{y\,dx - x\,dy}{y^2} = 2\,dy$
$\Rightarrow d\left(\frac{x}{y}\right) = 2\,dy$
Integrate both the side
$\Rightarrow \frac{x}{y} = 2y + c$
using $f(-1) = 1$, we get
$c = 1$
$\Rightarrow \frac{x}{y} = 2y + 1$
put $y = 1$, we get $f(a) = 3$

Question235

If y(x) is the solution of the differential equation$(x + 2)\frac{dy}{dx} = x^2 + 4x - 9,\ x \neq -2$ and $y(0) = 0$, then $y(-4)$is equal to : [Online April 10, 2015]

Options:

A. 0

B. 2

C. 1

D. -1

Answer: A

Solution:

$(x+2)\frac{dy}{dx} = x^2 + 4x - 9x \neq -2$
$\frac{dy}{dx} = \frac{x^2+4x-9}{x+2}$
$dy = \frac{x^2+4x-9}{x+2}dx$
$\int dy = \int \frac{x^2+4x-9}{x+2}dx$
$y = \int \left(x+2-\frac{13}{x+2}\right)dx$
$y = \int (x+2)dx - 13\int \frac{1}{x+2}dx$
$y = \frac{x^2}{2} + 2x - 13\log\left|x+2\right| + c$
Given that $y = (0) = 0$
$0 = -13\log 2 + c$
$y = \frac{x^2}{2} + 2x - 13\log|x+2| + 13\log 2$
$y(-4) = 8 - 8 - 13\log 2 + 13\log 2 = 0$

Question236

Let y(x) be the solution of the differential equation $(x\log x)\frac{dy}{dx} + y = 2x\log x, (x \geq 1)$. Then y(e) is equal to: [2015]

Options:

A. 2

B. 2e

C. e

D. 0

Answer: A

Solution:

Given, $\frac{dy}{dx} + \left(\frac{1}{x\log x}\right)y = 2$
$I.F. = e^{\int \frac{1}{x\log x}dx} = e^{\log(\log x)} = \log x$
$y.\log x = \int 2\log x\, d\,x + c$
$y\log x = 2[x\log x - x] + c$
Put $x = 1, y.0 = -2 + c$
$c = 2$
Put $x = e$
$y\log e = 2e(\log e - 1) + c$
$y(e) = c = 2$

Question237

If the differential equation representing the family of all circles touching x - axis at the origin is$(x^2 - y^2)\frac{dy}{dx} = g(x)y$, then g(x) equals:
[Online April 9, 2014]

Options:

A. $\frac{1}{2}x$

B. $2x^2$

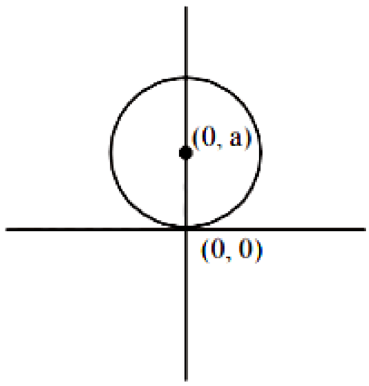
C. $2x$

D. $\frac{1}{2}x^2$

Answer: C

Solution:

Since family of all circles touching x-axis at the origin



∴ Eqn is $(x)^2 + (y - a)^2 = a^2$
where $(0, a)$ is the centre of circle.
$\Rightarrow x^2 + y^2 + a^2 - 2ay = a^2$
$\Rightarrow x^2 + y^2 - 2ay = 0$ ....(1)
Differentiate both side w.r.t 'x', we get
$2x + 2y\frac{dy}{dx} - 2a\frac{dy}{dx} = 0$
$\Rightarrow x + y\frac{dy}{dx} - a\frac{dy}{dx} = 0$
$\Rightarrow \frac{x + y\frac{dy}{dx}}{\frac{dy}{dx}} = a$
Put value of $'a'$ in eqn (1), we get

$$x^2+y^2-2y\left[\frac{y\frac{dy}{dx}+x}{\frac{dy}{dx}}\right]=0$$

$$\Rightarrow (x^2+y^2)\frac{dy}{dx}-2y^2\frac{dy}{dx}-2xy=0$$

$$\Rightarrow (x^2+y^2-2y^2)\frac{dy}{dx}=2xy$$

$$\Rightarrow (x^2-y^2)\frac{dy}{dx}=2xy \equiv g(x)y$$

Hence, $g(x)=2x$

Question238

Let the population of rabbits surviving at time t be governed by the differential equation $\frac{dp(t)}{dt}=\frac{1}{2}p(t)-200$.
If p(0) = 100, then p(t) equals:
[2014]

Options:

A. $600-500e^{t/2}$

B. $400-300e^{-t/2}$

C. $400-300e^{t/2}$

D. quad $300-200e^{-t/2}$

Answer: C

Solution:

Given differential equation is

$$\frac{dp(t)}{dt}=\frac{1}{2}p(t)-200$$

By separating the variable, we get

$$dp(t)=\left[\frac{1}{2}p(t)-200\right]dt$$

$$\Rightarrow \frac{dp(t)}{\frac{1}{2}p(t)-200}=dt$$

Integrate on both the sides,

$$\int \frac{d(p(t))}{\frac{1}{2}p(t)-200}=\int dt$$

Let $\frac{1}{2}p(t)-200=s \Rightarrow \frac{dp(t)}{2}=ds$

So, $\int \frac{d\,p(t)}{\left(\frac{1}{2}p(t)-200\right)} = \int d\,t$

$\Rightarrow \int \frac{2d\,s}{s} = \int d\,t \Rightarrow 2\log s = t + c$

$\Rightarrow 2\log\left(\frac{p(t)}{2} - 200\right) = t + c$

$\Rightarrow \frac{p(t)}{2} - 200 = e^{\frac{t}{2}}k$

Using given condition $p(t) = 400 - 300e^{t/2}$

Question239

If the general solution of the differential equation $y' = \frac{y}{x} + \Phi\left(\frac{x}{y}\right)$, for some function Φ, is given by $y \ln|cx| = x$, where c is an arbitrary constant, then $\Phi(2)$ is equal to: [Online April 11, 2014]

Options:

A. 4

B. $\frac{1}{4}$

C. -4

D. $-\frac{1}{4}$

Answer: D

Solution:

Given $\frac{d\,y}{d\,x} = \frac{y}{x} + \phi\left(\frac{y}{x}\right)$ .....(1)

Let $\left(\frac{y}{x}\right) = v$ so that $y = xv$

or $\frac{d\,y}{d\,x} = x\frac{d\,v}{d\,x} + v$ ......(2)

from (1) & (2), $x\frac{d\,v}{d\,x} + v = v + \phi\left(\frac{1}{v}\right)$

or, $\frac{d\,v}{\phi\left(\frac{1}{v}\right)} = \frac{d\,x}{x}$

Integrating both sides, we get

$\int \frac{d\,x}{x} = \int \frac{d\,v}{\phi\left(\frac{1}{v}\right)} \Rightarrow \ln x + c = \int \frac{d\,v}{\phi\left(\frac{1}{v}\right)}$ (where c being constant of integration)

But, given $y=\frac{x}{\ln|cx|}$ is the general solution

so that $\frac{x}{y}=\frac{1}{v}=\ln\left|cx\right|=\int\frac{dv}{\phi\left(\frac{1}{v}\right)}$

Differentiating w.r.t v both sides, we get

$\phi\left(\frac{1}{v}\right)=\frac{-1}{v^2}\Rightarrow\phi\left(\frac{x}{y}\right)=-\frac{y^2}{x^2}$

when $\frac{x}{y}=2$ i.e. $\phi(2)=-\left(\frac{y}{x}\right)^2=-\left(\frac{1}{2}\right)^2=\left(\frac{-1}{4}\right)$

Question240

If $\frac{dy}{dx}+y\tan x=\sin 2x$ and $y(0)=1$, then $y(\pi)$ is equal to: [Online April 19, 2014]

Options:

A. 1

B. -1

C. -5

D. 5

Answer: C

Solution:

$\frac{dy}{dx}+y\tan x=\sin 2x$

$I.F=e^{\int\tan x\,dx}=e^{-\log\cos x}=\sec x$

Required solution is

$y(\sec x)=\int\sin 2x\sec x\,dx+c$

$y(\sec x)=\int\frac{2\sin x\cos x}{\cos x}dx+c$

$y(\sec x)=2\int\sin x\,dx+c$

$y(\sec x)=-2\cos x+c$ ....(1)

Given $y(0)=1$

∴ put $x=0$ and $y=1$, we get

$1(\sec 0)=-2\cos 0+c$

$\Rightarrow c=1+2\Rightarrow c=3$

∴ from eqn (1), we have

$y\sec x=-2\cos x+3$

To find $y(\pi)$, put $x=\pi$ in eqn (2), we get

$y(\sec\pi)=-2\cos\pi+3$

$y=-2(-1)(-1)+3(-1)=-2-3=-5$

Question241

The general solution of the differential equation, $\sin 2x\left(\frac{dy}{dx}-\sqrt{\tan x}\right)-y=0$, is

[Online April 12, 2014]

Options:

A. $y\sqrt{\tan x}=x+c$

B. $y\sqrt{\cot x}=\tan x+c$

C. $y\sqrt{\tan x}=\cot x+c$

D. $y\sqrt{\cot x}=x+c$

Answer: D

Solution:

Given, $\sin 2x\left(\frac{dy}{dx}-\sqrt{\tan x}\right)-y=0$

or, $\frac{dy}{dx}=\frac{y}{\sin 2x}+\sqrt{\tan x}$

or, $\frac{dy}{dx}-y\operatorname{cosec} 2x=\sqrt{\tan x}$ ......(1)

Now, integrating factor (I.F) $=e^{\int -\operatorname{cosec} 2x}$

or, $I.F=e^{-\frac{1}{2}\log|\tan x|}=e^{\log(\sqrt{\tan x})^{-1}}$

$=\frac{1}{\sqrt{\tan x}}=\sqrt{\cot x}$

Now, general solution of eq. (1) is written as

$y(I.F)=\int Q(I.F)dx+c$

$\therefore y\sqrt{\cot x}=\int\sqrt{\tan x}.\sqrt{\cot x}dx+c$

$\therefore y\sqrt{\cot x}=\int 1.dx+c$

$\therefore y\sqrt{\cot x}=x+c$

Question242

Statement-1: The slope of the tangent at any point P on a parabola, whose axis is the axis of x and vertex is at the origin, is inversely proportional to the ordinate of the point P.

Statement-2: The system of parabolas $y^2=4ax$ satisfies a differential equation of degree 1 and order 1.

[Online April 9, 2013]

Options:

A. Statement-1 is true; Statement-2 is true; Statement-2 is a correct explanation for statement-1.

B. Statement-1 is true; Statement-2 is true; Statement-2 is not a correct explanation for statement-1.

C. Statement-1 is true; Statement-2 is false.

D. Statement-1 is false; Statement-2 is true.

Answer: B

Solution:

Statement -1: $y^2 = \pm 4ax$

$\Rightarrow \frac{dy}{dx} = \pm 2a \cdot \frac{1}{y} \Rightarrow \frac{dy}{dx} \propto \frac{1}{y}$

Statement - 2: $y^2 = 4ax \Rightarrow 2y\frac{dy}{dx} = 4a$

Thus both statements are true but statement- 2 is not a correct explanation for statement-1.

Question243

At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is [2013]

Options:

A. 2500

B. 3000

C. 3500

D. 4500

Answer: C

Solution:

Given, Rate of change is $\frac{dP}{dx} = 100 - 12\sqrt{x}$

$\Rightarrow dP = (100 - 12\sqrt{x})dx$

By integrating

$\int dP = \int (100 - 12\sqrt{x})dx$
$P = 100x - 8x^{3/2} + C$
Given when $x = 0$ then $P = 2000$
$\Rightarrow C = 2000$
Now when $x = 25$ then
$P = 100 \times 25 - 8 \times (25)^{\frac{3}{2}} + 2000 \ = 4500 - 1000$
$\Rightarrow P = 3500$

Question244

If a curve passes through the point $\left(2, \frac{7}{2}\right)$ and has slope $\left(1 - \frac{1}{x^2}\right)$ at any point (x, y) on it, then the ordinate of the point on the curve whose abscissa is -2 is : [Online April 23, 2013]

Options:

A. $-\frac{3}{2}$

B. $\frac{3}{2}$

C. $\frac{5}{2}$

D. $-\frac{5}{2}$

Answer: A

Solution:

Slope $= \frac{dy}{dx} = 1 - \frac{1}{x^2}$
$\Rightarrow \int dy = \int \left(1 - \frac{1}{x^2}\right) dx$
$\Rightarrow y = x + \frac{1}{x} + C$, which is the equation of the curve since curve passes through the point $\left(2, \frac{7}{2}\right)$
$\therefore \frac{7}{2} = 2 + \frac{1}{2} + C \Rightarrow C = 1$
$\therefore y = x + \frac{1}{x} + 1$
when $x = -2$, then $y = -2 + \frac{1}{-2} + 1 = \frac{-3}{2}$

Question245

Consider the differential equation :

$\frac{dy}{dx} = \frac{y^3}{2(xy^2 - x^2)}$

Statement-1: The substitution $z = y^2$ transforms the above equation into a first order homogenous differential equation.

Statement-2: The solution of this differential equation is $y^2 e^{-y^2/x} = C$ [Online April 22, 2013]

Options:

A. Both statements are false.

B. Statement- 1 is true and statement- 2 is false.

C. Statement- 1 is false and statement- 2 is true.

D. Both statements are true.

Answer: D

Solution:

Given differential equation is
$\frac{dy}{dx} = \frac{y^3}{2(xy^2 - x^2)}$
By substituting $z = y^2$, we get diff. eqn. as
$\frac{dz}{dx} = \frac{2z^2}{2(xz - x^2)} = \frac{z^2}{xz - x^2}$
Now, $\frac{dx}{dz} = \frac{x}{z} - \frac{x^2}{z^2} = \frac{x}{z}\left[1 - \frac{x}{z}\right] \approx F\left(\frac{x}{z}\right)$
Hence, statement- 1 is true.
Now, $y^2 e^{-y^2/x} = C$ satisfies the given diff. equation
∴ It is the solution of given diff. equation.
Thus, statement- 2 is also true.

Question246

The equation of the curve passing through the origin and satisfying the differential equation$(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2$ is [Online April 25, 2013]

Options:

A. $(1+x^2)y = x^3$

B. $3(1+x^2)y = 2x^3$

C. $(1+x^2)y = 3x^3$

D. $3(1+x^2)y = 4x^3$

Answer: D

Solution:

Given differential equation is

$(1+x^2)\frac{dy}{dx} + 2xy = 4x^2$

$\Rightarrow \frac{dy}{dx} + \left(\frac{2x}{1+x^2}\right)y = \frac{4x^2}{1+x^2}$

This is linear diff. equation

$I.F = e^{\int \frac{2x}{1+x^2}dx} = e^{\log(1+x^2)} = 1+x^2$

∴ Solution is

$y(1+x^2) = \int \frac{4x^2}{1+x^2} \times 1 + x^2 + C$

$\Rightarrow y(1+x^2) = \frac{4x^3}{3} + C$

⇒ Required curve is

$3y(1+x^2) = 4x^3 (\because C = 0)$

Question247

Statement 1: The degrees of the differential equations $\frac{dy}{dx} + y^2 = x$ and $\frac{d^2y}{dx^2} + y = \sin x$ are equal.

Statement 2: The degree of a differential equation, when it is a polynomial equation in derivatives, is the highest positive integral power of the highest order derivative involved in the differential equation, otherwise degree is not defined.

[Online May 12, 2012]

Options:

A. Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

B. Statement 1 is false, Statement 2 is true.

C. Statement 1 is true, Statement 2 is false.

D. Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

Answer: D

Solution:

Statement - 1

Given differential equations are $\frac{dy}{dx} + y^2 = x$ and $\frac{d^2y}{dx^2} + y = \sin x$

Their degrees are 1 .
Both have equal degree.
Also, Statement -2 is the correct explanation for Statement -1

Question248

The population p(t) at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is:
[2012]

Options:

A. $2 \ln 18$

B. $\ln 9$

C. $\frac{1}{2} \ln 18$

D. $\ln 18$

Answer: A

Solution:

Given differential equation is

$\frac{dp(t)}{dt} = 0.5p(t) - 450$

$\Rightarrow \frac{dp(t)}{dt} = \frac{1}{2}p(t) - 450$

$\Rightarrow \frac{dp(t)}{dt} = \frac{p(t) - 900}{2}$

$\Rightarrow 2\frac{dp(t)}{dt} = -[900 - p(t)]$

$\Rightarrow 2\frac{dp(t)}{900 - p(t)} = -dt$

Integrate both the side, we get

$-2\int \frac{dp(t)}{900 - p(t)} = \int dt$

Let $900 - p(t) = u$

$\Rightarrow -d\,p(t) = d\,u$
$2\int \frac{d\,u}{u} = \int d\,t \Rightarrow 2\ln u = t + c$ ......(i)
$\Rightarrow 2\ln[900 - p(t)] = t + c$
Given $t = 0,\ p(0) = 850$
$2\ln(50) = c$
Putting in (i)
$\Rightarrow 2\left[\ln\left(\frac{900 - p(t)}{50}\right)\right] = t$
$\Rightarrow 900 - p(t) = 50e^{\frac{t}{2}}$
$\Rightarrow p(t) = 900 - 50e^{\frac{t}{2}}$
$\text{let}\ p(t_1) = 0$
$0 = 900 - 50e^{\frac{t_1}{2}}\ \therefore t_1 = 2\ln 18$

Question249

Let y(x) be a solution of $\frac{(2+\sin x)}{(1+y)}\frac{d\,y}{d\,x} = \cos x$. If y(0) = 2,then $y\left(\frac{\pi}{2}\right)$ equals [Online May 7, 2012]

Options:

A. $\frac{5}{2}$

B. 2

C. $\frac{7}{2}$

D. 3

Answer: C

Solution:

Given differential equation is
$\frac{(2+\sin x)}{(1+y)} \cdot \frac{d\,y}{d\,x} = \cos x$
which can be rewritten as
$\frac{d\,y}{1+y} = \frac{\cos x}{2+\sin x}d\,x$
Integrate both the sides, we get
$\int \frac{d\,y}{1+y} = \int \frac{\cos x\,d\,x}{2+\sin x}$
$\Rightarrow \log(1+y) = \log(2+\sin x) + \log C$
$\Rightarrow 1 + y = C(2+\sin x)$
Given $y(0) = 2$

$\Rightarrow 1+2=C[2+\sin 0] \Rightarrow C=\frac{3}{2}$

Now, $y\left(\frac{\pi}{2}\right)$ can be found as

$1+y=\frac{3}{2}\left(2+\sin\frac{\pi}{2}\right) \Rightarrow 1+y=\frac{9}{2}$

$\Rightarrow y=\frac{7}{2}$

Hence, $y\left(\frac{\pi}{2}\right)=\frac{7}{2}$

Question250

The integrating factor of the differential equation$(x^2-1)\frac{dy}{dx}+2xy=x$ is [Online May 26, 2012]

Options:

A. $\frac{1}{x^2-1}$

B. x^2-1

C. $\frac{x^2-1}{x}$

D. $\frac{x}{x^2-1}$

Answer: B

Solution:

Given differential equation is $(x^2-1)\frac{dy}{dx}+2xy=x$

$\Rightarrow \frac{dy}{dx}+\frac{2x}{x^2-1}\cdot y=\frac{x}{x^2-1}$

This is in linear form.

Integrating factor $=e^{\int \frac{2x}{x^2-1}dx}=e^{\int \frac{dt}{t}}$ where $t=x^2-1$

$=e^{\log t}=x^2-1$

Hence, required integrating factor $=x^2-1$

Question251

The general solution of the differential equation $\frac{dy}{dx}+\frac{2}{x}y=x^2$ is [Online May 19, 2012]

Options:

A. $y = cx^{-3} - \frac{x^2}{4}$

B. $y = cx^{3} - \frac{x^2}{4}$

C. $y = cx^{2} + \frac{x^3}{5}$

D. $y = cx^{-2} + \frac{x^3}{5}$

Answer: D

Solution:

Given differential equation is

$\frac{dy}{dx} + \frac{2}{x} . y = x^2$

This is of the linear form.

$\therefore P = \frac{2}{x}, Q = x^2$

$I.F = e^{\int \frac{2}{x} dx} = e^{\log x^2} = x^2$

Solution is

$y . x^2 = \int x^2 . x^2 dx + c = \frac{x^5}{5} + c$

$y = \frac{x^3}{5} + cx^{-2}$

Question252

The curve that passes through the point (2, 3), and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact is given by: [2011RS]

Options:

A. quad $2y - 3x = 0$

B. $y = \frac{6}{x}$

C. $x^2 + y^2 = 13$

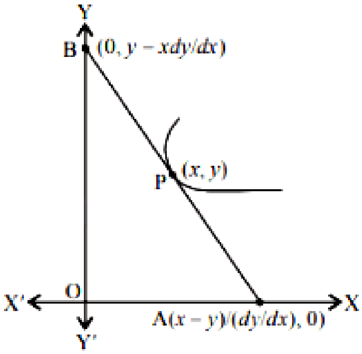
D. $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 2$

Answer: B

Solution:

Equation of tangent at P

$Y - y = \frac{dy}{dx}(X - x)$



X -intercept $= x - \frac{y}{dy/dx}$

Y -intercept $= y - \frac{xdy}{dx}$

Since P is mid-point of A and B

$x - \frac{y}{dy} = 2x$ and $y - \frac{xdy}{dx} = 2y$

$\Rightarrow \frac{-y}{\frac{dy}{dx}} = x$ and $\frac{-xdy}{dx} = y$

$\Rightarrow \frac{dx}{x} = -\frac{dy}{y}$

$\ln y = -\ln x + \ln c$

$y = \frac{c}{x}$

Since the above line passes through the point (2,3) .

$\therefore c = 6$

Hence $y = \frac{6}{x}$ is the required equation.

Question253

Let I be the purchase value of an equipment and V (t) be the value after it has been used for t years. The value V (t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T - t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value V (T) of the equipment is
[2011]

Options:

A. $I - \frac{kT^2}{2}$

B. $I-\frac{k(T-t)^2}{2}$

C. e^{-kT}

D. $T^2-\frac{1}{k}$

Answer: A

Solution:

$$\frac{dV(t)}{dt}=-k(T-t)$$
$$\Rightarrow\int dV(t)=-k\int(T-t)dt$$
$$V(t)=\frac{k(T-t)^2}{2}+c$$
at $t=0, V(t)=I$
$$I=\frac{kT^2}{2}+c$$
$$\Rightarrow c=I-\frac{kT^2}{2}$$
$$\Rightarrow V(t)=I+\frac{k}{2}(t^2-2tT)$$
$$V(T)=I+\frac{k}{2}(T^2-2T^2)=I-\frac{k}{2}T^2$$

Question254

If $\frac{dy}{dx}=y+3>0$ and $y(0)=2$, then $y(\ln 2)$ is equal to: [2011]

Options:

A. 5

B. 13

C. -2

D. 7

Answer: D

Solution:

$$\frac{dy}{dx}=y+3\Rightarrow\int\frac{dy}{y+3}=\int dx$$
$$\Rightarrow \ln|y+3|=x+c$$
Given $y(0)=2, \therefore \ln 5=c$

$\Rightarrow \ln|y+3| = x + \ln 5$
Put $x = \ln 2$, then $\ln|y+3| = \ln 2 + \ln 5$
$\Rightarrow \ln|y+3| = \ln 10$
$\therefore y+3 = \pm 10 \Rightarrow y = 7, -13$

Question255

Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by:
[2011RS]

Options:

A. $4 - \frac{2}{y} - \frac{e^{\frac{1}{y}}}{e}$

B. $3 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$

C. $1 + \frac{1}{y} - \frac{e^{\frac{1}{y}}}{e}$

D. $1 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$

Answer: C

Solution:

$\frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$
It is linear differential eqn.
$I.F. = e^{\int \frac{1}{y^2} dy} = e^{-\frac{1}{y}}$
So $x.e^{-\frac{1}{y}} = \int \frac{1}{y^3} e^{-\frac{1}{y}} dy$
Let $\frac{-1}{y} = t$
$\Rightarrow \frac{1}{y^2} dy = dt$
$\Rightarrow I = -\int te^t dt = e^t - te^t = e^{-\frac{1}{y}} + \frac{1}{y} e^{-\frac{1}{y}} + c$
$\Rightarrow xe^{-\frac{1}{y}} = e^{-\frac{1}{y}} + \frac{1}{y} e^{-\frac{1}{y}} + c$

$\Rightarrow x = 1 + \frac{1}{y} + c \cdot e^{1/y}$

Given $y(1) = 1$

$\therefore c = -\frac{1}{e}$

$\Rightarrow x = 1 + \frac{1}{y} - \frac{1}{e} \cdot e^{1/y}$

Question256

Solution of the differential equation $\cos x\, dy = y(\sin x - y)dx$, $0 < x < \frac{\pi}{2}$ is [2010]

Options:

A. $y \sec x = \tan x + c$

B. $y \tan x = \sec x + c$

C. $\tan x = (\sec x + c)y$

D. $\sec x = (\tan x + c)y$

Answer: D

Solution:

$\cos x\, dy = y(\sin x - y)dx$

$\frac{dy}{dx} = y \tan x - y^2 \sec x$

$\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y}\tan x = -\sec x$ ....(i)

Let $\frac{1}{y} = t \Rightarrow -\frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

Putting in (i)

$-\frac{dt}{dx} - t \tan x = -\sec x$

$\Rightarrow \frac{dt}{dx} + (\tan x)t = \sec x$

I.F. $= e^{\int \tan x\, dx} = e^{\log|\sec x|} = \sec x$

Solution : $t \sec x = \int \sec x \sec x\, dx$

$\Rightarrow \frac{1}{y}\sec x = \tan x + c$

Question257

The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1, and c_2 are arbitrary constants, is
[2009]

Options:

A. $y'' = y'y$

B. $yy'' = y'$

C. $yy'' = (y')^2$

D. $y' = y^2$

Answer: C

Solution:

We have $y = c_1 e^{c_2 x}$
Differentiate it w.r.t. to x
$\Rightarrow y' = c_1 c_2 e^{c_2 x} = c_2 y$
$\Rightarrow \frac{y'}{y} = c_2$ Differentiate it w.r.t. to x
$\Rightarrow \frac{y''y - (y')^2}{y^2} = 0 \Rightarrow y''y = (y')^2$

Question258

The differential equation of the family of circles with fixed radius 5 units and centre on the line y = 2 is
[2009]

Options:

A. $(x-2)y'^2 = 25-(y-2)^2$

B. $(y-2)y'^2 = 25-(y-2)^2$

C. $(y-2)^2 y'^2 = 25-(y-2)^2$

D. $(x-2)^2 y'^2 = 25-(y-2)^2$

Answer: C

Solution:

Let the centre of the circle be $(h, 2)$
∴ Equation of circle is
$(x-h)^2+(y-2)^2=25$ ......(1)
Differentiating with respect to x, we get
$2(x-h)+2(y-2)\frac{dy}{dx}=0$
$\Rightarrow x-h=-(y-2)\frac{dy}{dx}$
Substituting in equation (1) we get
$(y-2)^2\left(\frac{dy}{dx}\right)^2+(y-2)^2=25$
$\Rightarrow (y-2)^2(y')^2=25-(y-2)^2$

Question259

The solution of the differential equation $\frac{dy}{dy}=\frac{x+y}{x}$ satisfying the condition $y(1)=1$ is
[2008]

Options:

A. $y=\ln x+x$

B. $y=x\ln x+x^2$

C. $y=xe^{(x-1)}$

D. $y=x\ln x+x$

Answer: D

Solution:

$\frac{dy}{dx}=\frac{x+y}{x}=1+\frac{y}{x}$
It is homogeneous differential eqn.
Putting $y=vx$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$
we get $v+x\frac{dv}{dx}=1+v \Rightarrow \int\frac{dx}{x}=\int dv$
$\Rightarrow v=\ln x+c \Rightarrow y=x\ln x+cx$
As $y(1)=1$
$\therefore c=1$ So solution is $y=x\ln x+x$

Question260

The differential equation of all circles passing through the origin and having their centres on the x -axis is
[2007]

Options:

A. $y^2 = x^2 + 2xy\frac{dy}{dx}$

B. $y^2 = x^2 - 2xy\frac{dy}{dx}$

C. $x^2 = y^2 + xy\frac{dy}{dx}$

D. $x^2 = y^2 + 3xy\frac{dy}{dx}$

Answer: A

Solution:

General equation of circles passing through origin and having their centres on the x -axis is
$x^2 + y^2 + 2gx = 0$ ......(i)
On differentiating w.r.t x, we get
$2x + 2y.\frac{dy}{dx} + 2g = 0 \Rightarrow g = -\left(x + y\frac{dy}{dx}\right)$
Putting in (i)
$x^2 + y^2 + 2\left\{-\left(x + y\frac{dy}{dx}\right)\right\}.x = 0$
$\Rightarrow x^2 + y^2 - 2x^2 - 2x\frac{dy}{dx}.y = 0$
$\Rightarrow y^2 = x^2 + 2xy\frac{dy}{dx}$

Question261

The normal to a curve at P(x, y) meets the x-axis at G. If the distance of G from the origin is twice the abscissa of P, then the curve is a
[2007]

Options:

A. circle

B. hyperbola

C. ellipse

D. parabola.

Answer: B

Solution:

Equation of normal at $P(x, y)$ is

$Y - y = -\frac{dx}{dy}(X - x)$

Coordinate of G at X axis is $(X, 0)$ (let)

$\therefore 0 - y = -\frac{dx}{dy}(X - x)$

$\Rightarrow y\frac{dy}{dx} = X - x$

$\Rightarrow X = x + y\frac{dy}{dx}$

$\therefore$ Co-ordinate of $G\left(x + y\frac{dy}{dx}, 0\right)$

Given distance of G from origin $=$ twice of the abscissa of P

$\because$ distance cannot be $-ve$, therefore abscissa x should be +ve

$\therefore x + y\frac{dy}{dx} = 2x \Rightarrow y\frac{dy}{dx} = x$

$\Rightarrow y\,dy = x\,dx$

On Integrating, we have $\frac{y^2}{2} = \frac{x^2}{2} + c_1$

$\Rightarrow x^2 - y^2 = -2c_1$

$\therefore$ the curve is a hyperbola

Question262

The differential equation whose solution is $Ax^2 + By^2 = 1$ where A and B are arbitrary constants is of
[2006]

Options:

A. second order and second degree

B. first order and second degree

C. first order and first degree

D. second order and first degree

Answer: D

Solution:

$Ax^2 + By^2 = 1$ ......(i)

Differentiate w.r.t. to x

$Ax + By\frac{dy}{dx} = 0$ ......(ii)

Again differentiate w.r. to x

$A+By\frac{d^2y}{dx^2}+B\left(\frac{dy}{dx}\right)^2=0$ ......(iii)

From (ii) and (iii)

$x\left\{-By\frac{d^2y}{dx^2}-B\left(\frac{dy}{dx}\right)^2\right\}+By\frac{dy}{dx}=0$

Dividing both sides by $-B$, we get

$xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

Therefore order 2 and degree 1 .

Question263

The differential equation representing the family of curves$y^2=2c(x+\sqrt{c})$, where $c>0$, is a parameter, is of order and degree as follows: [2005]

Options:

A. order 1, degree 2

B. order 1, degree 1

C. order 1, degree 3

D. order 2, degree 2

Answer: C

Solution:

$y^2=2c(x+\sqrt{c})$ .... (i)Differentite it w.r. to x

$2yy'=2c.1$ or $yy'=c$ .... (ii)

[On putting value of c from (ii) in (i)]

$\Rightarrow y^2=2yy'(x+\sqrt{yy'})$

On simplifying, we get

$(y-2xy')^2=4yy^3$ .... (iii)

Hence equation (iii) is of order 1 and degree 3 .

Question264

If $x\frac{dy}{dx}=y(\log y-\log x+1)$, then the solution of the equation is [2005]

Options:

A. $y \log\left(\frac{x}{y}\right) = cx$

B. $x \log\left(\frac{y}{x}\right) = cy$

C. $\log\left(\frac{y}{x}\right) = cx$

D. $\log\left(\frac{x}{y}\right) = cy$

Answer: C

Solution:

$\frac{x\,dy}{dx} = y(\log y - \log x + 1)$

$\frac{dy}{dx} = \frac{y}{x}\left(\log\left(\frac{y}{x}\right) + 1\right)$

Put $y = vx$

$\frac{dy}{dx} = v + \frac{x\,dv}{dx} \Rightarrow v + \frac{x\,dv}{dx} = v(\log v + 1)$

$\frac{x\,dv}{dx} = v \log v \Rightarrow \int \frac{dv}{v \log v} = \int \frac{dx}{x}$

Put $\log v = z$

$\frac{1}{v} dv = dz \Rightarrow \int \frac{dz}{z} = \int \frac{dx}{x}$

$\ln z = \ln x + \ln c$

$x = cx$ or $\log v = cx$ or $\log\left(\frac{y}{x}\right) = cx$.

Question265

The differential equation for the family of circle $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is [2004]

Options:

A. $(x^2 + y^2)y' = 2xy$

B. $2(x^2 + y^2)y' = xy$

C. $(x^2 - y^2)y' = 2xy$

D. $2(x^2 - y^2)y' = xy$

Answer: C

Solution:

$x^2+y^2-2ay=0......(1)$
Differentiate w.r.t. to x

$2x+2y\frac{dy}{dx}-2a\frac{dy}{dx}=0 \Rightarrow a=\frac{x+yy'}{y'}$

Putting in (1) we get, $x^2+y^2-2\left(\frac{x+yy'}{y'}\right)y=0$

$\Rightarrow (x^2+y^2)y'-2xy-2y^2y'=0$

$\Rightarrow (x^2-y^2)y'=2xy$

Question266

Solution of the differential equation $y\,dx+(x+x^2y)\,dy=0$ is [2004]

Options:

A. $\log y=Cx$

B. $-\frac{1}{xy}+\log y=C$

C. $\frac{1}{xy}+\log y=C$

D. $-\frac{1}{xy}=C$

Answer: B

Solution:

$y\,dx+(x+x^2y)\,dy=0$

$\Rightarrow \frac{dx}{dy}=-\frac{x}{y}-x^2 \Rightarrow \frac{dx}{dy}+\frac{x}{y}=-x^2$

It is Bernoulli form. Divide by x^2

$x^{-2}\frac{dx}{dy}+x^{-1}\left(\frac{1}{y}\right)=-1$

put $x^{-1}=t, -x^{-2}\frac{dx}{dy}=\frac{dt}{dy}$ we get,

$-\frac{dt}{dy}+t\left(\frac{1}{y}\right)=-1 \Rightarrow \frac{dt}{dy}-\left(\frac{1}{y}\right)t=1$

It is linear differential eqn. in t.

$I.F=e^{\int -\frac{1}{y}dy}=e^{-\log y}=y^{-1}$

$\therefore$ Solution is $t(y^{-1})=\int (y^{-1})dy+C$

$\Rightarrow \frac{1}{x}\cdot\frac{1}{y}=\log y+C \Rightarrow \log y-\frac{1}{xy}=C$

Question267

The degree and order of the differential equation of the family of all parabolas whose axis is x - axis, are respectively.
[2003]

Options:

A. 2, 3

B. 2, 1

C. 1, 2

D. 3, 2.

Answer: C

Solution:

$y^2 = 4a(x-h)$
Differentiating $2yy_1 = 4a \Rightarrow yy_1 = 2a$
Again differentiating, we get $\Rightarrow y_1{}^2 + yy_2 = 0$
Degree $= 1$, order $= 2$

Question268

The solution of the differential equation $(1+y^2)+\left(x-e^{\tan^{-1}y}\right)\frac{dy}{dx}=0$, is [2003]

Options:

A. $xe^{2\tan^{-1}y} = e^{\tan^{-1}y} + k$

B. $(x-2) = ke^{2\tan^{-1}y}$

C. $2xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + k$

D. $xe^{\tan^{-1}y} = \tan^{-1}y + k$

Answer: C

Solution:

$$(1+y^2)+\left(x-e^{\tan^{-1}y}\right)\frac{dy}{dx}=0$$

$$\Rightarrow \frac{dx}{dy}+\frac{x}{(1+y^2)}=\frac{e^{\tan^{-1}y}}{(1+y^2)}$$

It is form of linear differential equation.

$$I.F=e^{\int \frac{1}{(1+y^2)}dy}=e^{\tan^{-1}y}$$

$$x\left(e^{\tan^{-1}y}\right)=\int \frac{e^{\tan^{-1}y}}{1+y^2}e^{\tan^{-1}y}dy$$

$$x\left(e^{\tan^{-1}y}\right)=\frac{e^{2\tan^{-1}y}}{2}+C\ \left[\because \int e^{2x}dx=\frac{e^{2x}}{2}\right]$$

$$\therefore 2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+k$$

Question269

The order and degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^{2/3}=4\frac{d^3y}{dx^3}$ are [2002]

Options:

A. $\left(1,\frac{2}{3}\right)$

B. (3,1)

C. (3,3)

D. (1,2)

Answer: C

Solution:

$$\left(1+3\frac{dy}{dx}\right)^2=\left(\frac{4d^3y}{dx^3}\right)^3$$

$$\Rightarrow \left(1+3\frac{dy}{dx}\right)^2=16\left(\frac{d^3y}{dx^3}\right)^3$$

Order $= 3$, degree 3

Question270

The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$ [2002]

Options:

A. $\frac{e^{-2x}}{4}$

B. $\frac{e^{-2x}}{4} + cx + d$

C. $\frac{1}{4}e^{-2x} + cx^2 + d$

D. $\frac{1}{4}e^{-4x} + cx + d$

Answer: B

Solution:

$\frac{d^2y}{dx^2} = e^{-2x}$; on integration $\frac{dy}{dx} = \frac{e^{-2x}}{-2} + c$;

Again integrate we get $y = \frac{e^{-2x}}{4} + cx + d$
